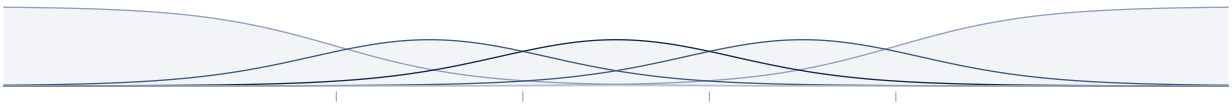
Where Does Criticism End?*

A Latent-Variable Map of Antisemitic Attitudes

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Abstract

Whether, and when, negative attitudes toward Israel reflect hostility toward Jews is sharply contested, and the competing definitions of antisemitism embed empirical claims alongside normative ones. This methods paper develops tools to test those empirical claims without adjudicating between the definitions. A bifactor graded response model separates a general factor, shared by classic stereotypes about Jews and by Israel-related statements, from factors specific to each domain. For each Israel-related item, the boundary estimand is the share of its common variance due to the general factor; items whose share reaches a pre-registered threshold form a boundary set, and each item is reported as in, out or undetermined. We prove that this decomposition is identified by heterogeneity, not by the correlation between domains: some domain must contain at least four items whose general-factor shares are not all equal save one. With three items per domain, or with proportional loadings, a curve of parameter values fits equally well. We derive the first-order bias in group comparisons when differential item functioning (DIF) is ignored, and how anchors that are not invariant fail. We give the variance cost and optimal design of list experiments, a joint model that identifies item-level social-desirability bias, and multilevel regression and poststratification on the latent trait, and we show when stereotype-count indices misrank groups. A simulation study on synthetic survey data with known ground truth checks these results with the limited-information estimator that the plan adopts as primary. It shows that convergence and agreement across starting values need not reveal an unidentified decomposition, and that cross-loadings of trope-based Israel items and acquiescence move the shares up for some items and down for others unless they are modelled; the primary model therefore includes an acquiescence factor. It also exposed a flaw in our own DIF screen, which missed non-uniform DIF. A pre-registered survey design and analysis plan, revised in light of the simulation, and an ethics protocol specify the empirical study. We report no empirical estimates.

Keywords item response theory; bifactor models; explained common variance; measurement invariance; differential item functioning; list experiments; social desirability; multilevel regression and poststratification; antisemitism.

*__Status.__ Methods paper; simulation study on synthetic survey data; no empirical estimates. We have fielded no survey and fitted no model to real data. Every number in Section 7 is computed from synthetic data with known ground truth; the survey of Section 8 is pre-registered and has not been fielded. Survey results published by others are quoted only as their authors report them. The companion papers (Kornstein Lab for Antisemitism Research, 2026a,b) study recorded incidents; this paper studies attitudes.

1 Introduction

Whether, and when, negative attitudes toward Israel reflect hostility toward Jews is a contested question with practical consequences. The answer shapes campus policy, law and community safety. In 2017 the European Parliament called on the Member States and the institutions of the European Union to adopt and apply the working definition of antisemitism of the International Holocaust Remembrance Alliance (IHRA) (European Parliament, 2017); in October 2024 the European Commission reported that 25 Member States had adopted or endorsed it (European Commission, 2024). In the United States, a 2019 executive order directs the agencies that enforce Title VI of the Civil Rights Act, which prohibits discrimination on the basis of race, color or national origin in federally funded programmes, to consider the IHRA definition and, where they are useful as evidence of discriminatory intent, its contemporary examples. The order also bars the agencies from diminishing or infringing any right protected by federal law or the First Amendment, and its statement of policy refers to harassment in schools and on university and college campuses (Executive Office of the President, 2019); a 2025 order reaffirmed it (Executive Office of the President, 2025). Several of the IHRA examples concern Israel (International Holocaust Remembrance Alliance, 2016). Two later texts, the Jerusalem Declaration on Antisemitism (Jerusalem Declaration on Antisemitism, 2021) and the Nexus Document (Nexus Task Force, 2021), offer other formulations. The three texts draw the line around Israel-related speech in different places (Section 2.1). The same line runs through incident statistics. The incident lists studied in the first paper of this series differ most in how they treat Israel- or Zionism-related expression (Kornstein Lab for Antisemitism Research, 2026a), so counts of incidents, and the security planning that rests on them, inherit a choice of definition.

Normative and empirical claims. A definition is a normative instrument. It says which speech and conduct ought to be called antisemitic, by whom and with what consequences, and no survey can settle that. But the competing definitions also embed empirical claims alongside the normative ones. Take a particular statement about Israel. One claim is that endorsing it typically expresses the same hostility toward Jews that drives endorsement of classic antisemitic stereotypes. The competing claim is that, in the population at large, endorsing it mainly expresses a distinct attitude toward a state and its policies. Each is a claim about how attitudes are structured in a population, statement by statement, and either can be right for some statements and wrong for others. Measurement can test the structural part of each claim: whether responses to the statement share their variance with responses to the stereotypes or are distinct from them. What a distinct attitude is, political disagreement or a different form of hostility, is interpretation, not estimate. This paper builds the tools for that test. It does not choose between the definitions, and it is designed so that its results would be equally usable, and equally open to checking, by people on every side of the debate.

A latent-variable map. We state the empirical question in a bifactor graded response model (Figure 1). Respondents rate two kinds of statements: classic stereotypes about Jews, and statements about Israel, its policies and its supporters. A general factor G loads on every item. Two specific factors absorb what is left: S, which the stereotype items share only with each other, and I, which the Israel-related items share only with each other. For each Israel-related item, the share of its common variance that is due to G, an item-level explained common variance ECV_j, places the item on a map. Near 0, the item's common variance is what it shares with the other Israel-related items alone; near 1, it is what it shares, through G, with the classic stereotypes. The *boundary set* $\mathcal{B}_\tau$ collects the items whose share reaches a threshold τ fixed before any data are seen, and sampling uncertainty sorts each item into *in*, *out* or *undetermined* (Figure 2). The map is drawn item by item, and the same rule applies to every item. Earlier studies suggest that the relation between the two domains is not uniform. Beattie (2017) reports that some criticisms of the

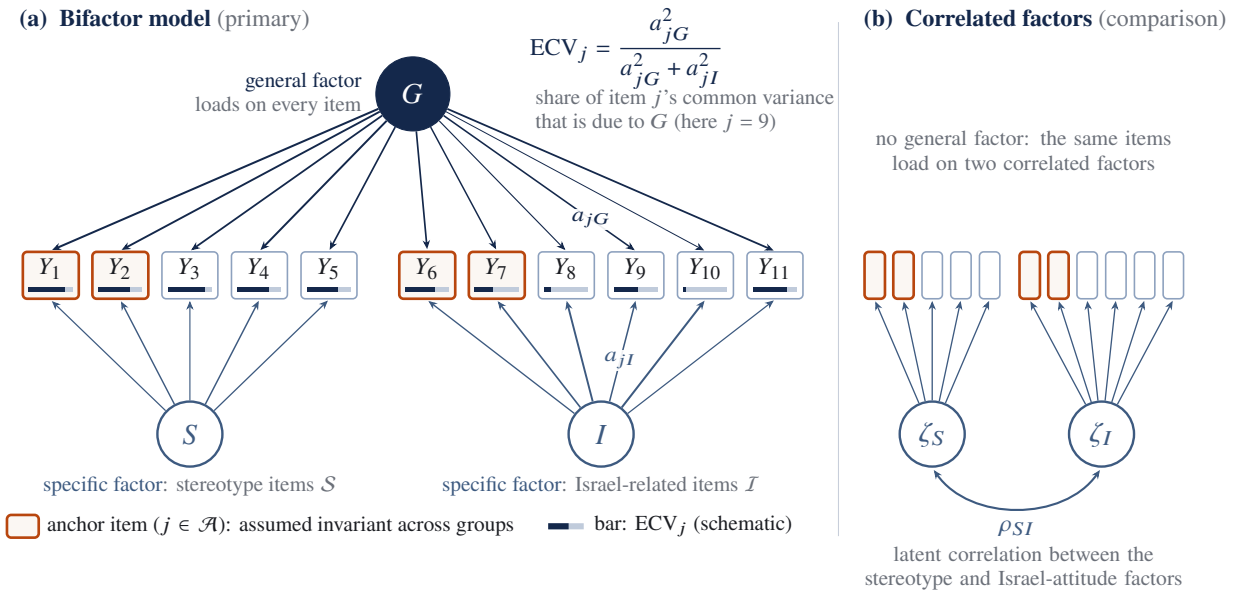


Figure 1. **The measurement model.** *(a)* The bifactor graded response model of Section 3. Every item loads on the general factor G, which carries the variance that all items share. Each item also loads on exactly one specific factor: S for the classic-stereotype items $\mathcal{S}$ and I for the Israel-related items $\mathcal{I}$. Responses are ordinal, with $\mathbb{P}(Y_{ij} \geq c \mid \boldsymbol{\theta}_i) = F(a_{jG}\theta_{iG} + a_{js}\theta_{is} - b_{jc})$ for item j's specific factor s, and the three factors are uncorrelated (Equation (3.3)). The boundary estimand of Definition 3.1 is the share of an item's common variance that is due to G, ECV_j (shown for item 9); the bar in each box shows it, and arrow widths grow with the slopes. Anchor items (orange outline, $\mathcal{A}$) are assumed to function identically in every group and carry group comparisons when other items show differential item functioning (Proposition 4.6). *(b)* The pre-registered comparison model: the same items, no general factor, and two correlated domain factors ζ_S and ζ_I, whose correlation ρ_{SI} is the second boundary estimand. Every such model is also a bifactor model with loadings proportional within each domain, so it identifies ρ_{SI} but not the item-level shares, which can lie anywhere in $[\rho^2_{SI}, 1]$ (Proposition 3.2). Schematic: the loadings are illustrative values, not estimates or simulation inputs.

Israeli government are more reliable indicators of antisemitic sentiment than others, and Staetsky (2017) that the two sets of attitudes occur both separately and together, with more intense anti-Israel attitudes more likely to be accompanied by antisemitic ones. A correlated-factors model, with no general factor, supplies a second, coarser estimand: the latent correlation ρ_{SI} between the stereotype and the Israel-attitude factors. On its own, that correlation cannot tell a common latent cause from distinct but correlated attitudes: every correlated-factors model is also a bifactor model in which the Israel-related items' share of general-factor variance can be anywhere between ρ^2_{SI} and one (Proposition 3.2). The general factor has no meaning of its own. It is defined by the items that load on it, which is why the item bank and its balance across the disputed categories are fixed in advance (Section 8.2).

What this paper does, and does not, do. This is a methods paper. We set out the measurement model and its boundary estimands, prove when they are identified and what goes wrong when the model's assumptions fail, develop the corrections for social desirability and the small-area estimators, evaluate them in a simulation study on synthetic survey data with known ground truth, and pre-register a survey design and analysis plan. We report *no* empirical estimates. We have fielded no survey and fitted no model to real data, and nothing here should be read as a finding that any attitude toward Israel is, or is not, antisemitic, or

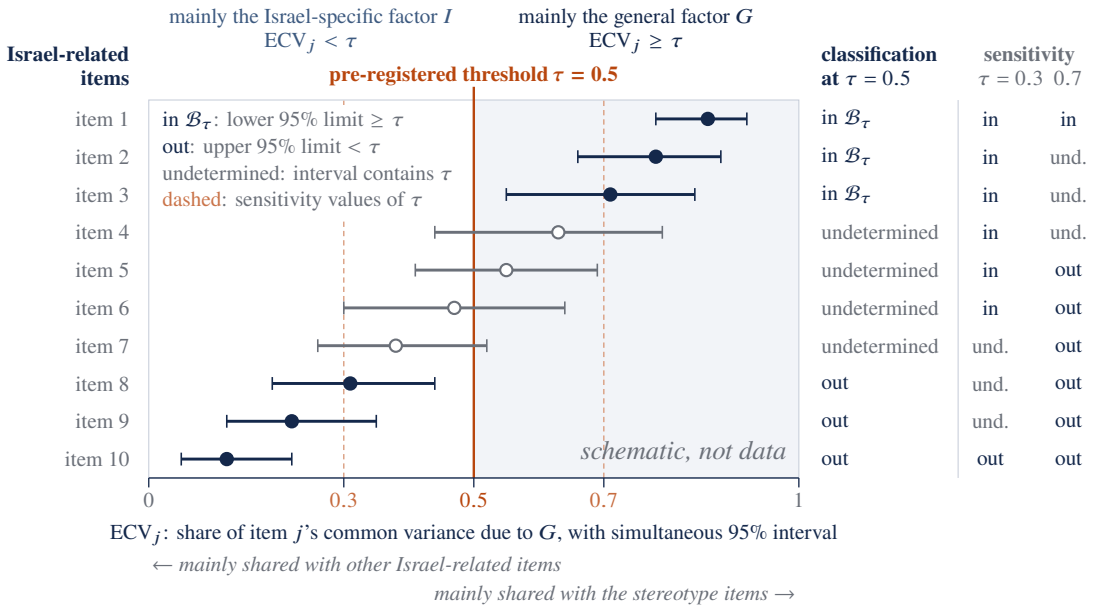


Figure 2. A boundary map (schematic, not data). What the primary pre-registered estimands E1 and E2 (Section 8.6) would look like when reported. Each row is one Israel-related item, placed by its item-level explained common variance ECV_j (Definition 3.1): the share of the item's common variance that it shares, through the general factor G, with the classic-stereotype items, rather than with the other Israel-related items alone, through the specific factor I (Figure 1). Bars are simultaneous 95% intervals (Proposition 4.8(b)). An item is *in* the boundary set $\mathcal{B}_\tau$ when its whole interval lies at or above the pre-registered threshold $\tau = 0.5$, *out* when it lies wholly below, and *undetermined* when it contains τ; the same rule is applied to every item, whatever its content. The right-hand columns repeat the classification at the sensitivity values $\tau = 0.3$ and 0.7. Positions and intervals are made up for illustration; the items are generic and carry no content, and nothing here is an estimate or a simulation result. The map describes response patterns in a surveyed population. It does not classify any utterance, speaker or context (Section 9.3).

about the attitudes of any group. Survey results published by others are quoted only as their authors report them. Notation is collected in Table 1.

Contributions. Fitting such a model is easy; reading it correctly is not. Four obstacles stand in the way, identification, invariance, social desirability and aggregation, and contributions C2–C4 take them in turn. Throughout we bring established psychometric and survey methods to a contested question and adapt them to it. Classical results are attributed where they are used. Most of the formal results specialise known theory to the two-domain bifactor design and to the boundary estimands, and we claim novelty for none of them beyond that specialisation and the explicit forms given.

C1 Measurement model and boundary estimands. A multidimensional graded response model (Birnbaum, 1968; Lord, 1980; Samejima, 1969) with a bifactor structure (Gibbons and Hedeker, 1992; Reise, 2012): a general factor and specific factors for classic stereotypes and for Israel-related attitudes (Section 3). Within it we define three boundary estimands (Definition 3.1): the item-level explained common variance ECV_j (Rodriguez et al., 2016) of each Israel-related item, overall and by group; the latent correlation ρ_{SI} between the stereotype and Israel-attitude factors in a correlated-factors model; and the boundary set $\mathcal{B}_\tau$ for a pre-registered threshold τ, reported with a three-way classification that shows indeterminacy instead of hiding it. We state what the estimands would and would not mean

Table 1. Notation. Symbols used throughout. Item parameters carry a superscript (g) when they are allowed to differ across groups.

Respondents, groups and items	
$i = 1, \dots, n$	respondent
$g \in \{0, \dots, m\}$; w_g	group (ideology, cohort, country, ...), $g = 0$ the reference; group share
$j \in \mathcal{J}$; $Y_{ij} \in \{0, \dots, C_j\}$	item; ordinal response of respondent i
$\mathcal{S}$, $\mathcal{I}$; $s(j)$	stereotype items; Israel-related items; item j's domain
$\mathcal{A} \subset \mathcal{S} \cup \mathcal{I}$	anchor items, assumed invariant across groups
Measurement model	
$\boldsymbol{\theta}_i = (\theta_{iG}, \theta_{iS}, \theta_{iI})$	general factor G; specific factors S and I
$\boldsymbol{\theta}_i \mid g \sim \mathrm{N}_3(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)$	latent law in group g; $\boldsymbol{\Sigma}_g$ diagonal, $\boldsymbol{\mu}_0 = \mathbf{0}$, $\boldsymbol{\Sigma}_0 = \boldsymbol{I}_3$
a_{jG}, a_{js}	slopes on G and on item j's specific factor $s = s(j)$
b_{jc}	threshold for category c of item j
F	link: logistic Λ or normal Φ (normal ogive)
$\boldsymbol{\lambda}_j$, h_j^2	standardised loadings; communality
$\boldsymbol{a}_j^{(g)}$, $b_{jc}^{(g)}$	group-specific parameters: DIF
lev_j	leverage of anchor j (Proposition 4.6)
ζ_S, ζ_I; α_j	correlated domain factors; their loadings
Boundary estimands	
ECV_j, $\mathrm{ECV}_j^{(g)}$	share of item j's common variance due to G; in group g
$\varkappa_j = \mathrm{logit}\,\mathrm{ECV}_j$	scale on which the boundary is tested
ρ_{SI}	latent domain correlation of ζ_S and ζ_I
τ; $\mathcal{B}_\tau$	pre-registered threshold; boundary set $\{j \in \mathcal{I} : \mathrm{ECV}_j \geq \tau\}$
Sensitive questions	
T, C, D	list-treatment, list-control and direct arms
J	number of control items in a list (Section 5 only)
π; π^D	prevalence of private agreement; of direct agreement
$\mathrm{SDB} = \pi - \pi^D$	social-desirability bias
π^H; δ_j	hidden-endorsement rate; threshold shift of item j
ϕ_c, ϕ_f	ceiling and floor suppression probabilities
Small areas and count indices	
k; N_k	poststratification cell; its population count
r; $w_k = N_k / N_r$	area; cell k's share of its area
ω_{fu}^2	variance of a batch of random effects
$\bar{\boldsymbol{\theta}}_r$	poststratified area mean (Equation (6.2))
U_{i+}; $P_g(q)$	number of the M stereotype statements agreed with; share of group g with $U_{i+} \geq q$

(Section 3.5), and show that the correlated-factors model identifies ρ_{SI} but only bounds the item-level shares (Proposition 3.2; Schmid and Leiman, 1957).

C2 Identification and invariance. A bifactor model can be fitted to data that do not identify it. The decomposition is identified by heterogeneity, not correlation: some domain must contain at least four items whose general-factor shares are not all equal save one. With three items per domain, or proportional loadings, a curve of parameter values fits equally well and moves the boundary ECVs; with two Israel-related items their identified sets are given in closed form (Theorem 4.2). Respondents who differ in ideology, age or country may also read a statement, about Israel or about Jews, differently. When such differential item functioning (DIF) is ignored, the first-order bias in group comparisons of latent means and of the group-specific $\mathrm{ECV}_j^{(g)}$ is an information-weighted projection of the DIF onto the loadings. Uniform DIF proportional to the general loadings is indistinguishable from a true difference in the general factor, and non-uniform DIF in some items moves the reported shares of all items (Theorem 4.5). Invariant anchor items restore the comparison under partial invariance (Byrne et al., 1989). DIF in the anchors themselves shifts every comparison they carry, undetectably when there are only three of them, and we bound the shift by the anchors' leverage (Proposition 4.6). Anchors drawn from the stereotype items alone also identify comparisons of the general factor and

of the boundary (Remark 4.7). They assume that those anchors function identically across groups, which is testable only in part (Proposition 4.6(b)), and with fewer anchors the comparison is more exposed to their DIF, so the plan uses them as a sensitivity analysis. Tests and confidence sets for $\mathcal{B}_\tau$ justify the three-way classification (Proposition 4.8).

C3 Social-desirability correction. Direct questions on these topics may be answered strategically, and not necessarily in the same direction in every group. List experiments (Glynn, 2013; Miller, 1984) reduce what an individual answer reveals. We give their exact variance and its cost relative to direct questioning, which no choice of control items can remove without making some answers fully revealing (Proposition 5.1). A joint model of direct and list responses identifies item-level social-desirability bias in split and paired designs, with and without the monotonicity assumption of Aronow et al. (2015), and decomposes it into hidden and false endorsement where the design allows (Theorem 5.3). We derive the budget-optimal allocation of respondents across direct and list arms and the choice of control list under floor and ceiling effects (Blair and Imai, 2012; Theorem B.5). Treated as a shift of an item's thresholds, social desirability biases prevalences but not the boundary estimands; it affects latent-mean comparisons only if the shift differs between groups, in which case it is a form of DIF, and a list experiment identifies the shift (Proposition 5.5).

C4 Small-area estimation and indices. Policy questions are often local, while samples are national. We adapt multilevel regression and poststratification (Gelman and Little, 1997; Park et al., 2004) to the latent traits, fitted jointly with the measurement model, with an explicit poststratification identity for area means (Equation (6.2)) that extends to area covariances and area-level boundary summaries (Section 6.1). Scoring respondents first and averaging the scores afterwards attenuates area differences by an explicit factor (Proposition 6.2). Indices that report the share of respondents agreeing with at least q of M stereotype statements rank groups correctly under invariance and stochastic dominance, but unequal dispersion, uniform DIF or non-uniform DIF can reverse the ranking (Propositions 6.3 and B.8). We make no claim about the results of any real index.

C5 A simulation study. In synthetic surveys with known parameters we assess recovery of the boundary estimands, the bias from ignoring DIF, the failure caused by a misspecified anchor, the accuracy of the list-experiment correction and of the optimal design, the accuracy of small-area estimates, how often a stereotype-count index ranks two groups in the wrong order when its items carry DIF, and how the plan detects and corrects two misspecifications that move general-factor shares, some up and some down: cross-loadings of trope-based Israel items, handled by a check and a cross-loading model for the whole trope-based block, and acquiescence, handled by an acquiescence factor in the primary model (Section 7). The fits use a limited-information estimator, which the plan adopts as primary, and the small-area experiment starts from a score of the general factor rather than from item responses. The results agree with the theory in the designs studied. They also show that convergence and agreement across starting values need not reveal an unidentified decomposition: agreement across starts exposed the two-item case but neither the proportional nor the near-proportional case. Anchoring in the stereotype items alone was two to three times as sensitive as the default anchors to the same DIF on those items. And they exposed a flaw in our own draft plan: a DIF screen that also required a minimum expected-score difference missed non-uniform DIF. The plan was revised accordingly, and the revised screen was simulated again.

C6 A pre-registered survey design and analysis plan. An item bank described by construct, with no claim of validation; the list-experiment module; the sampling frame and the poststrata; the estimands, the threshold τ and decision rules fixed before fieldwork; and an ethics protocol (Sections 8 and 9).

What the method cannot settle. Two limits are built into the approach. First, *the method does not answer normative questions.* An estimate of ECV_j says how responses to a statement covary with responses to classic stereotypes in a surveyed population. Whether speech with that content should be called antisemitic, restricted, disciplined or protected is a question of values, law and context. Evidence about covariance can inform that question, but it cannot answer it, and a high or low ECV_j is compatible with more than one normative position. Second, *the method does not diagnose individuals.* The estimands describe populations. An item in $\mathcal{B}_\tau$ does not show that a particular person who endorses it is hostile toward Jews, or that a particular utterance with similar content is antisemitic in its context; an item outside $\mathcal{B}_\tau$ is not thereby shown to be acceptable in every context. Definitions classify speech and conduct in context, often with reference to intent (Section 2.1); our estimands do neither, and Section 9.3 turns these limits into commitments about how results are reported.

Related work and roadmap. Methodological sources are attributed in Sections 3 to 6 where they are used; Section 2 reviews the three definitions, the main survey programmes and prior evidence relating the two attitude domains. This paper is the third in a series. The first estimates the number of antisemitic hate crimes that no list records (Kornstein Lab for Antisemitism Research, 2026a); the second models surges in recorded incidents after salient external events (Kornstein Lab for Antisemitism Research, 2026b). Both study incidents. This one studies attitudes, which no incident list records. Section 3 defines the measurement model and the boundary estimands, Section 4 develops identification and invariance, Section 5 treats social desirability and list experiments, and Section 6 small-area estimation and count indices. Section 7 reports the simulation study, Section 8 the pre-registered plan and Section 9 the ethics protocol. Section 10 discusses what a boundary map could and could not show. Supplementary results and proofs are in Appendices A and B, and simulation details in Appendix C.

2 Background: definitions, surveys and the boundary question

When does an attitude about Israel express hostility toward Jews, and when is it a political attitude distinct from such hostility? Part of this *boundary question* is normative: what ought to count as antisemitism. Part is empirical: how attitudes about Jews and attitudes about Israel are in fact structured in a population. This section reviews the three definitions that dominate current debate (Section 2.1), the main survey programmes that measure antisemitic attitudes (Section 2.2), and prior empirical work relating attitudes toward Israel to antisemitic attitudes (Section 2.3). It then explains why measurement invariance (Section 2.4) and indirect questioning (Section 2.5) are needed to answer the empirical part.

We follow two conventions. First, we do not adjudicate between definitions. We describe each in its own terms, and we identify the empirical premises that the definitions appear to rest on, because those premises are what our methods can test. Second, every survey figure quoted below is attributed to the organisation or authors that published it. None of them is an input to the simulation study of Section 7, and we have fielded no survey.

2.1 Three definitions and the claims embedded in them

The IHRA working definition (2016). On 26 May 2016 the Plenary of the International Holocaust Remembrance Alliance (IHRA), meeting in Bucharest, adopted a "non-legally binding working definition"

of antisemitism (International Holocaust Remembrance Alliance, 2016). Its core reads: "Antisemitism is a certain perception of Jews, which may be expressed as hatred toward Jews." A second sentence states that rhetorical and physical manifestations of antisemitism are directed toward Jewish or non-Jewish individuals or their property, and toward Jewish community institutions and religious facilities. The text then states that the examples which follow "may serve as illustrations" to guide IHRA in its work. It says that manifestations might include the targeting of the State of Israel, conceived as a Jewish collectivity. It adds: "However, criticism of Israel similar to that leveled against any other country cannot be regarded as antisemitic." Eleven contemporary examples follow. They are introduced as instances that could, "taking into account the overall context", count as antisemitism. Several concern Israel. They include denying the Jewish people their right to self-determination (for example, by claiming that the existence of a State of Israel is a racist endeavour); applying double standards by requiring of Israel a behaviour not expected or demanded of any other democratic nation; using symbols and images associated with classic antisemitism to characterise Israel or Israelis; drawing comparisons of contemporary Israeli policy to that of the Nazis; and holding Jews collectively responsible for actions of the State of Israel (International Holocaust Remembrance Alliance, 2016).

The Jerusalem Declaration on Antisemitism (2021). The Jerusalem Declaration on Antisemitism (JDA) was published in March 2021 by scholars working, in its own words, in antisemitism studies and related fields, including Jewish, Holocaust, Israel, Palestine and Middle East studies. It had 210 initial signatories (Jerusalem Declaration on Antisemitism, 2021). It defines antisemitism as "discrimination, prejudice, hostility or violence against Jews as Jews (or Jewish institutions as Jewish)". Fifteen guidelines follow in three groups: general guidelines (A); examples relating to Israel and Palestine that, "on the face of it", are antisemitic (B); and examples that, on the face of it, are not antisemitic, whether or not one approves of the view or action (C). Group B includes applying the symbols, images and negative stereotypes of classical antisemitism to the State of Israel, holding Jews collectively responsible for Israel's conduct, and denying the right of Jews in Israel to exist and flourish, collectively and individually, as Jews. Group C includes supporting Palestinian rights, criticising or opposing Zionism as a form of nationalism, evidence-based criticism of Israel as a state, and boycott, divestment and sanctions as forms of political protest. It also states that it is not antisemitic, "in and of itself", to compare Israel with other historical cases, including settler-colonialism or apartheid. The JDA says that context, including the intention behind an utterance, a pattern of speech over time and the identity of the speaker, can matter (Jerusalem Declaration on Antisemitism, 2021).

The Nexus Document (2021). The Nexus Document was produced by the Nexus Task Force, initially hosted by the Knight Program on Media and Religion at the University of Southern California's Annenberg School. It was published in February 2021 and updated in June 2024 (Nexus Task Force, 2021). It states that "antisemitism consists of anti-Jewish beliefs, attitudes, actions or systemic conditions". It then sets out when Israel-related expression is and is not antisemitic. Expression is antisemitic, for example, when it promotes myths, stereotypes or attitudes about Zionism or Israel that derive from or reinforce antisemitic accusations and tropes. As a general rule, criticism of Zionism and Israel, opposition to Israel's policies, and nonviolent political action directed at Israel or its policies "should not, as such, be deemed antisemitic". Treating Israel differently from other countries because it is a Jewish state is listed as antisemitic, but paying disproportionate attention to Israel, or treating Israel differently from other countries, is not in itself treated as proof of antisemitism (Nexus Task Force, 2021).

Table 2. Three definitions of antisemitism, summarised from their own texts. "Israel-related content" lists only provisions quoted or paraphrased in Section 2.1; it is not a complete inventory.

Document	Issuer; date	Core definition	Israel-related content
IHRA working definition	IHRA Plenary, Bucharest; 26 May 2016; non-legally binding	"A certain perception of Jews, which may be expressed as hatred toward Jews"	Eleven illustrative examples, several concerning Israel (self-determination, double standards, classic imagery, Nazi comparisons, collective responsibility), applicable "taking into account the overall context"; criticism similar to that of any other country is not antisemitic
Jerusalem Declaration (JDA)	Scholars; March 2021; 210 initial signatories	"Discrimination, prejudice, hostility or violence against Jews as Jews (or Jewish institutions as Jewish)"	Guidelines 6–10: examples antisemitic on the face of it (e.g. classic stereotypes applied to Israel, collective responsibility); guidelines 11–15: examples not antisemitic on the face of it (e.g. opposing Zionism as nationalism, BDS, comparisons with settler-colonialism or apartheid)
Nexus Document	Nexus Task Force; February 2021, updated June 2024	"Anti-Jewish beliefs, attitudes, actions or systemic conditions"	Antisemitic when drawing on antisemitic myths, stereotypes or tropes, or when treating Israel differently because it is a Jewish state; criticism, opposition to policies and nonviolent political action "should not, as such, be deemed antisemitic"; disproportionate attention not in itself proof

Agreement and disagreement. Table 2 summarises the three texts. They share some ground. All three list the transfer of classic antisemitic imagery or tropes to Israel among their examples of antisemitism, and all three list, in similar terms, holding Jews collectively responsible for Israel's conduct. All three state, in different words and with different qualifications, that criticism of Israel is not in itself antisemitic; the IHRA text refers to criticism "similar to that leveled against any other country". They differ on several categories of Israel-related speech. These include statements about Zionism and about the legitimacy of a Jewish state; double standards and disproportionate attention; comparisons of Israel with Nazi Germany, with apartheid or with settler-colonialism; and boycott campaigns. The IHRA text lists some of these among its contextual examples. The JDA lists some of them among its examples that are not antisemitic on the face of it. The Nexus Document treats several as not antisemitic as such, but antisemitic when they draw on antisemitic tropes. They also differ in how much weight they give to context and intent (International Holocaust Remembrance Alliance, 2016; Jerusalem Declaration on Antisemitism, 2021; Nexus Task Force, 2021).

Normative and empirical components. Much of this disagreement is normative: it concerns which speech acts ought to be called antisemitic, by whom, and with what consequences. Survey data cannot settle it. But each position also rests partly on claims about how attitudes are distributed in a population. One such claim is that endorsement of a given Israel-related statement typically reflects the same hostility toward Jews that drives endorsement of classic stereotypes. The competing claim is that, in the population at large, endorsement of that statement mainly reflects an attitude distinct from that hostility, such as opposition to a state's policies. These are claims about covariance structure, and they can be stated as restrictions on a measurement model. Definition 3.1 gives the item-level estimand that separates them, and Sections 3 and 4 give the conditions under which it is identified. The estimand describes response patterns in a population; it does not classify any utterance, speaker or context (Section 9).

2.2 Survey measurement of antisemitic attitudes

Stereotype-count indices (ADL). The Anti-Defamation League (ADL) measures antisemitic attitudes with a fixed battery of eleven statements expressing negative stereotypes about Jews. In its Global 100 survey (2014), respondents who answered "probably true" to at least six of the eleven were classified as harbouring antisemitic attitudes (Anti-Defamation League, 2014). ADL's United States surveys use the same eleven tropes, which it reports having measured since 1964, and the same threshold of six or more (Anti-Defamation League, 2023). Its 2023 report on the United States also included statements about Israel. ADL describes belief in anti-Jewish tropes and anti-Israel sentiment as distinct but overlapping in substantial ways (Anti-Defamation League, 2023). Indices of this kind are transparent and comparable over time when the battery is held fixed. Their group comparisons, however, rest on an assumption that each statement means the same thing, and has the same threshold for agreement, in every group compared. Proposition 6.3 shows what happens to a count index when that assumption fails. We make no claim about the results of any real index.

The elastic view (JPR). The Institute for Jewish Policy Research (JPR), in partnership with the Community Security Trust, published a study of attitudes towards Jews and Israel in Great Britain in 2017 (Staetsky, 2017). It drew on a face-to-face sample and an online-panel sample. The report proposes an "elastic view" of antisemitism: as an attitude, antisemitism exists at different levels of intensity, so no single figure captures its level in a society. The report distinguishes *counting antisemites*, which requires a threshold, from *measuring antisemitism*, which tracks how widely antisemitic ideas are diffused. It measures anti-Israel attitudes

separately from antisemitic attitudes. It reports that the two sets of attitudes exist both separately and together, and that the greater the intensity of anti-Israel attitudes, the more likely they are to be accompanied by antisemitic attitudes (Staetsky, 2017). The elastic view is close in spirit to a latent-trait model, in which a person's position on a continuum, rather than membership of a category, is the primary quantity. Keeping the two attitude domains separate and then relating them is the correlated-factors approach of Section 3.

Other general-population surveys. The American Jewish Committee (AJC) runs parallel annual surveys of American Jews and of the general public. For its 2024 report, both were conducted by SSRS. The general-public survey used the SSRS Opinion Panel, a probability-based panel (American Jewish Committee, 2024a,b). In academic research, Allington et al. (2022) proposed the Generalised Antisemitism (GeAs) scale. It is designed to measure antisemitism as expressed in relation both to Jews and to Israel, and its authors describe it as consistent with the IHRA working definition. Hersh and Royden (2023) surveyed about 3,500 US adults with two tests: one of overt antisemitic attitudes and one of double standards applied to Jews. They report that, on their measures, overt antisemitic attitudes were more common on the right than on the left, especially among young adults on the far right. On the double-standards test, they report a double standard against Jews on the left but not on the right, where respondents nevertheless attributed more responsibility to Jews for Israel's conduct than respondents on the left did (Hersh and Royden, 2023). Whether items about Israel measure antisemitism is itself disputed (Section 2.1), and the empirical part of that dispute is what a test of the measurement model can inform.

Surveys of Jewish people. A different family of surveys measures the experiences and perceptions of Jewish people, among them the surveys of the EU Agency for Fundamental Rights (FRA), whose third survey, published in July 2024, covered 13 Member States (European Union Agency for Fundamental Rights, 2018, 2024), and the Pew Research Center's *Jewish Americans in 2020* (Pew Research Center, 2021). They estimate how antisemitism is perceived and experienced by its targets, a different estimand from the distribution of attitudes in the general population, and we do not use them as measures of the latter. They inform which Israel-related statements Jewish respondents regard as hostile, one input to item selection alongside the categories on which the definitions disagree and the review of an advisory group with differing views (Sections 8.2 and 9.5), and they are a reminder that respondents exposed to our items include Jewish people (Section 9).

2.3 Prior evidence relating the two attitude domains

Several studies have related measures of attitudes toward Israel, in their authors' terms often anti-Israel attitudes, to measures of antisemitic attitudes.

- Kaplan and Small (2006) analysed a survey of 500 respondents in each of ten European countries. After controlling for many potentially confounding factors, they found that anti-Israel sentiment consistently predicted the probability that a respondent held antisemitic views. They noted that their analysis makes no assumption about, and does not establish, the direction of causation.
- Cohen et al. (2009) proposed that antisemitism is often expressed as hostility toward Israel and tested this with experiments that manipulated mortality salience. In their first study, reminders of mortality increased antisemitism and reduced support for Israel, but only when participants believed that their true attitudes could be detected (a "bogus pipeline" condition), which the authors interpret as evidence that social desirability can mask such attitudes.

- Beattie (2017) studied US participants. He examined how knowledge of conflicts around the world, antisemitism and Islamophobia relate to criticism of the Israeli and Saudi governments. He reports that some criticisms of the Israeli government are more reliable indicators of antisemitic sentiment than others.
- The JPR study reports that the two sets of attitudes also occur separately, and a gradient: more intense anti-Israel attitudes are more likely to be accompanied by antisemitic attitudes (Staetsky, 2017). ADL describes trope belief and anti-Israel sentiment as distinct but overlapping (Anti-Defamation League, 2023).

Taken together, these studies report positive associations between the two attitude domains. Two of them report that the strength of the association depends on the content of the Israel-related measure (Beattie, 2017; Staetsky, 2017). The studies differ in population, items, design and model, so their estimates are not directly comparable, and we do not pool them. Three features limit what they can say about the boundary question. First, a positive correlation between two scales is compatible both with a common latent cause and with distinct but correlated attitudes; the item-level decomposition of Definition 3.1 is needed to tell these apart, and Theorem 4.2 states when it can. Second, comparisons across ideological, age or national groups presuppose that items function equivalently in those groups (Section 2.4). Third, direct questions on both topics may be subject to social-desirability bias, and Cohen et al. (2009) offer experimental evidence that it can matter (Section 2.5).

2.4 Why measurement invariance matters

A comparison of attitude scores between groups is a comparison of the latent attitude only if the items relate to it in the same way in every group. Formally, measurement invariance holds when the conditional distribution of the item responses given the latent variable does not depend on group membership (Meredith, 1993; Millsap, 2011). In the usual hierarchy, configural invariance requires the same pattern of loadings, metric invariance equal loadings, scalar invariance also equal thresholds, and strict invariance also equal residual variances (Davidov et al., 2014; Meredith, 1993; Steenkamp and Baumgartner, 1998). Comparing latent means requires at least scalar invariance for the items that carry the comparison. Comparing the boundary estimands, which are functions of loadings, requires invariant loadings on both the general and the specific factor. Item-level violations are called differential item functioning (DIF): uniform DIF shifts an item's thresholds, non-uniform DIF changes its discrimination. Standard tests are the Mantel–Haenszel procedure (Holland and Thayer, 1988) and likelihood-ratio tests within an item response model (Thissen et al., 1993). When some items are non-invariant, group comparisons can still be identified through invariant *anchor* items (partial invariance; Byrne et al., 1989) or by the alignment method (Asparouhov and Muthén, 2014); fit-index changes are an imperfect guide (Chen, 2007).

Israel-related items, and some stereotype items, are natural candidates for DIF. A statement about a contested political dispute, or about the power or loyalty of a group, can attract agreement for reasons tied to a respondent's political identity, religious commitments or generational experience, and those reasons need not run through the general factor. That is a hypothesis to be tested, not a finding. If such DIF exists and is ignored, it biases group comparisons of latent means and of the boundary estimands (Theorem 4.5). Group comparisons then require anchors, which must themselves be invariant (Proposition 4.6). The same issue affects stereotype-count indices: the share of a group agreeing with at least q of M statements need not reflect the underlying attitude when invariance fails (Proposition B.8).

2.5 Why list experiments

Respondents may answer sensitive questions in the way they believe is socially acceptable rather than truthfully. Misreporting on sensitive questions is well documented, and self-administration tends to reduce it (Kreuter et al., 2008; Tourangeau and Yan, 2007). For questions about Jews and Israel the direction of the bias need not be the same for everyone. Under the social reference theory of Blair et al. (2020), sensitivity bias depends on whom respondents imagine they are answering to and what they believe that reference group expects. Different groups may therefore under-report different responses, which is itself a form of group-dependent measurement error.

The list experiment, or item count technique (Miller, 1984), reduces what an individual answer reveals. Respondents are randomly assigned to a control group, which sees a list of J non-sensitive items, or a treatment group, which sees the same list plus the sensitive item. Each respondent reports only *how many* items apply to them, not which. Under the assumptions of no design effect (adding the sensitive item does not change answers to the control items) and no liars (respondents in the treatment group answer the sensitive item truthfully within the count), the difference in mean counts between the two groups estimates the prevalence of the sensitive response (Blair and Imai, 2012; Glynn, 2013). Multivariate regression estimators allow the prevalence to depend on covariates (Corstange, 2009; Imai, 2011). Blair and Imai (2012) propose a test for design effects and models for floor and ceiling effects, the strategic misreporting that arises when a truthful count at an extreme of the scale would reveal, or appear to reveal, the answer to the sensitive item. Under a monotonicity assumption that rules out false admissions, combining direct and list responses is more efficient than the list experiment alone (Aronow et al., 2015). Randomised response (Blair et al., 2015; Warner, 1965) reduces disclosure by a different mechanism.

The reduction has a cost. List-experiment estimates are much noisier than direct estimates, because the control items add variance, so choosing between them is a bias–variance trade-off (Blair et al., 2020; Glynn, 2013); Proposition 5.1 and Theorem B.5 quantify the cost and minimise it. List experiments have been used to study prejudice, including racial attitudes in the United States (Imai, 2011; Kuklinski et al., 1997a,b). Evidence on the size of the bias counsels caution: a meta-analysis of list experiments found that sensitivity biases are typically smaller than ten percentage points and are approximately zero in some domains (Blair et al., 2020). A correction for social desirability is therefore an empirical question for each item and population, and the design of Section 8 treats it as one.

3 A bifactor model of antisemitic attitudes

This section sets up the measurement model on which the rest of the paper builds. It is a multidimensional graded response model with a bifactor structure, estimated on two item domains: classic antisemitic stereotypes and Israel-related attitudes. We introduce a competing correlated-factors model, define the boundary estimands of Definition 3.1, and state what they would and would not mean. We close by showing how the two models are related (Proposition 3.2), because that relation determines what a correlation between the two domains can and cannot reveal about the boundary.

Definitions of antisemitism differ in whether, and which, Israel-related attitudes they treat as antisemitic (Section 2.1). Part of that disagreement is normative and cannot be settled by data. Part of it rests on claims about how attitudes are structured in a population. The model turns that second part into measurement claims that can be tested, and it leaves the normative part to readers. The claim that endorsement of an Israel-related statement X reflects the same latent hostility as classic stereotypes becomes the claim that X has a large general-factor share; the claim that X expresses an attitude distinct from that hostility becomes

the claim that most of its systematic variance is specific to the Israel-related domain. Both claims are restrictions on one model, tested with the same machinery and the same pre-registered threshold. A finding in either direction is informative, and neither presupposes which definition is correct. The model does not classify any statement, speaker or context, and no estimate in it is a verdict on whether a particular attitude is antisemitic. The model does not say what the domain-specific attitude is; calling it political disagreement or a distinct form of hostility is interpretation, not estimate.

3.1 Items, responses and the graded response model

Respondents are indexed by $i = 1, \dots, n$, and each belongs to one of a finite set of groups $g \in \{0, 1, \dots, m\}$, such as ideological, age or national groups, with $g = 0$ the reference group. Items are indexed by $j \in \mathcal{J} = \mathcal{S} \cup \mathcal{I}$. The set $\mathcal{S}$ contains statements of classic antisemitic stereotypes and $\mathcal{I}$ contains Israel-related statements; the two sets are disjoint, and $|\mathcal{J}|$ is the number of items. Response $Y_{ij} \in \{0, 1, \dots, C_j\}$ is ordinal, with $C_j \geq 1$ (for example a five-point agreement scale has $C_j = 4$). Existing survey programmes measure both domains, usually with separate batteries (Section 2.2); here they are modelled jointly.

Each respondent carries a latent vector $\boldsymbol{\theta}_i = (\theta_{iG}, \theta_{iS}, \theta_{iI})^\top$: a general factor G and two specific factors, S for the stereotype domain and I for the Israel-related domain. For item j let $s(j) = S$ if $j \in \mathcal{S}$ and $s(j) = I$ if $j \in \mathcal{I}$. Item j has a loading vector $\boldsymbol{a}_j = (a_{jG}, a_{jS}, a_{jI})^\top$ with the *bifactor pattern*

$$a_{jI} = 0 \ \ (j \in \mathcal{S}), \qquad a_{jS} = 0 \ \ (j \in \mathcal{I}), \tag{3.1}$$

so that every item loads on G and on its own specific factor only. We write $a_{js} = a_{j,s(j)}$ for the specific loading and $\boldsymbol{A}$ for the $|\mathcal{J}| \times 3$ matrix with rows $\boldsymbol{a}_j^\top$. The graded response model (Samejima, 1969) with group-specific item parameters is

$$\mathbb{P}\big(Y_{ij} \geq c \mid \boldsymbol{\theta}_i,\, g_i = g\big) = F\big(\boldsymbol{a}_j^{(g)\top}\boldsymbol{\theta}_i - b_{jc}^{(g)}\big), \qquad c = 1, \dots, C_j, \tag{3.2}$$

with ordered thresholds $b_{j1}^{(g)} < \cdots < b_{jC_j}^{(g)}$, where F is either the logistic distribution function $\Lambda(x) = 1/(1+e^{-x})$ or the standard normal distribution function Φ (the normal-ogive model; Lord, 1980). Category probabilities are differences of adjacent terms in (3.2), with $b_{j0}^{(g)} = -\infty$ and $b_{j,C_j+1}^{(g)} = +\infty$. Responses are conditionally independent given $\boldsymbol{\theta}_i$ and g_i (local independence), and the latent vector has a group-specific normal distribution with orthogonal factors,

$$\boldsymbol{\theta}_i \mid g_i = g \ \sim \ \mathrm{N}_3\big(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g\big), \qquad \boldsymbol{\Sigma}_g = \mathrm{diag}\big(\sigma_{gG}^2, \sigma_{gS}^2, \sigma_{gI}^2\big), \qquad \boldsymbol{\mu}_0 = \mathbf{0}, \ \ \boldsymbol{\Sigma}_0 = \boldsymbol{I}_3. \tag{3.3}$$

The reference normalisation $\boldsymbol{\mu}_0 = \mathbf{0}$, $\boldsymbol{\Sigma}_0 = \boldsymbol{I}_3$ fixes the location and scale of each factor. Orthogonality of G, S and I is what makes the decomposition below a decomposition; with correlated factors the share of an item's variance due to G is not defined without further conventions. Full-information estimation of this model reduces to two-dimensional integrals, because given θ_G the two domains are independent (Gibbons and Hedeker, 1992; Gibbons et al., 2007); the multiple-group version with group-specific latent means and variances is that of Cai et al. (2011).

Latent responses. Equivalently, item j in group g has a continuous latent response and an observed category,

$$Y_{ij}^* = \boldsymbol{a}_j^{(g)\top}\boldsymbol{\theta}_i + \varepsilon_{ij}, \qquad Y_{ij} = \sum_{c=1}^{C_j} \mathbf{1}\Big\{Y_{ij}^* > b_{jc}^{(g)}\Big\}, \tag{3.4}$$

where the ε_{ij} are independent of each other and of $\boldsymbol{\theta}_i$, with distribution function F and variance v_F ($v_\Phi = 1$, $v_\Lambda = \pi^2/3$). Since F is symmetric, $\mathbb{P}(Y_{ij} \geq c \mid \boldsymbol{\theta}_i) = \mathbb{P}(\varepsilon_{ij} > b_{jc} - \boldsymbol{a}_j^\top\boldsymbol{\theta}_i) = F(\boldsymbol{a}_j^\top\boldsymbol{\theta}_i - b_{jc})$, which is

(3.2). Under the normal-ogive link, (3.4) is a factor model for discretised Gaussian variables, and marginal maximum likelihood in the item response model coincides with maximum likelihood in that factor model (Takane and de Leeuw, 1987). In the reference group we write

$$\boldsymbol{\lambda}_j = \frac{\boldsymbol{a}_j}{(v_F + \|\boldsymbol{a}_j\|^2)^{1/2}}, \qquad h_j^2 = \|\boldsymbol{\lambda}_j\|^2 = \lambda_{jG}^2 + \lambda_{js}^2, \qquad \psi_j = 1 - h_j^2 > 0, \tag{3.5}$$

for the standardised loadings, the communality and the uniqueness of item j: λ_{jf} is the correlation between Y_j^* and θ_f.

Invariance and differential item functioning. Item j is *invariant* if $\boldsymbol{a}_j^{(g)} = \boldsymbol{a}_j$ and $b_{jc}^{(g)} = b_{jc}$ for every g and c, so that the conditional law of Y_{ij} given $\boldsymbol{\theta}_i$ does not depend on the group. This is measurement invariance in the sense of Meredith (1993) and Millsap (2011), applied item by item. Otherwise item j shows differential item functioning (DIF) in group g, with

$$\Delta\boldsymbol{a}_j^{(g)} = \boldsymbol{a}_j^{(g)} - \boldsymbol{a}_j, \qquad \Delta b_{jc}^{(g)} = b_{jc}^{(g)} - b_{jc}. \tag{3.6}$$

DIF is *uniform* if $\Delta\boldsymbol{a}_j^{(g)} = \mathbf{0}$ and *non-uniform* if $\Delta\boldsymbol{a}_j^{(g)} \neq \mathbf{0}$. For simplicity we let uniform DIF shift all thresholds of an item by a common amount $\Delta b_j^{(g)}$; category-specific shifts are handled by the same arguments with one intercept per threshold. Throughout, $\boldsymbol{a}_j^{(g)}$ keeps the pattern (3.1) (configural invariance). The differences (3.6) refer to a latent metric common to all groups. That metric is itself identified only through items assumed invariant (Proposition 4.6).

3.2 The bifactor model and the general factor

The bifactor model (3.1)–(3.3) goes back to Schmid and Leiman (1957) and, in item response form, to Gibbons and Hedeker (1992); Reise (2012) reviews its use in measurement. Its general factor is, by construction, the dimension that the stereotype items and the Israel-related items have in common. It is *not* defined by the stereotype items alone. Interpreting G as "hostility toward Jews as Jews" is a construct-validity claim. It rests on the content of the item pool, above all of the stereotype items, and it is external to the model. If the item pool changes, G changes with it. A variant that builds the asymmetry into the model, the bifactor-$(S-1)$ model of Eid et al. (2017), is discussed in Remark 4.4.

3.3 The correlated-factors model

The competing model has no general factor. Each item loads on one of two correlated domain factors,

$$Y_{ij}^* = \alpha_j\, \zeta_{i,s(j)} + \varepsilon_{ij}, \qquad \begin{pmatrix}\zeta_{iS}\\ \zeta_{iI}\end{pmatrix} \,\Big|\, g_i = 0 \;\sim\; \mathrm{N}_2\!\left(\mathbf{0}, \begin{pmatrix}1 & \rho_{SI}\\ \rho_{SI} & 1\end{pmatrix}\right), \tag{3.7}$$

with the thresholds and link of (3.2)–(3.4). Here ζ_S is the stereotype factor, ζ_I the Israel-attitude factor and ρ_{SI} their correlation, corrected for measurement error. In group g the factors have means $\boldsymbol{\mu}_g$ and covariance matrix $\boldsymbol{\Sigma}_g$, with correlation $\rho_{SI}^{(g)}$. The sign of ρ_{SI} refers to an orientation of each factor, fixed by requiring a designated marker item of each domain to load positively.

3.4 The boundary estimands

Definition 3.1 (Boundary estimands). Let g be a group and let item j have $\boldsymbol{a}_j^{(g)} \neq \mathbf{0}$.

(i) The *item-level explained common variance* of item j in group g is

$$\mathrm{ECV}_j^{(g)} = \frac{\big(a_{jG}^{(g)}\big)^2 \sigma_{gG}^2}{\big(a_{jG}^{(g)}\big)^2 \sigma_{gG}^2 + \big(a_{js}^{(g)}\big)^2 \sigma_{g,s(j)}^2}, \qquad \mathrm{ECV}_j = \mathrm{ECV}_j^{(0)} = \frac{a_{jG}^2}{a_{jG}^2 + a_{js}^2}. \tag{3.8}$$

The boundary estimands are ECV_j and $\mathrm{ECV}_j^{(g)}$ for $j \in \mathcal{I}$.

(ii) The *latent domain correlation* is ρ_{SI} of the correlated-factors model (3.7), and $\rho_{SI}^{(g)}$ in group g.

(iii) For a threshold $\tau \in (0, 1)$ fixed in advance (Section 8), the *boundary set* is

$$\mathcal{B}_\tau = \{j \in \mathcal{I} : \mathrm{ECV}_j \geq \tau\}, \qquad \mathcal{B}_\tau^{(g)} = \{j \in \mathcal{I} : \mathrm{ECV}_j^{(g)} \geq \tau\}. \tag{3.9}$$

Write $\eta_j^{(g)} = \boldsymbol{a}_j^{(g)\top}\boldsymbol{\theta}$ for the systematic part of the latent response (3.4). Because the factors are orthogonal, each of the following equals $\mathrm{ECV}_j^{(g)}$:

$$\frac{\mathrm{Var}\big(a_{jG}^{(g)}\theta_G \mid g\big)}{\mathrm{Var}\big(\eta_j^{(g)} \mid g\big)} = \mathrm{Corr}^2\big(\theta_G, \eta_j^{(g)} \mid g\big) = 1 - \mathrm{Corr}^2\big(\theta_{s(j)}, \eta_j^{(g)} \mid g\big), \qquad \mathrm{ECV}_j = \frac{\lambda_{jG}^2}{h_j^2}. \tag{3.10}$$

Thus ECV_j is the share of item j's common (non-error) variance that is due to G, or equivalently the squared correlation between G and the item's systematic component. It does not depend on the link or on v_F, and it is unchanged by reflecting a factor. It is the item-level counterpart of the explained common variance and of the related omega-type indices for a whole scale (Reise, 2012; Rodriguez et al., 2016; Zinbarg et al., 2005). A further reading uses the logit,

$$\operatorname{logit} \mathrm{ECV}_j^{(g)} = 2\log\big|a_{jG}^{(g)}\big| - 2\log\big|a_{js}^{(g)}\big| + \log \sigma_{gG}^2 - \log \sigma_{g,s(j)}^2, \tag{3.11}$$

which is additive in log loadings and log factor variances; Theorem 4.5 and Proposition 4.8 use this form. The group-specific index $\mathrm{ECV}_j^{(g)}$ differs from ECV_j even for an invariant item whenever $\sigma_{gG}^2/\sigma_{g,s(j)}^2 \neq 1$. The boundary is therefore a property of an item *in a population*, and a difference between groups in $\mathrm{ECV}_j^{(g)}$ need not involve DIF.

3.5 What the estimands mean, and what they do not

What they would mean. A value of ECV_j near one for an Israel-related item says that, in the population and within this item pool, the systematic part of responses to item j varies almost entirely along the dimension shared with the classic stereotypes, and that little of it is shared only with other Israel-related items. A value near zero says the reverse: the item's systematic variance is mostly specific to the Israel-related domain. Intermediate values describe items whose responses reflect both. The boundary set $\mathcal{B}_\tau$ collects the items for which the first description holds to at least the pre-registered degree τ. The correlation ρ_{SI} answers a coarser question: whether people who score high on the stereotype dimension tend to score high on the Israel-attitude dimension. It says nothing about individual items.

What they would not mean. Five readings are *not* licensed.

(a) *No statement about individuals.* All three estimands are properties of a joint distribution. A large ECV_j does not show that a respondent who endorses item j holds a high position on G, and so does not show that the respondent is antisemitic. In the latent-response model the correlation between θ_G and Y_j^* is $\lambda_{jG} = h_j\,\mathrm{ECV}_j^{1/2}$. This is less than one even when $\mathrm{ECV}_j = 1$, because every item carries unique variance, and the observed category is a coarser signal still. Likewise, in the correlated-factors model the probability that a person in the upper tail of one domain, $\zeta_I > z$, is also beyond z on the other domain is $\Phi_2(-z, -z; \rho_{SI})/\Phi(-z)$, where Φ_2 is the bivariate normal distribution function. This is strictly less than one whenever $|\rho_{SI}| < 1$, and it tends to zero as z grows.[1] Treating a population association as a property of each person is the individualistic counterpart of the ecological fallacy. Area-level associations, such as those between the small-area estimates of Section 6, need not equal individual-level ones either (Robinson, 1950).

(b) *No statement about prevalence.* The estimands describe covariance, not level. An item can have ECV_j near one and be rarely endorsed, or near zero and be widely endorsed. Membership of $\mathcal{B}_\tau$ says nothing about how many people hold an attitude.

(c) *Relative to an item pool and a population.* The general factor is what the items share (Section 3.2). Adding or removing stereotype items can change G and every ECV_j. Response styles shared by all items, such as acquiescence to agree-keyed statements, also load on G unless they are modelled, as acquiescence is in the primary model of the plan (Section 8.1). $\mathrm{ECV}_j^{(g)}$ is specific to group g (Section 3.4).

(d) *No causal statement.* A common factor summarises covariance. A large ECV_j is compatible with a single underlying attitude, with common causes of distinct attitudes, and with shared method variance. The model alone does not distinguish these explanations.

(e) *No normative statement.* Neither a large nor a small ECV_j settles whether endorsing item j is antisemitic. One definition may treat a statement as antisemitic because of its content, whatever its covariance with stereotypes. Another may treat a statement as legitimate political speech even if its responses covary strongly with stereotypes. What the estimands can do is test the empirical premises that such positions invoke (Section 2.1).

3.6 Bifactor and correlated-factors representations

The two models are closely related. The Schmid–Leiman transformation (Schmid and Leiman, 1957) turns a model with correlated first-order factors into a bifactor model whose loadings are proportional within each domain. Yung et al. (1999) showed that higher-order factor models without direct effects are bifactor models constrained in this way. With only two domains the transformation is not unique, and this matters for the boundary.

Proposition 3.2 (Bifactor and correlated-factors representations). *Fix a link F and thresholds (b_{jc}).*

(a) Let $(\alpha_j)_{j\in\mathcal{J}}$ and $\rho_{SI} \in [-1, 1]$ be the parameters of a correlated-factors model (3.7). For any $\gamma_S, \gamma_I \in [-1, 1]$ with $\gamma_S\gamma_I = \rho_{SI}$, the bifactor model with

$$a_{jG} = \alpha_j\gamma_{s(j)}, \qquad a_{js} = \alpha_j\big(1 - \gamma_{s(j)}^2\big)^{1/2} \tag{3.12}$$

[1] Given $\zeta_I = x$, $\zeta_S \sim \mathrm{N}(\rho x, 1 - \rho^2)$. Given $\zeta_I > z$, $\zeta_I - z = O_p(1/z)$, so the conditional probability behaves like $1 - \Phi\big((1-\rho)z/(1-\rho^2)^{1/2}\big) \to 0$ for $\rho < 1$.

induces the same joint law of the latent responses $(Y_j^*)_{j\in\mathcal{J}}$*, and hence of* $(Y_j)_{j\in\mathcal{J}}$*, as the correlated-factors model. Its item-level ECVs are constant within each domain:* $\mathrm{ECV}_j = \gamma^2_{s(j)}$.

(b) Conversely, a bifactor model whose loading vectors (a_{jG}, a_{js}) *lie on a common line through the origin within each domain is of the form (3.12). Under the normal-ogive link, suppose a bifactor model has at least three items with non-zero specific loadings in each domain and at least one item with a non-zero general loading in each domain. If it is observationally equivalent to some correlated-factors model, then its loading vectors are proportional within each domain.*

(c) Consequently, under the normal-ogive link, let $\rho_{SI} \neq 0$ *and let each domain have at least three items with non-zero loadings. The bifactor representations of the correlated-factors model that satisfy the conditions in (b) form the curve* $\{\gamma_S\gamma_I = \rho_{SI},\ \gamma_S^2 < 1,\ \gamma_I^2 < 1\}$. *Along it* ECV_j*,* $j \in \mathcal{I}$*, takes every value in the open interval* $(\rho^2_{SI}, 1)$*, and* $\mathrm{ECV}_k = \rho^2_{SI}/\mathrm{ECV}_j$ *for* $k \in \mathcal{S}$*. The correlated-factors model identifies* ρ_{SI} *but not the boundary ECVs: among such representations it implies only the sharp bounds* $\rho^2_{SI} < \mathrm{ECV}_j < 1$ *for* $j \in \mathcal{I}$*. The end points correspond to a vanishing specific factor. Degenerate representations, in which a specific factor has fewer than three non-zero loadings, can lie outside these bounds, because a lone specific loading only adds to an item's diagonal variance.*

Proof sketch. For (a), the vector $(\gamma_S\theta_G + (1-\gamma_S^2)^{1/2}\theta_S,\ \gamma_I\theta_G + (1-\gamma_I^2)^{1/2}\theta_I)$ is bivariate normal with unit variances and correlation $\gamma_S\gamma_I = \rho_{SI}$. It therefore has the law of (ζ_S, ζ_I), and the latent responses inherit this. For the converse in (b), the observationally equivalent correlated-factors model forces the cross-domain block of the polychoric correlation matrix to have rank one. This makes the general loadings of each domain proportional to its correlated-factors loadings, and the one-factor structure within each domain then forces the specific loadings to be proportional too. Part (c) follows from (a) and (b). The full proof is in Appendix A.6. □

Proposition 3.2 sharpens the point made in Section 2.3. A correlation between the two domains, however precisely estimated, cannot by itself tell a common latent cause from correlated but distinct attitudes. Every correlated-factors model is also a bifactor model, and in that bifactor model the Israel-related items' general-factor share can be anywhere between ρ^2_{SI} and one. For example, it can be arbitrarily close to one, with almost no Israel-specific variance, and the stereotype items then carry the domain-specific variance. The conventional resolution for two first-order factors constrains $\gamma_S = \gamma_I = \rho^{1/2}_{SI}$ and so reports $\mathrm{ECV}_j = \rho_{SI}$ for every item. That is a normalisation, not an estimate. What separates the two descriptions is departure from proportionality: items within a domain whose signed ratios of general to specific loading differ. Theorem 4.2 makes this precise.

4 Identification and invariance

This section asks when the boundary estimands of Definition 3.1 are determined by the distribution of the responses, and what happens to group comparisons when items function differently across groups. We proceed in four steps.

(i) Under the normal-ogive link, identification reduces to a factor-analytic problem for the polychoric correlation matrix (Lemma 4.1). With two specific factors, and under mild conditions, the bifactor parameters are identified if and only if, in at least one domain, the items differ in their ratio of general to specific loading in a specific sense. When this fails in both domains, the ECVs of at least $|\mathcal{I}| - 1$

Israel-related items are not identified. With only two Israel-related items, the boundary ECVs are only partially identified, with sharp bounds (Theorem 4.2).

(ii) Ignoring DIF biases latent mean differences and group-specific boundary estimands. The first-order bias is an information-weighted projection of the DIF onto the loading matrix (Theorem 4.5). DIF that is proportional to the loadings is indistinguishable from a difference in latent means.

(iii) With invariant anchor items, group comparisons are identified. A single anchor with DIF contaminates every comparison, with a bound in terms of the anchors' leverage (Proposition 4.6).

(iv) A test of whether an Israel-related item lies in the boundary set, and confidence sets for $\mathcal{B}_\tau$ (Proposition 4.8).

Proofs are in Appendix A. Some ingredients are classical: identification of factor models from their covariance structure (Anderson and Rubin, 1956; Bollen, 1989); the polychoric formulation of the normal-ogive model (Olsson, 1979; Takane and de Leeuw, 1987); partial invariance (Byrne et al., 1989); and pseudo-true values of misspecified maximum likelihood (White, 1982). We attribute related results where they are stated. Throughout, *observationally equivalent* means inducing the same probability law of the response vector $\boldsymbol{Y} = (Y_j)_{j\in\mathcal{J}}$ in every group, and *identified up to reflection* means determined up to changing the sign of a factor, that is, of a whole column of $\boldsymbol{A}$. All ECVs are invariant under reflections.

4.1 Reduction to the polychoric correlation matrix

Lemma 4.1 (Latent-response reduction). *Under the normal-ogive link, let $\boldsymbol{\theta} \sim \mathrm{N}_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ in some group. Then $\boldsymbol{Y}$ has the law of the discretisation, at the thresholds (b_{jc}), of $\boldsymbol{Y}^* \sim \mathrm{N}_{|\mathcal{J}|}(\boldsymbol{A}\boldsymbol{\mu}, \boldsymbol{A}\boldsymbol{\Sigma}\boldsymbol{A}^\top + \boldsymbol{I}_{|\mathcal{J}|})$. Let $s_j = (1 + \boldsymbol{a}_j^\top \boldsymbol{\Sigma} \boldsymbol{a}_j)^{1/2}$. Two parameter values induce the same law of $\boldsymbol{Y}$ if and only if they induce the same standardised thresholds $\bar{b}_{jc} = (b_{jc} - \boldsymbol{a}_j^\top \boldsymbol{\mu})/s_j$ and the same correlation matrix $\boldsymbol{R}$ of $\boldsymbol{Y}^*$, with $R_{jk} = \boldsymbol{a}_j^\top \boldsymbol{\Sigma} \boldsymbol{a}_k/(s_j s_k)$ for $j \neq k$. In the reference group $R_{jk} = \boldsymbol{\lambda}_j^\top \boldsymbol{\lambda}_k$ and $\bar{b}_{jc} = b_{jc}(1 - h_j^2)^{1/2}$, and $(\boldsymbol{a}_j, b_{j\cdot}) \mapsto (\boldsymbol{\lambda}_j, \bar{b}_{j\cdot})$ is a bijection onto $\{h_j^2 < 1\}$, with inverse $\boldsymbol{a}_j = \boldsymbol{\lambda}_j/\psi_j^{1/2}$ and $b_{jc} = \bar{b}_{jc}/\psi_j^{1/2}$.*

In the reference group, therefore, the item parameters are identified if and only if the standardised loading matrix $\boldsymbol{\Lambda} = (\boldsymbol{\lambda}_j^\top)_{j\in\mathcal{J}}$ is determined, within the pattern (3.1), by the off-diagonal entries of $\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top$. The diagonal carries no information, because it is absorbed by the uniquenesses. For a domain $\mathcal{D} \in \{\mathcal{S}, \mathcal{I}\}$ and items $k, l \in \mathcal{D}$ define the 2×2 determinant

$$D_{kl} = \lambda_{kG}\lambda_{ls} - \lambda_{ks}\lambda_{lG}. \tag{4.1}$$

It vanishes exactly when the loading vectors $(\lambda_{kG}, \lambda_{ks})$ and $(\lambda_{lG}, \lambda_{ls})$ are proportional, that is, when the two items have the same ECV and the same sign of $\lambda_G\lambda_s$. A reverse-keyed item, whose loadings both change sign, is therefore proportional to its counterpart, whereas two items with equal ECV and specific loadings of opposite sign relative to their general loadings are not. We use three conditions.

Assumption A1 (Specific factors). Each domain contains at least three items with a non-zero specific loading.

Assumption A2 (A general factor in both domains). Each domain contains at least one item with a non-zero general loading.

Assumption A3 (Heterogeneity within a domain). In at least one domain $\mathcal{D}$ there are four distinct items $k_1, k_2, k_3, k_4 \in \mathcal{D}$ with $D_{k_1k_4}D_{k_2k_3} \neq 0$.

Assumption A3 needs at least four items in one domain. In a domain with at least four items, all with non-zero loading vectors, it fails exactly when all of that domain's items, with at most one exception, lie on a single line through the origin of the loading plane (Lemma A.4). It is therefore a condition on heterogeneity: some items of the domain must differ from the others in the signed ratio $\lambda_{kG}/\lambda_{ks}$ of general to specific loading, that is, in ECV or in the sign of $\lambda_{kG}\lambda_{ks}$.

4.2 Identification of the boundary estimands

Theorem 4.2 (Identification). *Consider the normal-ogive bifactor model in the reference group, with* $\boldsymbol{a}_j \neq \mathbf{0}$ *for every item.*

(a) Sufficiency. *Under Assumptions A1 to A3, the item parameters* $(\boldsymbol{a}_j, b_j)_{j\in\mathcal{J}}$ *are identified up to reflection. In particular every* ECV_j *and every boundary set* $\mathcal{B}_\tau$ *is identified.*

(b) Necessity of heterogeneity. *Suppose Assumptions A1 and A2 hold but Assumption A3 fails in both domains. This happens, for instance, when each domain has three items, or when the loadings are proportional within each domain. Then there are an open interval* $T \ni 1$ *and a continuous injective curve* $t \mapsto (\boldsymbol{A}(t), \boldsymbol{b}(t))$, $t \in T$, *of observationally equivalent parameter values, with* $(\boldsymbol{A}(1), \boldsymbol{b}(1)) = (\boldsymbol{A}, \boldsymbol{b})$, *along which*

$$\lambda_{kG}(t) = t\,\lambda_{kG} \quad (k \in \mathcal{S}), \qquad \lambda_{jG}(t) = \lambda_{jG}/t \quad (j \in \mathcal{I}). \tag{4.2}$$

If all loadings of the Israel-related items are non-zero, then $\mathrm{ECV}_j(t)$ *is non-constant on* T *for at least* $|\mathcal{I}| - 1$ *items* $j \in \mathcal{I}$, *and for all of them when* $|\mathcal{I}| = 3$ *or the loadings are proportional within* $\mathcal{I}$. *These ECVs are not identified, and neither is* $\mathcal{B}_\tau$ *for* τ *in a non-degenerate interval.*

(c) Two Israel-related items. *Suppose* $\mathcal{I} = \{j_1, j_2\}$, *the domain* $\mathcal{S}$ *contains at least three items with non-zero specific loadings and satisfies the condition of Assumption A3,* $(\lambda_{j_1G}, \lambda_{j_2G}) \neq \mathbf{0}$, *and* $\varpi = \lambda_{j_1I}\lambda_{j_2I} \neq 0$. *Then all general loadings, all parameters of the stereotype items, and* ϖ *are identified, but the specific loadings of* j_1, j_2 *are identified only through their product. The identified set of* ECV_{j_1} *is the open interval*

$$\left(\lambda^2_{j_1G},\ \frac{\lambda^2_{j_1G}(1-\lambda^2_{j_2G})}{\lambda^2_{j_1G}(1-\lambda^2_{j_2G}) + \varpi^2}\right), \tag{4.3}$$

and every value in it is attained by an observationally equivalent parameter value. The same holds for j_2 *with the roles exchanged. The two ECVs are not separately free: the joint identified set is the curve*

$$\left\{\left(\frac{\lambda^2_{j_1G}}{\lambda^2_{j_1G} + q^2},\ \frac{\lambda^2_{j_2G}}{\lambda^2_{j_2G} + \varpi^2/q^2}\right) :\ \frac{\varpi^2}{1-\lambda^2_{j_2G}} < q^2 < 1 - \lambda^2_{j_1G}\right\}, \tag{4.4}$$

along which ECV_{j_1} *falls and* ECV_{j_2} *rises as* q^2 *grows, and* (4.3) *is its projection on the first coordinate. If instead* $\mathcal{S}$ *fails the condition of Assumption A3, the curve of part (b) is present as well, and the general loadings of the Israel-related items then trade off against their specific loadings.*

(d) Correlated factors. *In the normal-ogive correlated-factors model* (3.7), ρ_{SI} *is identified whenever each factor has at least two items with non-zero loadings. The loadings are identified if, in addition, each factor has at least three such items or* $\rho_{SI} \neq 0$.

Proof sketch. By Lemma 4.1 it suffices to study $\boldsymbol{\Lambda}$ given $\operatorname{offdiag}(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top)$. The cross-domain block equals $\boldsymbol{\lambda}_{\mathcal{S}G}\boldsymbol{\lambda}_{\mathcal{I}G}^\top$. It has rank one by Assumption A2, so any equivalent $\tilde{\boldsymbol{\Lambda}}$ has $\tilde{\boldsymbol{\lambda}}_{\mathcal{S}G} = t\boldsymbol{\lambda}_{\mathcal{S}G}$ and $\tilde{\boldsymbol{\lambda}}_{\mathcal{I}G} = \boldsymbol{\lambda}_{\mathcal{I}G}/t$ for a scalar t. Within a domain, the specific loadings must then reproduce

$$\operatorname{offdiag}\big(\boldsymbol{\lambda}_s\boldsymbol{\lambda}_s^\top + \varsigma\,\boldsymbol{\lambda}_G\boldsymbol{\lambda}_G^\top\big) = \operatorname{offdiag}\big(\tilde{\boldsymbol{\lambda}}_s\tilde{\boldsymbol{\lambda}}_s^\top\big), \qquad \varsigma = 1 - t^2 \text{ in } \mathcal{S}, \quad \varsigma = 1 - t^{-2} \text{ in } \mathcal{I}.$$

A tetrad of a rank-one matrix vanishes. A direct computation gives the tetrad of the left-hand side as $\varsigma D_{k_1k_4}D_{k_2k_3}$, so Assumption A3 forces $\varsigma = 0$, hence $t = 1$. The one-factor structure of the remaining specific loadings, with at least three non-zero entries (Assumption A1), then pins them down up to sign. This proves (a). If every tetrad coefficient vanishes in both domains, the left-hand side has one-factor structure for every small ς. Solving for $\tilde{\boldsymbol{\lambda}}_s$ explicitly gives the curve in (b). In (c), $t = 1$ is forced by $\mathcal{S}$, and the Israel block contributes the single equation $\lambda_{j_1I}\lambda_{j_2I} = \varpi$, constrained only by positivity of the two uniquenesses, which yields (4.4) and its projection (4.3). Part (d) uses $\rho_{SI}^2 = R_{kj}R_{lm}/(R_{kl}R_{jm})$ for $k, l \in \mathcal{S}$ and $j, m \in \mathcal{I}$. Details are in Appendix A.3. □

Remark 4.3 (Relation to the literature). Rotational indeterminacy and the three-indicator conditions for factor models are classical (Anderson and Rubin, 1956; Bollen, 1989), and part (d) is an instance of the two- and three-indicator rules. For bifactor models, Fang et al. (2021) study identifiability with continuous and with probit binary responses and give a broad characterisation, and Green and Yang (2018) describe empirical underidentification of bifactor models in practice. We do not rely on either. Theorem 4.2 is a self-contained treatment of the two-specific-factor design used here, with its own proof. It adds an explicit tetrad form of the heterogeneity condition, the explicit curve of equivalent parameters with its consequences for the boundary ECVs, and the sharp identified set when the Israel-related domain has only two items.

Three further remarks are in Appendix A.4: what identifies the boundary, and what follows for design and analysis (Remark A.5); the logistic link, for which the conditions of Theorem 4.2 are assumed, not proved, to suffice (Remark A.6); and the case in which no Israel-related item loads on G (Remark A.7).

Remark 4.4 (The bifactor-$(S-1)$ alternative). In the bifactor-$(S-1)$ model of Eid et al. (2017), the stereotype items load on G only, so G is by definition the stereotype factor. For $j \in \mathcal{I}$, ECV_j is then the share of item j's common variance that it shares with that factor. This model is identified under weaker conditions: three stereotype items with non-zero loadings determine $\boldsymbol{\lambda}_{\mathcal{S}G}$, the cross-domain block then gives $\boldsymbol{\lambda}_{\mathcal{I}G}$, and three non-zero Israel-specific loadings give $\boldsymbol{\lambda}_{\mathcal{I}I}$. The price is that the model assumes the stereotype items have no variance specific to their domain. Whether G should be anchored in one domain or defined as what both domains share is itself part of the substantive question. We use the symmetric model as primary and report the $(S-1)$ variant as a sensitivity analysis.

4.3 Differential item functioning

DIF is usually screened with the Mantel–Haenszel procedure (Holland and Thayer, 1988) or with likelihood-ratio tests of item parameters within the item response model (Thissen et al., 1993). Both condition on a matching score or an anchor set, which Section 4.4 examines. Here we ask what DIF does to group comparisons when it goes undetected and is ignored. Consider two groups: the reference group $g = 0$, whose item parameters $(\boldsymbol{A}, \boldsymbol{b})$ are identified by Theorem 4.2, and a focal group $g = 1$ with latent parameters $\boldsymbol{\mu} = \boldsymbol{\mu}_1$, $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_1$, and $\boldsymbol{\ell} = (\log\sigma_{1G}^2, \log\sigma_{1S}^2, \log\sigma_{1I}^2)^\top$. In group 1 the items have DIF $(\Delta\boldsymbol{a}_j, \Delta b_j)$, with $\Delta\boldsymbol{A}$ the matrix with rows $\Delta\boldsymbol{a}_j^\top$. The analyst ignores the DIF. A *reference-linked* analysis holds the item parameters at their reference values and estimates $\boldsymbol{\xi} = (\boldsymbol{\mu}, \boldsymbol{\ell})$ by maximum likelihood from the group-1 data under the invariant model. This is also the limit of concurrent calibration when the focal group is

small relative to the reference group; Remark A.9 treats concurrent calibration in general, including the equal-sized groups of the simulation. The estimator converges to the pseudo-true value $\boldsymbol{\xi}^*$ that minimises the Kullback–Leibler divergence from the true group-1 law (White, 1982). We need two pieces of notation.

Item coordinates and their information. Perturb the group-1 linear predictor of item j as

$$\boldsymbol{a}_j^\top \boldsymbol{\theta} - b_{jc} \;\longmapsto\; \boldsymbol{a}_j^\top \boldsymbol{\theta} - b_{jc} + \iota_j + \beta_{jG}(\theta_G - \mu_G) + \beta_{js}(\theta_{s(j)} - \mu_{s(j)}), \tag{4.5}$$

an intercept shift ι_j and two slope changes acting on the centred factors. Let $\boldsymbol{x} = (\boldsymbol{\iota}, \boldsymbol{\beta}) \in \mathbb{R}^{|\mathcal{J}|} \times \mathbb{R}^{2|\mathcal{J}|}$, and let $p_1(\boldsymbol{y}; \boldsymbol{x})$ be the resulting probability of response pattern $\boldsymbol{y}$ in group 1, with $\boldsymbol{\theta} \sim \mathrm{N}_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Let $\boldsymbol{H} = \mathbb{E}\{\nabla_{\boldsymbol{x}} \log p_1(\boldsymbol{Y}; \boldsymbol{0}) \nabla_{\boldsymbol{x}} \log p_1(\boldsymbol{Y}; \boldsymbol{0})^\top\}$ be the Fisher information of these coordinates at $\boldsymbol{x} = \boldsymbol{0}$, partitioned into blocks $\boldsymbol{H}_{\iota\iota}$, $\boldsymbol{H}_{\iota\beta}$ and $\boldsymbol{H}_{\beta\beta}$. Two exact facts make $\boldsymbol{H}$ the right metric:

- the invariant model with focal parameters $\boldsymbol{\xi}' = (\boldsymbol{\mu}', \boldsymbol{\ell}')$ is the point $\boldsymbol{\iota} = \boldsymbol{A}(\boldsymbol{\mu}' - \boldsymbol{\mu})$, $\beta_{jf} = a_{jf}\{e^{(\ell'_f - \ell_f)/2} - 1\}$. Its Jacobian at $\boldsymbol{\xi}' = \boldsymbol{\xi}$ is $\boldsymbol{L} = \mathrm{diag}(\boldsymbol{A}, \boldsymbol{B})$, where $\boldsymbol{B} \in \mathbb{R}^{2|\mathcal{J}| \times 3}$ has the entry $a_{jf}/2$ in row (j, f) and column f, and zeros elsewhere;
- the true group-1 law with DIF is $p_1(\cdot\,; \boldsymbol{x}_\Delta)$, with

$$\boldsymbol{x}_\Delta = (\boldsymbol{d}, \boldsymbol{\Delta}_{\mathrm{sl}}), \qquad d_j = \Delta\boldsymbol{a}_j^\top \boldsymbol{\mu} - \Delta b_j, \qquad \boldsymbol{\Delta}_{\mathrm{sl}} = (\Delta a_{jG}, \Delta a_{js})_{j \in \mathcal{J}}, \tag{4.6}$$

  because $(\boldsymbol{a}_j + \Delta\boldsymbol{a}_j)^\top \boldsymbol{\theta} - b_{jc} - \Delta b_j = \boldsymbol{a}_j^\top \boldsymbol{\theta} - b_{jc} + d_j + \Delta\boldsymbol{a}_j^\top(\boldsymbol{\theta} - \boldsymbol{\mu})$.

Non-uniform DIF thus acts both as a slope change and, when $\boldsymbol{\mu} \neq \boldsymbol{0}$, as an intercept shift $\Delta\boldsymbol{a}_j^\top \boldsymbol{\mu}$. Finally, let $\Pi_{\boldsymbol{A}} = \boldsymbol{A}(\boldsymbol{A}^\top \boldsymbol{H}_{\iota\iota} \boldsymbol{A})^{-1} \boldsymbol{A}^\top \boldsymbol{H}_{\iota\iota}$ and $\Pi_{\boldsymbol{B}} = \boldsymbol{B}(\boldsymbol{B}^\top \boldsymbol{H}_{\beta\beta} \boldsymbol{B})^{-1} \boldsymbol{B}^\top \boldsymbol{H}_{\beta\beta}$, and define the efficient weights

$$\boldsymbol{W}_\iota = \boldsymbol{H}_{\iota\iota} - \boldsymbol{H}_{\iota\beta}\boldsymbol{B}(\boldsymbol{B}^\top \boldsymbol{H}_{\beta\beta}\boldsymbol{B})^{-1}\boldsymbol{B}^\top \boldsymbol{H}_{\beta\iota}, \qquad \boldsymbol{W}_\beta = \boldsymbol{H}_{\beta\beta} - \boldsymbol{H}_{\beta\iota}\boldsymbol{A}(\boldsymbol{A}^\top \boldsymbol{H}_{\iota\iota}\boldsymbol{A})^{-1}\boldsymbol{A}^\top \boldsymbol{H}_{\iota\beta}.$$

Theorem 4.5 (Bias from ignoring DIF). *Let $F \in \{\Lambda, \Phi\}$. Suppose that in the invariant model for group 1 with reference item parameters, $\boldsymbol{\xi}$ is identified, lies in the interior of a compact parameter set, and has non-singular Fisher information $\boldsymbol{L}^\top \boldsymbol{H} \boldsymbol{L}$. Then for all DIF in a neighbourhood of zero the pseudo-true value $\boldsymbol{\xi}^*$ of the reference-linked analysis is unique, and the following hold.*

(a) First-order bias. $\boldsymbol{\xi}^* - \boldsymbol{\xi} = (\boldsymbol{L}^\top \boldsymbol{H} \boldsymbol{L})^{-1} \boldsymbol{L}^\top \boldsymbol{H} \boldsymbol{x}_\Delta + O(\|\Delta\|^2)$. *Equivalently,*

$$\boldsymbol{\mu}^* - \boldsymbol{\mu} = (\boldsymbol{A}^\top \boldsymbol{W}_\iota \boldsymbol{A})^{-1} \boldsymbol{A}^\top \big\{\boldsymbol{W}_\iota\, \boldsymbol{d} + \boldsymbol{H}_{\iota\beta}(\boldsymbol{I} - \Pi_{\boldsymbol{B}})\boldsymbol{\Delta}_{\mathrm{sl}}\big\} + O(\|\Delta\|^2), \tag{4.7}$$

$$\boldsymbol{\ell}^* - \boldsymbol{\ell} = (\boldsymbol{B}^\top \boldsymbol{W}_\beta \boldsymbol{B})^{-1} \boldsymbol{B}^\top \big\{\boldsymbol{W}_\beta\, \boldsymbol{\Delta}_{\mathrm{sl}} + \boldsymbol{H}_{\beta\iota}(\boldsymbol{I} - \Pi_{\boldsymbol{A}})\boldsymbol{d}\big\} + O(\|\Delta\|^2). \tag{4.8}$$

Since $\boldsymbol{\mu}_0 = \boldsymbol{0}$, the estimated difference in general-factor means, μ_G^*, is biased by the first coordinate of (4.7).

(b) Orthogonal and Gaussian cases. *If $\boldsymbol{H}_{\iota\beta} = \boldsymbol{0}$, then $\boldsymbol{\mu}^* - \boldsymbol{\mu} = (\boldsymbol{A}^\top \boldsymbol{H}_{\iota\iota} \boldsymbol{A})^{-1} \boldsymbol{A}^\top \boldsymbol{H}_{\iota\iota} \boldsymbol{d}$ and $\boldsymbol{\ell}^* - \boldsymbol{\ell} = (\boldsymbol{B}^\top \boldsymbol{H}_{\beta\beta} \boldsymbol{B})^{-1} \boldsymbol{B}^\top \boldsymbol{H}_{\beta\beta} \boldsymbol{\Delta}_{\mathrm{sl}}$, up to $O(\|\Delta\|^2)$. This holds in particular when every item's group-1 thresholds are symmetric about its group-1 mean predictor, $b_{jc} - \boldsymbol{a}_j^\top \boldsymbol{\mu} = -(b_{j,C_j+1-c} - \boldsymbol{a}_j^\top \boldsymbol{\mu})$. Uniform DIF then biases only the latent means, and non-uniform DIF biases them only through $\Delta\boldsymbol{A}\boldsymbol{\mu}$, which vanishes when the groups have equal latent means. If the latent responses (3.4) were observed (the linear-Gaussian model), then for every Δ, not only to first order,*

$$\boldsymbol{\mu}^* - \boldsymbol{\mu} = (\boldsymbol{A}^\top \boldsymbol{A})^{-1} \boldsymbol{A}^\top \boldsymbol{d}. \tag{4.9}$$

(c) Boundary estimands. *For every item j, the group-1 ECV reported by the analysis, $\widehat{\mathrm{ECV}}_j^{(1)}$ (computed from $\boldsymbol{a}_j$ and $\boldsymbol{\ell}^*$), and the true $\mathrm{ECV}_j^{(1)}$ (computed from $\boldsymbol{a}_j + \Delta\boldsymbol{a}_j$ and $\boldsymbol{\ell}$) satisfy exactly*

$$\operatorname{logit}\widehat{\mathrm{ECV}}_j^{(1)} - \operatorname{logit}\mathrm{ECV}_j^{(1)} = (\ell^*_G - \ell_G) - (\ell^*_{s(j)} - \ell_{s(j)}) - 2\log\left|1 + \frac{\Delta a_{jG}}{a_{jG}}\right| + 2\log\left|1 + \frac{\Delta a_{js}}{a_{js}}\right|, \quad (4.10)$$

with $\boldsymbol{\ell}^* - \boldsymbol{\ell}$ given by (4.8).

(d) Manufacturing and masking. *(i) For every $\boldsymbol{v} \in \mathbb{R}^3$, the group-1 law with latent mean $\boldsymbol{\mu}$ and uniform DIF $\Delta b_j = -\boldsymbol{a}_j^\top \boldsymbol{v}$ on all items equals the law with latent mean $\boldsymbol{\mu} + \boldsymbol{v}$ and no DIF, for either link. The invariant analysis then reports $\boldsymbol{\mu} + \boldsymbol{v}$ and fits exactly. Taking $\boldsymbol{v} = (\beta, 0, 0)^\top$ manufactures a general-factor difference β where there is none, or masks a true difference when $\beta = -\mu_G$. (ii) If DIF is uniform and confined to $\mathcal{I}$, and the stereotype loading vectors are not all proportional, then exact confounding is possible only with μ_I, through $\Delta\boldsymbol{b}_{\mathcal{I}} \propto \boldsymbol{a}_{\mathcal{I}I}$. This leaves μ_G and every ECV unbiased. For other patterns the first-order bias of μ_G is the first coordinate of (4.7), a linear functional of $\Delta\boldsymbol{b}_{\mathcal{I}}$. In the linear-Gaussian case of (b) this functional is non-zero whenever the Israel-related loading vectors are not all proportional. (iii) By (4.10), non-uniform DIF moves the logit of the reported ECV of every item, including DIF-free ones, by $(\ell^*_G - \ell_G) - (\ell^*_{s(j)} - \ell_{s(j)})$, an amount common to all items of the same domain. At the same time it hides the change in the DIF item's own ECV, which can move items into or out of $\mathcal{B}_\tau^{(1)}$ in either direction.*

Proof sketch. A general lemma (Lemma A.8) shows that, for a smooth model $\boldsymbol{x}(\boldsymbol{\xi})$ on a finite outcome space and a true law at $\boldsymbol{x}_\Delta$, the KL projection satisfies $\boldsymbol{\xi}^* - \boldsymbol{\xi} = (\boldsymbol{L}^\top \boldsymbol{H} \boldsymbol{L})^{-1} \boldsymbol{L}^\top \boldsymbol{H} \boldsymbol{x}_\Delta + O(\|\boldsymbol{x}_\Delta\|^2)$. This is the implicit function theorem applied to the expected score, with the information identity supplying the derivatives. The two bulleted facts are exact, and (4.7)–(4.8) are the partitioned form of this projection. In (b) the symmetry condition makes the intercept scores odd and the slope scores even under the reflection $y_j \mapsto C_j - y_j$, so their cross-information vanishes. In the linear-Gaussian case the generalised least-squares projection with covariance $\boldsymbol{A}\boldsymbol{\Sigma}'\boldsymbol{A}^\top + \boldsymbol{I}$ does not depend on $\boldsymbol{\Sigma}'$, by the Woodbury identity. Part (c) is (3.11) evaluated at the two parameter values, and (d)(i) is the identity $\boldsymbol{a}_j^\top \boldsymbol{\theta} - b_{jc} + \boldsymbol{a}_j^\top \boldsymbol{v} = \boldsymbol{a}_j^\top (\boldsymbol{\theta} + \boldsymbol{v}) - b_{jc}$. Details are in Appendix A.7. □

The structure of (4.7) is the main message. Up to the correction from non-uniform DIF, the bias of the latent means is an information-weighted regression of the item-level DIF intercepts d_j on the item loadings $\boldsymbol{a}_j$. DIF whose pattern across items resembles a column of $\boldsymbol{A}$ is read as a difference in latent means. DIF that is orthogonal to the columns of $\boldsymbol{A}$, in the $\boldsymbol{W}_\iota$ metric, is not read that way and appears as misfit instead. Part (d)(i) is the extreme case of this. A uniform shift proportional to the general loadings is indistinguishable from a difference in the general factor, in any sample size. This is the classical confounding of DIF with impact, here made exact for the bifactor model. For the boundary question, part (d)(ii) shows that DIF on Israel-related items alone cannot be mistaken one-for-one for a difference in general hostility when the stereotype items are invariant, but it can still leak into μ_G. Part (d)(iii) shows that it distorts the group-specific boundary even for items without DIF. Equation (4.10) is exact, so the only approximation in (c) is the first-order expression for $\boldsymbol{\ell}^* - \boldsymbol{\ell}$.

4.4 Anchors and partial invariance

Let $\mathcal{A} \subseteq \mathcal{J}$ be a set of anchor items assumed invariant, and let $\boldsymbol{A}_{\mathcal{A}}$ be the corresponding rows of $\boldsymbol{A}$. An *anchored analysis* fixes the anchors' parameters at their reference values. It estimates $(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ together

with free group-1 parameters, within the pattern (3.1), for every non-anchor item. This is partial invariance in the sense of Byrne et al. (1989).

Proposition 4.6 (Anchors).

(a) Identification. *Under the normal-ogive link, suppose the anchors are invariant, each has at least three response categories ($C_j \geq 2$), and $\boldsymbol{A}_{\mathcal{A}}$ has full column rank three. Equivalently, the anchors include an item with a non-zero specific loading in each domain and, within at least one domain, two items whose loading vectors are not proportional. Two anchors per domain do not suffice if both pairs are proportional. Then $\boldsymbol{\mu}_1$, $\boldsymbol{\Sigma}_1$ (any positive definite matrix) and all group-1 item parameters are identified. Hence so are the DIF parameters of every non-anchor item and every group-1 boundary estimand. Without anchors, $\boldsymbol{\mu}_1$ is not identified: for every $\boldsymbol{v}$, the map $(\boldsymbol{\mu}_1, b_{jc}^{(1)}) \mapsto (\boldsymbol{\mu}_1 + \boldsymbol{v},\ b_{jc}^{(1)} + \boldsymbol{a}_j^{(1)\top}\boldsymbol{v})$ leaves the group-1 law unchanged.*

(b) Undetectable anchor DIF. *For either link, suppose the anchors carry uniform DIF $\Delta\boldsymbol{b}_{\mathcal{A}} = -\boldsymbol{A}_{\mathcal{A}}\boldsymbol{v}$ for some $\boldsymbol{v} \in \mathbb{R}^3$. This is always the case when $|\mathcal{A}| = 3$ and $\boldsymbol{A}_{\mathcal{A}}$ is invertible. Then the group-1 law equals that of a model with invariant anchors, latent mean $\boldsymbol{\mu}_1 + \boldsymbol{v}$, the same $\boldsymbol{\Sigma}_1$, and non-anchor thresholds $b_{kc}^{(1)} + \boldsymbol{a}_k^{(1)\top}\boldsymbol{v}$. The anchored analysis fits exactly, shifts every latent-mean comparison by $\boldsymbol{v}$, and attributes spurious uniform DIF $\boldsymbol{a}_k^{(1)\top}\boldsymbol{v}$ to every non-anchor item k.*

(c) General anchor DIF. *For either link, suppose the anchors carry uniform DIF $\Delta\boldsymbol{b}_{\mathcal{A}}$, and write $\boldsymbol{d}_{\mathcal{A}} = -\Delta\boldsymbol{b}_{\mathcal{A}}$. Under the regularity conditions of Theorem 4.5 for the anchored model, its pseudo-true latent mean satisfies*

$$\boldsymbol{\mu}_1^* - \boldsymbol{\mu}_1 = (\boldsymbol{A}_{\mathcal{A}}^\top \boldsymbol{W} \boldsymbol{A}_{\mathcal{A}})^{-1} \boldsymbol{A}_{\mathcal{A}}^\top \boldsymbol{W} \boldsymbol{d}_{\mathcal{A}} + O(\|\boldsymbol{d}_{\mathcal{A}}\|^2). \tag{4.11}$$

Here $\boldsymbol{W} \succeq 0$ is the efficient information for the anchors' intercepts: the Schur complement of their block of the Fisher information with respect to all other parameters of the anchored model. The regularity conditions make $\boldsymbol{A}_{\mathcal{A}}^\top \boldsymbol{W} \boldsymbol{A}_{\mathcal{A}}$ non-singular. Let $\boldsymbol{M} = \boldsymbol{W}^{1/2}\boldsymbol{A}_{\mathcal{A}}$ and let $\Pi = \boldsymbol{M}(\boldsymbol{M}^\top\boldsymbol{M})^{-1}\boldsymbol{M}^\top$. Then

$$\|\boldsymbol{\mu}_1^* - \boldsymbol{\mu}_1\|_2 \leq \frac{\|\Pi \boldsymbol{W}^{1/2} \boldsymbol{d}_{\mathcal{A}}\|_2}{\sigma_{\min}(\boldsymbol{M})} + O(\|\boldsymbol{d}_{\mathcal{A}}\|^2) \leq \frac{\|\boldsymbol{W}^{1/2} \boldsymbol{d}_{\mathcal{A}}\|_2}{\sigma_{\min}(\boldsymbol{M})} + O(\|\boldsymbol{d}_{\mathcal{A}}\|^2). \tag{4.12}$$

If only anchor j has DIF, the middle term equals $|\Delta b_j|\ \mathrm{lev}_j^{1/2}/\sigma_{\min}(\boldsymbol{M})$, where $\mathrm{lev}_j = (\boldsymbol{W}^{1/2}\Pi\boldsymbol{W}^{1/2})_{jj} \leq W_{jj}$ is the anchor's leverage. For non-uniform DIF on anchors, the first-order bias of $(\boldsymbol{\mu}_1, \boldsymbol{\ell}_1)$ is given by (4.7)–(4.8), with $(\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{H})$ replaced by $\boldsymbol{A}_{\mathcal{A}}$, $\boldsymbol{B}_{\mathcal{A}}$ and the efficient information of the anchors' coordinates.

Proof sketch. For (a), each anchor's two known thresholds determine the mean and standard deviation of its group-1 latent response, and the polychoric correlations then give the anchors' latent covariance matrix. Full column rank of $\boldsymbol{A}_{\mathcal{A}}$ recovers $\boldsymbol{\mu}_1$ and $\boldsymbol{\Sigma}_1$. Each non-anchor item is then a regression on a known latent distribution, identified through its correlations with the anchors. Part (b) is the identity behind Theorem 4.5(d)(i), applied to the anchors, with the non-anchor thresholds absorbing the shift. Part (c) follows from Lemma A.8 after profiling out the non-anchor parameters, which the anchored model leaves free. The bound uses $\boldsymbol{M}(\boldsymbol{\mu}_1^* - \boldsymbol{\mu}_1) = \Pi\boldsymbol{W}^{1/2}\boldsymbol{d}_{\mathcal{A}}$ to first order. Details are in Appendix A.8. □

Three features of Proposition 4.6 matter in practice. First, only the component of the anchors' DIF that the anchored model cannot detect, $\Pi\boldsymbol{W}^{1/2}\boldsymbol{d}_{\mathcal{A}}$, moves the latent means at first order. The detectable

component appears as misfit. With exactly three anchors nothing is detectable, by part (b). Second, the error does not stay with the faulty anchor. It shifts every group comparison of latent means and the metric in which every non-anchor item's DIF and every group-specific ECV is computed. In the exact case (b) every non-anchor item k acquires spurious uniform DIF $\boldsymbol{a}_k^{(1)\top}\boldsymbol{v}$, the projection of its loadings on the shift; its sign can differ between items unless $\boldsymbol{v}$ is a shift of the general factor alone and the general loadings share a sign. Third, the bound (4.12) is a sensitivity analysis. A stated maximum DIF on the anchors translates into a maximum bias of every comparison, and the bound is smaller for many anchors with strong, linearly independent loadings, which make $\sigma_{\min}(\boldsymbol{M})$ large.

Remark 4.7 (Anchors from the stereotype domain only). Researchers may be unwilling to assume that any Israel-related item is invariant across, say, ideological groups. If all anchors are stereotype items, $\boldsymbol{A}_{\mathcal{A}}$ has rank at most two, so Proposition 4.6(a) does not apply as stated. Suppose, however, that $\boldsymbol{\Sigma}_1$ is diagonal, the stereotype anchors have at least three categories and non-proportional loading vectors, and at least three Israel-related items have non-zero I-loadings in group 1. Then μ_{1G}, μ_{1S}, σ^2_{1G}, σ^2_{1S}, all group-1 general loadings, and the products $a^{(1)}_{jI}\sigma_{1I}$ for $j \in \mathcal{I}$ are identified, and hence so is every group-1 boundary ECV. What is not identified is μ_{1I}, σ^2_{1I}, the thresholds of the Israel-related items, and their I-loadings separately: the group-1 law is unchanged by a two-parameter family of transformations of these quantities (Appendix A.8). With proportional stereotype anchors only one linear combination of μ_{1G} and μ_{1S} is identified. Comparisons of the general factor and of the boundary can therefore be anchored entirely in the stereotype domain. Comparisons of the level of Israel-related attitudes cannot. This is an identification result, not a recommendation: with fewer anchors, the comparisons are more exposed to DIF on the anchors that remain. In the simulation, the same DIF on two stereotype anchors biased the stereotype-anchored comparison of G two to three times as much as the default anchor set did (Section 7.3), so the plan reports it as a sensitivity analysis (Section 8.3).

Data-driven anchor selection, and the limit that all such procedures share, are discussed in Remark A.10.

4.5 Testing the boundary

Membership of $\mathcal{B}_\tau$ is a statement about a parameter, so it should be decided by a test with stated error rates rather than by comparing a point estimate with τ. Let $\widehat{\boldsymbol{\vartheta}}$ be the marginal maximum likelihood estimator of the reference-group item parameters from n respondents, or any other consistent and asymptotically normal estimator with a consistent variance estimator, such as the two-step limited-information estimator used in the simulation (Section 7.1); the proof uses nothing else. For $j \in \mathcal{I}$ write $\varkappa_j = \operatorname{logit}\mathrm{ECV}_j = 2\log|a_{jG}| - 2\log|a_{jI}|$, let $\widehat{\varkappa}_j$ be its plug-in estimate, and let $\widehat{\mathrm{se}}_j$ be its delta-method standard error, computed from the inverse observed information.

Proposition 4.8 (Boundary tests and confidence sets). *Suppose the conditions of Theorem 4.2(a) hold, the true parameter is interior with non-singular Fisher information, and $a_{jG}a_{jI} \neq 0$ for $j \in \mathcal{I}$. Fix $\tau \in (0,1)$ and $\alpha \in (0, \frac{1}{2})$, and let $z_{1-\alpha}$ be the standard normal quantile.*

(a) The test that rejects $H_0^{(j)} : \mathrm{ECV}_j \geq \tau$ in favour of $\mathrm{ECV}_j < \tau$ when $(\widehat{\varkappa}_j - \operatorname{logit}\tau)/\widehat{\mathrm{se}}_j < -z_{1-\alpha}$ has pointwise asymptotic size α. The rejection probability tends to α at $\mathrm{ECV}_j = \tau$, to 0 when $\mathrm{ECV}_j > \tau$, and to 1 when $\mathrm{ECV}_j < \tau$. The signed-root likelihood-ratio test is asymptotically equivalent. At $\mathrm{ECV}_j = \tau$ the likelihood-ratio statistic for $H_0^{(j)}$ converges to the mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ (Chernoff, 1954; Self and Liang, 1987).

(b) *Let* $[L_j, U_j]$, $j \in \mathcal{I}$, *be asymptotically simultaneous level-*$(1-\alpha)$ *confidence intervals for* ECV_j, *for instance Bonferroni-adjusted Wald intervals for* $\varkappa_j$ *transformed by* expit. *Then the inner and outer sets* $\widehat{\mathcal{B}}^{\mathrm{in}}_\tau = \{j : L_j \geq \tau\}$ *and* $\widehat{\mathcal{B}}^{\mathrm{out}}_\tau = \{j : U_j \geq \tau\}$ *satisfy* $\liminf_n \mathbb{P}(\widehat{\mathcal{B}}^{\mathrm{in}}_\tau \subseteq \mathcal{B}_\tau \subseteq \widehat{\mathcal{B}}^{\mathrm{out}}_\tau) \geq 1-\alpha$ *at every parameter value satisfying the conditions above.*

(c) *The plug-in set* $\{j : \widehat{\mathrm{ECV}}_j \geq \tau\}$ *is not consistent for* $\mathcal{B}_\tau$*: an item with* $\mathrm{ECV}_j = \tau$ *belongs to it with probability tending to* $\frac{1}{2}$. *At the boundary* $\mathrm{ECV}_j = 1$ *(*$a_{jI} = 0$*) the delta method degenerates, and* $n(1 - \widehat{\mathrm{ECV}}_j)$ *converges in law to* $V_{jI}\chi^2_1/a^2_{jG}$, *where* V_{jI} *is the asymptotic variance of* $n^{1/2}\widehat{a}_{jI}$.

Proof sketch. Under the conditions of Theorem 4.2(a), $n^{1/2}(\widehat{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta})$ is asymptotically normal, and $\varkappa_j$ is smooth with non-zero gradient $(2/a_{jG}, -2/a_{jI})$ in the coordinates (a_{jG}, a_{jI}). Part (a) is then the delta method together with the standard one-sided boundary asymptotics. Part (b) holds on the event that all intervals cover. Part (c) follows from the symmetry of the normal limit at $\mathrm{ECV}_j = \tau$ and from a second-order expansion at $a_{jI} = 0$. Details are in Appendix A.9. □

Part (b) gives a pre-registrable decision rule (Section 8). Items in $\widehat{\mathcal{B}}^{\mathrm{in}}_\tau$ are, with stated confidence, in the boundary set. Items outside $\widehat{\mathcal{B}}^{\mathrm{out}}_\tau$ are, with the same confidence, not in it. The items in between are reported as undetermined at the achieved sample size, not forced to one side. The same construction applies to $\mathcal{B}^{(g)}_\tau$ under the conditions of Proposition 4.6(a), and to differences $\mathrm{ECV}^{(1)}_j - \mathrm{ECV}^{(0)}_j$.

Remark 4.9 (The guarantees are pointwise). The size and coverage statements of Proposition 4.8 hold at each fixed parameter value that satisfies Theorem 4.2(a). They are not uniform over that set. As the parameter approaches the non-identified set of Theorem 4.2(b), the products $D_{k_1k_4}D_{k_2k_3}$ tend to zero and the information along the curve of equivalent parameters vanishes. For every item whose ECV moves along that curve, the asymptotic variance of $\widehat{\varkappa}_j$ then diverges, and the sample size at which the normal approximation becomes accurate grows without bound. At any fixed n, the Wald test can have size far above α, and the sets of part (b) can bracket $\mathcal{B}_\tau$ with probability far below $1-\alpha$. The simulation shows Wald coverage well below nominal in a near-proportional design (Section 7.2). The decision rule should therefore be applied only when the data pass the identification checks of the analysis plan, in particular the numerical identification diagnostic with its registered cut-off (rule D2 in Section 8.7). Otherwise every item should be reported as undetermined. For the intervals themselves, profile-likelihood intervals for $\varkappa_j$, which widen along a nearly flat ridge instead of relying on local curvature, or bootstrap intervals are preferable to Wald intervals near the non-identified set.

5 Sensitive items: list experiments and social-desirability bias

The measurement model of Section 3 takes direct answers at face value. For items whose endorsement is socially sanctioned, a direct answer can differ from the answer a respondent would give if no one could attribute it to them (Section 2.5). The direction of that difference need not be the same for every item or every group. This section develops four tools for measuring it. Proposition 5.1 gives the exact variance of the list-experiment estimator and characterises its cost relative to direct questioning. Theorem 5.3 combines direct and list responses to identify item-level social-desirability bias and the rate of hidden endorsement, and says what is lost without a monotonicity assumption. Theorem B.5 allocates a budget across the arms of the experiment and chooses the control list; it is summarised in Section 5.4 and stated in full in Appendix B.4. Proposition 5.5 links the list experiment to the latent-variable model by correcting the

threshold of the sensitive item. Proofs, and the supplementary results to which this section points, are in Appendix B.

We build on established results and attribute them where they are used. The item count technique is due to Miller (1984). Its identifying assumptions, the difference-in-means estimator and its regression extensions are developed by Imai (2011), Blair and Imai (2012) and Glynn (2013), with an earlier multivariate estimator by Corstange (2009). The test for design effects and the models for floor and ceiling effects are those of Blair and Imai (2012). The recommendation to use negatively correlated control items, and sample-size formulas for the list experiment, are in Glynn (2013). The combined estimator for a design in which every list respondent also answers the direct question, its monotonicity assumption and its placebo tests are due to Aronow et al. (2015). Blair et al. (2014) combine list experiments with endorsement experiments in a joint regression framework, which is the closest precedent for the link to the measurement model in Section 5.5. Randomised response (Blair et al., 2015; Warner, 1965) reduces disclosure by a different mechanism and is not treated here. Our contributions are the characterisation of the designs that attain the direct-question benchmark (Proposition 5.1(b)); an explicit statement, building on Aronow et al. (2015), that in the paired design hidden and false endorsement are separately identified without monotonicity, together with the efficiency of their combined estimator (Theorem 5.3); the allocation rules and the comparative statics of the list choice (Theorem B.5); and the threshold-shift link to the item response model (Proposition 5.5).

5.1 The item count technique

Fix one sensitive statement; it is an item of the battery of Section 3, coded as agreement. Let $Z_i \in \{0, 1\}$ be respondent i's truthful agreement, the answer i would give if the answer could not be attributed to i, and let $\pi = \mathbb{P}(Z_i = 1)$. A *control list* consists of J non-sensitive statements with indicators $W_{i1}, \dots, W_{iJ} \in \{0, 1\}$ and count $W_i = \sum_{j=1}^{J} W_{ij}$. (In Sections 5.1 to 5.4, J is the number of control statements and j indexes them, not the battery items of Section 3; from Section 5.5 on, j again indexes items of the battery. Likewise, in this section and in Appendix B, W_i is a control count, B a budget, L a loss factor and π a prevalence; they are unrelated to the matrices $\boldsymbol{W}$, $\boldsymbol{B}$ and $\boldsymbol{L}$ of Section 4.) Respondents are randomised to three arms. In the *list-control* arm C, respondents see the J control statements. In the *list-treatment* arm T, they see the same statements plus the sensitive one. In both, they report only how many statements they agree with, Y_i. In the *direct* arm D, the sensitive statement is asked directly. We write $D_i \in \{0, 1\}$ for the direct answer, $\pi^D = \mathbb{P}(D_i = 1)$ for the direct prevalence, and

$$\mathrm{SDB} = \pi - \pi^D$$

for the item's social-desirability bias. Arm sizes are $n_{\mathsf{T}}, n_{\mathsf{C}}, n_{\mathsf{D}}$. We consider two designs. In the *split* design, list-arm respondents are never asked the sensitive statement directly. In the *paired* design, every respondent answers the direct question, list-arm respondents after the list. The plan of Section 8.4 is a split design with a paired subsample.

Assumption A4 (List experiment). Let $W_i(a)$ be the control count respondent i would report in arm $a \in \{\mathsf{C}, \mathsf{T}\}$, $\tilde{Z}_i$ the contribution of the sensitive statement to i's count in arm T, and $D_i(a)$ the direct answer i would give in arm a.

(i) *Sampling and randomisation.* Respondents are drawn independently from the target population, and arms are assigned by complete randomisation with fixed sizes, independently of all potential responses.

(ii) *No design effect.* $W_i(\mathsf{T}) = W_i(\mathsf{C}) =: W_i$.

(iii) *No liars.* $\tilde{Z}_i = Z_i$. Hence $Y_i = W_i$ in arm C and $Y_i = W_i + Z_i$ in arm T.

(iv) *Treatment independence (paired design).* $D_i(\mathsf{D}) = D_i(\mathsf{C}) = D_i(\mathsf{T}) =: D_i$, and asking the direct question does not change the list answer.

Parts (ii) and (iii) are the assumptions of Blair and Imai (2012); part (iv) is the treatment-independence assumption of Aronow et al. (2015). Under Assumption A4(i)–(iii) the difference in means

$$\widehat{\pi} = \overline{Y}_{\mathsf{T}} - \overline{Y}_{\mathsf{C}} \tag{5.1}$$

is unbiased for π, because randomisation makes the law of (W_i, Z_i) the same in both arms. If arms are assigned by independent draws, as in Section 8.4, every statement below holds conditionally on the realised arm sizes.

5.2 The variance cost of indirect questioning

Write $p_j = \mathbb{P}(W_{ij} = 1)$, $c_{jk} = \operatorname{Cov}(W_{ij}, W_{ik})$ and $\gamma_j = \operatorname{Cov}(W_{ij}, Z_i)$. The variance of the count in each list arm is

$$\sigma^2_{\mathsf{C}} = \operatorname{Var}(W_i) = \sum_{j=1}^{J} p_j(1-p_j) + \sum_{j \neq k} c_{jk}, \qquad \sigma^2_{\mathsf{T}} = \operatorname{Var}(W_i + Z_i) = \sigma^2_{\mathsf{C}} + \pi(1-\pi) + 2\sum_{j=1}^{J} \gamma_j. \tag{5.2}$$

Proposition 5.1 (Variance of the list experiment). *Under Assumption A4(i)–(iii), with $n_{\mathsf{T}}, n_{\mathsf{C}} \geq 1$, the estimator (5.1) is unbiased and its exact variance is*

$$\operatorname{Var}(\widehat{\pi}) = \frac{\sigma^2_{\mathsf{T}}}{n_{\mathsf{T}}} + \frac{\sigma^2_{\mathsf{C}}}{n_{\mathsf{C}}}. \tag{5.3}$$

Let $n = n_{\mathsf{T}} + n_{\mathsf{C}}$ and $r = n_{\mathsf{T}}/n$, and compare with the variance $\pi(1-\pi)/n$ of truthful direct questioning of n respondents.

(a) Efficiency loss. *The loss factor $L(r) = n \operatorname{Var}(\widehat{\pi})/\{\pi(1-\pi)\} = \{\sigma^2_{\mathsf{T}}/r + \sigma^2_{\mathsf{C}}/(1-r)\}/\{\pi(1-\pi)\}$ is minimised at $r^* = \sigma_{\mathsf{T}}/(\sigma_{\mathsf{T}} + \sigma_{\mathsf{C}})$, where*

$$L^* = \frac{(\sigma_{\mathsf{T}} + \sigma_{\mathsf{C}})^2}{\pi(1-\pi)}. \tag{5.4}$$

(b) Attaining the benchmark costs privacy. *If $0 < \pi < 1$, then $L^* \geq 1$, with equality if and only if W_i or $W_i + Z_i$ is almost surely constant, in which case the count in one of the two arms reveals Z_i. The converse fails: a count can reveal Z_i while L^* is large.*

(c) Correlation structure and list length. *L^* is strictly increasing in every c_{jk} and every γ_j. If $\sum_j \gamma_j \geq 0$, then $L^* \geq 1 + 4\sigma^2_{\mathsf{C}}/\{\pi(1-\pi)\}$. For exchangeable control items, with $p_j = p$ and $c_{jk} = \rho\, p(1-p)$,*

$$\sigma^2_{\mathsf{C}} = Jp(1-p)\{1 + (J-1)\rho\}, \tag{5.5}$$

where $\rho \geq -1/(J-1)$ is necessary but not sufficient for binary items (for $p = 1/2$ and odd J the sharp bound is $\rho \geq -1/J$). With uncorrelated items L^* grows like $4Jp(1-p)/\{\pi(1-\pi)\}$ for large J, and negative correlation among control items slows that growth.

The control items cost precision in two ways: through their own variance σ^2_{C}, which enters both arms, and through their covariance with the sensitive item, which enters the treatment arm. Part (b) says that no

choice of control items brings the cost down to that of direct questioning without making some answers fully revealing. It is a statement about the boundary case only, not a measure of disclosure. If W_i is uniform on $\{0, 2\}$ and independent of Z_i, with $\pi = 0.2$, the parity of the treatment count reveals Z_i, yet $L^* \approx 27$. If $W_i = 1 - Z_i + 2U_i$ with U_i an independent fair coin, the control count reveals Z_i, and again $L^* \approx 27$. Part (c) is the variance half of the case for negatively correlated control items made by Glynn (2013); Theorem B.5(d) adds the floor-and-ceiling half. For illustration, take $J = 4$ independent control items with $p_j = 1/2$, a sensitive item with $\pi = 0.2$ that is independent of them, and the optimal split. Then $\sigma^2_{\mathsf{C}} = 1$, $\sigma^2_{\mathsf{T}} = 1.16$ and $L^* \approx 27$: the list experiment needs about 27 times the sample of a truthful direct question for the same precision. For $J = 1, 2, 3$ the factor is 8.1, 14.4 and 20.7. Arranging the four control items as two pairs with within-pair correlation -0.6, and no correlation between pairs, gives $\sigma^2_{\mathsf{C}} = 0.4$ and $L^* \approx 12$. A finite-population version of (5.3) is in Remark B.1.

Remark 5.2 (When indirect questioning pays). The direct estimator $\overline{D}$ has bias $-$SDB and variance $\pi^D(1 - \pi^D)/n$; the list estimator is unbiased with variance $L^*\pi(1 - \pi)/n$. In mean squared error the list experiment is preferable exactly when $\mathrm{SDB}^2 > \{L^*\pi(1 - \pi) - \pi^D(1 - \pi^D)\}/n$. In the four-item illustration with $n = 2{,}000$ and π^D close to π, this requires $|\mathrm{SDB}|$ above about 0.046, and with $n = 6{,}000$ above about 0.026. This is the bias–variance trade-off discussed by Blair et al. (2020). It is why the plan of Section 8 treats the size of the bias as an empirical question for each item.

5.3 A joint model of direct and list responses

The difference $\pi - \pi^D$ is a net quantity. Some respondents who agree with the statement deny it when asked directly (*hidden endorsement*); others agree when asked directly although they would not agree in private (*false endorsement*), for example through acquiescence or inattentive responding. Define

$$\pi^H = \mathbb{P}(Z_i = 1, D_i = 0), \qquad \pi^F = \mathbb{P}(Z_i = 0, D_i = 1), \qquad \text{so that} \qquad \mathrm{SDB} = \pi^H - \pi^F.$$

For $d \in \{0, 1\}$ let $P_d = \mathbb{P}(D_i = d)$ and $\Delta_d = \mathbb{P}(Z_i = 1 \mid D_i = d)$. In the paired design we also need the variances of the list counts within strata of the direct answer, $v_{\mathsf{T}d} = \mathrm{Var}(W_i + Z_i \mid D_i = d)$ and $v_{\mathsf{C}d} = \mathrm{Var}(W_i \mid D_i = d)$. Means within strata are written $\overline{Y}_{ad}$ (arm a, direct answer d), and $\widehat{P}_d$ is the share of all $n = n_{\mathsf{D}} + n_{\mathsf{C}} + n_{\mathsf{T}}$ respondents who answer d to the direct question.

Assumption A5 (Monotonicity). $\mathbb{P}(Z_i = 0, D_i = 1) = 0$: no respondent agrees with the statement when asked directly unless they would agree in private.

Assumption A5 is the "no false confessions" assumption of Aronow et al. (2015). It is plausible when the social pressure on an item runs against agreement. It can fail through acquiescence, inattention or provocation. And under the social reference theory of Blair et al. (2020) the pressure may run in different directions in different groups, so the assumption should be justified item by item and, where possible, tested (Corollary B.3).

Theorem 5.3 (Joint identification and estimation of social-desirability bias). *Suppose Assumption A4(i)–(iii) hold.*

(a) Split design. *π, π^D and* SDB *are identified, and*

$$\widehat{\mathrm{SDB}}_{\mathrm{S}} = \overline{Y}_{\mathsf{T}} - \overline{Y}_{\mathsf{C}} - \overline{D}_{\mathsf{D}}, \qquad \mathrm{Var}\big(\widehat{\mathrm{SDB}}_{\mathrm{S}}\big) = \frac{\sigma^2_{\mathsf{T}}}{n_{\mathsf{T}}} + \frac{\sigma^2_{\mathsf{C}}}{n_{\mathsf{C}}} + \frac{\pi^D(1 - \pi^D)}{n_{\mathsf{D}}}, \tag{5.6}$$

is unbiased; the variance is exact. The hidden-endorsement rate is not identified: the sharp identified set for π^H *is*

$$\big[\max\{0, \pi - \pi^D\},\ \min\{\pi, 1 - \pi^D\}\big]. \tag{5.7}$$

Under Assumption A5 the set collapses to the point $\pi^H = \text{SDB}$.

(b) Paired design, without monotonicity. *If also Assumption A4(iv) holds, then for each* d *with* $P_d > 0$,

$$\Delta_d = \mathbb{E}(Y_i \mid \mathsf{T}, D_i = d) - \mathbb{E}(Y_i \mid \mathsf{C}, D_i = d), \tag{5.8}$$

so $\pi^H = P_0\Delta_0$, $\pi^F = P_1(1 - \Delta_1)$, $\pi = P_0\Delta_0 + P_1\Delta_1$ *and* SDB *are all identified, without Assumption A5. The post-stratified estimator* $\widehat{\text{SDB}}_{\text{PS}} = \sum_d \widehat{P}_d(\overline{Y}_{\mathsf{T}d} - \overline{Y}_{\mathsf{C}d}) - \widehat{P}_1$ *is consistent. If* $n_a/n \to r_a > 0$ *for* $a \in \{\mathsf{C}, \mathsf{T}\}$, *it is asymptotically normal with*

$$n\,\text{AVar}\big(\widehat{\text{SDB}}_{\text{PS}}\big) = (1 + \Delta_0 - \Delta_1)^2 P_0 P_1 + \sum_{d\in\{0,1\}} P_d\Big(\frac{v_{\mathsf{T}d}}{r_{\mathsf{T}}} + \frac{v_{\mathsf{C}d}}{r_{\mathsf{C}}}\Big). \tag{5.9}$$

(c) Paired design with monotonicity: the combined estimator. *If Assumption A4(iv) and Assumption A5 hold, then* $\Delta_1 = 1$, $\pi^F = 0$ *and* $\text{SDB} = \pi^H = P_0\Delta_0$. *Let*

$$\widehat{\pi}^H = \widehat{P}_0\,\big(\overline{Y}_{\mathsf{T}0} - \overline{Y}_{\mathsf{C}0}\big), \tag{5.10}$$

set to 0 *if no respondent in* T *or in* C *answers* $D_i = 0$. *Then:*

(i) $|\mathbb{E}\widehat{\pi}^H - \pi^H| \le P_1^{n_{\mathsf{T}}} + P_1^{n_{\mathsf{C}}}$;

(ii) if $n_a/n \to r_a > 0$ *for* $a \in \{\mathsf{C}, \mathsf{T}\}$, *then* $\sqrt{n}(\widehat{\pi}^H - \pi^H) \Rightarrow \text{N}(0, V_H)$ *with*

$$V_H = \Delta_0^2 P_0 P_1 + P_0\Big(\frac{v_{\mathsf{T}0}}{r_{\mathsf{T}}} + \frac{v_{\mathsf{C}0}}{r_{\mathsf{C}}}\Big); \tag{5.11}$$

(iii) with probability tending to one, $\widehat{\pi}^H$ *is the maximum likelihood estimator of* π^H *in the model defined by Assumptions A4 and A5. At every interior point of that model (all probabilities* $\mathbb{P}(W_i = w, Z_i = z, D_i = d)$ *that the model allows are positive),* V_H *is the Cramér–Rao bound, and no regular estimator has smaller asymptotic variance;*

(iv) imposing monotonicity gains $n\,\text{AVar}(\widehat{\text{SDB}}_{\text{PS}}) - V_H = P_1 v_{\mathsf{C}1}(1/r_{\mathsf{T}} + 1/r_{\mathsf{C}}) \ge 0$.

(d) What fails without monotonicity. *In the paired design under Assumption A4(i)–(iv) alone,* $\widehat{\pi}^H \to \pi^H = \text{SDB} + \pi^F$ *in probability. As an estimator of* SDB *it is therefore biased upward by the false-endorsement rate* π^F. *The combined prevalence estimator* $\widehat{P}_1 + \widehat{\pi}^H$ *of Aronow et al. (2015) converges to* $\pi + \pi^F$. *The post-stratified estimator of part (b) remains consistent. In the split design, Assumption A5 is what turns the interval* (5.7) *into a point.*

Three features of the theorem shape the design of Section 8. First, SDB itself never needs monotonicity; only its decomposition into hidden and false endorsement does, and only in the split design. Second, pairing is what identifies the decomposition: within each stratum of the direct answer the list experiment measures the truthful prevalence, so the share of direct deniers who agree in private (Δ_0) and the share of direct endorsers who do not ($1 - \Delta_1$) are separately estimable. Both ingredients are in Aronow et al. (2015): Δ_1 is the target of their first placebo test, and $P_0\Delta_0$ is the second term of their combined estimator. Part (b) makes explicit that, read this way, they identify π^H and π^F without monotonicity. Third, under

monotonicity, respondents who endorse directly contribute no list variance to $\widehat{\pi}^{H}$, because their treatment–control contrast is known to equal one. This is the source of the efficiency gain of Aronow et al. (2015); part (c)(iii) shows that no regular estimator does better with the same data. The assumptions have testable implications, among them the design-effect test of Blair and Imai (2012) and the placebo tests of Aronow et al. (2015) (Corollary B.3); passing them is necessary for the assumptions, not sufficient. The theorem holds conditionally on discrete covariates, and a parametric joint model in the spirit of Imai (2011) is in Remark B.4. If social pressure on an item runs towards agreement for some respondents, Assumption A5 is replaced by its mirror image, and the roles of hidden and false endorsement are exchanged (Remark B.4).

5.4 Optimal design

Theorem B.5, stated with its proof in Appendix B.4, allocates a budget across the arms and chooses the control list, with floor and ceiling effects (Blair and Imai, 2012) modelled by the probabilities ϕ_{c} and ϕ_{f} that a treated respondent with $Z_i = 1$ suppresses a count at the ceiling or the floor. Four messages shape the plan. In the split design the optimal allocation is of Neyman type, with arm sizes proportional to the square root of each arm's variance divided by its cost. In the paired design a direct-only arm hardly ever pays for precision, because its contribution is second order in the hidden-endorsement rate; it is fielded only to test treatment independence. Larger budgets select control lists that are noisier but less exposed to floor and ceiling effects. And negatively correlated control items lower both the variance and the exposure $e_\ell = \mathbb{P}(W_i \in \{0, J\})$ of a list, but can make the count more informative about the sensitive answer. With $\pi = 0.2$, the two negatively correlated pairs of Section 5.2 cut L^* from about 27 to 12, but they double the mutual information between the treatment count and Z_i, from 0.11 to 0.24 bits, and raise the share of treated respondents whose posterior $\mathbb{P}(Z_i = 1 \mid Y_i, \mathsf{T})$ is at least 0.5 from 11% to 30%. Neither the variance of the control count nor its exposure measures disclosure. The plan therefore bounds two disclosure measures directly, $I(Y_i; Z_i \mid \mathsf{T})$ and that posterior share (Section 8.4). Pairing every list respondent with the direct question is cheaper still (Corollary B.6), but it risks Assumption A4(iv), which is why the plan keeps the split design as primary.

5.5 Linking the list experiment to the measurement model

The list experiment measures a single binary indicator; the measurement model of Section 3 describes the graded responses to all items. We connect them by treating social desirability as a shift of the sensitive item's thresholds between private and direct answering. Let item j be sensitive, with systematic part $\eta_{ij} = \boldsymbol{a}_j^{\top}\boldsymbol{\theta}_i = a_{jG}\theta_{iG} + a_{js}\theta_{is}$ and latent response $Y^*_{ij} = \eta_{ij} + \varepsilon_{ij}$ as in (3.4), and let the list statement correspond to agreement, the event that the response reaches category c^*.

Definition 5.4 (Threshold-shift model of social desirability). For a shift $\delta_j \geq 0$, the *private* response, the one a respondent gives when the answer cannot be attributed to them, is $\tilde{Y}_{ij} = \sum_{c=1}^{C_j} \mathbf{1}\left\{Y^*_{ij} > b_{jc}\right\}$, and the *direct* response is $Y_{ij} = \sum_{c=1}^{C_j} \mathbf{1}\left\{Y^*_{ij} > b_{jc} + \delta_j\right\}$, with the same latent response Y^*_{ij} and a link F as in (3.2). The list experiment measures $Z_{ij} = \mathbf{1}\left\{\tilde{Y}_{ij} \geq c^*\right\}$, and $D_{ij} = \mathbf{1}\left\{Y_{ij} \geq c^*\right\}$.

The same latent response drives both answers; only the cut-points differ. In the language of Section 4, social desirability is *uniform differential item functioning between modes* of administration, private and direct.

Proposition 5.5 (Correcting the threshold of a sensitive item). *Under Definition 5.4:*

(a) The direct responses follow the graded response model of Section 3 with loadings (a_{jG}, a_{js}) and thresholds $b^D_{jc} = b_{jc} + \delta_j$. Assumption A5 holds, $\mathbb{P}(Z_{ij} = 1 \mid \boldsymbol{\theta}_i) = F(\eta_{ij} - b_{jc^*})$, and the hidden-endorsement curve is

$$\pi^H_j(\boldsymbol{\theta}_i) = F(\eta_{ij} - b_{jc^*}) - F(\eta_{ij} - b_{jc^*} - \delta_j), \qquad \mathrm{SDB}_j = \mathbb{E}\,\pi^H_j(\boldsymbol{\theta}_i). \tag{5.12}$$

(b) Identification. *Suppose the parameters of the measurement model for the direct responses, including the law of $\boldsymbol{\theta}_i$ (given X_i, if covariates are used), are identified (Theorem 4.2), and a list experiment for item j satisfies Assumption A4(i)–(iii). Then δ_j is identified as the unique root of*

$$\mathbb{E}\,F\big(\eta_{ij} - b^D_{jc^*} + \delta\big) = \pi_j, \tag{5.13}$$

and so are the private thresholds $b_{jc} = b^D_{jc} - \delta_j$, the curve $\pi^H_j(\cdot)$, and $\mathrm{SDB}_j(x)$ for every covariate value x. If the root is negative, Definition 5.4 is rejected.

(c) Boundary estimands. *If δ_j is common to all groups, the direct item is invariant whenever the private item is, and ECV_j, $\mathrm{ECV}^{(g)}_j$ and the boundary sets of Definition 3.1 are the same whether computed from private or direct responses. If δ_j differs between groups, the direct item has uniform DIF of size $\delta^{(g)}_j - \delta^{(g')}_j$. Ignoring it has the consequences for latent means and group-specific estimands given in Theorem 4.5, and a list experiment in each group identifies $\delta^{(g)}_j$ whenever the group's latent distribution is identified (Proposition 4.6).*

Part (c) is the substantive point. A uniform shift common to all groups moves levels, not structure: it biases prevalences, and differences in prevalence between groups, but leaves latent means and the boundary estimands of Definition 3.1 unchanged. A shift that differs between groups is uniform DIF, and its effects are those of Theorem 4.5. The structural estimands are threatened only by a shift that acts differently on the general and specific parts of η_{ij}, for example pressure that suppresses the part of agreement that covaries with the stereotype items but not the part specific to the Israel domain. Separating such a shift requires two slopes, one for each part of η_{ij}. Without covariates a list experiment identifies a single scalar per item. With covariates the uniform shift is overidentified, and one common rescaling of η_{ij} is identified (Proposition B.7), but separate rescalings would need covariates whose associations with θ_{iG} and θ_{is} differ sharply. We do not pursue this and flag it as a limitation.

For estimation, the list responses enter the likelihood of the measurement model. For respondent i in a list arm,

$$L_i = \int \Big\{\prod_{k \neq j} f(y_{ik} \mid \boldsymbol{\theta})\Big\} \times \sum_{z \in \{0,1\}} q_z\big(y_i - z\mathbf{1}\{A_i = \mathsf{T}\}; x_i, \psi\big)\, F\{\eta_j(\boldsymbol{\theta}) - b_{jc^*}\}^z \big[1 - F\{\eta_j(\boldsymbol{\theta}) - b_{jc^*}\}\big]^{1-z}\, \Pi(\mathrm{d}\boldsymbol{\theta} \mid x_i), \tag{5.14}$$

where $f(y_{ik} \mid \boldsymbol{\theta})$ are the graded-response probabilities of the other items, A_i is the arm, q_z is the control-count model of (B.2), $\eta_j(\boldsymbol{\theta}) = a_{jG}\theta_G + a_{js}\theta_s$, and $\Pi(\cdot \mid x)$ is the latent distribution given covariates. Direct-arm respondents contribute the usual likelihood with thresholds $b_{jc} + \delta_j$. Equation (5.14) assumes that the control count is independent of $\boldsymbol{\theta}_i$ given (Z_{ij}, X_i); part (b) does not need this, because it uses only the first moment of the list response. With several sensitive items, the shifts δ_j can be given a common hierarchical prior, which borrows strength across items when each list experiment alone is imprecise.

6 Small-area estimation and the fragility of stereotype indices

National estimates of the latent factors say little about variation between states, regions or campuses, and area samples are usually too small for direct estimation. This section adapts multilevel regression and poststratification (MRP) to the latent traits of Section 3 (Section 6.1), shows why the obvious two-step shortcut, poststratifying individual trait scores, is biased (Proposition 6.2), and then turns to a simpler and widely used summary, the share of respondents who agree with at least a given number of stereotype statements. Proposition 6.3 states when group differences in such a share have the same sign as differences in the latent trait, and gives constructions in which they do not. Proofs, and the results for non-invariant items, are in Appendix B.

MRP was introduced by Gelman and Little (1997) and developed for state-level opinion by Park et al. (2004). Comparisons with disaggregation (Lax and Phillips, 2009), models with deep interactions (Ghitza and Gelman, 2013), applications to non-probability samples (Wang et al., 2015) and the relation between weighting and regression (Gelman, 2007) shape the choices below. Estimating latent-trait distributions in subpopulations from item responses, rather than from individual scores, is the logic of the plausible-values methodology of Mislevy (1991).

6.1 MRP on the latent trait

Let the population be partitioned into cells $k \in \mathcal{K}$, cross-classifications of demographic variables and an area $r(k)$, with known population counts N_k. For an area r, let $\mathcal{K}_r = \{k : r(k) = r\}$, $N_r = \sum_{k \in \mathcal{K}_r} N_k$ and $w_k = N_k / N_r$. The measurement model is the graded response model of Section 3, with item parameters common to all cells (or partially invariant, with DIF modelled as in Section 4). The structural model is a multilevel latent regression that replaces the group distributions (3.3) by cell distributions: for respondent i in cell $k[i]$,

$$\boldsymbol{\theta}_i \mid k[i] = k \;\sim\; \mathrm{N}_3(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \qquad \mu_{kf} = x_k^{\top}\beta_f + \sum_{u=1}^{U_f} \alpha^{(u)}_{f,u[k]} + z_{r(k)}^{\top}\gamma_f, \qquad \alpha^{(u)}_{f,\cdot} \sim \mathrm{N}(0, \omega^2_{fu}), \tag{6.1}$$

for each factor $f \in \{G, S, I\}$. Here x_k are cell-level fixed effects, the $\alpha^{(u)}$ are batches of exchangeable random effects (area, demographic groups, and their interactions, following Ghitza and Gelman, 2013), z_r are area-level predictors (Park et al., 2004), and $\boldsymbol{\Sigma}_k$ is the within-cell covariance, diagonal in the bifactor model and subject to the normalisations of Section 3. The item responses and (6.1) are fitted jointly, so the measurement uncertainty of each $\boldsymbol{\theta}_i$ is propagated into the cell parameters.

The poststratified area mean of the latent traits is

$$\bar{\boldsymbol{\theta}}_r = \frac{\sum_{k \in \mathcal{K}_r} N_k \, \mathbb{E}[\boldsymbol{\theta}_i \mid k]}{\sum_{k \in \mathcal{K}_r} N_k} = \sum_{k \in \mathcal{K}_r} w_k \boldsymbol{\mu}_k. \tag{6.2}$$

The same weights carry every estimand that is linear in the cell distributions. For a function φ, $\mathbb{E}_r\varphi(\boldsymbol{\theta}) = \sum_k w_k \mathbb{E}[\varphi(\boldsymbol{\theta}_i) \mid k]$; for example, the share of area r above a pre-registered cut-off t on G is $\sum_k w_k \Phi\{(\mu_{kG} - t)/\Sigma^{1/2}_{k,GG}\}$. The area covariance of the latent traits is

$$\boldsymbol{\Sigma}_r = \sum_{k \in \mathcal{K}_r} w_k \left\{ \boldsymbol{\Sigma}_k + (\boldsymbol{\mu}_k - \bar{\boldsymbol{\theta}}_r)(\boldsymbol{\mu}_k - \bar{\boldsymbol{\theta}}_r)^{\top} \right\}. \tag{6.3}$$

Inference proceeds draw by draw. Each posterior draw $(\boldsymbol{\mu}_k^{(s)}, \boldsymbol{\Sigma}_k^{(s)})_k$ gives a draw $\bar{\boldsymbol{\theta}}_r^{(s)}$ of (6.2), and equal-tailed quantiles of the draws give credible intervals. Nonlinear area estimands are computed per draw

from (6.2) and (6.3), never by plugging posterior means into a nonlinear function. Uncertainty in the N_k, for example when they come from survey-based population tables, is propagated by drawing the N_k jointly with the model parameters.

Remark 6.1 (Estimands and the finite population). For a finite population, the realised area mean of $\boldsymbol{\theta}$ equals $\sum_k w_k \bar{\boldsymbol{\theta}}_k^{\text{pop}}$ exactly, where $\bar{\boldsymbol{\theta}}_k^{\text{pop}}$ is the realised mean in cell k; (6.2) replaces $\bar{\boldsymbol{\theta}}_k^{\text{pop}}$ by its superpopulation expectation $\boldsymbol{\mu}_k$. The difference has standard deviation of order $N_r^{-1/2}$ and is negligible for census-scale cells. The same identity, with realised cell covariances, gives (6.3).

Area-level boundary estimands. Under measurement invariance the loadings, and hence ECV_j, are common to all areas. What varies between areas is the distribution of the latent traits. Two area-level summaries are informative. The first is the latent correlation $\rho_{SI}^{(r)}$ of the correlated-factors model (3.7), obtained from the area covariance of (ζ_S, ζ_I) computed as in (6.3). The second is the area analogue of the population-specific index $\text{ECV}_j^{(g)}$ of Definition 3.1,

$$\text{ECV}_j^{(r)} = \frac{a_{jG}^2 \, \Sigma_{r,GG}}{a_{jG}^2 \, \Sigma_{r,GG} + a_{js}^2 \, \Sigma_{r,s(j)s(j)}}, \tag{6.4}$$

which equals ECV_j when both factor variances in the area are one. By (6.3), $\boldsymbol{\Sigma}_r$ has a between-cell component even when the factors are uncorrelated within cells, so the off-diagonal entries of $\boldsymbol{\Sigma}_r$ need not vanish, and (6.4) leaves them out by convention. Area-level factor covariances therefore mix individual-level structure with the composition of the area. They should not be read as statements about how attitudes are organised within individuals (Section 3.5). If loadings may differ between areas, area-specific loadings require a multiple-group model with partial invariance (Proposition 4.6). This is feasible only for large areas, which is why the plan of Section 8 estimates boundary estimands by region only.

Social-desirability correction in small areas. List experiments are too imprecise to estimate SDB_j area by area (Proposition 5.1). Under Proposition 5.5, the national or regional shift δ_j enters the item's thresholds, and the poststratified prevalence of private agreement is $\sum_k w_k \mathbb{E}\{F(\eta_{ij} - b_{jc^*}) \mid k\}$, with $\eta_{ij} = \boldsymbol{a}_j^\top \boldsymbol{\theta}_i$. The assumption that δ_j does not vary between areas within a region is untestable at the area level and is reported as such.

6.2 Scoring first and poststratifying later attenuates differences

A common shortcut is to compute a score $\tilde{\theta}_i$ for each respondent, typically the posterior mean (EAP) under a single population prior, and to run MRP on the scores. The next result shows that this attenuates area differences.

Proposition 6.2 (Two-step attenuation). *Suppose $\theta_i \mid k \sim \text{N}(\mu_k, \sigma^2)$, that the items yield a statistic $\hat{\theta}_i$ with $\hat{\theta}_i \mid \theta_i \sim \text{N}(\theta_i, s^2)$, and that scores are computed as posterior means under the common prior $\text{N}(m_0, \sigma_0^2)$. Then*

$$\mathbb{E}[\tilde{\theta}_i \mid k] = m_0 + \varrho \, (\mu_k - m_0), \qquad \varrho = \frac{\sigma_0^2}{\sigma_0^2 + s^2} < 1, \tag{6.5}$$

so the two-step estimator of $\bar{\theta}_r$ converges to $m_0 + \varrho(\bar{\theta}_r - m_0)$, where $\bar{\theta}_r$ is the scalar version of (6.2), and every difference between areas is shrunk by the factor ϱ, at every cell size n_k. In the joint model (6.1) the posterior mean of μ_k is consistent as $n_k \to \infty$.

The Gaussian setting makes the factor explicit; in the graded response model the information, and hence the attenuation, varies with θ, so the bias is not even a constant multiple. The joint model also shrinks small cells and areas towards the regression prediction through partial pooling, but that shrinkage reflects the data available for each cell and vanishes as n_k grows. The two-step attenuation does not vanish: it persists however large the cells are. Including the cell or area in the scoring prior removes the bias for the corresponding contrasts. Fitting the latent regression jointly does this for all of them, and is the approach of Mislevy (1991).

6.3 When do count indices compare groups correctly?

Let $U_{ij} \in \{0, 1\}$ indicate agreement with stereotype statement $j = 1, \dots, M$, let $U_{i+} = \sum_j U_{ij}$ be the number of statements respondent i agrees with, and consider the index $\mathbf{1}\{U_{i+} \geq q\}$, "agrees with at least q of M statements". For a group g, let $P_g(q) = \mathbb{P}(U_{i+} \geq q \mid g)$. We use a unidimensional trait θ_i with law Γ_g and mean μ_g in group g; the multidimensional case is discussed after the proposition. The items follow

$$\mathbb{P}(U_{ij} = 1 \mid \theta_i, g) = F\big(a_j^{(g)}\theta_i - b_j^{(g)}\big), \tag{6.6}$$

with local independence given (θ_i, g) and F a continuous, strictly increasing distribution function. *Full invariance* means $a_j^{(1)} = a_j^{(2)}$ and $b_j^{(1)} = b_j^{(2)}$ for all j. A *location family* means $\Gamma_g(t) = \Gamma_0(t - \mu_g)$ for a common Γ_0. We write $\Gamma_1 \geq_{\mathrm{st}} \Gamma_2$ for first-order stochastic dominance, which holds, for instance, when the two laws have a monotone likelihood ratio.

Proposition 6.3 (Count indices under invariance). *(a)* Sufficient condition for sign agreement. *Under full invariance with all $a_j > 0$, if $\Gamma_1 \geq_{\mathrm{st}} \Gamma_2$ then $P_1(q) \geq P_2(q)$ for every $q \in \{1, \dots, M\}$, with strict inequality for every such q if $\Gamma_1 \neq \Gamma_2$. In a location family,* $\operatorname{sign}\{P_1(q) - P_2(q)\} = \operatorname{sign}(\mu_1 - \mu_2)$ *for every q.*

(b) Mean ordering alone is not enough. *Under full invariance, let $\Gamma_g = \mathrm{N}(\mu_g, \sigma_g^2)$ with $\mu_1 > \mu_2$ and $\sigma_1 < \sigma_2$. For every $q \in \{1, \dots, M\}$ there are invariant item parameters with $P_1(q) < P_2(q)$, and others with $P_1(q) > P_2(q)$. Specifically, let all statements have a common slope a and common location $t \neq t_\times = (\mu_1\sigma_2 - \mu_2\sigma_1)/(\sigma_2 - \sigma_1)$, so that $b_j = at$. Then, for all sufficiently large a, $P_1(q) < P_2(q)$ if $t > t_\times$ and $P_1(q) > P_2(q)$ if $t < t_\times$. For moderate slopes the sign can also vary with q.*

Part (a) is the positive result. Invariance and stochastic dominance together guarantee that the whole distribution of the count is ordered, $U_{i+} \mid 1 \geq_{\mathrm{st}} U_{i+} \mid 2$, so every threshold q ranks the groups the same way. Part (b) shows that invariance alone is not enough: the index compares tails, not means, and a group with a higher mean but less dispersion can have fewer members above a high threshold. When invariance fails, the index can mislead further (Proposition B.8 in Appendix B.6). A uniform shift of all thresholds by $a_j(\mu_1 - \mu_2)$ is observationally equivalent to equal latent means, the index counterpart of Theorem 4.5(d), and slightly larger shifts reverse the sign of every index comparison. Differences in discrimination alone, with identical thresholds, can make the low-threshold and high-threshold versions of the index disagree about which group has more agreement. And because the index is nonlinear in the latent mean, two groups starting from different baselines can have their latent changes ranked in the wrong order even under full invariance; Example B.9 gives numerical instances. These are properties of any index of this form. We make no claim about the results of any particular index or survey, and the constructions are not calibrated to any. Whether an index's group comparisons are affected is an empirical question about its items in the groups compared, and it is answered by testing invariance (Sections 4 and 8).

Several factors. In the bifactor model a stereotype statement loads on G and on S. If all loadings are non-negative and the latent vectors are ordered by the usual multivariate stochastic order, Proposition 6.3(a) extends with the same proof, since $\mathbb{P}(U_{i+} \geq q \mid \boldsymbol{\theta})$ is then non-decreasing in each coordinate of $\boldsymbol{\theta}$. Without such an ordering, an index of stereotype statements responds to G and S together. Two groups can then differ in the index because they differ in the specific stereotype factor, with no difference in G. This matters for the boundary question: an index built from classic stereotypes is not, by itself, a measure of the general factor.

6.4 Practical guidance

(1) *Model the latent trait jointly.* Fit the measurement model and the multilevel latent regression (6.1) together and poststratify draw by draw (Equation (6.2)). Do not poststratify individual scores (Proposition 6.2).

(2) *Test invariance across the cells and areas being compared.* Poststratification averages cell means; if items function differently across cells, the cell means are on different scales. Use the DIF screens and anchors of Section 4, or model DIF explicitly.

(3) *Poststratify the estimand of interest.* Means, tail shares and covariances each have their own poststratification (Equations (6.2) and (6.3)). Report cut-off-based shares only for pre-registered cut-offs. Describe them as shares of a population above a point on a latent scale, not as counts of a type of person.

(4) *Check the model where data allow.* Compare MRP with design-based estimates in areas large enough to support both, and report the posterior shrinkage of each area towards the model prediction.

(5) *Treat count indices as descriptive.* Report the whole distribution of U_{i+} by group, not one threshold. Crossing survival curves, $\mathbb{P}(U_{i+} \geq q \mid 1) - \mathbb{P}(U_{i+} \geq q \mid 2)$ changing sign in q, refute the conjunction of invariance and stochastic dominance (Proposition 6.3(a)). Report index comparisons between groups only alongside invariance diagnostics, and do not read differences in index changes between groups with different baselines as differences in latent change (Proposition B.8(e)).

7 Simulation study

All data in this section are synthetic. Every dataset is drawn from a model whose parameters we chose, so the truth is known. No survey responses are used, and nothing here is a finding about the attitudes of any population. Every number is taken from `results/sim_results.json`, which `code/run_experiments.py` writes. Appendix C gives the design table (Table 9), configurations, seeds, run times, code hashes, Monte Carlo standard errors (MCSE) and further results. We label the experiments X1 to X9. The first five were called E1 to E5 in an earlier version; we renamed them because those labels collided with the estimands (E1)–(E9) of the plan (Section 8.6). Their seeds and results are unchanged.

7.1 Design and estimation

X1 tests identification and recovery (Theorem 4.2 and Propositions 3.2 and 4.8) and X9 the heterogeneity test of rule D2(ii). X2 tests DIF and anchors (Theorem 4.5 and Proposition 4.6), X2b DIF on the stereotype anchors (Proposition 4.6(c), Remark 4.7) and X7 the revised DIF screen of the plan. X6 adds two misspecifications that the plan must guard against. X3 tests list experiments (Proposition 5.1

and Theorems 5.3 and B.5) and X8 the identification of the threshold shift (Proposition 5.5). X4 tests small-area estimation (Equation (6.2) and Proposition 6.2) and X5 count indices (Propositions 6.3 and B.8). X1, X2, X2b and X7 generate data from the normal-ogive bifactor graded response model of Section 3 with five categories. The eight stereotype items have item-level ECVs between 0.45 and 0.90 and communalities between 0.50 and 0.60, so the stereotype domain satisfies Assumption A3. The Israel-related items I1, . . . , I6 have ECVs 0.25, 0.75, 0.40, 0.60, 0.15 and 0.85 and communality 0.5; a design with $|\mathcal{I}|$ items uses the first $|\mathcal{I}|$. With $\tau = 0.5$ the true boundary set is {I2} for $|\mathcal{I}| \in \{2, 3\}$ and {I2, I4, I6} for $|\mathcal{I}| = 6$. X6 and X9 use the registered sizes of Table 6: 10 stereotype items, 8 ID items, 6 IT items and $n = 6{,}000$.

Estimation. All fits use the limited-information estimator that the plan adopts as primary (Section 8.7). Thresholds and polychoric correlations are estimated in two steps (Olsson, 1979), with an asymptotic covariance matrix $\mathbf{\Gamma}$ from per-respondent influence functions. The bifactor model is fitted by weighted least squares with the diagonal weights $\operatorname{diag}(\mathbf{\Gamma})^{-1}$, with sandwich standard errors, from a default start and random starts. For each ECV_j we form a logit-scale Wald interval with the sandwich standard error, and in X1 also a percentile bootstrap interval from $B = 100$ resamples. Under the normal-ogive model this estimator targets the same parameters as maximum likelihood (Lemma 4.1), but its finite-sample behaviour can differ (Section 7.7).

7.2 Recovery and identification (X1, X9)

X1 compares five designs with $n \in \{500, 1000, 3000\}$. The designs with $|\mathcal{I}| = 6$ and $|\mathcal{I}| = 3$ are identified by Theorem 4.2(a). The design with $|\mathcal{I}| = 2$ is the case of Theorem 4.2(c): its two ECVs, 0.25 and 0.75, have the identified sets $(0.125, 0.625)$ and $(0.375, 0.875)$ of (4.3), and both contain τ. In the *proportional* design every stereotype item has ECV 0.70 and $|\mathcal{I}| = 3$, so Theorem 4.2(b) applies. The *near-proportional* design has stereotype ECVs between 0.65 and 0.75 and $|\mathcal{I}| = 3$; it is identified, but weakly (Remark A.5). The *identification diagnostic* is the ratio of the smallest to the largest singular value of the weighted Jacobian of the fitted correlations, zero when the model is not locally identified. At the population values it is 0.094 and 0.068 in the identified designs, 0.0060 in the near-proportional design and zero to machine precision in the other two. Table 3 and Figure 3 give the results. With $R = 40$, as in the proportional and near-proportional designs, the MCSE of a coverage or a classification rate is about 7 percentage points (4.7 to 7.9 for the item-level and set coverages), so differences of less than about 15 points between those cells are not resolved.

Identified designs. With $|\mathcal{I}| = 6$ the pooled bias of $\widehat{\mathrm{ECV}}_j$ is at most 0.006 and the RMSE falls from 0.082 to 0.035 as n grows from 500 to 3,000. With $|\mathcal{I}| = 3$ the RMSE is 0.141, 0.074 and 0.046; the bias of +0.038 at $n = 500$ comes with fits at a slope bound in 6% of replications. Wald intervals cover in 93.2 to 96.3% of cases and bootstrap intervals in 91.0 to 97.0%. No item is classified on the wrong side of τ, except in 0.2% of cases for $|\mathcal{I}| = 6$ at $n = 500$, and the share classified correctly rises from 66.2% to 91.0% with $|\mathcal{I}| = 6$ and from 39.0% to 77.0% with $|\mathcal{I}| = 3$. Items near τ stay undetermined longest: at $n = 3{,}000$ with $|\mathcal{I}| = 6$, I3 (ECV 0.40) and I4 (0.60) are classified correctly in 67% and 79% of replications and every other item in 100%. Comparing point estimates with τ instead would misclassify 5.2%, 2.7% and 0.2% of the items, with no indication of which (Proposition 4.8(c)). With Bonferroni-adjusted Wald intervals, the rule of the plan, the inner and outer sets bracket $\mathcal{B}_{0.5}$ in 100% of replications with $|\mathcal{I}| = 6$ and in 98 to 100% with $|\mathcal{I}| = 3$ (column Set), at the price of more undetermined items (43.2%, 65.0% and 82.3% correct with $|\mathcal{I}| = 6$).

Table 3. X1, recovery of the boundary estimands (synthetic data; $R = 100$ for $|\mathcal{I}| = 6, 3$, 50 for $|\mathcal{I}| = 2$, 40 otherwise, where the MCSE of a coverage is about 7 points). Bias and RMSE of $\widehat{\mathrm{ECV}}_j$ are pooled over $j \in \mathcal{I}$. Cov. B and Cov. W: coverage (%) of nominal 95% percentile-bootstrap ($B = 100$) and Wald intervals (logit scale, sandwich standard errors). Corr., Undet., Wrong: per-item classification (%) as in, out or undetermined from the bootstrap interval, against the true $\mathcal{B}_{0.5}$. Set: probability (%) that $\widehat{\mathcal{B}}^{\mathrm{in}}_{0.5} \subseteq \mathcal{B}_{0.5} \subseteq \widehat{\mathcal{B}}^{\mathrm{out}}_{0.5}$ with Bonferroni-adjusted Wald intervals (Proposition 4.8(b)). $\hat{\rho}_{SI}$: correlated-factors estimate against its pseudo-true value; Cov.: sandwich interval. †Not identified (Theorem 4.2(b),(c)); the entries describe the optimiser, not the estimand.

		$\widehat{\mathrm{ECV}}_j$, $j \in \mathcal{I}$				Boundary set $\mathcal{B}_{0.5}$ (%)				$\hat{\rho}_{SI}$	
Design	n	Bias	RMSE	Cov. B	Cov. W	Corr.	Undet.	Wrong	Set	RMSE	Cov.
$\|\mathcal{I}\| = 6$	500	+0.005	0.082	95	95	66	34	0	100	0.034	96
	1000	+0.006	0.062	91	93	73	27	0	100	0.028	94
	3000	+0.004	0.035	92	96	91	9	0	100	0.016	94
$\|\mathcal{I}\| = 3$	500	+0.038	0.141	95	94	39	61	0	98	0.042	97
	1000	+0.002	0.074	97	96	57	43	0	100	0.030	96
	3000	+0.004	0.046	95	96	77	23	0	100	0.019	95
$\|\mathcal{I}\| = 3$, near-prop. S	500	−0.005	0.202	92	73	38	62	0	88	0.044	90
	1000	+0.020	0.192	92	57	48	52	0	72	0.034	92
	3000	+0.023	0.168	88	58	63	35	2	72	0.021	90
$\|\mathcal{I}\| = 2$†	500	+0.032	0.177	89	81	45	55	0	98	0.060	96
	1000	+0.021	0.105	89	86	73	27	0	100	0.035	94
	3000	+0.001	0.106	65	62	88	12	0	98	0.028	90
$\|\mathcal{I}\| = 3$, prop. S†	500	−0.037	0.182	91	71	44	56	0	90	0.044	95
	1000	−0.050	0.180	91	54	40	59	1	88	0.028	98
	3000	−0.047	0.168	60	25	70	28	2	82	0.017	95

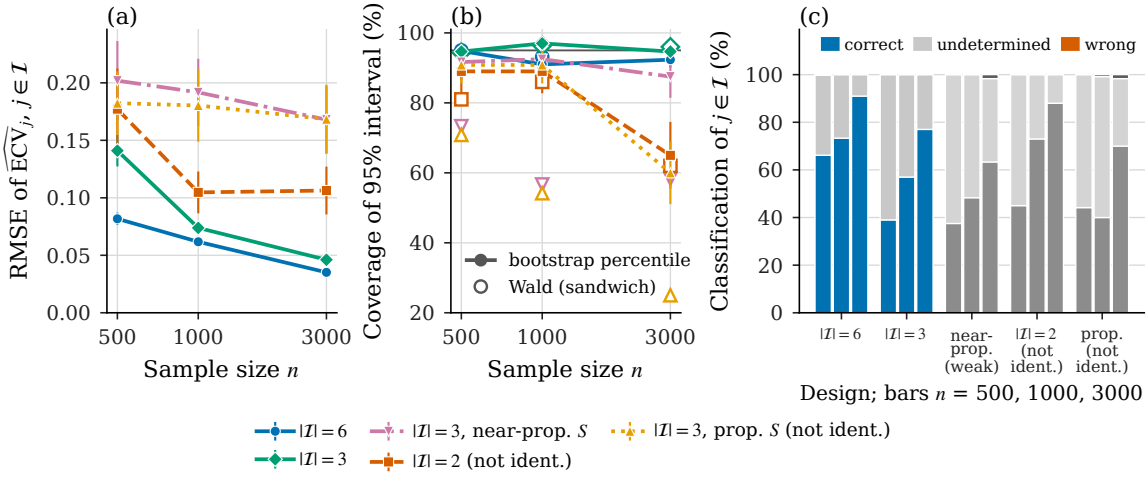


Figure 3. X1, recovery of the item-level ECV and of the boundary set $\mathcal{B}_{0.5}$ (synthetic data). (a) RMSE of $\widehat{\mathrm{ECV}}_j$, pooled over the Israel-related items (±2 MCSE bars). (b) Coverage of nominal 95% intervals: percentile bootstrap with $B = 100$ (filled markers) and Wald intervals on the logit scale with sandwich standard errors (open markers, "Wald (sandwich)"). (c) Classification of each Israel-related item from its bootstrap interval, against the true boundary set; bars are $n = 500$, 1,000 and 3,000. The designs in grey are not identified ($|\mathcal{I}| = 2$ and proportional S; Theorem 4.2(b),(c)) or only weakly identified (near-proportional S). Their "correct" shares, up to 88%, describe where the optimiser stops, not information in the data.

Non-identified and weakly identified designs. With $|\mathcal{I}| = 2$ the diagnostic is below 0.01 in every replication, and the five starts reach the same objective value, within a relative 10^{-6}, at points whose ECVs differ by 0.19 to 0.22 on average. The reported estimate is whichever point wins on rounding error, so its "correct" classifications carry no information from the data: both identified sets contain τ, and bootstrap coverage falls from 89% to 65% as n grows. In the proportional design sampling noise tilts the flat population objective, all starts reach the same point and the optimiser reports convergence in 92.5 to 97.5% of replications, yet the ECVs are not identified. The diagnostic is small (median 0.028, 0.027 and 0.0013, against 0.061 to 0.101 in the identified designs), 35 to 47.5% of fits put a slope at its bound, the RMSE stays near 0.17 to 0.18, and Wald coverage falls from 71% to 25%. The near-proportional design degrades continuously, as Remark A.5 predicts. Its starts agree, but the median diagnostic is 0.012 to 0.016, its RMSE is 0.202, 0.192 and 0.168 (against 0.141, 0.074 and 0.046 for the identified design with the same Israel-related items), Wald coverage is 57 to 73% and 1.7% of items are misclassified at $n = 3{,}000$. The estimates wander along the near-flat curve from sample to sample, not from start to start. Agreement across starts therefore exposed the two-item case but neither the proportional nor the near-proportional case; the identification diagnostic flagged all three. The correlated-factors $\hat{\rho}_{SI}$ is recovered against its pseudo-true value (0.589 to 0.621) in every design, including the non-identified ones (RMSE 0.016 to 0.060, coverage 90 to 97.5%), so a precise $\hat{\rho}_{SI}$ says nothing about the boundary (Theorem 4.2(d)).

Testing heterogeneity (X9). Rule D2(ii) asks the data whether Assumption A3 holds before any ECV is reported. By Lemmas A.3 and A.4, the assumption fails in a domain $\mathcal{D}$ exactly when every polychoric tetrad among its items vanishes. Its failure is therefore the null hypothesis of a single *joint test*, the Wald test of a non-redundant set of $|\mathcal{D}|(|\mathcal{D}|-3)/2$ tetrads (35 for the stereotype domain, 77 for the Israel-related one; Appendix C.3). X9 compares it with the *intersection–union test*, which accepts heterogeneity only if the joint test of $\mathcal{D} \setminus \{k\}$ rejects for every $k \in \mathcal{D}$, and with the *earlier rule*, one of the 3,633 within-domain tetrads significant after a Bonferroni adjustment (Table 4). At $n = 6{,}000$ the joint test has size 5.0 to 6.5% in each domain under both null configurations, the rule built on it (either domain, each at 2.5%) rejects in 5.2 to 5.8%, and it rejects in every replication in the heterogeneous and two-lines configurations. At $n = 1{,}500$ it over-rejects in the Israel-related domain (11.0 and 14.5%), so at smaller samples it should be calibrated by simulation. The intersection–union test is valid but very conservative (size at most 2.5%), because all its component hypotheses are true whenever Assumption A3 fails. It also tests a stronger property, heterogeneity that survives the removal of any one item: when heterogeneity rests on two items on a second line, it rejects in only 3.0 and 3.5% of replications at $n = 6{,}000$, against 100% for the joint test. The earlier rule is not powerless at the registered sizes (100% in the heterogeneous and two-lines configurations at $n = 6{,}000$), but in the stereotype domain at $n = 1{,}500$ its power is 52.0% against 82.5% for the joint test. No test detects the near-proportional configuration (rejection rates 6.2 to 8.0% at $n = 6{,}000$); there the guard is the diagnostic of rule D2(iii).

7.3 Differential item functioning and anchors (X2, X2b, X7)

X2 uses the six-item design in a reference and a focal group of 1,500 each, with focal latent means $(\mu_G, \mu_S, \mu_I) = (0.3, 0, 0.3)$. In the focal group, I1 and I2 have uniform DIF, with thresholds lowered by κ standard deviations of the latent response, and I3 has non-uniform DIF, with its general slope multiplied by $1 - 1.25\kappa$; its focal ECV falls from 0.40 to 0.273, 0.143 and 0.040 at $\kappa = 0.2$, 0.4 and 0.6. *Ignore DIF* fits the invariant model to both groups (Remark A.9); the *oracle* frees exactly the DIF items; the *anchored* fits hold an anchor set invariant and free all other items, with the correct set S1–S4, I5, I6 or one or two contaminated anchors (I1 for I5; I1, I2 for I5, I6). The *drafted screen* adds a Wald test of equal slopes and

Table 4. X9, tests of within-domain heterogeneity (Assumption A3) at the registered sizes (synthetic data; $R = 400$ at $n = 6{,}000$, 200 at $n = 1{,}500$). Rejection rates (%) per domain at the 5% level of the joint tetrad test, the intersection–union test (IUT) and the earlier Bonferroni rule over 3,633 tetrads; D2(ii): heterogeneity accepted in at least one domain, with the joint and intersection–union tests at 2.5% per domain. [0]Assumption A3 fails in both domains, so the rates are sizes. Two lines: two items of each domain lie on a second line through the origin of the loading plane.

		Stereotype domain ($\lvert\mathcal{S}\rvert = 10$)			Israel domain ($\lvert\mathcal{I}\rvert = 14$)			D2(ii): either domain		
Configuration	n	Joint	IUT	Earlier	Joint	IUT	Earlier	Joint	IUT	Earlier
Proportional[0]	1500	9.0	0.5	0.5	11.0	1.0	4.0	13.0	0.0	4.5
	6000	5.0	0.0	0.8	6.0	0.2	4.2	5.8	0.2	5.0
All but one proportional[0]	1500	8.0	0.0	0.0	14.5	2.5	2.0	12.5	1.5	2.0
	6000	5.5	0.0	0.5	6.5	1.0	4.8	5.2	0.5	5.0
Two lines	1500	69.5	4.5	68.5	54.0	2.5	70.0	75.0	4.0	92.5
	6000	100.0	3.0	100.0	100.0	3.5	100.0	100.0	3.2	100.0
Near-proportional	1500	12.0	0.0	2.0	14.0	1.5	5.0	13.0	0.5	7.0
	6000	8.0	0.2	1.0	6.2	0.5	4.0	7.0	0.0	5.0
Heterogeneous	1500	82.5	30.0	52.0	100.0	99.5	100.0	100.0	100.0	100.0
	6000	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

thresholds for each non-anchor item, Benjamini–Hochberg at $q = 0.05$, a flag only if the expected item scores also differ by at least 0.25 at the focal mean of G, and a refit with flagged items free. The *revised screen* of the plan (Section 8.3) drops the effect-size requirement; X7 runs it end to end on the same 400 data sets. *Stereotype-only anchors* use S1–S4 and fix the focal μ_I and σ_I^2 (Remark 4.7). Table 5 gives the results at $\kappa = 0.4$; Figure 5 and Table 11 in Appendix C.6 give them across κ.

Ignoring DIF and anchors. DIF confined to the Israel-related items is absorbed mainly by μ_I, whose bias is +0.098, +0.243 and +0.756 at the three non-zero values of κ. It leaks into μ_G much less (+0.029, +0.036, −0.001), consistent with Theorem 4.5(d)(ii). The ECVs move more: the focal ECV of I3 is overstated by 0.091, 0.199 and 0.381, and at $\kappa = 0.4$ the focal ECVs of the DIF-free items I4, I5 and I6 move by +0.055, +0.044 and +0.030, as Theorem 4.5(d)(iii) predicts. With the correct anchors, and for the oracle, the bias of both mean differences and of both ECVs of I3 is at most 0.010 in absolute value, and coverage is 92 to 100%. One contaminated anchor biases $\hat{\mu}_G$ by −0.093, −0.170 and −0.222, with coverage falling to 3%; two bias it by +0.083, +0.149 and +0.227. The error does not stay with the faulty anchor (Proposition 4.6): the screen flags DIF-free items at rates of 29.8% and 42.9% at $\kappa = 0.4$, against 0.0% with the correct anchors. The leave-one-anchor-out range of $\hat{\mu}_G$ is 0.043 with the correct anchors and 0.133 and 0.099 with one and two contaminated ones. Stereotype-only anchors give $\hat{\mu}_G$ with bias between −0.011 and 0.000 and coverage 94 to 97%, at the cost of a larger RMSE (0.069 to 0.082, against 0.049 to 0.061). In X2 that protection holds by construction, because only Israel-related items have DIF; X2b removes that feature.

The DIF screen, drafted and revised (X7). With the correct anchors the drafted screen never flags a DIF-free item, but it flags I3 in 0%, 0% and 1% of replications at $\kappa = 0.2$, 0.4 and 0.6: a change in the general slope alone barely moves the expected score at the focal mean. It then treats I3 as invariant and reproduces the bias it is meant to prevent: at $\kappa = 0.4$ the ECV of I3 is biased by −0.109 in the reference group and +0.146 in the focal group. We therefore revised the plan so that the multiplicity-adjusted test

Table 5. X2 and X7, group comparisons under DIF at $\kappa = 0.4$ (synthetic data; 1,500 per group; $R = 100$, the same data sets in every row). Bias, RMSE and coverage (%, sandwich 95% interval) of the focal-minus-reference latent mean differences, and bias of the reference and focal ECV of I3, the item with non-uniform DIF (true values 0.40 and 0.143). Drafted screen: Benjamini–Hochberg and an expected-score difference of at least 0.25 (X2); revised screen: Benjamini–Hochberg alone, the rule of the plan (X7).

	$\mu_G^{(1)} - \mu_G^{(0)} = 0.3$			$\mu_I^{(1)} - \mu_I^{(0)} = 0.3$			ECV_{I3} bias	
Strategy	Bias	RMSE	Cov.	Bias	RMSE	Cov.	Ref.	Focal
Ignore DIF (full invariance)	+0.036	0.074	82	+0.243	0.255	3	−0.072	+0.199
Oracle (true DIF items free)	−0.006	0.047	98	+0.008	0.062	95	+0.009	−0.002
Correct anchors, free baseline	−0.008	0.049	100	+0.010	0.065	95	+0.010	−0.002
Correct anchors, drafted screen + refit	+0.007	0.046	98	−0.032	0.064	94	−0.109	+0.146
Correct anchors, revised screen + refit (X7)	−0.006	0.047	98	+0.008	0.061	95	+0.009	−0.002
1 contaminated anchor	−0.170	0.179	22	+0.744	0.751	0	+0.047	+0.014
1 contaminated, drafted screen + refit	−0.117	0.129	48	+0.688	0.696	0	+0.058	+0.021
1 contaminated, revised screen + refit (X7)	−0.123	0.134	42	+0.703	0.710	0	+0.054	+0.018
2 contaminated anchors	+0.149	0.191	54	+0.669	0.682	0	+0.003	−0.005
2 contaminated, drafted screen + refit	+0.054	0.101	82	+0.753	0.760	0	−0.008	−0.010
2 contaminated, revised screen + refit (X7)	+0.070	0.126	77	+0.739	0.748	0	−0.007	−0.009
Stereotype-only anchors (S1-S4)	−0.011	0.069	97	not identified			+0.014	−0.003

alone decides which items are freed, and re-simulated the revised screen end to end (X7). Its flags coincide with the Benjamini–Hochberg flags of X2 in every replication. With the correct anchors it frees I1 and I2 in 94 and 98% of replications at $\kappa = 0.2$ and in 100% at $\kappa \geq 0.4$, and I3 in 41%, 100% and 100%. It flags DIF-free items at mean rates of 0.7%, 3.6%, 2.8% and 1.6% at $\kappa = 0$, 0.2, 0.4 and 0.6, which costs precision but not bias. Downstream, $\hat{\mu}_G$ has bias between −0.006 and 0.000 and coverage 95 to 98% at every κ, and the focal ECV of I3 is biased by +0.042, −0.002 and +0.002, against +0.082, +0.146 and +0.190 for the drafted screen. The remaining bias at $\kappa = 0.2$ comes from the 59% of replications in which I3 is not flagged (+0.077 there): small non-uniform DIF stays hard to detect with 1,500 per group. With contaminated anchors the revised screen does not help ($\hat{\mu}_G$ biased by −0.123 and +0.070 at $\kappa = 0.4$); only the choice of anchors addresses that.

DIF on the stereotype anchors (X2b). X2b gives two of the four stereotype anchors, S1 and S2, uniform DIF of $\kappa_S \in \{0.1, 0.2, 0.3\}$ in the same direction, with or without the Israel-related DIF of X2 at $\kappa_I = 0.4$ (Table 10). At $\kappa_S = 0.1$ the mean bias of $\hat{\mu}_G$ is +0.025 and +0.031 with the default anchors S1–S4, I5, I6 (coverage 96 and 89%) and +0.073 and +0.080 with stereotype-only anchors (coverage 84 and 75%). At $\kappa_S = 0.2$ it is +0.047 and +0.069 against +0.259 and +0.410 (probability limits of the latter +0.254 and +0.266), and at $\kappa_S = 0.3$ the stereotype-only fit breaks down (probability limits +2.521 and +2.634) while 21 to 22% of the default fits fail to converge. Stereotype-only anchoring is therefore not a safe default. It drops the assumption that Israel-related anchors are invariant at the price of resting everything on four stereotype items, and the same DIF on those items biases $\hat{\mu}_G$ two to three times as much as with the six default anchors (to first order, +0.058 against +0.026 per 0.1 of κ_S), and more than linearly as κ_S grows. The contamination leaves traces: at $\kappa_S = 0.1$ and $\kappa_I = 0$ the leave-one-anchor-out range of $\hat{\mu}_G$ is 0.165 (default) and 0.464 (stereotype-only), against 0.043 with clean anchors, and the revised screen flags DIF-free stereotype items in up to 11% of replications (57% at $\kappa_S = 0.2$), which also reduces the bias of the default-anchor comparison to +0.006 and +0.013. The bound (4.12), evaluated for this estimator at a

stated maximum DIF δ on every anchor (Appendix C.3), is valid but wide. At $\delta = 0.1$ the largest first-order bias of $\hat{\mu}_G$ is 0.345 with the default anchors and 0.442 with stereotype-only anchors, more than the true difference of 0.3, and the norm bound is 0.707 and 0.763; at $\delta = 0.2$, the value the plan registers, all of these double. The worst case has anchor DIF of opposite signs on anchors with different ECVs, which the model reads as a shift of G against S; if all anchors' DIF has the same sign, the first-order bias of $\hat{\mu}_G$ is 0.104 and 0.099 at $\delta = 0.1$. The pattern of X2b lies well inside the bound (+0.026 and +0.058 to first order at $\kappa_S = 0.1$), but for stereotype-only anchors the probability limit exceeds the first-order value from $\kappa_S = 0.2$ on, so the linear bound is not conservative there.

7.4 Cross-loadings and acquiescence (X6)

X6 adds two misspecifications at the registered sizes, with 8 ID items with ECVs from 0.10 to 0.70 and 6 trope-based IT items with ECVs from 0.35 to 0.85 (Appendix C.2). In (a) every IT item also loads on S with standardised loading $c \in \{0.1, 0.2, 0.3\}$. In (b) an acquiescence factor adds a slope $+\alpha$ to every positively worded item and $-\alpha$ to every reverse-keyed item, $\alpha \in \{0.25, 0.40, 0.55\}$ (standardised loadings 0.16 to 0.36), with the registered keying (four reverse-keyed items, two in S and two in ID, none in IT) or balanced keying (half of every block). The true values are the content ECVs of each item's own domain. Besides the standard bifactor fit we evaluate five remedies: a model with free loadings of all IT items on S; a method factor with one free slope and the instrument's keying, the random-intercept model of Maydeu-Olivares and Coffman (2006); a refit with $\mathcal{I}$ the ID items alone; the two-stage rule that the plan registers (Section 8.7), which tests the mean residual correlation between IT and S items and, if the test rejects, uses the model with cross-loadings; and a one-item check proposed in an earlier draft, which tests the 60 residual correlations between IT and S items and fits a cross-loading for each IT item in turn. Figure 4 shows the main results and Table 12 gives them in full. Without misspecification no estimator has a bias above 0.006 and all cover in 94 to 95% of cases.

Cross-loading. The standard fit credits the covariance between IT and S items to G. The ECVs of the IT items rise by 0.018, 0.051 and 0.089 on average at $c = 0.1$, 0.2 and 0.3, most for those with low ECV (IT1, true 0.35, by 0.141 at $c = 0.3$), and IT3 (true 0.45) is classified in $\mathcal{B}_{0.5}$ in 78% of replications at $c = 0.3$. The shift is not all towards "in". The whole general factor tilts towards S: the stereotype ECVs rise by 0.11 to 0.25 and the ID ECVs fall by 0.041 to 0.080 on average (ID8, true 0.70, by 0.141). Coverage of the intervals for the Israel-related ECVs falls to 63, 26 and 11%. The model with cross-loadings removes the bias (no item above 0.007; coverage 93 to 95%); the ID-only refit gives unbiased ID ECVs but none for the IT items; the method factor does nothing here. The plan's two-stage rule inherits the accuracy of the cross-loading model: the mean-residual test rejects in 100, 100 and 85% of replications at $c = 0.1$, 0.2 and 0.3 and in 5% without cross-loadings, and the bias of every Israel-related ECV is at most 0.005, 0.007 and 0.021. The one-item check flags at least one IT item in every replication with $c \geq 0.1$, almost always through a one-item cross-loading, and under its rule the share of IT items on the wrong side of τ at $c = 0.3$ falls from 14.2% to 1.5% (IT3 classified in: from 78% to 9%). It achieves this by declaring items undetermined (48.5% of the IT items), not by correcting them: a cross-loading fitted for one item at a time leaves those of the others in place, so the one-item model still overstates the general-factor share of IT3 by 0.083. Without misspecification it flags at least one IT item in 26% of replications (19.8 against 19.0% of IT items undetermined). The plan therefore fits the cross-loadings of the whole IT block.

Acquiescence. With the registered keying, acquiescence acts like a second, nearly uniform general factor. The standard fit inflates the ECVs of positively worded Israel-related items with low ECV, by up to 0.030,

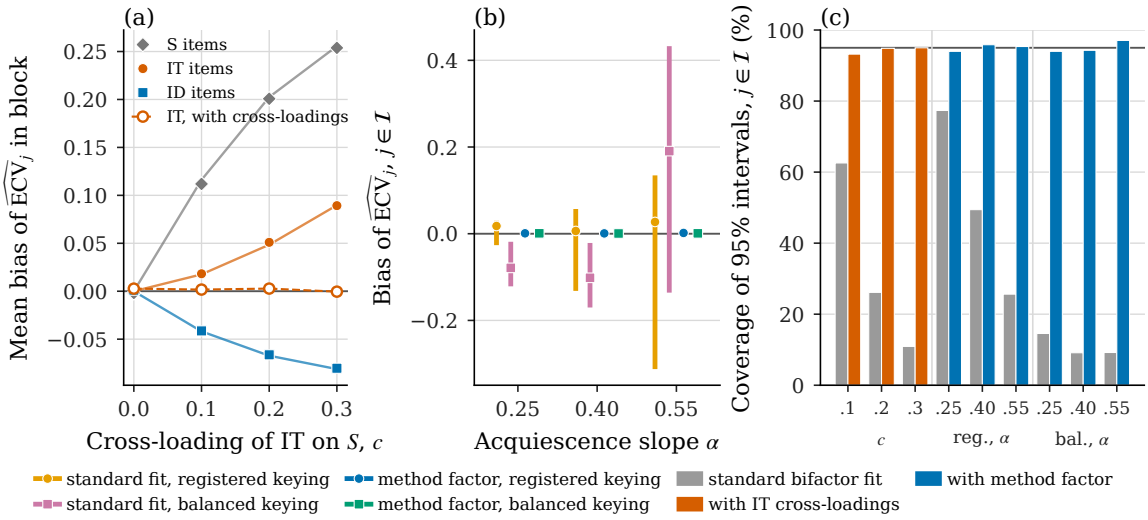


Figure 4. X6, two misspecifications at the registered sizes (synthetic data; 10 S, 8 ID and 6 IT items; n = 6,000; R = 100 per cell). (a) Cross-loading of the IT items on S with standardised loading c: mean bias of $\widehat{\mathrm{ECV}}_j$ by block in the standard bifactor fit (filled markers; lines are probability limits) and for the IT items in the model with cross-loadings (open markers). (b) Acquiescence with slope α and the registered (4 reverse-keyed items) or balanced keying: range (bar) and mean (marker) over the 14 Israel-related items of the bias of $\widehat{\mathrm{ECV}}_j$, in the standard fit and with a one-parameter method factor. (c) Coverage of the 95% Wald intervals for the Israel-related ECVs in the standard fit (grey) and with the remedy for each misspecification: cross-loadings in the first three cells, the method factor in the others.

0.057 and 0.135 at the three values of α, and deflates those of the two reverse-keyed ID items, by up to 0.026, 0.132 and 0.312. Block means move by at most 0.05 and items rarely cross τ (at most 0.6% wrongly classified), but coverage falls to 77, 49 and 26%. Balanced keying does not rescue the standard fit: it lowers every Israel-related ECV (block means −0.064 to −0.128 at $\alpha \leq 0.40$) and raises the stereotype ECVs by 0.14 to 0.15, and at α = 0.55 the fit breaks down, with 45% of the Israel-related items on the wrong side of τ. The one-parameter method factor removes the bias under both keyings (no item above 0.006; coverage 94 to 97%; its slope is estimated without bias). Without acquiescence it costs little (10.7 against 10.5% of Israel-related items undetermined), and when the acquiescence slopes vary between items by a factor of three, which one slope cannot represent, it still reduces the largest bias from 0.112 to 0.043. The one-item check reacts to acquiescence too (an IT item flagged in 42, 98 and 100% of replications with the registered keying) but cannot correct it, and the ID-only refit does not help. One caveat concerns rule D2(iii). The method slope enters the correlations only through its square, so its information vanishes without acquiescence and the diagnostic of the method-factor model is then zero. Computed over the content parameters it is 0.082 to 0.105 (median by cell), close to the 0.090 of the standard fit without misspecification, so the plan, whose primary model includes the method factor, applies the rule to the content parameters.

7.5 List experiments (X3, X8)

X3 simulates one sensitive item under the threshold-shift model of Definition 5.4, with standardised loading 0.6 on G, π = 0.30 and π^D = 0.20, so SDB = 0.10 and monotonicity holds; a second scenario adds false endorsement π^F = 0.02. Control items have prevalence 1/2 and come in independent pairs with within-pair correlation ρ, and n = 3,000 (Figure 6 and Tables 14 and 15 in Appendix C.6). In the split

design the estimator (5.6) is unbiased in every cell, its Monte Carlo SD matches the exact formula to within 0.0008, and its coverage is 94.3 to 95.5%. The Neyman allocation of Theorem B.5(a) reduces the SD from 0.0487 to 0.0456 with four independent controls, a pilot-based allocation attains almost all of the gain, and negatively correlated pairs ($\rho = -0.6$) reduce it further, to 0.0331 (Proposition 5.1(c)). In the paired design the optimal allocation leaves the direct-only arm empty; the post-stratified and combined estimators have SDs 0.0249 and 0.0226, against 0.0339 for the split design. With false endorsement, $\widehat{\mathrm{SDB}}_{\mathrm{PS}}$ stays unbiased while the combined estimator, read as an estimator of SDB, is biased by +0.0206, the false-endorsement rate, and covers in 85.3% of replications (Theorem 5.3(d)). Across $J \in \{2, \dots, 6\}$, $\rho \in \{0, -0.6\}$ and floor and ceiling probabilities $\phi_c = \phi_f \in \{0, 0.5\}$, the simulated bias is within 0.0027 of (B.5) in all 20 cells. With $J = 2$ and $\rho = 0$ the exposure $\mathbb{P}(W_i \in \{0, J\})$ is 0.5 and the bias −0.073; negative correlation cuts them to 0.2 and −0.031. With floor and ceiling effects the MSE-optimal list is $J = 2$ for $n \le 1{,}000$, $J = 4$ for $3{,}000 \le n \le 10^5$ and $J = 6$ for $n \ge 3 \times 10^5$, always with $\rho = -0.6$, and at $n = 3{,}000$ the simulated MSEs pick the same list as the theory (Theorem B.5(c),(d)). The simulation does not measure disclosure.

Identifying the threshold shift (X8). X8 links the list experiment to the measurement model as in Proposition 5.5(b). The two-group bifactor model is fitted to the direct arm, and the shift δ_j of the sensitive item (I4 of the six-item design) is the root of (5.13) in each group, with the group's latent law from the anchored model. Two groups of 3,000 with latent means $\mathbf{0}$ and $(0.5, 0, 0.3)$ are each split equally into a direct arm and two list arms (Table 13). With a common shift of 0.4 latent-response standard deviations, $\hat{\delta}_j$ is unbiased (bias +0.0004 and −0.0034, MCSE 0.009 and 0.006), its SD is 0.153 and 0.109 in the two groups, its standard error is accurate (0.149 and 0.109), and 95% intervals cover in 94.3 and 95.7% of replications; the list experiment contributes 81 to 84% of the variance. The test of a common shift (Proposition 5.5(c), Proposition B.7(d)) rejects in 3.0% of replications when the shift is common, in 58% when it is 0.4 in one group and 0.8 in the other, and in 14% when the private response has its slope halved (Proposition B.7(e)), in which case $\hat{\delta}_j$ in the second group is biased by −0.149. The identification of Proposition 5.5(b) works at this sample size, but a list experiment detects only large departures from a common uniform shift.

7.6 Small-area estimation and count indices (X4, X5)

X4 draws a new synthetic population of 50 areas in each replication, with differential non-response, and each respondent contributes an unbiased score of G, so it does not test the joint measurement and latent-regression model that Section 6 recommends (Figure 7 and Table 16). MRP has RMSE 0.168, 0.133 and 0.105 at $n = 1{,}000$, 3,000 and 6,000, against 0.342, 0.232 and 0.177 for the unweighted direct mean, which non-response also biases by about −0.1; at $n = 3{,}000$ its RMSE is 0.204 against 0.602 in areas with fewer than five respondents, and its coverage is 92.0 to 94.7%. The two-step shortcut of Proposition 6.2 has slopes on the true area deviations of 0.637, 0.704 and 0.762, each equal to the estimated shrinkage factor (0.845, 0.844, 0.846) times the MRP slope to within 0.001, and coverage 84.0 to 86.0%.

X5 computes the index $P_g(q) = \mathbb{P}(U_{i+} \ge q \mid g)$ for the eight stereotype items exactly, for two groups that differ only in their mean of G, with each item's agreement threshold shifted in the second group by an independent normal amount with standard deviation s (Figure 8 and Table 17). Under invariance the sign of $P_1(q) - P_0(q)$ always agrees with $\Delta\mu_G$ (Proposition 6.3(a)). With $s = 0.1$ the sign is reversed in 20.1 to 21.1% of configurations when $|\Delta\mu_G| = 0.05$ and never when $|\Delta\mu_G| \ge 0.2$; with $s = 0.3$, in 39.4 to 41.3% at $|\Delta\mu_G| = 0.05$ and 4.2 to 7.8% at 0.3. With $\Delta\mu_G = 0.2$ and full invariance, a second group whose standard deviation of G is 0.6 has the lower index for every $q \ge 4$ (Proposition 6.3(b)), and random

non-uniform DIF makes the sign depend on q in 25.9 to 63.6% of configurations (Proposition B.8(d)). This concerns synthetic items only.

7.7 What the simulation does and does not show

In the designs studied, the formal results describe finite-sample behaviour. Heterogeneity within a domain identifies the boundary estimands, which are recovered with small bias and close to nominal Wald coverage at $n \geq 1{,}000$; two Israel-related items or proportional stereotype loadings leave them unidentified, and near-proportional loadings degrade them badly. Agreement across starts exposed the two-item case but neither the proportional nor the near-proportional case; the identification diagnostic flagged all three. A joint tetrad test detects heterogeneity at the registered sizes with close to nominal size, but no test separates near-proportional from proportional loadings. Ignored DIF and contaminated anchors bias comparisons as Theorem 4.5 and Proposition 4.6 predict. Stereotype-only anchoring protects comparisons of G against DIF on Israel-related items but is two to three times as sensitive as the default anchors to DIF on the stereotype anchors. The revised DIF screen, re-simulated end to end, removes the bias the drafted screen left. A cross-loading of trope-based items on S and acquiescence both distort the boundary estimands, not always towards "in", and a cross-loading model and a one-parameter method factor remove the respective biases. The list-experiment formulas hold to Monte Carlo precision and a list experiment identifies the threshold shift; MRP improves on direct estimation in small areas; count indices misrank groups under modest DIF. The simulation changed the plan in several places: the DIF screen, simultaneous intervals for the boundary set, the identification rule, the misspecification checks and the anchor sensitivity analysis (Section 8).

It does not show the following. (i) *Full-information fits*: the full-information and Bayesian fits the plan reports as sensitivity analyses target the same parameters, but their finite-sample behaviour can differ, and the logistic link is not simulated (Remark A.6). (ii) *Exact interval calibration*: the bootstrap with $B = 100$ undercovers slightly in some cells. (iii) *Estimates in non-identified designs*: there the point estimates are optimiser artefacts. (iv) *General DIF*: X2, X2b and X7 have two groups of equal size and a few DIF patterns; the Mantel–Haenszel cross-check and alignment are not simulated, and the anchor bound is evaluated at first order and at population values. (v) *Other misspecification*: X6 covers one cross-loading pattern and acquiescence; local dependence, other response styles, non-normal latent distributions and correlated specific factors are not simulated (Section 10). (vi) *Violated list-experiment assumptions*: X3 and X8 have no design effects and no liars (Assumption A4), and do not measure disclosure. (vii) *The joint latent MRP model*: X4 starts from an unbiased score of G. Above all, the simulation contains no information about real attitudes. Its parameters were chosen to test the methods; they are not estimates of anything and are not calibrated to any survey.

8 A pre-registered survey design and analysis plan

This section is the design and analysis plan that we will register, time-stamped and with the commit hash of the analysis code, before any respondent is interviewed (Nosek et al., 2018). It fixes in advance the item bank structure, the groups for invariance testing, the list-experiment module, the sample and poststratification frame, the estimands, the threshold τ, the decision rules and the model checks. Any departure will be reported as a deviation, with its reason. The estimators are those of Sections 3 to 6, implemented and tested on synthetic data in Section 7. No survey has been fielded. The items described below are *constructs and selection criteria*, not a validated instrument. Item wording will be developed through the pre-testing steps of Section 8.2 and registered only once it is final.

8.1 Population, constructs and model

The target population is the adult household population of one country. The plan is written for the United States because the poststratification frame of Section 8.5 is available there; Section 8.3 describes the cross-national extension. The measurement model is the bifactor graded response model of Section 3, with a general factor G and specific factors S (classic stereotypes) and I (Israel-related attitudes) (Gibbons and Hedeker, 1992; Reise, 2012; Samejima, 1969). Responses are on a five-point agreement scale, so $C_j = 4$ for every attitude item. The primary model adds an acquiescence factor (Maydeu-Olivares and Coffman, 2006): a person-level tendency to agree, orthogonal to G, S and I, with one loading common to all attitude items, entered with a positive sign for agree-keyed items and a negative sign for reverse-keyed items, which are scored in the direction of their construct. ECV_j is computed from the content loadings as in Definition 3.1. The identification results of Section 4 are stated without this factor; its effect on recovery was checked by simulation. Without such a factor, acquiescence loads on G and moves the boundary estimands (Section 3.5); in the simulation the factor removed that bias, whereas balancing the keying without it did not (Section 8.7). The correlated-factors model, with factors S and I and no general factor, and a unidimensional model are fitted alongside it as pre-registered comparison models.

8.2 Item bank

Table 6 lists the item blocks. Every block is described by construct; none of the content descriptions below is a questionnaire item.

Selection criteria. Candidate items enter the pilot only if they meet five criteria.

(i) *One construct per item.* Each attitude item is written to load on G and on exactly one specific factor. Double-barrelled statements (for example, a statement about a policy combined with a trope) are excluded, because they would make the item-level estimand of Definition 3.1 uninterpretable.

(ii) *Coverage of the disputed categories.* The ID block includes at least one item for each category of Israel-related speech on which Section 2.1 records disagreement. The IT block draws on categories that more than one definition treats as antisemitic. Neither block is chosen to make the boundary set larger or smaller.

(iii) *Definitional tags, recorded but not used in estimation.* Before fielding, each Israel-related item is tagged with how the text of each definition treats its content category, or "unclear". Tags are assigned independently by advisory-group members who hold different views of the definitions (Section 9), and disagreements are recorded rather than resolved. Tags are metadata. They let readers relate the estimated boundary set to any definition, but no estimand depends on them.

(iv) *Pre-testing.* Cognitive interviews with respondents who differ in ideology, age and religious background check that each item is understood as intended. Items that are understood differently by different groups are revised or dropped before the pilot; this is the cheapest stage at which to remove DIF.

(v) *Identification.* Theorem 4.2 needs at least four items in one domain whose general-factor shares are not all equal save one, and the joint tetrad test of rule D2(ii) in Section 8.7 needs no more (its intersection–union diagnostic needs five). Both domains exceed these counts, with room to drop items at the pilot stage. The pilot also checks the loading conditions of Theorem 4.2, including heterogeneity of general-factor shares within a domain (Assumption A3). A block whose specific-factor loadings are near zero in the pilot, or whose items have nearly equal general-factor shares, is redesigned before

Table 6. Item bank structure. Counts are targets for the main survey; the pilot fields about 1.5 times as many candidates per block. $\mathcal{S}$, $\mathcal{I}$ and $\mathcal{A}$ are the item sets of Section 3.

Block	Model role	Items	Construct and content (generic)
S: classic stereotypes	$\mathcal{S}$; loads on G and S	10	Statements attributing to Jews as a group excessive power or influence (economic, political, media), disloyalty, conspiratorial intent, or responsibility for historical misfortunes; statements that minimise or deny the Holocaust. Content categories follow those used in prior batteries (Section 2.2) without reusing their wording.
ID: Israel, disputed categories	$\mathcal{I}$; loads on G and I	8	Statements in the categories of Israel-related speech on which the definitions of Section 2.1 disagree: positions on specific policies or actions of Israeli governments and of the Israeli state, on Zionism, on boycotts, and on comparisons with other historical cases. Selected to span those categories.
IT: Israel, classic tropes	$\mathcal{I}$; loads on G and I	6	Israel-related statements that invoke classic antisemitic tropes, for example claims of secret control by Israel or its supporters, or collective responsibility of Jews for Israel's conduct. Selected from categories that more than one definition treats as antisemitic.
A: anchors	$\mathcal{A} \subset \mathcal{S} \cup \mathcal{I}$	6	Items expected, on substantive grounds and pilot evidence, to function equivalently across the groups of Table 7: four from S and two from the Israel-related blocks, so that every factor has anchors. Counted within their own blocks.
C: comparison states	auxiliary	16	For each ID item, parallel statements about a comparable policy or case of each of two pre-registered comparison states. Not in the measurement model.
R: reverse-keyed	$\mathcal{S} \cup \mathcal{I}$	4	Favourably worded items, two in S and two in ID, which separate the acquiescence factor of the primary model from content (Section 8.1). Counted within their own blocks.
Q: data quality	auxiliary	3	Instructed-response attention checks, placed in separate blocks.
K: knowledge and covariates	auxiliary	–	Factual knowledge of the Israeli–Palestinian conflict and of other conflicts; ideology; religiosity and affiliation; birth year; region; demographics for poststratification.

the main survey; the simulation shows that near-proportional loadings make the boundary estimands unreliable at realistic sample sizes (Section 7.2).

Pilot. A pilot sample, which is not used in any confirmatory analysis, screens candidates. Items are dropped for severe floor or ceiling effects, for high item non-response, or for DIF flagged in the pilot by both screening methods of Section 8.3. The pilot also supplies the variance inputs for Theorem B.5. The final item set, the anchor set $\mathcal{A}$ and all thresholds are frozen in the registration after the pilot and before the main fieldwork.

Administration. The questionnaire is self-administered online, which tends to elicit more reporting of sensitive information than interviewer administration (Kreuter et al., 2008). The list-experiment module (Section 8.4) comes first, before any direct question about Jews or Israel. The order of the S and Israel-related blocks is randomised, and items are randomised within blocks. Block order is recorded and treated as a grouping variable for the invariance checks, so that priming between the two domains can be detected.

Table 7. Pre-registered groupings for invariance and DIF testing. Groups with fewer than $n_{\min} = 300$ respondents are merged with an adjacent group or omitted, as fixed in the registration.

Grouping	Categories	Rationale
Ideology	Self-placement, collapsed to left / centre / right	Political identity may shape responses to Israel-related items, and to some stereotype items, independently of G (Section 2.4)
Cohort	Ages 18–29, 30–44, 45–64, 65+	Generational differences in the salience and framing of both domains
Religiosity	Attendance at religious services: never; less than monthly; monthly or more	Religious commitments may affect both domains
Region	Four census regions	Required for the small-area model (Section 8.5)
Block order	S first / Israel-related first	Detects priming between domains
Country (extension)	Countries in a cross-national replication	Translation and context; anchoring vignettes considered (King et al., 2004)

8.3 Groups for invariance testing

Table 7 lists the pre-registered groupings. Each is tested separately; intersections are not tested, because their cells are too small.

For each grouping the procedure is fixed.

1. Fit the configural multiple-group bifactor model, then constrain loadings (metric) and then thresholds (scalar) (Meredith, 1993; Millsap, 2011). Changes in fit are reported with the caveats of Chen (2007); they do not by themselves decide invariance.

2. Screen every item for DIF with two methods: Wald tests of equal slopes and thresholds in the multiple-group fit of the primary estimator, with the anchor set held invariant, the limited-information counterpart of the likelihood-ratio tests of Thissen et al. (1993) and the test used in the simulation (Section 7.3); and the Mantel–Haenszel procedure with the total score on the remaining items as the matching variable (Holland and Thayer, 1988). Multiplicity is controlled by the Benjamini–Hochberg procedure at a false discovery rate of 0.05 within each grouping (Benjamini and Hochberg, 1995). An item is flagged if its Wald test is significant after this adjustment. The Mantel–Haenszel procedure, which is sensitive mainly to uniform DIF, is a cross-check, and disagreements between the two methods are reported. The size of each flagged item's DIF is a secondary screen. We report the difference in expected item score between the group-specific and the reference item parameters, averaged in absolute value over the group's latent distribution, and describe flagged items with a difference below 0.25 scale points as having small DIF; they are freed all the same. An earlier draft flagged an item only if it was also at least 0.25 scale points apart at the group's mean of G. In the simulation that requirement almost never flagged an item with non-uniform DIF, whose expected scores differ little near the group mean, and the procedure then reproduced the bias it was meant to remove (Section 7.3). The revised screen was re-simulated end to end on the same data sets (Section 7). With the correct anchors it freed the item with non-uniform DIF in 41% of replications at the smallest DIF and in every replication at larger DIF; that item's focal ECV was then biased by +0.042 at the smallest DIF and by at most 0.002 in absolute value at larger DIF, and the bias of the difference in means of G was at most 0.006 in absolute value.

3. Fit the partial-invariance model with flagged items freed and the anchors constrained (Byrne et al., 1989). Group comparisons of latent means and of ECV_j are reported from this model. Their identification rests on Proposition 4.6. Comparisons of G and of the group-specific ECVs are also reported, for every grouping, from a model anchored in the four stereotype anchors alone (Remark 4.7). That model need not assume that any Israel-related item functions identically across groups; it assumes that the stereotype anchors do, which is testable only in part (Proposition 4.6(b)). It is a sensitivity analysis, not the primary comparison. Resting on four anchors instead of six, it is more exposed to DIF on them: in the simulation, DIF of 0.1 on two stereotype anchors biased its difference in means of G by +0.073 and +0.080, against +0.025 and +0.031 with the default anchors, and at 0.3 it broke down (Section 7.3). Every anchored comparison of latent means is therefore reported with the bound (4.12), evaluated at the estimated information and maximised over all uniform anchor DIF of at most $\delta = 0.2$ standard deviations of each anchor's latent response, the smallest Israel-item DIF in the simulation (Section 7.3); the bound at 0.4 is reported alongside. The bound is wide. In the simulated design, at $\delta = 0.2$ it allows a first-order bias of the difference in means of G of up to 0.69 with the default anchors and 0.89 with stereotype-only anchors, against a true difference of 0.3; if all anchors' DIF has the same sign, the largest bias is about 0.2 for both. As a further check on the anchors, each anchor is in turn freed and the comparisons recomputed; the spread of these leave-one-anchor-out estimates is reported.
4. As a sensitivity analysis, refit with the alignment method (Asparouhov and Muthén, 2014), which does not require a pre-selected anchor set.

Items flagged for non-uniform DIF have group-specific loadings, so their ECV_j is reported by group (Section 8.6). Theorem 4.5 gives the bias that the analysis would incur if it ignored flagged DIF; we report that bias, evaluated at the estimates, as a diagnostic.

8.4 The list-experiment module

Sensitive items. The module covers four sensitive items drawn from the final item bank: two from S and two from the Israel-related blocks (one ID, one IT). For each sensitive item j, every respondent is randomised independently to one of the three arms of Section 5: *direct* (D), in which item j appears in the direct battery; *list control* (C); or *list treatment* (T). Respondents in the two list arms are not asked item j directly in the main questionnaire. For them, item j is missing by design, and since assignment is random the missingness is ignorable in the measurement model. The prevalence π_j targeted by the list is agreement with item j, coded as the top two categories of the five-point scale. The direct-arm counterpart is π_j^D, and $\mathrm{SDB}_j = \pi_j - \pi_j^D$ as in Section 5.

Control lists. Each sensitive item has its own list of J control items (J as in Section 5). J and the arm probabilities (r_D, r_C, r_T) are chosen by Theorem B.5, with $J \in \{3, 4, 5\}$ and the inputs (control-item prevalences and their covariance) estimated in the pilot. Control items are low in sensitivity and unrelated to Jews or Israel. Following Glynn (2013), each list includes negatively correlated pairs of control items. This reduces the variance of the count and makes it rare for a respondent to agree with all or none of the control items, which limits ceiling and floor effects. It can also make the count more informative about the sensitive answer (Section 5.4), so every list must meet two registered disclosure bounds, evaluated at the pilot estimates of the control-count distribution and of the sensitive item's prevalence: the mutual information $I(Y_i; Z_i \mid \mathsf{T})$ between the treatment count and the sensitive answer is at most 0.15 bits, and at most 15% of treated respondents have posterior $\mathbb{P}(Z_i = 1 \mid Y_i, \mathsf{T}) \geq 0.5$. For comparison, four independent

control items of prevalence $1/2$ with $\pi = 0.2$ give 0.11 bits and 11%, and two pairs with within-pair correlation -0.6 give 0.24 bits and 30%. Within the lists that meet both bounds, J and the correlation are chosen by Theorem B.5(d); a list that fails either bound is replaced before the main survey.

Checks. For each list we report the test of Blair and Imai (2012) for design effects. The decision rule is in Section 8.7. We also estimate the floor-and-ceiling model of Blair and Imai (2012), and report the multivariate estimator of Imai (2011) with the covariates of Table 7. In addition, a random 20% of list-arm respondents are asked item j directly at the very end of the questionnaire. This supports the combined estimator and placebo tests of Aronow et al. (2015) as a secondary analysis. It does not enter the primary estimator, because a direct answer given after the list may be affected by it. For these respondents the file records the direct answer as well as the count, so the list gives them no additional protection; Section 9.2 sets out how these records are handled.

The cost of the module. Proposition 5.1 gives the variance of the difference-in-means estimator. As illustrative arithmetic, not a result, suppose that both count variances equal 1 ($\sigma_C^2 = \sigma_T^2 = 1$), that $\pi_j^D = 0.2$, and that the three arms are of equal size m; these are the inputs used for Theorem B.5(a) in Appendix B.4. Equal count variances assume that the control items covary slightly negatively with the sensitive item. With four independent control items of prevalence $1/2$ and $\pi_j = 0.2$, $\sigma_T^2 = 1.16$ instead (Section 5.2). Under the first assumption the variance of $\widehat{\mathrm{SDB}}_j$ is about $(2 + 0.16)/m$, and detecting $\mathrm{SDB}_j = 0.10$ with power 0.8 in a two-sided test at the 5% level requires $m \approx 1{,}700$, or about 5,100 respondents in total for that item. With $\sigma_T^2 = 1.16$ it requires $m \approx 1{,}820$. Detecting $\mathrm{SDB}_j = 0.05$ requires four times as many, about 20,000 respondents for each item, against a planned 6,000. A meta-analysis of list experiments found sensitivity biases typically below ten percentage points (Blair et al., 2020), so the module is not powered for the effects that are most plausible. SDB_j is therefore a secondary estimand (E5 in Section 8.6), with a registered precision target instead of a power target: at the planned sample size and the allocation of Theorem B.5, the 95% interval for each SDB_j should have a half-width of at most 0.07 (about 0.06 under the inputs above). Two features of the design reduce the cost. Each respondent serves in some arm for every sensitive item, so the four experiments share one sample. And the joint model of Theorem 5.3 borrows strength across items and from the measurement model. Equal allocation is not optimal; Theorem B.5 gives the allocation actually used.

8.5 Sampling, weighting and poststratification

Primary sample. The primary sample is drawn from a probability-based online panel recruited by address-based sampling, with a target of 6,000 completed interviews. The target will be revised in the registration from the power analysis of Section 8.8. Design weights are raked to population margins for age, sex, education, race and ethnicity, and region. These weights are used for the national estimands.

Optional opt-in supplement. For small-area estimation, a non-probability (opt-in) supplement may be added in under-sampled areas. Opt-in samples can be less accurate than probability samples, even after demographic weighting, and their accuracy varies widely across surveys and measures (Baker et al., 2013; Yeager et al., 2011). Model-based adjustment can nonetheless make them informative (Wang et al., 2015). The supplement therefore enters only the MRP model. It carries an indicator for sample source, and its estimates are compared with those from the probability sample in areas where both are available. If the two disagree by more than the pre-registered tolerance (twice the combined standard error of an area mean of G), the supplement is dropped.

Poststratification. The small-area estimand is the poststratified mean in Equation (6.2) (Gelman and Little, 1997; Park et al., 2004). Cells k are formed by state × age (4) × sex (2) × education (4) × race and ethnicity (4). Population counts N_k come from the most recent five-year American Community Survey tables available at registration. Ideology and religiosity are not in census tables. For estimates that condition on them, the cell table is extended with a joint distribution estimated from a large reference survey, and the uncertainty of that extension is propagated. State-level predictors are fixed at registration. Multilevel regression with poststratification performs well relative to disaggregation of national samples, especially when state samples are small (Lax and Phillips, 2009). Its limits are those of the model: the estimates depend on the predictors, and design-based and model-based weighting can diverge (Gelman, 2007). We therefore report a design-based estimate alongside every MRP estimate for areas large enough to have one.

Campus module (optional). Campus-level estimates require a separate frame of students. If such a module is fielded, its poststrata are institution type × enrolment characteristics, with counts from institutional enrolment data. Estimates for named institutions will not be published (Section 9).

8.6 Pre-registered estimands

Primary estimands.

(E1) For every Israel-related item $j \in \mathcal{I}$, the item-level explained common variance ECV_j of Definition 3.1, with a 95% interval.

(E2) The boundary set $\mathcal{B}_\tau = \{j \in \mathcal{I} : \mathrm{ECV}_j \geq \tau\}$ with $\tau = 0.5$, the point at which the general factor accounts for at least half of an item's common variance. The value of τ is fixed here, before any data are seen. Each item is classified three ways from simultaneous 95% intervals for all ECV_j, $j \in \mathcal{I}$: Bonferroni-adjusted Wald intervals on the logit scale (Proposition 4.8(b)). An item is *in* $\mathcal{B}_\tau$ if its lower limit is at least τ, *out* if its upper limit is below τ, and *undetermined* otherwise. With simultaneous intervals, the items classified in and the items not classified out bracket $\mathcal{B}_\tau$ with probability at least 0.95 in large samples, which pointwise intervals do not guarantee; in the simulation they did so in every replication with six Israel-related items (Section 7.2). Undetermined is a reportable outcome, not a failure.

(E3) The latent correlation between the stereotype factor and the Israel-attitude factor in the correlated-factors model, with a 95% interval.

(E4) Differences in the latent mean of G between the groups of Table 7, under the partial-invariance model, with stereotype-only anchoring as a sensitivity analysis (Section 8.3).

Secondary estimands.

(E5) For each sensitive item j, SDB_j from the joint model of Theorem 5.3, with the precision target of Section 8.4.

(E6) ECV_j and $\mathcal{B}_\tau$ by group, with the same simultaneous rule, for groupings in which loadings are not invariant.

(E7) MRP estimates of the mean of G by state and census region (Equation (6.2)); boundary estimands are estimated by region only, as a multiple-group model.

(E8) Within-person contrasts between each ID item and its comparison-state counterparts (block C), as a description of whether the same kind of statement is endorsed differently about Israel and about other

states. A contrast is not by itself evidence of a double standard: respondents may know more about one case, so contrasts are also reported conditional on the knowledge measures of block K (Beattie, 2017).

(E9) The proportion agreeing with at least q of the M S items (Section 6.3), computed for comparability with published indices, and reported by group only together with the invariance diagnostics that Proposition 6.3 shows to be necessary.

Sensitivity analyses. $\mathcal{B}_\tau$ is also reported for $\tau \in \{0.3, 0.7\}$; these do not change the primary classification. Other sensitivity analyses: exclusion of respondents who fail two or more attention checks (the primary analysis keeps them, because passing attention checks correlates with respondent characteristics; Berinsky et al., 2014); a refit in which $\mathcal{I}$ is the ID block alone, which removes the trope-based items most exposed to a cross-loading on S; the primary model without the acquiescence factor; group comparisons anchored in the stereotype items alone, with the bound (4.12) (Section 8.3); the alignment model; the bifactor-$(S-1)$ model of Eid et al. (2017), in which the general factor is defined by the stereotype items alone (Remark 4.4); percentile-bootstrap intervals with at least 1,000 resamples, as a check on the Wald intervals of E1 and E2; full-information and Bayesian fits (Section 8.7); and Schmid–Leiman and exploratory bifactor solutions (Reise, 2012; Schmid and Leiman, 1957).

8.7 Model checks and decision rules

Estimation and checks. The primary estimator is the two-step limited-information estimator validated in the simulation (Section 7.1): thresholds and polychoric correlations (Olsson, 1979), with an asymptotic covariance matrix from per-respondent influence functions, then weighted least squares with diagonal weights and sandwich standard errors, from a default start and at least 20 random starts. It is primary because it is the estimator whose recovery, coverage and identification diagnostic the simulation established; the diagnostic of rule D2(iii) and the power study of Section 8.8 are defined for it. Full-information marginal maximum likelihood, exploiting the dimension reduction of the bifactor structure (Gibbons and Hedeker, 1992), and a Bayesian fit with weakly informative priors (Gelman et al., 2013) are reported as sensitivity analyses. Under the normal-ogive model all three target the same parameters (Lemma 4.1), so a material disagreement between them is itself reported as a finding about the fit. Model adequacy is assessed with limited-information fit statistics, residual correlations between item pairs, and, for the Bayesian fit, posterior predictive checks of item-pair and sum-score distributions. The bifactor indices of Rodriguez et al. (2016) are reported for all three models: explained common variance, ω_H, percentage of uncontaminated correlations and construct replicability. Their thresholds are descriptive and do not enter the decision rules.

Cross-loadings and response style. Two misspecifications would move the ECVs of Israel-related items, though not all in the same direction: a loading of trope-based items on the stereotype factor, which a bifactor model without that loading credits to G, and acquiescence (Section 3.5). In the simulation (Section 7.4), a cross-loading of 0.3 raised the mean ECV of the IT items by 0.089 and lowered that of the ID items by 0.080, and with the registered keying acquiescence moved the ECVs of single Israel-related items by between +0.135 and −0.312, upwards for agree-keyed and downwards for reverse-keyed items. Acquiescence is modelled in the primary model (Section 8.1). In the simulation of both misspecifications at the registered item counts and $n = 6{,}000$, the acquiescence factor kept the bias of every Israel-related ECV below 0.007 under both the registered and a balanced keying, against biases of up to 0.43 without it; balancing the keying without the factor did not remove the bias. With acquiescence slopes that differ between items, which the factor does not model, the largest bias was 0.043, against 0.112 without it. For cross-loadings, the mean

residual polychoric correlation between the IT and the S items under the primary fit is tested one-sided at the 5% level. If the test rejects, the ECVs and classifications of the Israel-related items are taken from a model in which every IT item also loads on S, with the cross-loadings of the whole IT block fitted jointly, ECV_j still computed from the loadings on G and I, and the estimated cross-loadings reported. In the simulation, with the test computed from the bifactor fit, it rejected in 85 to 100% of replications when the IT items cross-loaded on S with standardised loadings of 0.1 to 0.3 (in 5% without cross-loadings), and this two-stage rule kept the bias of every Israel-related ECV below 0.021, against biases of up to 0.141 for the bifactor model alone. A check that fits a cross-loading for one IT item at a time, which an earlier draft proposed, is not used: at $c = 0.3$ it cut the share of IT items on the wrong side of τ from 14.2% to 1.5% only by leaving 48.5% of them undetermined, and its one-item fits stayed biased, because the other IT items' cross-loadings remain unmodelled (Section 7.4). The refit with $\mathcal{I}$ restricted to the ID block (Section 8.6) is the corresponding check that does not rely on the test.

Decision rules. The following rules are fixed in advance.

D1, structure. If the bifactor model fails the pre-registered fit criteria and the correlated-factors model passes them, we report E3 and no boundary set. If both fail, we report descriptive item statistics and the model comparison only.

D2, identification. Item-level ECVs and the boundary set are reported only if the data support the conditions of Theorem 4.2. (i) Each domain has at least three items whose specific loadings have 95% intervals that exclude zero (Assumption A1). (ii) In at least one domain $\mathcal{D}$ the items are heterogeneous in the sense of Assumption A3. By Lemmas A.3 and A.4, the assumption fails in $\mathcal{D}$ exactly when every polychoric tetrad $R_{k_1k_2}R_{k_3k_4} - R_{k_1k_3}R_{k_2k_4}$ among its items vanishes (under the bifactor model each equals $D_{k_1k_4}D_{k_2k_3}$). Its failure is therefore the null hypothesis of a single test, the simultaneous Wald test of a non-redundant set of $|\mathcal{D}|(|\mathcal{D}| - 3)/2$ tetrads of $\mathcal{D}$ (Bollen and Ting, 1993). It uses only the polychoric correlations and their asymptotic covariance, so it is regular whether or not the bifactor model is identified, and it replaces the Bonferroni rule over all quadruples of an earlier draft. Each domain is tested at the 2.5% level, so that the rule, which needs heterogeneity in only one domain, has level at most 5%. At $n = 6{,}000$ and the registered item counts the simulation found sizes close to nominal and power of one in the heterogeneous designs (Section 7); at a smaller sample size the critical values are calibrated by simulation, because the test over-rejects at $n = 1{,}500$ in the larger domain. The joint test needs only four items in $\mathcal{D}$, which give two non-redundant tetrads. We also report the intersection–union version (Berger, 1982), which accepts heterogeneity only if the same test rejects for $\mathcal{D} \setminus \{k\}$ for every $k \in \mathcal{D}$, and so needs five items. The null hypothesis for $\mathcal{D} \setminus \{k\}$, that all its tetrads vanish, says that all its items but at most one lie on one line, not that its loadings are proportional. The intersection–union version therefore tests a stronger property, heterogeneity that survives the removal of any one item, and is conservative; it is a robustness diagnostic and does not decide. Neither test detects near-proportional loadings, against which (iii) is the guard. (iii) The fit is numerically identified: the identification diagnostic of Section 7.2, the ratio of the smallest to the largest singular value of the weighted Jacobian of the fitted polychoric correlations of the primary fit with respect to its content loadings (on G, S and I), exceeds a cut-off fixed in the registration from the simulation of Section 8.8, and at least 20 random starts reach the same optimum with no loading at a bound. The slope of the acquiescence factor is left out of the diagnostic because it enters the correlations only through its square: without acquiescence its information vanishes and the diagnostic over all parameters is zero, whereas over the content loadings it was 0.082 to 0.105 in the simulation, close to the value for the model without the factor (Section 7.4). If (i) or (ii) fails, every ECV_j is reported as not identified, no boundary set

is reported, and E3 is reported instead; if only two Israel-related items carry specific variance, the identified sets of Theorem 4.2(c) are reported. If only (iii) fails, the ECVs are reported as weakly identified and every item is classified as undetermined. The interval-based classification of E2 is used only when (iii) holds, because the guarantees of Proposition 4.8 are pointwise and unreliable near the non-identified set (Remark 4.9). Agreement across starts is necessary, not sufficient: in the simulation it exposed the two-item case but neither the proportional nor the near-proportional case (Section 7.2).

D3, invariance. Group comparisons (E4, E6) are reported only for groupings in which the anchors that pass the DIF screen satisfy the rank condition of Proposition 4.6(a). There must be at least two such anchors in each domain, at least one with a non-zero specific loading in each domain, and, within at least one domain, two anchors whose loading vectors are not proportional, judged by a 95% interval for their determinant D_{kl} (4.1) that excludes zero. Two proportional anchors per domain do not suffice. If no Israel-related anchor passes but the stereotype anchors contain a non-proportional pair, comparisons of G and of the group-specific ECVs are reported, labelled as resting on the stereotype anchors alone, and comparisons of the Israel-attitude mean are not (Remark 4.7). Otherwise only within-group estimates and the alignment results are reported, labelled exploratory (Proposition 4.6). Whatever the outcome, the stereotype-anchored comparisons and the bound (4.12) of step 3 in Section 8.3 are reported alongside.

D4, design effects. If the design-effect test rejects at the 5% level for a list, that list's estimate is not used to correct the direct estimate; E5 for that item is reported as not estimable. If floor or ceiling effects are detected, the floor-and-ceiling model of Blair and Imai (2012) replaces the difference in means.

D5, completeness. Every primary estimand is reported for every item and group, whatever its direction or magnitude. Items are dropped after fielding only for the pre-registered quality reasons (item non-response above 10% or failed comprehension in the pilot), and dropped items are listed.

8.8 Power and precision

The sample size is set by Monte Carlo simulation, a practical approach to power analysis for latent-variable models of this size (Muthén and Muthén, 2002). We use the simulation code of Section 7, which implements the primary estimator, with item parameters set from the pilot. The registration will report three precision targets: the half-width of the 95% interval for each ECV_j at the national level; the power to detect DIF with an expected-score difference of 0.25 scale points in the smallest group of Table 7; and the half-width of the 95% interval for each SDB_j under the allocation of Theorem B.5 (target 0.07; Section 8.4). The same simulation fixes the cut-off of the identification diagnostic in rule D2(iii) and reports the power of the joint tetrad test of rule D2(ii), and of its intersection–union version, at the pilot parameters. The recovery results of Section 7 indicate which of these binds. If the budget cannot meet all three, the list module is reduced (fewer sensitive items) before the main sample is, because E1–E4 do not depend on it.

8.9 Replication package and transparency

The following will be released. The registration document, with the commit hash of the analysis code. The code for every estimator, the simulation study and the figures. The final item bank, including wording, block, definitional tags and the record of tag disagreements, released after fieldwork closes. Aggregate results for every estimand, and the poststratification table. A log of deviations from the plan. De-identified microdata will be deposited with a data archive under restricted access, because the data combine religion, political opinion and geography (Section 9). Synthetic microdata generated from the fitted model will be released openly, so that the code can be run without access to the restricted file.

Table 8. Timeline, in months from the start of the project.

Months	Phase	Output
1–2	Advisory group convened; candidate items drafted; tagging	Candidate bank; tag record
3–4	Cognitive interviews; ethics review	Revised bank; approval
5	Pilot fieldwork and screening	Pilot report (not confirmatory)
6	Item bank, anchors, J, allocation and thresholds frozen	Registration, with code hash
7–8	Main fieldwork	Raw data, held by panel provider
9	Data-quality checks; data freeze	Frozen analysis file
10–11	Confirmatory analysis as registered	Results for E1–E9; deviations log
12	Release	Paper, replication package, data deposit

Table 8 gives the timeline. Fieldwork dates will be recorded and reported with every estimate. Attitudes about Israel may move with events, so the estimates describe the population at the time of fieldwork and are not assumed to be stable.

9 Ethics and responsible interpretation

The survey of Section 8 asks people to respond to statements about Jews and about Israel, some of which many readers will find offensive and some of which express political positions held by large parts of the public. Its respondents include Jewish people, Israelis, Palestinians, Muslims and people of every political view. Its results will be read in a debate in which the stakes are high and the parties distrust each other's evidence. This section sets out the safeguards the plan commits to and the limits on what the results can be used to show.

9.1 Review, consent and debriefing

The protocol will be reviewed by an institutional review board before the pilot, and again before the main survey if the pilot leads to changes. The consent page states, before any question is shown, that the survey includes statements about Jews and about Israel, that some of these statements are offensive stereotypes and that respondents with different views may find others offensive or distressing, that respondents may skip any question or stop at any time without penalty, and that answers are confidential. Consent is recorded before the first item.

Two features of the design need explicit justification to the review board. First, respondents are shown stereotype statements, and Israel-related statements that some respondents will find offensive. These are needed to measure the attitudes they express, and the item bank uses the smallest number that identification and precision allow (Section 8.2). Second, the list experiment does not tell respondents which item on a list is of interest. This is indirect questioning, not deception: nothing false is stated, and the purpose of the technique is explained in the debriefing.

Every respondent who starts the survey, including those who stop early, sees a debriefing page. It states that the study measures how widely the statements about Jews and about Israel are endorsed and how endorsement of the two kinds of statement is related, and it applies one rule to every item: inclusion of any statement does not mean the researchers regard it as true, false, acceptable or unacceptable. It explains the list experiment, gives contact details for the research team and the review board, and lists sources of

support for respondents who found the content distressing. The pilot measures whether respondents report distress, and the debriefing is revised if it is inadequate.

9.2 Data protection

The survey data combine religion, political opinion, attitudes toward an ethnic and religious group, and geography. In the European Union, data revealing political opinions or religious beliefs are special categories of personal data (European Parliament and Council of the European Union, 2016); any cross-national extension will be designed under that regime, and we treat the US data with the same care. The research team does not receive names, addresses or contact details; these stay with the panel provider, and respondents are linked across waves only through a pseudonymous identifier. Geography is held at the level needed for the small-area model and no finer. The analysis file is stored encrypted, with access limited to named analysts. Microdata are released only under restricted access, and openly only in synthetic form (Section 8.9). Released tables apply small-cell suppression and are checked for disclosure by differencing against other released tables (Hundepool et al., 2012). Small-area results are released only as model-based estimates with intervals.

The list-experiment module needs two further safeguards. First, a list experiment reduces what an individual answer reveals; it does not make answers anonymous. A treated respondent who reports none or all of the statements on a list discloses the sensitive answer, and the negatively correlated control lists that make the estimator precise can make counts more informative about it (Section 5.4). The plan therefore bounds two disclosure measures for every list, the mutual information between the treatment count and the sensitive answer and the share of treated respondents whose posterior probability of agreement is at least 0.5, and replaces any list that exceeds either bound in the pilot (Section 8.4). Second, respondents in the paired subsample answer the sensitive item directly at the end of the questionnaire, so the file records their direct answer as well as their count, and the list gives them no additional protection. Their records are held under the same restricted access as the rest of the file, and neither the consent page nor the debriefing describes the list experiment as hiding individual answers.

9.3 Population associations are not individual diagnoses

The estimands of this paper describe populations. Four consequences follow, and every output will state them.

- A respondent's estimated position on G is a model-based summary with substantial uncertainty. It is used only to estimate population quantities. No individual score will be reported, and the methods are not a screening tool for individuals.
- An item's membership of the boundary set $\mathcal{B}_\tau$ means that, *in the surveyed population*, most of the common variance in responses to it is shared with responses to classic stereotypes. It does not mean that a particular person who endorses the item holds hostile attitudes toward Jews, or that a particular utterance with similar content is antisemitic in its context. Conversely, an item outside $\mathcal{B}_\tau$ is not thereby shown to be acceptable in every context. Definitions of antisemitism classify speech and conduct in context, often with reference to intent (Section 2.1); our estimands do not.
- Differences in mean G between ideological, religious, age or regional groups are differences between averages of overlapping distributions. They will always be reported with intervals and alongside the within-group spread, never as rankings of groups. They say nothing about any member of a group.

- Small-area estimates describe areas, not the people in them, and model-based estimates for small areas lean on the model. Estimates for named campuses will not be published (Section 8.5), because the risk of stigmatising a student body outweighs the value of the estimate.

These points apply to every group, including groups whose members are themselves frequent targets of prejudice. The analysis plan does not single out any religious or ethnic group for comparison; religiosity enters as a measure of practice, not of affiliation (Table 7).

9.4 Misuse in political debate

The results bear on a contested question, and any of them can be quoted selectively. An item placed in $\mathcal{B}_\tau$ can be cited as proof that a category of Israel-related speech is antisemitic; an item placed outside it can be cited as proof that it never is. Neither reading is supported. We commit to the following.

- *Publication whatever the results.* Every pre-registered estimand will be published, whichever position in the definitional debate it appears to support and including undetermined classifications and null results (decision rule D5).
- *No adjudication.* No output will describe the results as validating or refuting the IHRA working definition, the Jerusalem Declaration or the Nexus Document. Results will be presented item by item, with the definitional tags of Section 8.2, so that readers can relate them to any definition themselves.
- *Full context in every release.* Releases will state the population, the fieldwork dates, the threshold τ and the uncertainty of every classification. Plain-language summaries will say what the results do not show.
- *Correction.* Errors found after publication will be corrected publicly, in the paper and in the replication package.

9.5 Advisory input

Before items are drafted, we will convene an advisory group whose members hold different views on the definitions of Section 2.1. It will include representatives of Jewish communal organisations with differing positions on those definitions; Palestinian, Arab and Muslim community organisations; scholars who have argued for different definitions; civil-liberties and academic-freedom specialists; and survey methodologists. The group reviews the item bank for balance and wording, assigns the definitional tags independently, reviews the consent and debriefing texts, and comments on how results are communicated. It has no veto over the results, and its members are not asked to endorse them. Membership, funding sources and any relationship between the research team and organisations with a stake in the results will be disclosed in every output.

9.6 Neutrality

The Kornstein Lab exists to study antisemitism. Readers are entitled to ask whether that mission shapes the instrument or the analysis. We do not ask readers to rely on our intentions. The design builds in four protections. The threshold τ, the decision rules and the item bank are fixed before data collection (Section 8). The item bank covers every disputed category of Israel-related speech, and its balance is reviewed by advisers with opposing views (Section 9.5). The estimands are symmetric: the same quantity is computed for every item, and an item can fall in or out of $\mathcal{B}_\tau$ under the same rule. And the code, the synthetic data and the registered plan are public (Section 8.9), so that anyone can check that the analysis

follows the plan. The paper's aim is to make an empirical sub-question of the boundary debate answerable by evidence that people on every side can inspect, not to settle the debate.

10 Discussion

What a boundary map could show. The public argument about Israel-related speech is usually conducted as a single question: is this kind of speech about Israel antisemitic? The framework replaces it with many narrower questions, one for each statement in a pre-registered item bank, each with an answer that has an interval and stated assumptions. For each Israel-related statement, a boundary map would show how much of the common variance in responses to it is shared, through the general factor, with responses to classic stereotypes (Definition 3.1), and whether that share lies clearly above, clearly below or indeterminately around a threshold fixed in advance (Figure 2 and Proposition 4.8). It would show this only where the data identify it (Theorem 4.2), and it would say so where they do not. The correlated-factors model adds a coarser summary, the latent correlation of the two attitude domains. It is the quantity closest to the scale-level associations reported in earlier studies (Section 2.3), and it cannot by itself separate a common cause from correlated but distinct attitudes (Proposition 3.2); the item-level map can, where some domain has at least four items that differ in their general-factor shares.

Every outcome of such a map is informative to every side of the debate. Someone who holds that a category of Israel-related speech typically expresses hostility toward Jews learns whether, in the surveyed population, statements in that category covary with classic stereotypes as that view implies. Someone who holds that the category typically expresses an attitude distinct from that hostility learns the same thing about the opposite claim. A low share shows distinctness from the general factor; it does not show what the distinct attitude is, and calling it political disagreement or a different form of hostility is interpretation, not estimate. Both sides learn where the evidence is indeterminate at the sample size used, which is itself a finding.

Under partial invariance the map can also be compared across political, generational and national groups (Theorem 4.5 and Proposition 4.6). These comparisons can also be anchored in the stereotype items alone (Remark 4.7). Such an analysis need not assume Israel-related items function identically across groups; it assumes the stereotype anchors do, which is testable only in part (Proposition 4.6(b)). It is not a safe default. In the simulation, the same DIF on two stereotype anchors biased it two to three times as much as the default anchors, and it broke down at the largest DIF (Section 7.3). The plan therefore reports it as a sensitivity analysis, and every anchored comparison with the bound (4.12) at a stated maximum anchor DIF of 0.2 latent-response standard deviations (Section 8.3). In the simulated design that bound allows a first-order bias of 0.69 in the difference in means of G with the default anchors and 0.89 with stereotype-only anchors, against a true difference of 0.3, and about 0.2 if all anchors' DIF has the same sign. It measures how much a comparison rests on its anchors; it does not correct it. An item could sit inside the boundary set in one group and outside it in another. A finding of that kind would bear directly on whether one rule can be applied to speakers from every group, a question that the definitions answer differently through the weight they give to context (Section 2.1). The list-experiment module would show whether direct answers understate or overstate endorsement, item by item (Theorem 5.3). Under a uniform threshold shift, social desirability biases prevalences but leaves the boundary estimands unchanged (Proposition 5.5); it affects latent-mean comparisons only if the shift differs between groups, where it acts as DIF. The map is therefore robust to the simplest form of strategic answering, though not to a shift that acts differently on the general and the specific parts of an item's response. Multilevel regression and poststratification would

carry the latent trait to regions and states with calibrated intervals (Equation (6.2)), and Propositions 6.3 and B.8 explain why a stereotype-count index cannot do the same job without invariance diagnostics.

What it could not show. A boundary map is not a verdict. It does not say which speech ought to be called antisemitic, and it is compatible with more than one answer. A definition could reasonably treat a statement as antisemitic even if its ECV_j is low, for example because of what the statement does to the people it is about, or because definitions classify speech in context. A definition could equally decline to treat a statement as antisemitic even if its ECV_j is high, because covariance with stereotypes across a population does not establish what any particular speaker meant.

Nor is the map a diagnosis of anyone. ECV_j is a ratio of variances: it says how responses to an item move with the general factor across the population, not how many of the item's endorsers stand high on that factor (Section 3.5). An item can have a high share and still be endorsed by many respondents who stand low on the general factor, if its threshold is low. Nor is the map causal. A general factor summarises shared variance, and shared variance can have several sources: a common attitude, a common information environment, or a common response style. The design separates some of these, through knowledge measures, reverse-keyed items and comparison-state items (Section 8.2), but not all. Finally, the general factor means only what its items make it mean. We read it as hostility toward Jews as Jews because the stereotype items, whose content the definitions do not dispute, give G its interpretation; Israel items also shape it, since G is what both domains share (Section 3.2). If the items also share a broader disposition, such as a general readiness to endorse conspiratorial claims, the general factor absorbs it, and the boundary estimands inherit that ambiguity. This is why the stereotype battery, and not only the Israel-related items, is fixed before fieldwork and reviewed by advisers with opposing views (Section 9.5). Whether the general factor should be defined as what both domains share, as in our primary model, or anchored in the stereotype domain alone, as in the bifactor-$(S-1)$ variant reported as a sensitivity analysis (Remark 4.4), is itself part of the substantive question.

Limitations. We stress four; the limits on individual and normative readings are set out in Section 9.3.

Survey mode. The plan uses a self-administered online panel (Section 8.2). Mode affects what respondents are willing to report (Kreuter et al., 2008; Tourangeau and Yan, 2007), panels differ in coverage, and the assumptions of the list experiment, no design effect and no liars (Assumption A4), can be tested only in part (Corollary B.3). Estimates from one mode need not transfer to another, and the social-desirability correction is specific to the mode in which it is estimated.

Dependence on item wording. ECV_j is a property of an item, a particular wording answered by a particular population, and not of a category of speech. Two wordings of the same idea can load differently, and a statement that is double-barrelled, ambiguous or unfamiliar can land anywhere. The map generalises from items to categories only as far as the bank samples the wordings of each category, which is why the plan requires several items per disputed category and records how each definition treats each item (Section 8.2). A claim about a category resting on one item would be a claim about that item.

Translation across countries and cultures. A statement about Israel does not mean the same thing in every country, language or political culture. Translation can create differential item functioning, and scalar invariance is often hard to establish across countries (Davidov et al., 2014; Steenkamp and Baumgartner, 1998). A cross-national boundary map therefore needs a translation protocol, invariance testing of every item in every country, and, where invariance fails, anchoring vignettes (King et al., 2004) or alignment (Asparouhov and Muthén, 2014). Without these, a difference in boundary sets between countries could reflect translation rather than attitudes.

Latent-variable assumptions. Every result rests on a model: a graded response link, local independence given the factors, normal latent distributions, orthogonal factors in the bifactor model, and an assignment of each item to exactly one specific factor. Two misspecifications move the boundary estimands, though not all in one direction. If an Israel-related item, for example one that invokes a classic trope, also loads on the stereotype factor, a bifactor model without that cross-loading credits the shared variance to the general factor; and acquiescence to agree-keyed statements, if it is not modelled, loads on the general factor too. In the simulation a cross-loading raised the shares of the trope-based items and lowered those of the other Israel-related items, and acquiescence raised the shares of agree-keyed items and lowered those of reverse-keyed ones (Section 7.4). The primary model therefore includes an acquiescence factor, and the plan pre-registers a check for cross-loadings, with a cross-loading model for the whole trope-based block if the check rejects, and a refit on the items of the disputed categories alone (Section 8.7). Our identification results state what is identified under the model, not that the model is true. The decision rules compare the bifactor model with the correlated-factors and unidimensional models and report a boundary set only when the bifactor model passes pre-registered fit criteria (Section 8.7). The simulation study tests the estimators under the departures it simulates (Section 7), not under every departure that real data might present.

Extensions. Three extensions seem most valuable. First, *longitudinal invariance*. Attitudes about Israel may move with events, and so may the meaning of the items. Tracking a boundary map over time requires invariance across waves as well as groups, with anchor items that are themselves stable over time; an item that drifts across the boundary could be changing in meaning, not in the attitudes behind it. The event calendars of the second paper in this series (Kornstein Lab for Antisemitism Research, 2026b) offer a principled way to place waves before and after salient external events. Second, *conjoint designs*. The item-level estimand is observational: it describes how responses to fixed wordings covary. A conjoint design that randomises attributes of a statement, such as whether it refers to Israel or to another state, whether it invokes a classic trope, and how it is framed, would estimate the average causal effect of each attribute on endorsement, the average marginal component effect of Hainmueller et al. (2014). That would separate the content of a statement from its wording, and so address the dependence on wording described above. Third, *linkage to text*. The lab's other projects propose to measure antisemitism in online discourse, to evaluate detection algorithms and to trace narratives (Kornstein Lab for Antisemitism Research, 2026c,d,e). A boundary map could inform them: the definitional tags and estimated ECV_j of each item describe, for a population, how responses to Israel-related content covary with responses to stereotypes. That description can sit beside the labels that annotators assign under each definition; it cannot replace them, because it classifies no utterance (Section 9.3). In the other direction, narratives observed in text can suggest wordings for the item bank. Any linkage of survey respondents to their own online text would require separate consent and its own ethics review (Section 9).

Measurement as common ground. The debate over where criticism of Israel ends and antisemitism begins will not be settled by a survey. It turns on values, law and the experience of people on every side, and those are not measured by covariance. But the debate also contains empirical claims, and those can be tested in a way that does not depend on which side one is on. The threshold, the items, the decision rules and the code are fixed before the data are seen; the same rule is applied to every item; and every estimate is published whatever its direction. A result produced that way is harder to use selectively. It is a shared reference point that people who disagree about definitions can inspect, criticise and replicate. That is what measurement can offer a contested question: not an answer to it, but common ground on which to argue about it.

Declarations

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AI assistance. This paper was drafted with substantial assistance from AI language models (Anthropic's Claude), directed and reviewed by the Kornstein Lab; all derivations were checked numerically and by independent automated review, and all simulation numbers are produced by the released code.

Code and data availability. The code for the simulation study (the `attsim` Python package, the experiment driver `run_experiments.py` with its deterministic seeds, and the script that produces every figure and table of Section 7) is in the paper's repository, together with the simulation output (Appendix C.7). All data in this paper are synthetic and are regenerated by that code. No survey has been fielded and no real survey responses were used. The registration, item bank and data deposit of the planned survey are described in Section 8.9.

References

Daniel Allington, David Hirsh, and Louise Katz. The Generalised Antisemitism (GeAs) scale: A questionnaire instrument for measuring antisemitism as expressed in relation both to Jews and to Israel. *Journal of Contemporary Antisemitism*, 5(1):37–48, 2022. doi: 10.26613/jca/5.1.99.

American Jewish Committee. The state of antisemitism in America 2024: AJC's survey of the general public. Survey conducted by SSRS on the SSRS Opinion Panel, 10 October–25 November 2024, 2024a. URL https://www.ajc.org/AntisemitismReport2024/GeneralPublic. Released February 2025. Accessed November 2026.

American Jewish Committee. The state of antisemitism in America 2024: AJC's survey of American Jews. Survey conducted by SSRS, online and by telephone, 8 October–29 November 2024, 2024b. URL https://www.ajc.org/AntisemitismReport2024/AmericanJews. Released February 2025. Accessed November 2026.

T. W. Anderson and Herman Rubin. Statistical inference in factor analysis. In Jerzy Neyman, editor, *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability*, volume 5, pages 111–150, Berkeley, CA, 1956. University of California Press.

Anti-Defamation League. The ADL Global 100: An index of anti-semitism. Technical report, Anti-Defamation League, New York, 2014. URL https://www.adl.org/adl-global-100-index-antisemitism. Accessed November 2026.

Anti-Defamation League. Antisemitic attitudes in America: Topline findings. Technical report, Anti-Defamation League, New York, 2023. URL https://www.adl.org/resources/report/antisemitic-attitudes-america-topline-findings. Survey fielded in 2022. Accessed November 2026.

Peter M. Aronow, Alexander Coppock, Forrest W. Crawford, and Donald P. Green. Combining list experiment and direct question estimates of sensitive behavior prevalence. *Journal of Survey Statistics and Methodology*, 3(1): 43–66, 2015. doi: 10.1093/jssam/smu023.

Tihomir Asparouhov and Bengt Muthén. Multiple-group factor analysis alignment. *Structural Equation Modeling: A Multidisciplinary Journal*, 21(4):495–508, 2014. doi: 10.1080/10705511.2014.919210.

Reg Baker, J. Michael Brick, Nancy A. Bates, Mike Battaglia, Mick P. Couper, Jill A. Dever, Krista J. Gile, and Roger Tourangeau. Summary report of the AAPOR task force on non-probability sampling. *Journal of Survey Statistics and Methodology*, 1(2):90–143, 2013. doi: 10.1093/jssam/smt008.

Peter Beattie. Anti-Semitism and opposition to Israeli government policies: The roles of prejudice and information. *Ethnic and Racial Studies*, 40(15):2749–2767, 2017. doi: 10.1080/01419870.2016.1260751.

William Belzak and Daniel J. Bauer. Improving the assessment of measurement invariance: Using regularization to select anchor items and identify differential item functioning. *Psychological Methods*, 25(6):673–690, 2020. doi: 10.1037/met0000253.

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.

Roger L. Berger. Multiparameter hypothesis testing and acceptance sampling. *Technometrics*, 24(4):295–300, 1982.

Adam J. Berinsky, Michele F. Margolis, and Michael W. Sances. Separating the shirkers from the workers? Making sure respondents pay attention on self-administered surveys. *American Journal of Political Science*, 58(3):739–753, 2014. doi: 10.1111/ajps.12081.

Allan Birnbaum. Some latent trait models and their use in inferring an examinee's ability. In Frederic M. Lord and Melvin R. Novick, editors, *Statistical Theories of Mental Test Scores*. Addison-Wesley, Reading, MA, 1968. Part 5 (Chapters 17–20).

Graeme Blair and Kosuke Imai. Statistical analysis of list experiments. *Political Analysis*, 20(1):47–77, 2012. doi: 10.1093/pan/mpr048.

Graeme Blair, Kosuke Imai, and Jason Lyall. Comparing and combining list and endorsement experiments: Evidence from Afghanistan. *American Journal of Political Science*, 58(4):1043–1063, 2014. doi: 10.1111/ajps.12086.

Graeme Blair, Kosuke Imai, and Yang-Yang Zhou. Design and analysis of the randomized response technique. *Journal of the American Statistical Association*, 110(511):1304–1319, 2015. doi: 10.1080/01621459.2015.1050028.

Graeme Blair, Alexander Coppock, and Margaret Moor. When to worry about sensitivity bias: A social reference theory and evidence from 30 years of list experiments. *American Political Science Review*, 114(4):1297–1315, 2020. doi: 10.1017/S0003055420000374.

Kenneth A. Bollen. *Structural Equations with Latent Variables*. Wiley, New York, 1989.

Kenneth A. Bollen and Kwok-fai Ting. Confirmatory tetrad analysis. *Sociological Methodology*, 23, 1993.

Barbara M. Byrne, Richard J. Shavelson, and Bengt Muthén. Testing for the equivalence of factor covariance and mean structures: The issue of partial measurement invariance. *Psychological Bulletin*, 105(3):456–466, 1989. doi: 10.1037/0033-2909.105.3.456.

Li Cai, Ji Seung Yang, and Mark Hansen. Generalized full-information item bifactor analysis. *Psychological Methods*, 16(3):221–248, 2011. doi: 10.1037/a0023350.

Gregory Camilli. Origin of the scaling constant $d = 1.7$ in item response theory. *Journal of Educational and Behavioral Statistics*, 19(3):293–295, 1994. doi: 10.3102/10769986019003293.

Gregory L. Candell and Fritz Drasgow. An iterative procedure for linking metrics and assessing item bias in item response theory. *Applied Psychological Measurement*, 12(3):253–260, 1988. doi: 10.1177/014662168801200304.

Fang Fang Chen. Sensitivity of goodness of fit indexes to lack of measurement invariance. *Structural Equation Modeling: A Multidisciplinary Journal*, 14(3):464–504, 2007. doi: 10.1080/10705510701301834.

Yunxiao Chen, Chengcheng Li, Jing Ouyang, and Gongjun Xu. DIF statistical inference without knowing anchoring items. *Psychometrika*, 88(4):1097–1122, 2023. doi: 10.1007/s11336-023-09930-9.

Herman Chernoff. On the distribution of the likelihood ratio. *The Annals of Mathematical Statistics*, 25(3):573–578, 1954. doi: 10.1214/aoms/1177728725.

Florette Cohen, Lee Jussim, Kent D. Harber, and Gautam Bhasin. Modern anti-Semitism and anti-Israeli attitudes. *Journal of Personality and Social Psychology*, 97(2):290–306, 2009. doi: 10.1037/a0015338.

Daniel Corstange. Sensitive questions, truthful answers? Modeling the list experiment with LISTIT. *Political Analysis*, 17(1):45–63, 2009. doi: 10.1093/pan/mpn013.

Eldad Davidov, Bart Meuleman, Jan Cieciuch, Peter Schmidt, and Jaak Billiet. Measurement equivalence in cross-national research. *Annual Review of Sociology*, 40:55–75, 2014. doi: 10.1146/annurev-soc-071913-043137.

Michael Eid, Christian Geiser, Tobias Koch, and Moritz Heene. Anomalous results in G-factor models: Explanations and alternatives. *Psychological Methods*, 22(3):541–562, 2017. doi: 10.1037/met0000083.

European Commission. First progress report of the EU strategy on combating antisemitism and fostering Jewish life. Document COM(2024) 476 final, European Commission, Brussels, October 2024. URL https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52024DC0476.

European Parliament. Resolution of 1 June 2017 on combating anti-Semitism (2017/2692(RSP)). Texts adopted, P8_TA(2017)0243, 2017. URL https://www.europarl.europa.eu/doceo/document/TA-8-2017-0243_EN.html.

European Parliament and Council of the European Union. Regulation (EU) 2016/679 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation). Official Journal of the European Union, L 119, 4 May 2016, p. 1, 2016. URL https://eur-lex.europa.eu/eli/reg/2016/679/oj. Article 9: special categories of personal data. Accessed November 2026.

European Union Agency for Fundamental Rights. Experiences and perceptions of antisemitism: Second survey on discrimination and hate crime against Jews in the EU. Technical report, Publications Office of the European Union, Luxembourg, 2018. URL https://fra.europa.eu/en/publication/2018/experiences-and-perceptions-antisemitism-second-survey-discrimination-and-hate. Accessed November 2026.

European Union Agency for Fundamental Rights. Jewish people's experiences and perceptions of antisemitism. Third FRA survey on discrimination and hate crime against Jews, Publications Office of the European Union, Luxembourg, July 2024. URL https://fra.europa.eu/sites/default/files/fra_uploads/fra-2024-experiences-perceptions-antisemitism-survey_en.pdf. Published 11 July 2024. Accessed November 2026.

Executive Office of the President. Executive order 13899 of December 11, 2019: Combating anti-semitism. *Federal Register*, 84(241):68779, December 2019. URL https://www.federalregister.gov/documents/2019/12/16/2019-27217/combating-anti-semitism. Published 16 December 2019. Document 2019-27217.

Executive Office of the President. Executive order 14188 of January 29, 2025: Additional measures to combat anti-semitism. *Federal Register*, 90(21):8847, February 2025. URL https://www.federalregister.gov/documents/2025/02/03/2025-02230/additional-measures-to-combat-anti-semitism. Published 3 February 2025. Document 2025-02230.

Guanhua Fang, Jinxin Guo, Xin Xu, Zhiliang Ying, and Susu Zhang. Identifiability of bifactor models. *Statistica Sinica*, 31(5):2309–2330, 2021. doi: 10.5705/ss.202020.0386.

Andrew Gelman. Struggles with survey weighting and regression modeling. *Statistical Science*, 22(2):153–164, 2007. doi: 10.1214/088342306000000691.

Andrew Gelman and Thomas C. Little. Poststratification into many categories using hierarchical logistic regression. *Survey Methodology*, 23(2):127–135, 1997.

Andrew Gelman, John B. Carlin, Hal S. Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin. *Bayesian Data Analysis*. CRC Press, Boca Raton, FL, third edition, 2013. ISBN 978-1-4398-4095-5.

Yair Ghitza and Andrew Gelman. Deep interactions with MRP: Election turnout and voting patterns among small electoral subgroups. *American Journal of Political Science*, 57(3):762–776, 2013. doi: 10.1111/ajps.12004.

Robert D. Gibbons and Donald R. Hedeker. Full-information item bi-factor analysis. *Psychometrika*, 57(3):423–436, 1992. doi: 10.1007/BF02295430.

Robert D. Gibbons, R. Darrell Bock, Donald Hedeker, David J. Weiss, Eisuke Segawa, Dulal K. Bhaumik, David J. Kupfer, Ellen Frank, Victoria J. Grochocinski, and Angela Stover. Full-information item bifactor analysis of graded response data. *Applied Psychological Measurement*, 31(1):4–19, 2007.

Adam N. Glynn. What can we learn with statistical truth serum? Design and analysis of the list experiment. *Public Opinion Quarterly*, 77(S1):159–172, 2013. doi: 10.1093/poq/nfs070.

Samuel B. Green and Yanyun Yang. Empirical underidentification with the bifactor model: A case study. *Educational and Psychological Measurement*, 78(5):717–736, 2018.

Jens Hainmueller, Daniel J. Hopkins, and Teppei Yamamoto. Causal inference in conjoint analysis: Understanding multidimensional choices via stated preference experiments. *Political Analysis*, 22(1):1–30, 2014. doi: 10.1093/pan/mpt024.

Peter F. Halpin. Differential item functioning via robust scaling. *Psychometrika*, 89(3):796–821, 2024. doi: 10.1007/s11336-024-09957-6.

Eitan Hersh and Laura Royden. Antisemitic attitudes across the ideological spectrum. *Political Research Quarterly*, 76(2):697–711, 2023. doi: 10.1177/10659129221111081.

Paul W. Holland and Dorothy T. Thayer. Differential item performance and the Mantel–Haenszel procedure. In Howard Wainer and Henry I. Braun, editors, *Test Validity*, pages 129–145. Lawrence Erlbaum Associates, Hillsdale, NJ, 1988.

Anco Hundepool, Josep Domingo-Ferrer, Luisa Franconi, Sarah Giessing, Eric Schulte Nordholt, Keith Spicer, and Peter-Paul de Wolf. *Statistical Disclosure Control*. Wiley Series in Survey Methodology. Wiley, Chichester, 2012. doi: 10.1002/9781118348239.

Kosuke Imai. Multivariate regression analysis for the item count technique. *Journal of the American Statistical Association*, 106(494):407–416, 2011. doi: 10.1198/jasa.2011.ap10415.

Guido W. Imbens and Donald B. Rubin. *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge University Press, Cambridge, 2015. doi: 10.1017/CBO9781139025751.

International Holocaust Remembrance Alliance. Working definition of antisemitism. Non-legally binding working definition adopted by the IHRA Plenary, Bucharest, 26 May 2016, 2016. URL https://holocaustremembrance.com/resources/working-definition-antisemitism. Accessed November 2026.

Jerusalem Declaration on Antisemitism. The Jerusalem Declaration on Antisemitism. Declaration with preamble, definition, 15 guidelines and FAQ, March 2021. URL https://jerusalemdeclaration.org/. Accessed November 2026.

Edward H. Kaplan and Charles A. Small. Anti-Israel sentiment predicts anti-Semitism in Europe. *Journal of Conflict Resolution*, 50(4):548–561, 2006. doi: 10.1177/0022002706289184.

Gary King, Christopher J. L. Murray, Joshua A. Salomon, and Ajay Tandon. Enhancing the validity and cross-cultural comparability of measurement in survey research. *American Political Science Review*, 98(1):191–207, 2004. doi: 10.1017/S000305540400108X.

Julia Kopf, Achim Zeileis, and Carolin Strobl. Anchor selection strategies for DIF analysis: Review, assessment, and new approaches. *Educational and Psychological Measurement*, 75(1):22–56, 2015. doi: 10.1177/0013164414529792.

Kornstein Lab for Antisemitism Research. The dark figure: Estimating unreported antisemitic hate crime with Bayesian record linkage and multiple-systems estimation. Companion methods paper, first in the series, 2026a. November 2026.

Kornstein Lab for Antisemitism Research. Shock and contagion: Self-exciting point processes for surges in antisemitic incidents. Companion methods paper, second in the series, 2026b. November 2026.

Kornstein Lab for Antisemitism Research. Evaluating algorithms for antisemitism detection. Design-stage research proposal, 2026c. Reports no empirical results.

Kornstein Lab for Antisemitism Research. Measuring antisemitism in online discourse. Design-stage research proposal, 2026d. Reports no empirical results.

Kornstein Lab for Antisemitism Research. Mapping the spread of antisemitic narratives. Design-stage research proposal, 2026e. Reports no empirical results.

Frauke Kreuter, Stanley Presser, and Roger Tourangeau. Social desirability bias in CATI, IVR, and Web surveys: The effects of mode and question sensitivity. *Public Opinion Quarterly*, 72(5):847–865, 2008. doi: 10.1093/poq/nfn063.

James H. Kuklinski, Michael D. Cobb, and Martin Gilens. Racial attitudes and the "New South". *The Journal of Politics*, 59(2):323–349, 1997a. doi: 10.1017/S0022381600053470.

James H. Kuklinski, Paul M. Sniderman, Kathleen Knight, Thomas Piazza, Philip E. Tetlock, Gordon R. Lawrence, and Barbara Mellers. Racial prejudice and attitudes toward affirmative action. *American Journal of Political Science*, 41(2):402–419, 1997b. doi: 10.2307/2111770.

Jeffrey R. Lax and Justin H. Phillips. How should we estimate public opinion in the states? *American Journal of Political Science*, 53(1):107–121, 2009. doi: 10.1111/j.1540-5907.2008.00360.x.

Frederic M. Lord. *Applications of Item Response Theory to Practical Testing Problems*. Lawrence Erlbaum Associates, Hillsdale, NJ, 1980.

Alberto Maydeu-Olivares and Donna L. Coffman. Random intercept item factor analysis. *Psychological Methods*, 11 (4):344–362, 2006. doi: 10.1037/1082-989X.11.4.344.

William Meredith. Measurement invariance, factor analysis and factorial invariance. *Psychometrika*, 58(4):525–543, 1993. doi: 10.1007/BF02294825.

Judith Droitcour Miller. *A New Survey Technique for Studying Deviant Behavior*. PhD thesis, The George Washington University, 1984.

Roger E. Millsap. *Statistical Approaches to Measurement Invariance*. Routledge, New York, 2011.

Robert J. Mislevy. Randomization-based inference about latent variables from complex samples. *Psychometrika*, 56 (2):177–196, 1991. doi: 10.1007/BF02294457.

Linda K. Muthén and Bengt O. Muthén. How to use a Monte Carlo study to decide on sample size and determine power. *Structural Equation Modeling: A Multidisciplinary Journal*, 9(4):599–620, 2002. doi: 10.1207/S1532800 7SEM0904_8.

Nexus Task Force. The Nexus Document. Nexus Project; originally convened by the Knight Program on Media and Religion, USC Annenberg, February 2021. URL https://nexusproject.us/the-nexus-document/. Updated June 2024. Accessed November 2026.

Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018. doi: 10.1073/pnas.1708274114.

Ulf Olsson. Maximum likelihood estimation of the polychoric correlation coefficient. *Psychometrika*, 44(4):443–460, 1979.

David K. Park, Andrew Gelman, and Joseph Bafumi. Bayesian multilevel estimation with poststratification: State-level estimates from national polls. *Political Analysis*, 12(4):375–385, 2004. doi: 10.1093/pan/mph024.

Pew Research Center. Jewish Americans in 2020. Technical report, Pew Research Center, Washington, DC, May 2021. URL https://www.pewresearch.org/religion/2021/05/11/jewish-americans-in-2020/. Released 11 May 2021. Accessed November 2026.

R. L. Plackett. A reduction formula for normal multivariate integrals. *Biometrika*, 41(3–4):351–360, 1954. doi: 10.1093/biomet/41.3-4.351.

Steven P. Reise. The rediscovery of bifactor measurement models. *Multivariate Behavioral Research*, 47(5):667–696, 2012. doi: 10.1080/00273171.2012.715555.

W. S. Robinson. Ecological correlations and the behavior of individuals. *American Sociological Review*, 15(3): 351–357, 1950. doi: 10.2307/2087176.

Anthony Rodriguez, Steven P. Reise, and Mark G. Haviland. Evaluating bifactor models: Calculating and interpreting statistical indices. *Psychological Methods*, 21(2):137–150, 2016. doi: 10.1037/met0000045.

Fumiko Samejima. *Estimation of Latent Ability Using a Response Pattern of Graded Scores*. Number 17 in Psychometric Monograph. Psychometric Society, Richmond, VA, 1969.

John Schmid and John M. Leiman. The development of hierarchical factor solutions. *Psychometrika*, 22(1):53–61, 1957. doi: 10.1007/BF02289209.

Steven G. Self and Kung-Yee Liang. Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions. *Journal of the American Statistical Association*, 82(398):605–610, 1987. doi: 10.1080/01621459.1987.10478472.

L. D. Staetsky. Antisemitism in contemporary Great Britain: A study of attitudes towards Jews and Israel. Technical report, Institute for Jewish Policy Research, London, September 2017. URL https://www.jpr.org.uk/reports/antisemitism-contemporary-great-britain. In partnership with the Community Security Trust. Accessed November 2026.

Jan-Benedict E. M. Steenkamp and Hans Baumgartner. Assessing measurement invariance in cross-national consumer research. *Journal of Consumer Research*, 25(1):78–90, 1998. doi: 10.1086/209528.

Yoshio Takane and Jan de Leeuw. On the relationship between item response theory and factor analysis of discretized variables. *Psychometrika*, 52(3):393–408, 1987. doi: 10.1007/BF02294363.

David Thissen, Lynne Steinberg, and Howard Wainer. Detection of differential item functioning using the parameters of item response models. In Paul W. Holland and Howard Wainer, editors, *Differential Item Functioning*, pages 67–113. Lawrence Erlbaum Associates, Hillsdale, NJ, 1993.

Roger Tourangeau and Ting Yan. Sensitive questions in surveys. *Psychological Bulletin*, 133(5):859–883, 2007. doi: 10.1037/0033-2909.133.5.859.

Gerhard Tutz and Gunther Schauberger. A penalty approach to differential item functioning in Rasch models. *Psychometrika*, 80(1):21–43, 2015. doi: 10.1007/s11336-013-9377-6.

A. W. van der Vaart. *Asymptotic Statistics*. Number 3 in Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, Cambridge, 1998. doi: 10.1017/CBO9780511802256.

Wei Wang, David Rothschild, Sharad Goel, and Andrew Gelman. Forecasting elections with non-representative polls. *International Journal of Forecasting*, 31(3):980–991, 2015. doi: 10.1016/j.ijforecast.2014.06.001.

Stanley L. Warner. Randomized response: A survey technique for eliminating evasive answer bias. *Journal of the American Statistical Association*, 60(309):63–69, 1965. doi: 10.1080/01621459.1965.10480775.

Halbert White. Maximum likelihood estimation of misspecified models. *Econometrica*, 50(1):1–25, 1982. doi: 10.2307/1912526.

Carol M. Woods. Empirical selection of anchors for tests of differential item functioning. *Applied Psychological Measurement*, 33(1):42–57, 2009. doi: 10.1177/0146621607314044.

David S. Yeager, Jon A. Krosnick, LinChiat Chang, Harold S. Javitz, Matthew S. Levendusky, Alberto Simpser, and Rui Wang. Comparing the accuracy of RDD telephone surveys and Internet surveys conducted with probability and non-probability samples. *Public Opinion Quarterly*, 75(4):709–747, 2011. doi: 10.1093/poq/nfr020.

Yiu-Fai Yung, David Thissen, and Lori D. McLeod. On the relationship between the higher-order factor model and the hierarchical factor model. *Psychometrika*, 64(2):113–128, 1999. doi: 10.1007/BF02294531.

Richard E. Zinbarg, William Revelle, Iftah Yovel, and Wen Li. Cronbach's α, Revelle's β, and McDonald's ω_h: Their relations with each other and two alternative conceptualizations of reliability. *Psychometrika*, 70(1):123–133, 2005. doi: 10.1007/s11336-003-0974-7.

Appendix A Proofs for Sections 3–4

The appendix also contains five remarks on Section 4, placed next to the proofs they relate to (Appendices A.4, A.7 and A.8). Throughout, offdiag($\boldsymbol{M}$) denotes the off-diagonal entries of a square matrix $\boldsymbol{M}$, and two matrices are *off-diagonally equal* if their off-diagonal entries agree. For a domain $\mathcal{D} \in \{\mathcal{S}, \mathcal{I}\}$ we write $\boldsymbol{w} = \boldsymbol{\lambda}_{\mathcal{D}G}$ and $\boldsymbol{s} = \boldsymbol{\lambda}_{\mathcal{D}s}$ for the standardised general and specific loadings of its items, and we drop the superscript $\mathcal{D}$ when the domain is clear. Response patterns take finitely many values, so every expectation below is a finite sum. All models considered are C^∞ in their parameters, because F is smooth and the latent distributions are Gaussian.

A.1 Algebraic lemmas

Lemma A.1 (Rank-one factorisations). *If $\boldsymbol{u}\boldsymbol{v}^\top = \tilde{\boldsymbol{u}}\tilde{\boldsymbol{v}}^\top \neq \boldsymbol{0}$, then $\tilde{\boldsymbol{u}} = t\boldsymbol{u}$ and $\tilde{\boldsymbol{v}} = \boldsymbol{v}/t$ for a unique $t \neq 0$.*

Proof. Choose k, l with $u_k v_l \neq 0$; then $\tilde{u}_k \tilde{v}_l = u_k v_l \neq 0$. For every k', $\tilde{u}_{k'} = \tilde{u}_{k'} \tilde{v}_l / \tilde{v}_l = u_{k'} v_l / \tilde{v}_l$, so $\tilde{\boldsymbol{u}} = t\boldsymbol{u}$ with $t = v_l/\tilde{v}_l$. Likewise $\tilde{\boldsymbol{v}} = (u_k/\tilde{u}_k)\boldsymbol{v} = \boldsymbol{v}/t$. Uniqueness is clear because $\boldsymbol{u} \neq \boldsymbol{0}$. □

Lemma A.2 (One-factor off-diagonal structure). *Let $\boldsymbol{u} \in \mathbb{R}^p$ have at least three non-zero entries. If $\tilde{\boldsymbol{u}}\tilde{\boldsymbol{u}}^\top$ and $\boldsymbol{u}\boldsymbol{u}^\top$ are off-diagonally equal, then $\tilde{\boldsymbol{u}} = \pm\boldsymbol{u}$.*

Proof. Let $N = \{k : u_k \neq 0\}$, so $|N| \geq 3$. For $k \in N$ pick distinct $l, m \in N \setminus \{k\}$. Then $\tilde{u}_l \tilde{u}_m = u_l u_m \neq 0$ and $\tilde{u}_k^2 = (\tilde{u}_k \tilde{u}_l)(\tilde{u}_k \tilde{u}_m)/(\tilde{u}_l \tilde{u}_m) = u_k^2$. For $k \notin N$ and any $l \in N$, $\tilde{u}_k \tilde{u}_l = 0$ with $\tilde{u}_l \neq 0$, so $\tilde{u}_k = 0$. Finally, $\tilde{u}_k \tilde{u}_l = u_k u_l$ for $k, l \in N$ gives $\operatorname{sgn}(\tilde{u}_k/u_k) = \operatorname{sgn}(\tilde{u}_l/u_l)$, so the sign change is common to all entries. □

Lemma A.3 (Tetrads). *For $\boldsymbol{w}, \boldsymbol{s} \in \mathbb{R}^p$, $\varsigma \in \mathbb{R}$ and $\boldsymbol{M} = \boldsymbol{s}\boldsymbol{s}^\top + \varsigma \boldsymbol{w}\boldsymbol{w}^\top$, define $D_{kl} = w_k s_l - s_k w_l$. For distinct $k_1, \dots, k_4$,*

$$M_{k_1k_2} M_{k_3k_4} - M_{k_1k_3} M_{k_2k_4} = \varsigma\, D_{k_1k_4} D_{k_2k_3},$$

whereas the same tetrad of any rank-one matrix $\boldsymbol{u}\boldsymbol{u}^\top$ vanishes.

Proof. Write $1, \dots, 4$ for $k_1, \dots, k_4$. Expanding, the terms in ς^0 ($s_1 s_2 s_3 s_4$) and in ς^2 ($w_1 w_2 w_3 w_4$) cancel, leaving $\varsigma\,(s_1 s_2 w_3 w_4 + w_1 w_2 s_3 s_4 - s_1 s_3 w_2 w_4 - w_1 w_3 s_2 s_4)$. On the other hand, $D_{14} D_{23} = (w_1 s_4 - s_1 w_4)(w_2 s_3 - s_2 w_3) = w_1 w_2 s_3 s_4 - w_1 w_3 s_2 s_4 - s_1 s_3 w_2 w_4 + s_1 s_2 w_3 w_4$, which is the same bracket. For $\boldsymbol{u}\boldsymbol{u}^\top$ the tetrad is $u_1 u_2 u_3 u_4 - u_1 u_3 u_2 u_4 = 0$. □

Lemma A.4 (When heterogeneity fails). *Let a domain contain $p \geq 4$ items whose loading vectors (w_k, s_k) are all non-zero. There exist distinct $k_1, \dots, k_4$ with $D_{k_1k_4} D_{k_2k_3} \neq 0$ if and only if it is not the case that all items, with at most one exception, lie on one line through the origin.*

Proof. $D_{kl} = 0$ if and only if the two non-zero vectors are proportional, which is an equivalence relation; call its classes *lines*. Note that the condition concerns two disjoint non-proportional pairs, and relabelling within the quadruple is free. If one line contains all items but at most one, every pair of distinct items that avoids the exceptional item is proportional. Two disjoint pairs cannot both contain the exceptional item, so no quadruple works. Conversely, suppose no line contains $p - 1$ items. If two lines each contain at least two items, say k_1, k_2 on one and k_3, k_4 on the other, the pairs (k_1, k_3) and (k_2, k_4) are non-proportional. Otherwise every line but at most one is a singleton, and since no line has $p - 1 \geq 3$ items there are at least three lines. Pick x, y, z on three different lines and a fourth item u. If u is not on the line of x, the pairs (x, u) and (y, z) work; otherwise (u, y) and (x, z) do. □

A.2 Proof of Lemma 4.1

Under the normal-ogive link, (3.4) with $\varepsilon_j \sim \mathrm{N}(0,1)$ gives $\boldsymbol{Y}^* \sim \mathrm{N}_{|\mathcal{J}|}(\boldsymbol{A}\boldsymbol{\mu}, \boldsymbol{A}\boldsymbol{\Sigma}\boldsymbol{A}^\top + \boldsymbol{I}_{|\mathcal{J}|})$, and $\boldsymbol{Y}$ is its discretisation at the thresholds. Let $Z_j = (Y_j^* - \boldsymbol{a}_j^\top \boldsymbol{\mu})/s_j$. Then $\boldsymbol{Z} \sim \mathrm{N}_{|\mathcal{J}|}(\boldsymbol{0}, \boldsymbol{R})$ and $Y_j = \sum_c \mathbf{1}\{Z_j > \bar{b}_{jc}\}$, so the law of $\boldsymbol{Y}$ depends on the parameters only through $(\bar{\boldsymbol{b}}, \boldsymbol{R})$. Conversely, $\mathbb{P}(Y_j \geq c) = \Phi(-\bar{b}_{jc})$ determines $\bar{b}_{jc}$. For $j \neq k$, $\mathbb{P}(Y_j \geq 1, Y_k \geq 1) = \Phi_2(-\bar{b}_{j1}, -\bar{b}_{k1}; R_{jk})$, where $\Phi_2(\cdot,\cdot;\rho)$ is the standard bivariate normal distribution function with correlation ρ. Since $\partial\Phi_2(x,y;\rho)/\partial\rho = \phi_2(x,y;\rho) > 0$ for $|\rho| < 1$ (Plackett, 1954), and $|R_{jk}| < 1$ because the error variances are positive, this probability determines R_{jk}. In the reference group, $s_j^2 = 1 + \|\boldsymbol{a}_j\|^2$ and $h_j^2 = \|\boldsymbol{a}_j\|^2/(1+\|\boldsymbol{a}_j\|^2)$, so $s_j^{-2} = 1 - h_j^2$, $R_{jk} = \boldsymbol{a}_j^\top \boldsymbol{a}_k/(s_j s_k) = \boldsymbol{\lambda}_j^\top \boldsymbol{\lambda}_k$ and $\bar{b}_{jc} = b_{jc}(1-h_j^2)^{1/2}$. Given $\boldsymbol{\lambda}_j$ with $\|\boldsymbol{\lambda}_j\|^2 < 1$, the map $\boldsymbol{a}_j = \boldsymbol{\lambda}_j(1-\|\boldsymbol{\lambda}_j\|^2)^{-1/2}$ satisfies $1 + \|\boldsymbol{a}_j\|^2 = (1 - \|\boldsymbol{\lambda}_j\|^2)^{-1}$ and hence returns $\boldsymbol{\lambda}_j$. This establishes the bijection and its inverse. □

A.3 Proof of Theorem 4.2

By Lemma 4.1, two reference-group parameter values are observationally equivalent if and only if they share $\bar{\boldsymbol{b}}$ and $\mathrm{offdiag}(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top)$. Because the map to $(\boldsymbol{A}, \boldsymbol{b})$ is a bijection, it suffices to determine the set of $\tilde{\boldsymbol{\Lambda}}$ with the pattern (3.1), communalities below one, and $\tilde{\boldsymbol{\Lambda}}\tilde{\boldsymbol{\Lambda}}^\top$ off-diagonally equal to $\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top$. Fix such a $\tilde{\boldsymbol{\Lambda}}$.

Step 1: the cross-domain block. For $k \in \mathcal{S}$ and $j \in \mathcal{I}$, $(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top)_{kj} = \lambda_{kG}\lambda_{jG}$, because the specific factors do not overlap. Hence $\tilde{\boldsymbol{w}}^{\mathcal{S}}(\tilde{\boldsymbol{w}}^{\mathcal{I}})^\top = \boldsymbol{w}^{\mathcal{S}}(\boldsymbol{w}^{\mathcal{I}})^\top$. Under Assumption A2 this matrix is non-zero, and Lemma A.1 gives $t \neq 0$ with $\tilde{\boldsymbol{w}}^{\mathcal{S}} = t\boldsymbol{w}^{\mathcal{S}}$ and $\tilde{\boldsymbol{w}}^{\mathcal{I}} = \boldsymbol{w}^{\mathcal{I}}/t$. Reflecting G if necessary, we may take $t > 0$.

Step 2: within a domain. Within $\mathcal{S}$, $\tilde{\boldsymbol{w}}\tilde{\boldsymbol{w}}^\top + \tilde{\boldsymbol{s}}\tilde{\boldsymbol{s}}^\top$ is off-diagonally equal to $\boldsymbol{w}\boldsymbol{w}^\top + \boldsymbol{s}\boldsymbol{s}^\top$. By Step 1 this says that $\tilde{\boldsymbol{s}}\tilde{\boldsymbol{s}}^\top$ is off-diagonally equal to $\boldsymbol{s}\boldsymbol{s}^\top + \varsigma_{\mathcal{S}}\boldsymbol{w}\boldsymbol{w}^\top$ with $\varsigma_{\mathcal{S}} = 1 - t^2$. Within $\mathcal{I}$ the same holds with $\varsigma_{\mathcal{I}} = 1 - t^{-2}$.

Part (a). Let $\mathcal{D}$ be the domain in Assumption A3. By Lemma A.3, the tetrad of $\tilde{\boldsymbol{s}}\tilde{\boldsymbol{s}}^\top$ on $(k_1, \ldots, k_4)$ is zero and equals $\varsigma_{\mathcal{D}} D_{k_1k_4} D_{k_2k_3}$. Hence $\varsigma_{\mathcal{D}} = 0$, and since $t > 0$ both conditions $\varsigma_{\mathcal{S}} = 0$ and $\varsigma_{\mathcal{I}} = 0$ are equivalent to $t = 1$. So $\tilde{\boldsymbol{w}} = \boldsymbol{w}$ in both domains, and in each domain $\tilde{\boldsymbol{s}}\tilde{\boldsymbol{s}}^\top$ is off-diagonally equal to $\boldsymbol{s}\boldsymbol{s}^\top$. By Assumption A1 and Lemma A.2, $\tilde{\boldsymbol{s}} = \pm\boldsymbol{s}$, which is a reflection of the specific factor. Thus $\tilde{\boldsymbol{\Lambda}}$ equals $\boldsymbol{\Lambda}$ up to column signs, and so do the item parameters, by Lemma 4.1. The ECVs (3.8) depend on squared loadings only, so they and the sets $\mathcal{B}_\tau$ are identified.

Part (b): construction of the curve. We first show that in each domain $\mathcal{D}$ satisfying Assumption A1 but not Assumption A3 there is a continuous map $\varsigma \mapsto \boldsymbol{u}(\varsigma)$ on a neighbourhood of 0 with $\boldsymbol{u}(0) = \boldsymbol{s}$, such that $\boldsymbol{u}(\varsigma)\boldsymbol{u}(\varsigma)^\top$ is off-diagonally equal to $\boldsymbol{M}(\varsigma) = \boldsymbol{s}\boldsymbol{s}^\top + \varsigma\boldsymbol{w}\boldsymbol{w}^\top$.

- If $|\mathcal{D}| = 3$, Assumption A1 makes all three s_k non-zero. For $\{k,l,m\} = \mathcal{D}$ put $u_k(\varsigma) = \mathrm{sgn}(s_k)\{M_{kl}(\varsigma)M_{km}(\varsigma)/M_{lm}(\varsigma)\}^{1/2}$. For small $|\varsigma|$ each $M_{kl}(\varsigma)$ has the sign of $s_k s_l \neq 0$, so the ratio is positive and continuous, and it equals s_k^2 at $\varsigma = 0$. Moreover $u_k u_l = \mathrm{sgn}(s_k s_l)\,|M_{kl}| = M_{kl}$.
- If $|\mathcal{D}| \geq 4$, Lemma A.4 gives a unit vector (w^*, s^*) and scalars α_k with $(w_k, s_k) = \alpha_k(w^*, s^*)$ for all $k \in \mathcal{D}$ except at most one item o. Line items have $s_k = \alpha_k s^*$, so if $s^* = 0$ at most one item of $\mathcal{D}$ would have a non-zero specific loading, contradicting Assumption A1. Orient the line so that $s^* > 0$, put $r(\varsigma) = (s^{*2} + \varsigma w^{*2})^{1/2}$, which is positive near 0, and set $u_k = \alpha_k r(\varsigma)$ on the line and $u_o = (s^* s_o + \varsigma w^* w_o)/r(\varsigma)$. Then $u_k u_l = \alpha_k\alpha_l(s^{*2} + \varsigma w^{*2}) = M_{kl}$ and $u_k u_o = \alpha_k(s^* s_o + \varsigma w^* w_o) = M_{ko}$, and $\boldsymbol{u}(0) = \boldsymbol{s}$.

Now let both domains fail Assumption A3. For t near 1 define $\tilde{\boldsymbol{w}}^{\mathcal{S}} = t\boldsymbol{w}^{\mathcal{S}}$, $\tilde{\boldsymbol{w}}^{\mathcal{I}} = \boldsymbol{w}^{\mathcal{I}}/t$, $\tilde{\boldsymbol{s}}^{\mathcal{S}} = \boldsymbol{u}^{\mathcal{S}}(1-t^2)$ and $\tilde{\boldsymbol{s}}^{\mathcal{I}} = \boldsymbol{u}^{\mathcal{I}}(1-t^{-2})$. The cross-domain block is $\boldsymbol{w}^{\mathcal{S}}(\boldsymbol{w}^{\mathcal{I}})^\top$, as required. Within $\mathcal{S}$, $\tilde{\boldsymbol{w}}\tilde{\boldsymbol{w}}^\top + \tilde{\boldsymbol{s}}\tilde{\boldsymbol{s}}^\top$ is

off-diagonally equal to $t^2\boldsymbol{w}\boldsymbol{w}^\top + \boldsymbol{s}\boldsymbol{s}^\top + (1-t^2)\boldsymbol{w}\boldsymbol{w}^\top = \boldsymbol{w}\boldsymbol{w}^\top + \boldsymbol{s}\boldsymbol{s}^\top$, and similarly within $\mathcal{I}$. All communalities are continuous in t and below one at $t = 1$, so they stay below one on an open interval $T \ni 1$. Mapping back with Lemma 4.1, keeping $\bar{\boldsymbol{b}}$ fixed, gives a continuous curve of observationally equivalent $(\boldsymbol{A}(t), \boldsymbol{b}(t))$ satisfying (4.2). It is injective because $\boldsymbol{w}^{\mathcal{I}} \neq \boldsymbol{0}$.

Part (b): the ECVs move. Let $\varsigma = \varsigma_{\mathcal{I}} = 1 - t^{-2}$, which is strictly increasing in t, and suppose all loadings in $\mathcal{I}$ are non-zero. For $j \in \mathcal{I}$ the odds $\mathrm{ECV}_j/(1-\mathrm{ECV}_j)$ along the curve equal $O_j(\varsigma) = w_j^2(1-\varsigma)/u_j(\varsigma)^2$.

- If $|\mathcal{I}| = 3$, put $x_k = w_k/s_k \neq 0$. Then $u_j^2 = s_j^2(1+\varsigma x_j x_l)(1+\varsigma x_j x_m)/(1+\varsigma x_l x_m)$, and

$$O_j(\varsigma) = x_j^2 \, \frac{(1-\varsigma)(1+\varsigma x_l x_m)}{(1+\varsigma x_j x_l)(1+\varsigma x_j x_m)}.$$

  If O_j were constant on an interval, the two quadratic polynomials $(1-\varsigma)(1+\varsigma x_l x_m)$ and $(1+\varsigma x_j x_l)(1+\varsigma x_j x_m)$ would coincide. Their ς^2 coefficients are $-x_l x_m$ and $x_j^2 x_l x_m$, so $x_l x_m(1+x_j^2) = 0$, which is impossible. Hence every ECV_j is non-constant.

- If $|\mathcal{I}| \geq 4$, every item on the common line has $O_j(\varsigma) = w^{*2}(1-\varsigma)/(s^{*2}+\varsigma w^{*2})$, where $w^* \neq 0$ because $w_j \neq 0$. Its derivative is $-w^{*2}(s^{*2}+w^{*2})/(s^{*2}+\varsigma w^{*2})^2 < 0$, so the odds are strictly decreasing. At least $|\mathcal{I}| - 1$ items lie on the line, and all of them do when the loadings are proportional.

A continuous non-constant function of t on T has a range with non-empty interior. For τ in that interior, some equivalent parameter values place j in $\mathcal{B}_\tau$ and others do not.

Part (c). Steps 1–2 and the argument of part (a), applied to $\mathcal{S}$, give $t = 1$. Hence all general loadings are identified, and $\boldsymbol{\lambda}_{SS}$ is identified up to sign by Lemma A.2. The Israel domain contributes the single off-diagonal entry $R_{j_1 j_2} = \lambda_{j_1 G}\lambda_{j_2 G} + \lambda_{j_1 I}\lambda_{j_2 I}$, which identifies $\varpi = \lambda_{j_1 I}\lambda_{j_2 I}$. Every $(\tilde{\lambda}_{j_1 I}, \tilde{\lambda}_{j_2 I}) = (q, \varpi/q)$ that keeps both communalities below one is observationally equivalent, and nothing else is. The admissible set is $q^2 \in \big(\varpi^2/(1-\lambda_{j_2 G}^2),\ 1-\lambda_{j_1 G}^2\big)$, which is non-empty because it contains the true value. Since $\mathrm{ECV}_{j_1} = \lambda_{j_1 G}^2/(\lambda_{j_1 G}^2 + q^2)$ is continuous and strictly decreasing in q^2, its range is the open interval (4.3). Substituting q into both ECVs gives the joint set (4.4); $\mathrm{ECV}_{j_2} = \lambda_{j_2 G}^2/(\lambda_{j_2 G}^2 + \varpi^2/q^2)$ is strictly increasing in q^2, so the two coordinates move in opposite directions. If instead $\mathcal{S}$ fails the condition of Assumption A3, the construction of part (b) applies to $\mathcal{S}$. In $\mathcal{I}$, the single equation becomes $\tilde{\lambda}_{j_1 I}\tilde{\lambda}_{j_2 I} = R_{j_1 j_2} - \lambda_{j_1 G}\lambda_{j_2 G}/t^2$, which is solvable for t near one. The general loadings of j_1, j_2, which equal λ_{jG}/t, then vary together with their specific loadings.

Part (d). The correlated-factors model is the normal-ogive model with $\boldsymbol{\Sigma} = \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$, so Lemma 4.1 applies. Write $\tilde{\alpha}_j$ for the correlation of Y_j^* with its factor. Then $R_{kl} = \tilde{\alpha}_k\tilde{\alpha}_l$ within a domain and $R_{kj} = \rho\tilde{\alpha}_k\tilde{\alpha}_j$ across domains. For $k, l \in \mathcal{S}$ and $j, m \in \mathcal{I}$ with non-zero loadings, $R_{kj}R_{lm}/(R_{kl}R_{jm}) = \rho^2$. If $\rho \neq 0$, its sign is that of $R_{k_0 j_0}$ for the positively loading marker items k_0, j_0. With at least three non-zero loadings per factor, Lemma A.2 gives the loadings up to sign, and the markers fix the signs. With two, say $\mathcal{S} \supseteq \{k, l\}$, and $\rho \neq 0$, we have $\tilde{\alpha}_k^2 = R_{kl}R_{kj}/R_{lj}$. □

A.4 Further remarks on identification

Remark A.5 (What identifies the boundary). Taken together, Proposition 3.2 and Theorem 4.2 say that the bifactor decomposition is identified by heterogeneity, not by correlation. If every item of a domain has the same signed ratio of general to specific loading, then that domain carries no information about where its common variance comes from. If this holds in both domains, the bifactor model is observationally equivalent

to a whole curve of alternatives, and to a correlated-factors model when the proportionality is exact. The boundary ECVs are then bounded below by ρ_{SI}^2 and are otherwise unrestricted. Two consequences follow for design and analysis. First, the item bank should include, within at least one domain and preferably both, items that competing definitions place on different sides of the boundary (Section 8). That is where heterogeneity of general-factor shares is to be expected if the competing claims differ empirically. Second, identification degrades continuously as the products $D_{k_1k_4}D_{k_2k_3}$ shrink. With near-proportional loadings the likelihood is nearly flat along a curve like that of part (b), and estimated ECVs can then wander along it in finite samples. The simulation study shows this degradation in a near-proportional design (Section 7.2), which is why the analysis plan pairs the formal conditions with a numerical identification diagnostic (Section 8.7).

Remark A.6 (The logistic link). Lemma 4.1 is specific to the normal-ogive link. For the logistic link, two facts hold. The curves generated by proportional loadings (Proposition 3.2) leave the law of the linear predictors unchanged, so they are exact for every link. The remaining curves of Theorem 4.2(b)–(c) change the variances of the linear predictors. Under the logistic link they are therefore not exact equivalences. Any information that separates their points, however, comes only from the difference in shape between logistic and normal-ogive response functions. After rescaling the logistic argument by 1.702, these functions differ by less than 0.01 everywhere (Camilli, 1994). Identification along these directions would rest on the assumed shape of the link, not on the data's dependence structure. We therefore impose the conditions of Theorem 4.2 under both links and do not rely on identification by functional form. Our proof of sufficiency, Theorem 4.2(a), covers the normal-ogive link only. When the logistic link is used, we *assume* that the same conditions identify the model; this is an assumption, not a result of this paper.

Remark A.7 (No Israel-related item loads on G). Assumption A2 fails under the strongest version of the "distinct attitude" hypothesis, $\boldsymbol{\lambda}_{\mathcal{I}G} = \mathbf{0}$. The polychoric correlation matrix is then block diagonal. If $\mathcal{S}$ satisfies the condition of Assumption A3, the null value $\mathrm{ECV}_j = 0$, $j \in \mathcal{I}$, is identified, but the split of the stereotype domain between G and S is determined only up to a rotation. If $\mathcal{S}$ fails it, ECV_j for $j \in \mathcal{I}$ need not be identified at all. Whenever the stereotype block admits a one-factor representation with positive uniquenesses, as it does when its loadings are proportional, the general factor can be rotated into the Israel-related domain, and every value of $\mathrm{ECV}_j \in [0, 1]$ common to all $j \in \mathcal{I}$ is observationally equivalent (Appendix A.5). The Fisher information of the bifactor model is singular at $\boldsymbol{\lambda}_{\mathcal{I}G} = \mathbf{0}$, so likelihood-ratio tests of that hypothesis within the bifactor model are non-standard. The weaker hypothesis that the two domains share no variance, $\boldsymbol{\lambda}_{\mathcal{S}G}\boldsymbol{\lambda}_{\mathcal{I}G}^{\top} = \mathbf{0}$, is equivalent to a vanishing cross-domain block of the polychoric correlation matrix. It can be tested regularly on that block without fitting any factor model; within the correlated-factors model it is the hypothesis $\rho_{SI} = 0$.

A.5 Proof of the claims in Remark A.7

Let $\boldsymbol{w}^{\mathcal{I}} = \mathbf{0}$. Suppose first that $\mathcal{S}$ satisfies the condition of Assumption A3, and let $\tilde{\boldsymbol{\Lambda}}$ be equivalent. The cross-domain block vanishes, so $\tilde{\boldsymbol{w}}^{\mathcal{S}} = \mathbf{0}$ or $\tilde{\boldsymbol{w}}^{\mathcal{I}} = \mathbf{0}$. If $\tilde{\boldsymbol{w}}^{\mathcal{S}} = \mathbf{0}$, then $\boldsymbol{w}\boldsymbol{w}^{\top} + \boldsymbol{s}\boldsymbol{s}^{\top}$ on $\mathcal{S}$ would be off-diagonally equal to a rank-one matrix. Its tetrad would then vanish, but by Lemma A.3 with $\varsigma = 1$ the tetrad is $D_{k_1k_4}D_{k_2k_3} \neq 0$. Hence $\tilde{\boldsymbol{w}}^{\mathcal{I}} = \mathbf{0}$ and $\mathrm{ECV}_j = 0$ for all $j \in \mathcal{I}$. On $\mathcal{S}$, replacing $[\boldsymbol{w}^{\mathcal{S}}, \boldsymbol{s}^{\mathcal{S}}]$ by $[\boldsymbol{w}^{\mathcal{S}}, \boldsymbol{s}^{\mathcal{S}}]\boldsymbol{Q}$ for any 2×2 orthogonal $\boldsymbol{Q}$ leaves $\boldsymbol{w}\boldsymbol{w}^{\top} + \boldsymbol{s}\boldsymbol{s}^{\top}$ unchanged, so the split between G and S is determined only up to rotation.

Suppose instead that $\boldsymbol{w}\boldsymbol{w}^{\top} + \boldsymbol{s}\boldsymbol{s}^{\top}$ on $\mathcal{S}$ is off-diagonally equal to $\boldsymbol{u}\boldsymbol{u}^{\top}$ with all $u_k^2 < 1$. This holds for proportional loadings, with $u_k = \pm h_k$. Then for every $\varphi \in [0, \pi/2]$ the parameter value with $\tilde{\boldsymbol{w}}^{\mathcal{S}} = \mathbf{0}$,

$\tilde{\boldsymbol{s}}^{\mathcal{S}} = \boldsymbol{u}$, $\tilde{\boldsymbol{w}}^{\mathcal{I}} = \sin\varphi\, \boldsymbol{s}^{\mathcal{I}}$ and $\tilde{\boldsymbol{s}}^{\mathcal{I}} = \cos\varphi\, \boldsymbol{s}^{\mathcal{I}}$ is observationally equivalent, with $\mathrm{ECV}_j = \sin^2\varphi$ for every $j \in \mathcal{I}$, which takes every value in $[0, 1]$. Finally, along the smooth curve $\boldsymbol{Q}(\varphi)$ of rotations the law of $\boldsymbol{Y}$ is constant, so the score in the tangent direction vanishes identically, and that direction lies in the null space of the Fisher information. □

A.6 Proof of Proposition 3.2

Part (a). Put $\tilde{\zeta}_D = \gamma_D\theta_G + (1-\gamma_D^2)^{1/2}\theta_D$ for $D \in \{S, I\}$. With $\theta_G, \theta_S, \theta_I$ independent standard normal, $(\tilde{\zeta}_S, \tilde{\zeta}_I)$ is bivariate normal with unit variances and covariance $\gamma_S\gamma_I = \rho_{SI}$, so it has the law of (ζ_S, ζ_I). Under (3.12) the linear predictor of item j is $a_{jG}\theta_G + a_{js}\theta_{s(j)} = \alpha_j\tilde{\zeta}_{s(j)}$. Hence the vector of linear predictors has the same law as under the correlated-factors model. Since the errors are independent of it with the same distribution F, the latent responses, and therefore the responses, have the same law for every link. Finally, $\mathrm{ECV}_j = \alpha_j^2\gamma_{s(j)}^2/\{\alpha_j^2\gamma_{s(j)}^2 + \alpha_j^2(1-\gamma_{s(j)}^2)\} = \gamma_{s(j)}^2$ for $\alpha_j \neq 0$.

Part (b). If the loading vectors of domain D lie on the line spanned by $(\cos\phi_D, \sin\phi_D)$, write $(a_{jG}, a_{js}) = \alpha_j(\cos\phi_D, \sin\phi_D)$. After reflecting the specific factor if necessary, $\sin\phi_D \geq 0$. Then $\gamma_D = \cos\phi_D$ gives (3.12) with $\rho_{SI} = \gamma_S\gamma_I$.

For the converse under the normal-ogive link, let the bifactor model with standardised loadings $(\boldsymbol{w}, \boldsymbol{s})$ be observationally equivalent to a correlated-factors model with standardised loadings $\tilde{\boldsymbol{\alpha}}$ and correlation ρ. By Lemma 4.1 the two polychoric matrices are off-diagonally equal. Across domains, $\boldsymbol{w}^{\mathcal{S}}(\boldsymbol{w}^{\mathcal{I}})^\top = \rho\,\tilde{\boldsymbol{\alpha}}^{\mathcal{S}}(\tilde{\boldsymbol{\alpha}}^{\mathcal{I}})^\top$. The left side is non-zero, so $\rho \neq 0$, and Lemma A.1 gives $\boldsymbol{w}^{\mathcal{S}} = c\,\tilde{\boldsymbol{\alpha}}^{\mathcal{S}}$ for some $c \neq 0$. Within $\mathcal{S}$, $\boldsymbol{s}\boldsymbol{s}^\top$ is then off-diagonally equal to $(1-c^2)\tilde{\boldsymbol{\alpha}}\tilde{\boldsymbol{\alpha}}^\top$. Take three items k, l, m with non-zero s. Then $(1-c^2)\tilde{\alpha}_k\tilde{\alpha}_l = s_k s_l \neq 0$, and so on for the other pairs, which gives $s_k^2 = (s_k s_l)(s_k s_m)/(s_l s_m) = (1-c^2)\tilde{\alpha}_k^2$. Hence $1 - c^2 > 0$. Applying Lemma A.2 to $\boldsymbol{u} = (1-c^2)^{1/2}\tilde{\boldsymbol{\alpha}}$, which has at least three non-zero entries, yields $\boldsymbol{s} = \pm(1-c^2)^{1/2}\tilde{\boldsymbol{\alpha}}$. Therefore $(w_k, s_k) = \tilde{\alpha}_k\big(c, \pm(1-c^2)^{1/2}\big)$ for all $k \in \mathcal{S}$, which are proportional vectors. The same argument applies to $\mathcal{I}$.

Part (c). By (a), every point of $\{\gamma_S\gamma_I = \rho_{SI},\ |\gamma_S|, |\gamma_I| \leq 1\}$ is a bifactor representation. With $\rho_{SI} \neq 0$ and at least three non-zero α_j per domain, it satisfies the conditions of (b) exactly when $\gamma_S^2 < 1$ and $\gamma_I^2 < 1$: the specific loadings $\alpha_j(1-\gamma^2)^{1/2}$ are then non-zero, and the general loadings $\alpha_j\gamma$ are non-zero because $\gamma_S\gamma_I \neq 0$. Conversely, by (b) any representation satisfying these conditions has proportional loadings and is therefore of the form (3.12), with $\gamma_S\gamma_I$ equal to ρ_{SI} by Theorem 4.2(d). On the curve $\mathrm{ECV}_j = \gamma_I^2$ for $j \in \mathcal{I}$. The constraint $\gamma_S^2 = \rho_{SI}^2/\gamma_I^2 < 1$ forces $\gamma_I^2 > \rho_{SI}^2$, and every $v \in (\rho_{SI}^2, 1)$ is attained with $\gamma_I = v^{1/2}$ and $\gamma_S = \rho_{SI}v^{-1/2}$. The end points correspond to $\gamma_I^2 = 1$, where all I-loadings vanish, and $\gamma_S^2 = 1$, where all S-loadings do. For $k \in \mathcal{S}$, $\mathrm{ECV}_k = \gamma_S^2 = \rho_{SI}^2/\gamma_I^2$. Representations outside these conditions are not covered by (b). For example, if only one Israel-related item has a non-zero specific loading, that loading enters only the item's diagonal and is free, so its ECV is not tied to ρ_{SI}. □

A.7 Proof of Theorem 4.5

Lemma A.8 (First-order pseudo-true values). *Let $\mathcal{Y}$ be finite, and let $\{p(\cdot\,;\boldsymbol{x}) : \boldsymbol{x} \in U\}$, with $U \subseteq \mathbb{R}^k$ open and $\mathbf{0} \in U$, be a family of positive probability mass functions on $\mathcal{Y}$, three times continuously differentiable in $\boldsymbol{x}$. Let $\boldsymbol{H} = \sum_y p(y;\mathbf{0})\boldsymbol{\sigma}(y)\boldsymbol{\sigma}(y)^\top$, where $\boldsymbol{\sigma}(y) = \nabla_{\boldsymbol{x}} \log p(y;\mathbf{0})$. Let $\boldsymbol{x}(\cdot)$ be a C^3 map from an open set containing a compact $\Xi_c \subset \mathbb{R}^q$ into U, with $\boldsymbol{x}(\boldsymbol{\xi}_0) = \mathbf{0}$ for some $\boldsymbol{\xi}_0$ in the interior of Ξ_c and Jacobian $\boldsymbol{L}$ at $\boldsymbol{\xi}_0$. Let $K(\boldsymbol{\xi};\boldsymbol{x}) = \sum_y p(y;\boldsymbol{x}) \log\{p(y;\boldsymbol{x})/p(y;\boldsymbol{x}(\boldsymbol{\xi}))\}$, and suppose that (i) $\boldsymbol{\xi}_0$ is the unique minimiser of $K(\cdot\,;\mathbf{0})$ on Ξ_c, and (ii) $\boldsymbol{L}^\top\boldsymbol{H}\boldsymbol{L}$ is non-singular. Then there is a neighbourhood N of*

$\mathbf{0}$ *such that, for every* $\boldsymbol{x}_\Delta \in N$, $K(\cdot\,;\boldsymbol{x}_\Delta)$ *has a unique minimiser* $\boldsymbol{\xi}^*(\boldsymbol{x}_\Delta)$ *on* Ξ_c. *It lies in the interior of* Ξ_c, *and*

$$\boldsymbol{\xi}^*(\boldsymbol{x}_\Delta) = \boldsymbol{\xi}_0 + (\boldsymbol{L}^\top \boldsymbol{H} \boldsymbol{L})^{-1} \boldsymbol{L}^\top \boldsymbol{H}\, \boldsymbol{x}_\Delta + O(\|\boldsymbol{x}_\Delta\|^2).$$

Proof. K is jointly continuous on $\Xi_c \times \bar{N}_0$ for a compact neighbourhood $\bar{N}_0$ of $\mathbf{0}$, because it is a finite sum of continuous terms with positive arguments. Minimisers exist by compactness. Let $\boldsymbol{x}_n \to \mathbf{0}$ and let $\boldsymbol{\xi}_n$ minimise $K(\cdot\,;\boldsymbol{x}_n)$. Since $K(\boldsymbol{\xi}_n;\boldsymbol{x}_n) \le K(\boldsymbol{\xi}_0;\boldsymbol{x}_n) \to K(\boldsymbol{\xi}_0;\mathbf{0}) = 0$, every limit point $\bar{\boldsymbol{\xi}}$ satisfies $K(\bar{\boldsymbol{\xi}};\mathbf{0}) \le 0$, so $\bar{\boldsymbol{\xi}} = \boldsymbol{\xi}_0$ by (i). Hence minimisers converge to $\boldsymbol{\xi}_0$; for small $\boldsymbol{x}$ they are interior and solve $\boldsymbol{\Gamma}(\boldsymbol{\xi},\boldsymbol{x}) = \mathbf{0}$, where $\boldsymbol{\Gamma}(\boldsymbol{\xi},\boldsymbol{x}) = \nabla_{\boldsymbol{\xi}} K = -\boldsymbol{L}(\boldsymbol{\xi})^\top \sum_y p(y;\boldsymbol{x}) \nabla_{\boldsymbol{x}} \log p(y;\boldsymbol{x}(\boldsymbol{\xi}))$ and $\boldsymbol{L}(\boldsymbol{\xi}) = \partial \boldsymbol{x}/\partial \boldsymbol{\xi}^\top$. At $(\boldsymbol{\xi}_0,\mathbf{0})$ we have $\sum_y p\,\boldsymbol{\sigma} = \mathbf{0}$ and $-\sum_y p\,\nabla_{\boldsymbol{x}}^2 \log p = \boldsymbol{H}$, so $\partial \boldsymbol{\Gamma}/\partial \boldsymbol{\xi}^\top = \boldsymbol{L}^\top \boldsymbol{H} \boldsymbol{L}$. The weights depend on $\boldsymbol{x}$ through $\nabla_{\boldsymbol{x}} p = p\,\boldsymbol{\sigma}$, so $\partial \boldsymbol{\Gamma}/\partial \boldsymbol{x}^\top = -\boldsymbol{L}^\top \sum_y p\,\boldsymbol{\sigma}\boldsymbol{\sigma}^\top = -\boldsymbol{L}^\top \boldsymbol{H}$. $\boldsymbol{\Gamma}$ is C^2 and its $\boldsymbol{\xi}$-derivative is invertible by (ii). The implicit function theorem gives a unique C^2 solution $\boldsymbol{\xi}(\boldsymbol{x})$ near $(\boldsymbol{\xi}_0,\mathbf{0})$ with derivative $(\boldsymbol{L}^\top \boldsymbol{H} \boldsymbol{L})^{-1} \boldsymbol{L}^\top \boldsymbol{H}$ at $\mathbf{0}$. Every minimiser for small $\boldsymbol{x}$ lies in this neighbourhood and solves the equation, so it equals $\boldsymbol{\xi}(\boldsymbol{x})$. Taylor's theorem gives the expansion. □

The two exact facts. Let $\boldsymbol{D} = \operatorname{diag}(\sigma_G, \sigma_S, \sigma_I)$ and $\boldsymbol{z} = \boldsymbol{D}^{-1}(\boldsymbol{\theta} - \boldsymbol{\mu}) \sim \mathrm{N}_3(\mathbf{0}, \boldsymbol{I})$. Under the invariant model with focal parameters $(\boldsymbol{\mu}', \boldsymbol{\ell}')$, write $\boldsymbol{\theta}' = \boldsymbol{\mu}' + \boldsymbol{D}'\boldsymbol{z}$. The predictor of item j is then

$$\boldsymbol{a}_j^\top \boldsymbol{\theta}' - b_{jc} = \boldsymbol{a}_j^\top \boldsymbol{\theta} - b_{jc} + \boldsymbol{a}_j^\top (\boldsymbol{\mu}' - \boldsymbol{\mu}) + \sum_f a_{jf} \big(\sigma'_f/\sigma_f - 1\big)(\theta_f - \mu_f).$$

This is (4.5) with $\boldsymbol{\iota} = \boldsymbol{A}(\boldsymbol{\mu}' - \boldsymbol{\mu})$ and $\beta_{jf} = a_{jf}(e^{(\ell'_f - \ell_f)/2} - 1)$. Only $f \in \{G, s(j)\}$ contribute, by the pattern (3.1), and the Jacobian at $\boldsymbol{\xi}' = \boldsymbol{\xi}$ is $\operatorname{diag}(\boldsymbol{A}, \boldsymbol{B})$. The identity for $\boldsymbol{x}_\Delta$ was shown in the text; it is exact and linear in Δ.

Part (a). Identification of $\boldsymbol{\xi}$ in the invariant model is hypothesis (i) of Lemma A.8, and non-singularity of $\boldsymbol{L}^\top \boldsymbol{H} \boldsymbol{L}$ is (ii). The lemma, with $\boldsymbol{x}_\Delta$ linear in Δ, gives $\boldsymbol{\xi}^* - \boldsymbol{\xi} = (\boldsymbol{L}^\top \boldsymbol{H} \boldsymbol{L})^{-1} \boldsymbol{L}^\top \boldsymbol{H} \boldsymbol{x}_\Delta + O(\|\Delta\|^2)$. With $\boldsymbol{L} = \operatorname{diag}(\boldsymbol{A}, \boldsymbol{B})$, the first-order term $(\Delta\boldsymbol{\mu}, \Delta\boldsymbol{\ell})$ solves

$$\begin{pmatrix} \boldsymbol{P} & \boldsymbol{C} \\ \boldsymbol{C}^\top & \boldsymbol{Q} \end{pmatrix} \begin{pmatrix} \Delta\boldsymbol{\mu} \\ \Delta\boldsymbol{\ell} \end{pmatrix} = \begin{pmatrix} \boldsymbol{A}^\top (\boldsymbol{H}_{\iota\iota}\boldsymbol{d} + \boldsymbol{H}_{\iota\beta}\boldsymbol{\Delta}_{\mathrm{sl}}) \\ \boldsymbol{B}^\top (\boldsymbol{H}_{\beta\iota}\boldsymbol{d} + \boldsymbol{H}_{\beta\beta}\boldsymbol{\Delta}_{\mathrm{sl}}) \end{pmatrix},$$

with $\boldsymbol{P} = \boldsymbol{A}^\top \boldsymbol{H}_{\iota\iota} \boldsymbol{A}$, $\boldsymbol{C} = \boldsymbol{A}^\top \boldsymbol{H}_{\iota\beta} \boldsymbol{B}$ and $\boldsymbol{Q} = \boldsymbol{B}^\top \boldsymbol{H}_{\beta\beta} \boldsymbol{B}$. Eliminating $\Delta\boldsymbol{\ell} = \boldsymbol{Q}^{-1}\{\boldsymbol{B}^\top (\boldsymbol{H}_{\beta\iota}\boldsymbol{d} + \boldsymbol{H}_{\beta\beta}\boldsymbol{\Delta}_{\mathrm{sl}}) - \boldsymbol{C}^\top \Delta\boldsymbol{\mu}\}$ gives $(\boldsymbol{P} - \boldsymbol{C}\boldsymbol{Q}^{-1}\boldsymbol{C}^\top)\Delta\boldsymbol{\mu} = \boldsymbol{A}^\top \boldsymbol{W}_\iota \boldsymbol{d} + \boldsymbol{A}^\top \boldsymbol{H}_{\iota\beta}(\boldsymbol{I} - \Pi_{\boldsymbol{B}})\boldsymbol{\Delta}_{\mathrm{sl}}$, and $\boldsymbol{P} - \boldsymbol{C}\boldsymbol{Q}^{-1}\boldsymbol{C}^\top = \boldsymbol{A}^\top \boldsymbol{W}_\iota \boldsymbol{A}$. This is (4.7). The symmetric elimination gives (4.8). Both Schur complements are non-singular because $\boldsymbol{L}^\top \boldsymbol{H} \boldsymbol{L}$ is.

Part (b): symmetry. Write $\bar{\tau}_{jc} = b_{jc} - \boldsymbol{a}_j^\top \boldsymbol{\mu}$, so the hypothesis is $\bar{\tau}_{j,C_j+1-c} = -\bar{\tau}_{jc}$; with $\bar{\tau}_{j0} = -\infty$ and $\bar{\tau}_{j,C_j+1} = +\infty$ it holds for $c = 0, \dots, C_j{+}1$. Let $\boldsymbol{z} = \boldsymbol{\theta} - \boldsymbol{\mu}$, whose law is symmetric, and let $\omega_j = \boldsymbol{a}_j^\top \boldsymbol{z} + \iota_j + \boldsymbol{\beta}_j^\top \boldsymbol{z}$ be the perturbed predictor, so that $\mathbb{P}(Y_j = k \mid \boldsymbol{z}) = F(\omega_j - \bar{\tau}_{jk}) - F(\omega_j - \bar{\tau}_{j,k+1}) =: \pi_k(\omega_j)$. By symmetry of F,

$$\pi_{C_j - k}(\omega) = F(\omega + \bar{\tau}_{j,k+1}) - F(\omega + \bar{\tau}_{jk}) = F(-\omega - \bar{\tau}_{jk}) - F(-\omega - \bar{\tau}_{j,k+1}) = \pi_k(-\omega).$$

Let $\mathsf{R}\boldsymbol{y} = (C_j - y_j)_j$. Then $p_1(\mathsf{R}\boldsymbol{y};\boldsymbol{\iota},\boldsymbol{\beta}) = \mathbb{E} \prod_j \pi_{y_j}(-\omega_j)$. Since $-\omega_j = \boldsymbol{a}_j^\top(-\boldsymbol{z}) - \iota_j + \boldsymbol{\beta}_j^\top(-\boldsymbol{z})$ and $-\boldsymbol{z}$ has the law of $\boldsymbol{z}$, this equals $p_1(\boldsymbol{y};-\boldsymbol{\iota},\boldsymbol{\beta})$. At $\boldsymbol{x} = \mathbf{0}$, therefore, $p_1(\mathsf{R}\boldsymbol{y}) = p_1(\boldsymbol{y})$, $\nabla_\iota \log p_1(\mathsf{R}\boldsymbol{y}) = -\nabla_\iota \log p_1(\boldsymbol{y})$ and $\nabla_\beta \log p_1(\mathsf{R}\boldsymbol{y}) = \nabla_\beta \log p_1(\boldsymbol{y})$. Because R is a bijection of the pattern space, $\boldsymbol{H}_{\iota\beta} = -\boldsymbol{H}_{\iota\beta} = \mathbf{0}$. With $\boldsymbol{H}_{\iota\beta} = \mathbf{0}$, $\boldsymbol{W}_\iota = \boldsymbol{H}_{\iota\iota}$ and $\boldsymbol{W}_\beta = \boldsymbol{H}_{\beta\beta}$, and the formulas reduce as stated. The remaining claims are read off: $\boldsymbol{d}$ enters only the $\boldsymbol{\mu}$-equation, and $\boldsymbol{d} = \Delta\boldsymbol{A}\boldsymbol{\mu}$ when $\Delta\boldsymbol{b} = \mathbf{0}$.

Part (b): linear-Gaussian case. Suppose $\boldsymbol{Y}^*$ is observed. The truth is $\mathrm{N}_{|\mathcal{J}|}(\boldsymbol{A}\boldsymbol{\mu}+\boldsymbol{d}, \tilde{\boldsymbol{V}})$ and the model is $\mathrm{N}_{|\mathcal{J}|}(\boldsymbol{A}\boldsymbol{\mu}', \boldsymbol{V}')$ with $\boldsymbol{V}' = \boldsymbol{A}\boldsymbol{\Sigma}'\boldsymbol{A}^\top + v_F\boldsymbol{I}$. The Kullback–Leibler divergence equals $\frac{1}{2}\{\mathrm{tr}(\boldsymbol{V}'^{-1}\tilde{\boldsymbol{V}}) + (\boldsymbol{A}\boldsymbol{\mu} + \boldsymbol{d} - \boldsymbol{A}\boldsymbol{\mu}')^\top\boldsymbol{V}'^{-1}(\cdot) + \log\det\boldsymbol{V}' - \log\det\tilde{\boldsymbol{V}} - |\mathcal{J}|\}$. For fixed $\boldsymbol{\Sigma}' \succ 0$ it is minimised over $\boldsymbol{\mu}'$ by generalised least squares, $\boldsymbol{\mu}' = \boldsymbol{\mu} + (\boldsymbol{A}^\top\boldsymbol{V}'^{-1}\boldsymbol{A})^{-1}\boldsymbol{A}^\top\boldsymbol{V}'^{-1}\boldsymbol{d}$. Let $\boldsymbol{N} = \boldsymbol{A}^\top\boldsymbol{A}/v_F$ and $\boldsymbol{K} = (\boldsymbol{\Sigma}'^{-1} + \boldsymbol{N})^{-1}$. By the Woodbury identity $\boldsymbol{V}'^{-1} = v_F^{-1}\boldsymbol{I} - v_F^{-2}\boldsymbol{A}\boldsymbol{K}\boldsymbol{A}^\top$. Hence $\boldsymbol{A}^\top\boldsymbol{V}'^{-1} = (\boldsymbol{I} - \boldsymbol{N}\boldsymbol{K})\boldsymbol{A}^\top/v_F = \boldsymbol{\Sigma}'^{-1}\boldsymbol{K}\boldsymbol{A}^\top/v_F$ and $\boldsymbol{A}^\top\boldsymbol{V}'^{-1}\boldsymbol{A} = \boldsymbol{\Sigma}'^{-1}\boldsymbol{K}\boldsymbol{N}$. Therefore $(\boldsymbol{A}^\top\boldsymbol{V}'^{-1}\boldsymbol{A})^{-1}\boldsymbol{A}^\top\boldsymbol{V}'^{-1} = \boldsymbol{N}^{-1}\boldsymbol{A}^\top/v_F = (\boldsymbol{A}^\top\boldsymbol{A})^{-1}\boldsymbol{A}^\top$. This does not depend on $\boldsymbol{\Sigma}'$, so the joint minimiser has $\boldsymbol{\mu}^* = \boldsymbol{\mu} + (\boldsymbol{A}^\top\boldsymbol{A})^{-1}\boldsymbol{A}^\top\boldsymbol{d}$ for every Δ.

Part (c). Apply (3.11) to $(\boldsymbol{a}_j, \boldsymbol{\ell}^*)$ and to $(\boldsymbol{a}_j + \Delta\boldsymbol{a}_j, \boldsymbol{\ell})$ and subtract. The identity is exact whenever $a_{jG} + \Delta a_{jG}$ and $a_{js} + \Delta a_{js}$ are non-zero.

Part (d). (i) $\boldsymbol{a}_j^\top\boldsymbol{\theta} - b_{jc} - \Delta b_j = \boldsymbol{a}_j^\top(\boldsymbol{\theta}+\boldsymbol{v}) - b_{jc}$ when $\Delta b_j = -\boldsymbol{a}_j^\top\boldsymbol{v}$, and $\boldsymbol{\theta}+\boldsymbol{v} \sim \mathrm{N}(\boldsymbol{\mu}+\boldsymbol{v}, \boldsymbol{\Sigma})$. The invariant model at $(\boldsymbol{\mu}+\boldsymbol{v}, \boldsymbol{\ell})$ thus reproduces the law, the divergence is zero, and by identification $\boldsymbol{\xi}^* = (\boldsymbol{\mu}+\boldsymbol{v}, \boldsymbol{\ell})$. (ii) If $-\boldsymbol{A}\boldsymbol{v}$ vanishes on $\mathcal{S}$, then $v_G a_{kG} + v_S a_{kS} = 0$ for all $k \in \mathcal{S}$. Since the stereotype loading vectors span $\mathbb{R}^2$, $v_G = v_S = 0$ and $\Delta\boldsymbol{b}_{\mathcal{I}} = -v_I\boldsymbol{a}_{\mathcal{I}I}$. By (i) such DIF changes only μ_I, so $\boldsymbol{\ell}$ and all ECVs are unaffected. In the linear-Gaussian case the bias of μ_G for DIF supported on $\mathcal{I}$ is $\boldsymbol{e}_G^\top(\boldsymbol{A}^\top\boldsymbol{A})^{-1}\sum_{j\in\mathcal{I}}\boldsymbol{a}_j d_j$. It vanishes for all such $\boldsymbol{d}$ only if $\boldsymbol{e}_G^\top(\boldsymbol{A}^\top\boldsymbol{A})^{-1}$ is orthogonal to $\{(a_{jG}, 0, a_{jI}) : j \in \mathcal{I}\}$. If these vectors are not all proportional they span $\{\boldsymbol{e}_G, \boldsymbol{e}_I\}$, and orthogonality would give $\boldsymbol{e}_G^\top(\boldsymbol{A}^\top\boldsymbol{A})^{-1}\boldsymbol{e}_G = 0$, which contradicts positive definiteness. (iii) This is (4.10). □

Concurrent calibration.

Remark A.9 (Concurrent calibration). If the item parameters $\boldsymbol{\nu}$ are estimated from both groups under invariance, with group shares $w_g = n_g/n$, where n_g respondents are in group g and $n = n_0 + n_1$, then Lemma A.8 applies to $\boldsymbol{\vartheta} = (\boldsymbol{\nu}, \boldsymbol{\xi})$ with information $w_0\boldsymbol{\mathcal{F}}^{(0)} + w_1\boldsymbol{\mathcal{F}}^{(1)}$, where $\boldsymbol{\mathcal{F}}^{(g)}$ is the per-respondent information in group g. Because the DIF parameters enter only the group-1 law, the cross-information with the DIF is w_1 times its group-1 value. The first-order bias is again a projection, now in the pooled information metric and onto the span of both item and structural directions. The item parameters absorb part of the DIF, so the reference group's ECVs are also biased. As $w_1 \to 0$ the result reduces to Theorem 4.5. The "ignore DIF" fit of the simulation (Section 7.3) is concurrent calibration with equal group sizes, so its probability limits are the pseudo-true values of this remark, not those of Theorem 4.5. The qualitative conclusions of Theorem 4.5(d) carry over, because the exact confounding in (d)(i) does not depend on how the item parameters are estimated.

A.8 Proof of Proposition 4.6

Part (a). Let $m_j = \boldsymbol{a}_j^\top\boldsymbol{\mu}_1$ and $\tilde{s}_j = (1 + \boldsymbol{a}_j^\top\boldsymbol{\Sigma}_1\boldsymbol{a}_j)^{1/2}$ for an anchor j. By Lemma 4.1, $\mathbb{P}(Y_j \geq c \mid g = 1) = \Phi\{(m_j - b_{jc})/\tilde{s}_j\}$ for $c = 1, 2$, with $b_{j1} < b_{j2}$ known. Writing z_c for the two normal quantiles, $\tilde{s}_j = (b_{j2} - b_{j1})/(z_1 - z_2)$ and $m_j = b_{j1} + \tilde{s}_j z_1$. The bivariate margins identify the group-1 polychoric correlations, hence $\mathrm{Cov}(Y_j^*, Y_k^*) = R_{jk}^{(1)}\tilde{s}_j\tilde{s}_k$ for anchors $j \neq k$. Thus $\boldsymbol{m}_{\mathcal{A}} = \boldsymbol{A}_{\mathcal{A}}\boldsymbol{\mu}_1$ and $\boldsymbol{C}_{\mathcal{A}} = \boldsymbol{A}_{\mathcal{A}}\boldsymbol{\Sigma}_1\boldsymbol{A}_{\mathcal{A}}^\top + \boldsymbol{I}$ are identified. Full column rank gives $\boldsymbol{\mu}_1 = (\boldsymbol{A}_{\mathcal{A}}^\top\boldsymbol{A}_{\mathcal{A}})^{-1}\boldsymbol{A}_{\mathcal{A}}^\top\boldsymbol{m}_{\mathcal{A}}$ and $\boldsymbol{\Sigma}_1 = \boldsymbol{A}_{\mathcal{A}}^+(\boldsymbol{C}_{\mathcal{A}} - \boldsymbol{I})\boldsymbol{A}_{\mathcal{A}}^{+\top}$.

For the stated equivalent form of the rank condition, recall that each row of $\boldsymbol{A}_{\mathcal{A}}$ is $(a_{jG}, a_{jS}, 0)$ or $(a_{jG}, 0, a_{jI})$. If no anchor has a non-zero I-loading (respectively S-loading), a column of $\boldsymbol{A}_{\mathcal{A}}$ vanishes. If the anchors of each domain are proportional, the rows span at most two directions. In either case the rank is below three. Conversely, a non-proportional pair of stereotype anchors spans $\{(x, y, 0)\}$, and an Israel-related anchor with $a_{jI} \neq 0$ adds an independent third direction; the case of a non-proportional pair in $\mathcal{I}$ is symmetric.

For a non-anchor k with group-1 parameters $(\boldsymbol{a}_k, b_{kc})$, the correlations with the anchors identify the vector $\boldsymbol{A}_{\mathcal{A}}\boldsymbol{\Sigma}_1\boldsymbol{a}_k/\tilde{s}_k$, with entries $R^{(1)}_{jk}\tilde{s}_j$. Full column rank then identifies $\boldsymbol{q} = \boldsymbol{a}_k/\tilde{s}_k$. From $\tilde{s}_k^2 = 1 + \tilde{s}_k^2\,\boldsymbol{q}^\top\boldsymbol{\Sigma}_1\boldsymbol{q}$ we get $\boldsymbol{q}^\top\boldsymbol{\Sigma}_1\boldsymbol{q} < 1$, $\tilde{s}_k^2 = (1 - \boldsymbol{q}^\top\boldsymbol{\Sigma}_1\boldsymbol{q})^{-1}$ and $\boldsymbol{a}_k = \tilde{s}_k\boldsymbol{q}$. Finally $\mathbb{P}(Y_k \geq c \mid g = 1) = \Phi\{(\boldsymbol{a}_k^\top\boldsymbol{\mu}_1 - b_{kc})/\tilde{s}_k\}$ identifies b_{kc}. Without anchors, $\boldsymbol{a}_j^{(1)\top}\boldsymbol{\theta} - b_{jc}^{(1)} = \boldsymbol{a}_j^{(1)\top}(\boldsymbol{\theta} + \boldsymbol{v}) - (b_{jc}^{(1)} + \boldsymbol{a}_j^{(1)\top}\boldsymbol{v})$ for every item, which is the stated invariance.

Part (b). For anchors, $\boldsymbol{a}_j^\top\boldsymbol{\theta} - b_{jc} - \Delta b_j = \boldsymbol{a}_j^\top(\boldsymbol{\theta} + \boldsymbol{v}) - b_{jc}$. For non-anchors, $\boldsymbol{a}_k^{(1)\top}\boldsymbol{\theta} - b_{kc}^{(1)} = \boldsymbol{a}_k^{(1)\top}(\boldsymbol{\theta} + \boldsymbol{v}) - (b_{kc}^{(1)} + \boldsymbol{a}_k^{(1)\top}\boldsymbol{v})$. The two descriptions therefore induce the same law, and under (a) the anchored analysis converges to the second one.

Part (c). Let $\boldsymbol{z} = \boldsymbol{\Sigma}_1^{-1/2}(\boldsymbol{\theta} - \boldsymbol{\mu}_1)$. Reparametrise each non-anchor item by its predictor as a function of $\boldsymbol{z}$, $\boldsymbol{c}_k^\top\boldsymbol{z} - e_{kc}$. For fixed $(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ this is a bijection of its parameters, and it makes the non-anchor parameters $\boldsymbol{\nu} = (\boldsymbol{c}_k, e_{k\cdot})_{k\notin\mathcal{A}}$ variation-independent of $(\boldsymbol{\mu}, \boldsymbol{\ell})$. In the coordinates $(\boldsymbol{\iota}_{\mathcal{A}}, \boldsymbol{\beta}_{\mathcal{A}}, \boldsymbol{\nu})$, the anchors' perturbations (4.5) together with $\boldsymbol{\nu}$, the anchored model is $\boldsymbol{\iota}_{\mathcal{A}} = \boldsymbol{A}_{\mathcal{A}}(\boldsymbol{\mu}' - \boldsymbol{\mu})$, $\boldsymbol{\beta}_{\mathcal{A}}$ a function of $\boldsymbol{\ell}'$ with Jacobian $\boldsymbol{B}_{\mathcal{A}}$, and $\boldsymbol{\nu}$ free. Hence $\boldsymbol{L} = \operatorname{diag}(\boldsymbol{A}_{\mathcal{A}}, \boldsymbol{B}_{\mathcal{A}}, \boldsymbol{I})$. The truth is at $(\boldsymbol{d}_{\mathcal{A}}, \boldsymbol{0}, \boldsymbol{0})$ for uniform anchor DIF, or $(\boldsymbol{d}_{\mathcal{A}}, \boldsymbol{\Delta}_{\mathrm{sl},\mathcal{A}}, \boldsymbol{0})$ in general. By Lemma A.8, the first-order term solves the normal equations of the $\boldsymbol{H}$-weighted projection. Eliminating the free block $\boldsymbol{\nu}$ (and, for uniform DIF, the $\boldsymbol{\ell}$ block) replaces $\boldsymbol{H}$ by its Schur complement, as in the proof of Theorem 4.5(a). This gives (4.11) with $\boldsymbol{W}$ the efficient information of $\boldsymbol{\iota}_{\mathcal{A}}$, and the stated analogue of (4.7)–(4.8) for non-uniform DIF. For the bound, let $\boldsymbol{e} = (\boldsymbol{M}^\top\boldsymbol{M})^{-1}\boldsymbol{M}^\top\boldsymbol{W}^{1/2}\boldsymbol{d}_{\mathcal{A}}$ be the first-order term, where $\boldsymbol{W}^{1/2}$ is the symmetric square root. Then $\boldsymbol{M}\boldsymbol{e} = \Pi\boldsymbol{W}^{1/2}\boldsymbol{d}_{\mathcal{A}}$ and $\|\boldsymbol{M}\boldsymbol{e}\|_2 \geq \sigma_{\min}(\boldsymbol{M})\|\boldsymbol{e}\|_2$, which gives the first inequality; the second holds because Π is an orthogonal projection. For one contaminated anchor, $\boldsymbol{d}_{\mathcal{A}} = -\Delta b_j\boldsymbol{e}_j$ and $\|\Pi\boldsymbol{W}^{1/2}\boldsymbol{e}_j\|_2^2 = \boldsymbol{e}_j^\top\boldsymbol{W}^{1/2}\Pi\boldsymbol{W}^{1/2}\boldsymbol{e}_j = \mathrm{lev}_j$. Moreover $\boldsymbol{W}^{1/2}\Pi\boldsymbol{W}^{1/2} \preceq \boldsymbol{W}$, so $\mathrm{lev}_j \leq W_{jj}$. □

Proof of the claims in Remark 4.7. Let $\mathcal{A} \subseteq \mathcal{S}$, let $\bar{\boldsymbol{A}}_{\mathcal{A}}$ be the $|\mathcal{A}| \times 2$ matrix of the anchors' (G, S) loadings, which has rank two because the anchors are not all proportional, and let $\bar{\boldsymbol{\Sigma}} = \operatorname{diag}(\sigma_{1G}^2, \sigma_{1S}^2)$. Group-1 item parameters below carry no superscript. As in part (a), the anchors' group-1 latent means $\bar{\boldsymbol{A}}_{\mathcal{A}}(\mu_{1G}, \mu_{1S})^\top$ and covariance $\bar{\boldsymbol{A}}_{\mathcal{A}}\bar{\boldsymbol{\Sigma}}\bar{\boldsymbol{A}}_{\mathcal{A}}^\top + \boldsymbol{I}$ are identified, hence so are μ_{1G}, μ_{1S}, σ_{1G}^2 and σ_{1S}^2. Non-anchor stereotype items are then identified as in part (a), with $(\boldsymbol{A}_{\mathcal{A}}, \boldsymbol{\Sigma}_1)$ replaced by $(\bar{\boldsymbol{A}}_{\mathcal{A}}, \bar{\boldsymbol{\Sigma}})$. For $j \in \mathcal{I}$ let $\tilde{s}_j = \{1 + (a_{jG}\sigma_{1G})^2 + (a_{jI}\sigma_{1I})^2\}^{1/2}$. For an anchor k with $a_{kG} \neq 0$, the diagonal $\boldsymbol{\Sigma}_1$ gives $\mathrm{Cov}(Y_k^*, Y_j^*) = a_{kG}\sigma_{1G}^2 a_{jG}$, so $R^{(1)}_{kj}\tilde{s}_k$ identifies $x_j = a_{jG}/\tilde{s}_j$. For distinct $j, m \in \mathcal{I}$, $R^{(1)}_{jm} = \sigma_{1G}^2 x_j x_m + u_j u_m$ with $u_j = a_{jI}\sigma_{1I}/\tilde{s}_j$. Thus $\boldsymbol{u}\boldsymbol{u}^\top$ is identified off the diagonal, and with at least three non-zero u_j, Lemma A.2 gives $\boldsymbol{u}$ up to sign. Then $\tilde{s}_j^{-2} = 1 - \sigma_{1G}^2 x_j^2 - u_j^2$ identifies $\tilde{s}_j$, hence $a_{jG} = \tilde{s}_j x_j$, $a_{jI}\sigma_{1I} = \tilde{s}_j u_j$ and $\mathrm{ECV}_j^{(1)} = (a_{jG}\sigma_{1G})^2/\{(a_{jG}\sigma_{1G})^2 + (a_{jI}\sigma_{1I})^2\}$. For the non-identification, write $\theta_I = \mu_{1I} + \sigma_{1I}z_I$. The predictor of $j \in \mathcal{I}$ is $a_{jG}\theta_G + a_{jI}\sigma_{1I}z_I + a_{jI}\mu_{1I} - b_{jc}$, and θ_I enters no other item. Any $(\mu'_{1I}, \sigma'_{1I}, a'_{jI}, b'_{jc})$ with $a'_{jI}\sigma'_{1I} = a_{jI}\sigma_{1I}$ and $a'_{jI}\mu'_{1I} - b'_{jc} = a_{jI}\mu_{1I} - b_{jc}$ for all $j \in \mathcal{I}$ and c therefore induces the same law. This is a two-parameter family indexed by $(\mu'_{1I}, \sigma'_{1I})$. Finally, if the stereotype anchors are proportional, $\bar{\boldsymbol{A}}_{\mathcal{A}}$ has rank one. A shift $\boldsymbol{v} = (v_G, v_S, 0)$ of the latent mean orthogonal to its row space leaves every anchor unchanged and is absorbed by the free thresholds of the non-anchor items, as at the end of part (a), so only one linear combination of μ_{1G} and μ_{1S} is identified. □

Choosing anchors.

Remark A.10 (Choosing anchors). Because invariance of the anchors is an identifying assumption, anchors are usually chosen from the data. Every data-driven procedure replaces "these items are invariant" with a sparsity premise: most items, or most parameters, are invariant.

- *Rank-based selection and iterative purification* test items against provisional anchors, rank them by a DIF statistic, and re-link the metric using only the items not flagged (Candell and Drasgow, 1988; Kopf et al., 2015; Lord, 1980; Woods, 2009). These procedures are transparent and work well when DIF is sparse. Their starting point, however, is contaminated by whatever DIF exists, and anchors become harder to find as the share of DIF items grows (Kopf et al., 2015; Woods, 2009).
- *Regularised and robust methods* estimate all DIF effects jointly. They use a penalty that shrinks most of them to zero (Belzak and Bauer, 2020; Tutz and Schauberger, 2015), a minimal-ℓ_1 condition that identifies the latent distribution and supports inference without a known anchor set (Chen et al., 2023), or a redescending M-estimator of the linking parameters that treats DIF items as outliers (Halpin, 2024). They need no pre-specified anchors. Their answer depends on the penalty or the identifying condition, and inference after selection needs care.
- *Alignment* (Asparouhov and Muthén, 2014) starts from the configural model and chooses group means and variances to minimise a loss that favours a few large non-invariances over many small ones. It scales to many groups, such as countries, but rests on the same sparsity premise.

The common limit is exact. DIF in the column space of the loading matrix, the same direction for many items in proportion to their loadings, is observationally equivalent to a difference in latent means (Theorem 4.5(d)(i), Proposition 4.6(b)). No procedure can detect it, and sparsity-based procedures attribute it to the latent means by construction. For the boundary question this matters most where DIF is plausible for a whole domain at once, as for Israel-related items, and some stereotype items, across ideological groups. That is the case for anchoring in the stereotype domain (Remark 4.7) and for reporting the sensitivity bound (4.12).

A.9 Proof of Proposition 4.8

The response pattern takes finitely many values, and its distribution is a smooth function of $\boldsymbol{\vartheta}$. Under the conditions of Theorem 4.2(a), with $\boldsymbol{\vartheta}$ interior and Fisher information $\boldsymbol{\mathcal{F}}$ non-singular, the maximum likelihood estimator satisfies $n^{1/2}(\widehat{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}) \Rightarrow \mathrm{N}(\mathbf{0}, \boldsymbol{\mathcal{F}}^{-1})$, and the inverse observed information is consistent. The function $\varkappa_j$ is smooth near $\boldsymbol{\vartheta}$ when $a_{jG}a_{jI} \neq 0$, with gradient $(2/a_{jG}, -2/a_{jI})$ in (a_{jG}, a_{jI}). Hence $n^{1/2}(\widehat{\varkappa}_j - \varkappa_j) \Rightarrow \mathrm{N}(0, v_j)$ with $v_j > 0$ and $n\,\widehat{\mathrm{se}}_j^2 \to v_j$ in probability. The rest of the argument uses only these two facts, so it applies to any consistent, asymptotically normal estimator with a consistent variance estimator. All limits are taken at a fixed parameter value. The argument gives no uniformity as v_j diverges near the non-identified set (Remark 4.9).

Part (a). Write the statistic as $(\widehat{\varkappa}_j - \varkappa_j)/\widehat{\mathrm{se}}_j + (\varkappa_j - \operatorname{logit}\tau)/\widehat{\mathrm{se}}_j$. The first term converges to $\mathrm{N}(0,1)$, and the second tends to 0, $+\infty$ or $-\infty$ according as $\varkappa_j =, >$ or $<$ $\operatorname{logit}\tau$, because logit is increasing. The stated limits follow, and the supremum over the null of the limiting rejection probability is α. The signed root of the likelihood-ratio statistic for the smooth scalar restriction $\varkappa_j = \operatorname{logit}\tau$ differs from the Wald statistic by $o_p(1)$ under the null and local alternatives. For the one-sided null, the likelihood-ratio statistic is zero when $\widehat{\varkappa}_j \geq \operatorname{logit}\tau$; otherwise, with probability tending to one, the restricted maximum lies on $\{\varkappa_j = \operatorname{logit}\tau\}$. Its limit is therefore $Z^2\mathbf{1}\{Z < 0\}$ with $Z \sim \mathrm{N}(0,1)$, that is $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ (Chernoff, 1954; Self and Liang, 1987).

Part (b). On the event E that $L_j \leq \mathrm{ECV}_j \leq U_j$ for all $j \in \mathcal{I}$, every $j \in \widehat{\mathcal{B}}^{\mathrm{in}}_\tau$ has $\mathrm{ECV}_j \geq L_j \geq \tau$, and every $j \in \mathcal{B}_\tau$ has $U_j \geq \mathrm{ECV}_j \geq \tau$. Hence the inclusions hold on E and $\liminf \mathbb{P}(E) \geq 1 - \alpha$. For Bonferroni intervals, each two-sided interval for $\varkappa_j$ at level $1 - \alpha/|\mathcal{I}|$ has asymptotic coverage $1 - \alpha/|\mathcal{I}|$, monotone transformation by expit preserves coverage, and the union bound gives $\liminf \mathbb{P}(E) \geq 1 - \alpha$.

Part (c). At $\mathrm{ECV}_j = \tau$, $\mathbb{P}(\widehat{\mathrm{ECV}}_j \geq \tau) = \mathbb{P}\{n^{1/2}(\widehat{\varkappa}_j - \varkappa_j) \geq 0\} \to \frac{1}{2}$. At $a_{jI} = 0$, $n(1 - \widehat{\mathrm{ECV}}_j) = n\widehat{a}_{jI}^2/(\widehat{a}_{jG}^2 + \widehat{a}_{jI}^2)$, with $n\widehat{a}_{jI}^2 \Rightarrow V_{jI}\chi_1^2$ and a denominator converging to $a_{jG}^2 > 0$. Slutsky's theorem gives the limit. □

Appendix B Supplementary results and proofs for Sections 5–6

This appendix states the supplementary results to which Sections 5 and 6 point and proves every result of those sections. Throughout, $\mathcal{A} = (A_1, \ldots, A_n)$ is the vector of arm assignments. Under Assumption A4(i), the vectors (W_i, Z_i, D_i), $i = 1, \ldots, n$, are independent and identically distributed and independent of $\mathcal{A}$. Conditionally on $\mathcal{A}$, the respondents in each arm therefore form an independent random sample from the population, and samples in different arms are independent. We use this fact without further comment.

B.1 The variance of the list experiment

Remark B.1 (Finite-population variance). Condition on the n sampled respondents and take the in-sample prevalence $\pi_n = n^{-1}\sum_i Z_i$ as the estimand. Under complete randomisation, Neyman's formula (Imbens and Rubin, 2015, ch. 6) gives $\mathrm{Var}(\widehat{\pi} \mid \text{sample}) = S_{\mathsf{T}}^2/n_{\mathsf{T}} + S_{\mathsf{C}}^2/n_{\mathsf{C}} - \pi_n(1-\pi_n)/(n-1)$, where S_{T}^2 and S_{C}^2 are the in-sample variances (divisor $n-1$) of $W_i + Z_i$ and W_i. In a general experiment the last term, the variance of unit-level effects, is not identified, and the usual variance estimator is conservative. In a list experiment the unit-level effect is $Z_i \in \{0, 1\}$, so the term is a known function of π_n and can be estimated by $\widehat{\pi}(1-\widehat{\pi})/(n-1)$.

Proof of Proposition 5.1. Unbiasedness and (5.3). Under Assumption A4(ii)–(iii), $\overline{Y}_{\mathsf{T}}$ is the mean of n_{T} independent copies of $W_i + Z_i$ and $\overline{Y}_{\mathsf{C}}$ the mean of n_{C} independent copies of W_i, and the two means are independent. Hence $\mathbb{E}\widehat{\pi} = \mathbb{E}(W_i + Z_i) - \mathbb{E}W_i = \pi$ and $\mathrm{Var}(\widehat{\pi}) = \mathrm{Var}(W_i + Z_i)/n_{\mathsf{T}} + \mathrm{Var}(W_i)/n_{\mathsf{C}}$. Expanding $\mathrm{Var}(\sum_j W_{ij})$ and $\mathrm{Var}(\sum_j W_{ij} + Z_i)$ by bilinearity gives (5.2).

Part (a). Suppose first $\sigma_{\mathsf{T}}, \sigma_{\mathsf{C}} > 0$. The function $r \mapsto \sigma_{\mathsf{T}}^2/r + \sigma_{\mathsf{C}}^2/(1-r)$ is strictly convex on $(0, 1)$, and its derivative $-\sigma_{\mathsf{T}}^2/r^2 + \sigma_{\mathsf{C}}^2/(1-r)^2$ vanishes only at $r/(1-r) = \sigma_{\mathsf{T}}/\sigma_{\mathsf{C}}$, that is at r^*. There its value is $\sigma_{\mathsf{T}}(\sigma_{\mathsf{T}} + \sigma_{\mathsf{C}}) + \sigma_{\mathsf{C}}(\sigma_{\mathsf{T}} + \sigma_{\mathsf{C}}) = (\sigma_{\mathsf{T}} + \sigma_{\mathsf{C}})^2$, which gives (5.4). If one of the two variances is zero, (5.4) is the infimum of L, approached as r tends to 0 or 1.

Part (b). Write $U = (W_i + Z_i) - \mathbb{E}(W_i + Z_i)$ and $V = -(W_i - \mathbb{E}W_i)$, so that $U + V = Z_i - \pi$. By Minkowski's inequality in L^2,

$$\sqrt{\pi(1-\pi)} = \|U + V\|_2 \leq \|U\|_2 + \|V\|_2 = \sigma_{\mathsf{T}} + \sigma_{\mathsf{C}},$$

so $L^* \geq 1$. Equality holds if and only if $V = 0$ almost surely, or $U = cV$ almost surely for some $c \geq 0$. The first case is W_i constant. In the second, $Z_i - \pi = -(1+c)(W_i - \mathbb{E}W_i)$, so $W_i = \mathbb{E}W_i - (Z_i - \pi)/(1+c)$. Because $0 < \pi < 1$, W_i then takes two values that differ by $1/(1+c) \in (0, 1]$. Since W_i is integer-valued, $1/(1+c) = 1$, so $c = 0$ and $W_i + Z_i$ is constant. Conversely, if $W_i \equiv w_0$ then $\sigma_{\mathsf{C}} = 0$ and $\sigma_{\mathsf{T}}^2 = \pi(1-\pi)$; if $W_i + Z_i \equiv y_0$ then $\sigma_{\mathsf{T}} = 0$ and $\sigma_{\mathsf{C}}^2 = \pi(1-\pi)$. In both cases $L^* = 1$. In the first case the treatment count $w_0 + Z_i$ reveals Z_i, and in the second the control count $y_0 - Z_i$ does. The converse fails. If W_i is uniform on $\{0, 2\}$ and independent of Z_i, the parity of $W_i + Z_i$ reveals Z_i, but $\sigma_{\mathsf{C}}^2 = 1$ and $\sigma_{\mathsf{T}}^2 = 1 + \pi(1-\pi)$, so $L^* > 1$; for $\pi = 0.2$, $L^* \approx 27$.

Part (c). Each c_{jk} (which appears twice in the sum over $j \neq k$) and each γ_j enters σ_{C}^2 or σ_{T}^2 linearly with a positive coefficient, and neither enters with a negative one. Since $x \mapsto \sqrt{x}$ is strictly increasing, $\sigma_{\mathsf{T}} + \sigma_{\mathsf{C}}$ and hence L^* are strictly increasing in each. If $\sum_j \gamma_j \geq 0$ then $\sigma_{\mathsf{T}} \geq \sigma_{\mathsf{C}}$, and

$$(\sigma_{\mathsf{T}} + \sigma_{\mathsf{C}})^2 \geq \sigma_{\mathsf{T}}^2 + 3\sigma_{\mathsf{C}}^2 = 4\sigma_{\mathsf{C}}^2 + \pi(1-\pi) + 2\textstyle\sum_j \gamma_j \geq 4\sigma_{\mathsf{C}}^2 + \pi(1-\pi).$$

Dividing by $\pi(1-\pi)$ gives the bound. For exchangeable items, $\sigma_{\mathsf{C}}^2 = Jp(1-p) + J(J-1)\rho p(1-p)$, which is (5.5). The condition $\rho \geq -1/(J-1)$ is $\sigma_{\mathsf{C}}^2 \geq 0$, which is necessary. It is not sufficient for binary items. For $p = 1/2$ and odd J, the integer-valued W_i has mean $J/2$, so $\sigma_{\mathsf{C}}^2 \geq 1/4$, which gives $\rho \geq -1/J$. This bound is attained by drawing W_i uniformly from $\{(J-1)/2, (J+1)/2\}$ and endorsing a uniformly random subset of that size. For large J, $(\sigma_{\mathsf{T}} + \sigma_{\mathsf{C}})^2 = 4\sigma_{\mathsf{C}}^2 + O(1)$ when $\sum_j \gamma_j$ is bounded, which gives the growth rate stated for uncorrelated items.

Remark B.1. Condition on the sample, so that the (W_i, Z_i) are fixed and only $\mathcal{A}$ is random. Under Assumption A4(ii)–(iii) the potential responses are $Y_i(\mathsf{C}) = W_i$ and $Y_i(\mathsf{T}) = W_i + Z_i$, and the unit-level effect is $Y_i(\mathsf{T}) - Y_i(\mathsf{C}) = Z_i$. Neyman's formula for completely randomised experiments (Imbens and Rubin, 2015, ch. 6) gives $\mathrm{Var}(\widehat{\pi} \mid \text{sample}) = S_{\mathsf{T}}^2/n_{\mathsf{T}} + S_{\mathsf{C}}^2/n_{\mathsf{C}} - S_Z^2/n$, where $S_Z^2 = (n-1)^{-1}\sum_i (Z_i - \pi_n)^2$. Because $Z_i^2 = Z_i$, $\sum_i (Z_i - \pi_n)^2 = n\pi_n(1-\pi_n)$, so $S_Z^2/n = \pi_n(1-\pi_n)/(n-1)$. □

B.2 Proof of Theorem 5.3

We first record the identity SDB $= \pi^H - \pi^F$:

$$\pi - \pi^D = \{\mathbb{P}(Z_i = 1, D_i = 0) + \mathbb{P}(Z_i = 1, D_i = 1)\} - \{\mathbb{P}(Z_i = 0, D_i = 1) + \mathbb{P}(Z_i = 1, D_i = 1)\} = \pi^H - \pi^F.$$

Lemma B.2 (Stratified means in the paired design). *Under Assumption A4(i)–(iv), suppose $n_a/n \to r_a > 0$ for $a \in \{\mathsf{C}, \mathsf{T}\}$ and $0 < P_0 < 1$. For $a \in \{\mathsf{C}, \mathsf{T}\}$ and $d \in \{0, 1\}$, let $m_{ad} = \mathbb{E}(Y_i \mid a, D_i = d)$ and $v_{ad} = \mathrm{Var}(Y_i \mid a, D_i = d)$, so $m_{\mathsf{T}d} - m_{\mathsf{C}d} = \Delta_d$, $v_{\mathsf{T}d} = \mathrm{Var}(W_i + Z_i \mid D_i = d)$ and $v_{\mathsf{C}d} = \mathrm{Var}(W_i \mid D_i = d)$. Then $\sqrt{n}(\widehat{P}_0 - P_0)$, $\sqrt{n}(\overline{Y}_{ad} - m_{ad})$ for the four pairs (a, d) are jointly asymptotically normal and asymptotically independent, with asymptotic variances P_0P_1 and $v_{ad}/(r_aP_d)$.*

Proof. Within arm a, $\overline{Y}_{ad} = \{n_a^{-1}\sum_{i\in a} \mathbf{1}\{D_i = d\}\, Y_i\}/\{n_a^{-1}\sum_{i\in a}\mathbf{1}\{D_i = d\}\}$ is a ratio of two sample means of independent and identically distributed terms. By the multivariate central limit theorem and the delta method, $\sqrt{n_a}(\overline{Y}_{ad} - m_{ad}) = n_a^{-1/2}\sum_{i\in a}\mathbf{1}\{D_i = d\}\,(Y_i - m_{ad})/P_d + o_p(1)$. The summands for $d = 0$ and $d = 1$ have zero covariance, because $\mathbf{1}\{D_i = 0\}\,\mathbf{1}\{D_i = 1\} = 0$. Each also has zero covariance with $\mathbf{1}\{D_i = 0\} - P_0$, since $\mathbb{E}[\mathbf{1}\{D_i = d\}\,(Y_i - m_{ad})(\mathbf{1}\{D_i = 0\} - P_0)] = (\mathbf{1}\{d = 0\} - P_0)\mathbb{E}[\mathbf{1}\{D_i = d\}\,(Y_i - m_{ad})] = 0$. The estimator $\widehat{P}_0$ pools all arms, which have the same law of D_i by Assumption A4(iv), and $\sqrt{n}(\widehat{P}_0 - P_0) = n^{-1/2}\sum_{i=1}^n(\mathbf{1}\{D_i = 0\} - P_0)$. Arms are independent. The joint central limit theorem for the stacked influence terms, with $n_a = r_an + o(n)$, gives the claim, and the variance of $\mathbf{1}\{D_i = d\}\,(Y_i - m_{ad})/P_d$ is v_{ad}/P_d. □

Part (a). The three arm means are independent sample means with expectations $\mathbb{E}(W_i + Z_i)$, $\mathbb{E}W_i$ and π^D and variances $\sigma_{\mathsf{T}}^2/n_{\mathsf{T}}$, $\sigma_{\mathsf{C}}^2/n_{\mathsf{C}}$ and $\pi^D(1-\pi^D)/n_{\mathsf{D}}$, which proves unbiasedness and (5.6); π and π^D are identified as the limits of these means. In the split design, the law of the observed data depends on the joint law Q of (W_i, Z_i, D_i) only through the law of (W_i, Z_i) (list arms) and the law of D_i (direct arm). Let $t = \mathbb{P}(Z_i = 1, D_i = 1)$. The Fréchet–Hoeffding bounds for a 2×2 table with margins π and π^D give $\max(0, \pi + \pi^D - 1) \leq t \leq \min(\pi, \pi^D)$, and $\pi^H = \pi - t$, which yields (5.7). For sharpness, fix any t in this range. Define Q' by taking (Z_i, D_i) with $\mathbb{P}(Z_i = 1, D_i = 1) = t$ and margins π, π^D, which is a valid table because t lies in the Fréchet range, and then drawing W_i from its Q-conditional law given Z_i, independently of D_i. Then Q' and Q induce the same law of (W_i, Z_i) and of D_i, hence the same law of the observed data, and $\pi^H(Q') = \pi - t$. Every point of (5.7) is therefore compatible with the data. Under Assumption A5, $\mathbb{P}(Z_i = 0, D_i = 1) = 0$ forces $t = \pi^D$, and so $\pi^H = \pi - \pi^D = \text{SDB}$.

Part (b). By Assumption A4(iv), D_i is the same variable in every arm, and by (i) the assignment is independent of (W_i, Z_i, D_i). Hence $\mathbb{E}(Y_i \mid \mathsf{T}, D_i = d) = \mathbb{E}(W_i + Z_i \mid D_i = d)$ and $\mathbb{E}(Y_i \mid \mathsf{C}, D_i = d) =$

$\mathbb{E}(W_i \mid D_i = d)$, whose difference is $\mathbb{E}(Z_i \mid D_i = d) = \Delta_d$; this is (5.8). The formulas $\pi^H = P_0\Delta_0$, $\pi^F = P_1(1-\Delta_1)$ and $\pi = \sum_d P_d\Delta_d$ follow from the definitions, and P_d is identified from D_i in any arm. Consistency of $\widehat{\mathrm{SDB}}_{\mathrm{PS}} = g(\widehat{P}_0, \widehat{\Delta}_0, \widehat{\Delta}_1)$, with $g(P_0, \Delta_0, \Delta_1) = P_0\Delta_0 + (1-P_0)\Delta_1 - (1-P_0)$ and $\widehat{\Delta}_d = \overline{Y}_{\mathsf{T}d} - \overline{Y}_{\mathsf{C}d}$, follows from Lemma B.2. The partial derivatives of g are $\partial g/\partial P_0 = 1 + \Delta_0 - \Delta_1$ and $\partial g/\partial \Delta_d = P_d$. By Lemma B.2, $\sqrt{n}(\widehat{\Delta}_d - \Delta_d)$ has asymptotic variance $(v_{\mathsf{T}d}/r_{\mathsf{T}} + v_{\mathsf{C}d}/r_{\mathsf{C}})/P_d$, independently across d and of $\widehat{P}_0$. The delta method gives (5.9).

Part (c). Under Assumption A5, $D_i = 1$ implies $Z_i = 1$, so $\Delta_1 = 1$, $\pi^F = 0$ and $\mathrm{SDB} = \pi^H = P_0\Delta_0$.

(i) Let $\mathcal{G}$ collect $\mathcal{A}$ and all direct answers, and let E be the event that $n_{\mathsf{T}0} \geq 1$ and $n_{\mathsf{C}0} \geq 1$, where n_{a0} counts respondents in arm a with $D_i = 0$. Conditionally on $\mathcal{G}$, the list answers of respondents in arm a with $D_i = 0$ are independent draws from the law of Y_i given $(a, D_i = 0)$. Hence on E, $\mathbb{E}(\widehat{\pi}^H \mid \mathcal{G}) = \widehat{P}_0\Delta_0$, while $\widehat{\pi}^H = 0$ on E^c. Therefore $\mathbb{E}\widehat{\pi}^H = \Delta_0\mathbb{E}(\widehat{P}_0\mathbf{1}\{E\}) = \pi^H - \Delta_0\mathbb{E}(\widehat{P}_0\mathbf{1}\{E^c\})$, and

$$|\mathbb{E}\widehat{\pi}^H - \pi^H| \leq \mathbb{P}(E^c) \leq \mathbb{P}(n_{\mathsf{T}0} = 0) + \mathbb{P}(n_{\mathsf{C}0} = 0) = P_1^{n_{\mathsf{T}}} + P_1^{n_{\mathsf{C}}}.$$

(ii) Apply the delta method to $(P_0, \Delta_0) \mapsto P_0\Delta_0$ with Lemma B.2: the asymptotic variance is $\Delta_0^2 P_0 P_1 + P_0^2(v_{\mathsf{T}0}/r_{\mathsf{T}} + v_{\mathsf{C}0}/r_{\mathsf{C}})/P_0$, which is (5.11).

(iii) Under Assumptions A4 and A5 the parameter is the joint law Q of (W_i, Z_i, D_i) on $\mathcal{S} = \{0, \ldots, J\} \times \{0,1\}^2 \setminus \{(w, 0, 1) : w \leq J\}$. Reparametrise Q by $p_0 = Q(D_i = 0)$, $f(w, z) = Q(W_i = w, Z_i = z \mid D_i = 0)$ and $g(w) = Q(W_i = w \mid D_i = 1)$; under monotonicity $Z_i = 1$ on $\{D_i = 1\}$, so this is a bijection. Let $q_{\mathsf{C}}(y) = f(y, 0) + f(y, 1)$ and $q_{\mathsf{T}}(y) = f(y, 0) + f(y-1, 1)$ be the laws of the list count in the two arms within $\{D_i = 0\}$, with cumulative distribution functions F_{C} and F_{T}. By the computation that gives (B.1), $f(y, 1) = F_{\mathsf{C}}(y) - F_{\mathsf{T}}(y)$ and $f(y, 0) = F_{\mathsf{T}}(y) - F_{\mathsf{C}}(y-1)$. Conversely, any pair $(q_{\mathsf{C}}, q_{\mathsf{T}})$ of laws on $\{0, \ldots, J\}$ and $\{0, \ldots, J+1\}$ with $F_{\mathsf{C}}(y-1) \leq F_{\mathsf{T}}(y) \leq F_{\mathsf{C}}(y)$ for all y defines through these formulas a law f whose induced pair is $(q_{\mathsf{C}}, q_{\mathsf{T}})$: the two formulas sum to $q_{\mathsf{C}}(y)$, and $f(y, 0) + f(y-1, 1) = q_{\mathsf{T}}(y)$. So $f \leftrightarrow (q_{\mathsf{C}}, q_{\mathsf{T}})$ is a bijection onto the set $\mathcal{Q}$ cut out by these inequalities, and interior points $f > 0$ correspond to strict inequalities. The likelihood of the observed data is

$$p_0^{N_0}(1-p_0)^{N_1} \prod_y q_{\mathsf{C}}(y)^{n_{\mathsf{C}0}(y)} \prod_y q_{\mathsf{T}}(y)^{n_{\mathsf{T}0}(y)} \prod_y g(y)^{n_{\mathsf{C}1}(y)} g(y-1)^{n_{\mathsf{T}1}(y)},$$

where N_d counts all respondents with $D_i = d$ and $n_{ad}(y)$ counts those in arm a with $D_i = d$ and $Y_i = y$. It factorises over p_0, $(q_{\mathsf{C}}, q_{\mathsf{T}}) \in \mathcal{Q}$ and g, which vary independently. Without the constraint $\mathcal{Q}$, the maximisers of the first three factors are the empirical frequencies. At an interior point these converge almost surely to a point of the interior of $\mathcal{Q}$, so with probability tending to one they satisfy the constraint and are the constrained maximisers. Since $\pi^H = p_0 \sum_y y\{q_{\mathsf{T}}(y) - q_{\mathsf{C}}(y)\}$, invariance of maximum likelihood gives $\widehat{\pi}^H_{\mathrm{ML}} = \widehat{P}_0(\overline{Y}_{\mathsf{T}0} - \overline{Y}_{\mathsf{C}0}) = \widehat{\pi}^H$ on that event. Locally the model is a smooth parametric family with an open parameter set and non-singular Fisher information (a product of multinomials with positive cell probabilities). The arm samples are independent and identically distributed with sizes proportional to n, so the experiment is locally asymptotically normal. The maximum likelihood estimator of the smooth functional π^H then attains the Cramér–Rao bound, and by the convolution theorem no regular estimator has smaller asymptotic variance (van der Vaart, 1998, chs. 7–8). Its asymptotic variance is V_H by (ii).

(iv) Under Assumption A5, $\Delta_1 = 1$, so the first term of (5.9) equals the first term of (5.11). The difference is $P_1(v_{\mathsf{T}1}/r_{\mathsf{T}} + v_{\mathsf{C}1}/r_{\mathsf{C}})$, and $v_{\mathsf{T}1} = \mathrm{Var}(W_i + 1 \mid D_i = 1) = v_{\mathsf{C}1}$.

Part (d). Without Assumption A5, Lemma B.2 still gives $\widehat{P}_0 \to P_0$ and $\widehat{\Delta}_0 \to \Delta_0$, so $\widehat{\pi}^H \to P_0\Delta_0 = \pi^H = \mathrm{SDB} + \pi^F$. Also $\widehat{P}_1 + \widehat{\pi}^H \to \mathbb{P}(D_i = 1) + \mathbb{P}(Z_i = 1, D_i = 0) = \pi + \pi^F$, since $\mathbb{P}(D_i = 1) = \mathbb{P}(Z_i = 1, D_i = 1) + \pi^F$. Consistency of $\widehat{\mathrm{SDB}}_{\mathrm{PS}}$ was shown in (b), and the split-design statement is part (a). □

B.3 Testable implications and covariates

Corollary B.3 (Testable implications). *(a)* Design effects *(Blair and Imai, 2012). Under Assumption A4(i)–(iii), for $y = 0, \dots, J$,*

$$\begin{aligned} \mathbb{P}(W_i = y, Z_i = 1) &= \mathbb{P}(Y_i \le y \mid \mathsf{C}) - \mathbb{P}(Y_i \le y \mid \mathsf{T}), \\ \mathbb{P}(W_i = y, Z_i = 0) &= \mathbb{P}(Y_i \le y \mid \mathsf{T}) - \mathbb{P}(Y_i \le y - 1 \mid \mathsf{C}), \end{aligned} \tag{B.1}$$

and both must be non-negative. In the paired design under Assumption A4(iv), the same holds within each stratum $D_i = d$.

(b) Monotonicity. *Under Assumption A4 and Assumption A5 in the paired design, $\mathbb{P}(Y_i = y + 1 \mid \mathsf{T}, D_i = 1) = \mathbb{P}(Y_i = y \mid \mathsf{C}, D_i = 1)$ for every y. In particular $\Delta_1 = 1$, which is the first placebo test of Aronow et al. (2015).*

(c) Treatment independence. *Under Assumption A4(iv), the law of D_i is the same in D, C and T. Comparing T with C is the second placebo test of Aronow et al. (2015); the direct arm adds a comparison with respondents who never saw the list.*

The test of Blair and Imai (2012) estimates the $2(J + 1)$ proportions in (B.1) and rejects if any is significantly negative, with a Bonferroni correction. Like every test in the corollary, it detects only violations that change the law of the observed data, and design effects of opposite signs can cancel. Passing these tests is necessary for the assumptions, not sufficient.

Proof of Corollary B.3. (a) Under Assumption A4(i)–(iii), Y_i has the law of W_i in C and of $W_i + Z_i$ in T. The event $\{W_i + Z_i \le y\}$ is the disjoint union of $\{W_i \le y - 1\}$ and $\{W_i = y, Z_i = 0\}$, so $\mathbb{P}(Y_i \le y \mid \mathsf{T}) = \mathbb{P}(Y_i \le y - 1 \mid \mathsf{C}) + \mathbb{P}(W_i = y, Z_i = 0)$, which is the second identity in (B.1). Subtracting it from $\mathbb{P}(W_i = y) = \mathbb{P}(Y_i \le y \mid \mathsf{C}) - \mathbb{P}(Y_i \le y - 1 \mid \mathsf{C})$ gives the first. Under Assumption A4(iv) the assignment is independent of (W_i, Z_i, D_i), so the argument applies conditionally on $D_i = d$. (b) Under Assumption A5, $Z_i = 1$ on $\{D_i = 1\}$, so within that stratum $Y_i = W_i + 1$ in T and $Y_i = W_i$ in C, and W_i has the same conditional law in both arms. (c) Under Assumption A4(i) and (iv), D_i is a single variable independent of the assignment. □

Remark B.4 (Covariates and a parametric joint model). All statements in Theorem 5.3 hold conditionally on discrete pre-treatment covariates $X_i = x$, so they identify $\mathrm{SDB}(x)$, $\pi^H(x)$ and $\pi^F(x)$. A parametric joint model in the spirit of Imai (2011) writes, under Assumption A5,

$$\begin{gathered} \mathbb{P}(Z_i = 1 \mid x) = \mathrm{logit}^{-1}(x^\top \beta), \qquad \mathbb{P}(D_i = 1 \mid Z_i = 1, x) = \mathrm{logit}^{-1}(x^\top \zeta), \\ W_i \mid (Z_i = z, X_i = x) \sim q_z(\cdot\,; x, \psi), \end{gathered} \tag{B.2}$$

where q_z is a binomial or beta-binomial count model (Imai, 2011) and $\alpha(x) = \mathrm{logit}^{-1}(x^\top \zeta)$ is the *candour rate*, the probability that a private endorser endorses directly. Then $\mathrm{SDB}(x) = \pi(x)\{1 - \alpha(x)\}$. In the split design the direct arm identifies $\pi(x)\alpha(x)$ and the list arms identify $\pi(x)$, so β and ζ are identified when X_i contains an intercept and its support spans $\mathbb{R}^{\dim X_i}$. In the paired design the likelihood of (D_i, Y_i) is a finite mixture over Z_i with known structure. If social pressure on an item runs towards agreement for some respondents, Assumption A5 is replaced by its mirror image, $\mathbb{P}(Z_i = 1, D_i = 0) = 0$; the theorem then holds with the roles of $D_i = 0$ and $D_i = 1$ exchanged, and $\mathrm{SDB} = -\pi^F \le 0$.

B.4 Optimal design

A direct interview costs c_{D} and an interview in a list arm costs $c_{\mathsf{L}} \geq c_{\mathsf{D}}$; in the paired design c_{L} includes the direct question. The budget is $B = c_{\mathsf{D}} n_{\mathsf{D}} + c_{\mathsf{L}}(n_{\mathsf{C}} + n_{\mathsf{T}})$, and arm sizes are treated as continuous. For the paired design under Assumption A5 we write the variance (5.11) in absolute terms as $v_0/n + b_{\mathsf{T}}/n_{\mathsf{T}} + b_{\mathsf{C}}/n_{\mathsf{C}}$, with

$$v_0 = \Delta_0^2 P_0 P_1, \qquad b_{\mathsf{T}} = P_0 v_{\mathsf{T}0}, \qquad b_{\mathsf{C}} = P_0 v_{\mathsf{C}0}, \qquad \bar{b} = \sqrt{b_{\mathsf{T}}} + \sqrt{b_{\mathsf{C}}}.$$

Floor and ceiling effects (Blair and Imai, 2012) are modelled as follows: in arm T, a respondent with $Z_i = 1$ and $W_i = J$ reports J instead of $J + 1$ with probability ϕ_{c}, and a respondent with $Z_i = 1$ and $W_i = 0$ reports 0 instead of 1 with probability ϕ_{f}. For a candidate control list ℓ (its statements and hence J), let V_ℓ^* be B times the smallest variance attainable with ℓ, and let bias_ℓ be the resulting bias of the SDB estimator. The *exposure* of a list is $e_\ell = \mathbb{P}(W_i \in \{0, J\})$.

Theorem B.5 (Optimal allocation and choice of the control list). *(a)* Split design. *With $v_{\mathsf{D}} = \pi^D(1 - \pi^D)$, $v_{\mathsf{C}} = \sigma_{\mathsf{C}}^2$ and $v_{\mathsf{T}} = \sigma_{\mathsf{T}}^2$, the variance (5.6) is minimised subject to the budget by*

$$n_a = \frac{B}{K_{\mathrm{S}}} \sqrt{\frac{v_a}{c_a}} \quad (a \in \{\mathsf{D}, \mathsf{C}, \mathsf{T}\},\ c_{\mathsf{C}} = c_{\mathsf{T}} = c_{\mathsf{L}}), \qquad K_{\mathrm{S}} = \sum_a \sqrt{v_a c_a}, \qquad V^* = K_{\mathrm{S}}^2. \tag{B.3}$$

With equal costs the arm shares are proportional to $(\sqrt{v_{\mathsf{D}}}, \sigma_{\mathsf{C}}, \sigma_{\mathsf{T}})$.

(b) Paired design. *Let $c_+ = c_{\mathsf{L}} - c_{\mathsf{D}}$, the marginal cost of moving a respondent from the direct arm into a list arm. If $v_0 c_+ > c_{\mathsf{D}} \bar{b}^2$, the optimum uses all three arms:*

$$n = \frac{B}{K_{\mathrm{P}}} \sqrt{\frac{v_0}{c_{\mathsf{D}}}}, \quad n_{\mathsf{T}} = \frac{B}{K_{\mathrm{P}}} \sqrt{\frac{b_{\mathsf{T}}}{c_+}}, \quad n_{\mathsf{C}} = \frac{B}{K_{\mathrm{P}}} \sqrt{\frac{b_{\mathsf{C}}}{c_+}}, \quad K_{\mathrm{P}} = \sqrt{v_0 c_{\mathsf{D}}} + \bar{b}\sqrt{c_+}, \quad V^* = K_{\mathrm{P}}^2, \tag{B.4}$$

with $n_{\mathsf{D}} = n - n_{\mathsf{T}} - n_{\mathsf{C}} > 0$. Otherwise the direct arm is empty, $n_{\mathsf{T}} + n_{\mathsf{C}} = B/c_{\mathsf{L}}$, $n_{\mathsf{T}}/n_{\mathsf{C}} = \sqrt{b_{\mathsf{T}}/b_{\mathsf{C}}}$, and $V^* = c_{\mathsf{L}}(v_0 + \bar{b}^2)$.

(c) Choice of list. *Let $\mathcal{L}$ be a finite menu of control lists, and let $\ell(B)$ minimise $\mathrm{MSE}_\ell(B) = V_\ell^*/B + \mathrm{bias}_\ell^2$ over $\mathcal{L}$. Under the floor-and-ceiling model,*

$$\mathrm{bias}_\ell = -\pi\big\{\phi_{\mathrm{c}} \mathbb{P}(W_i = J \mid Z_i = 1) + \phi_{\mathrm{f}} \mathbb{P}(W_i = 0 \mid Z_i = 1)\big\} \tag{B.5}$$

in the split design, and the same expression with π replaced by π^H and the conditioning event by $\{Z_i = 1, D_i = 0\}$ in the paired design with estimator (5.10). If $B < B'$, then $V_{\ell(B')}^* \geq V_{\ell(B)}^*$ and $\mathrm{bias}_{\ell(B')}^2 \leq \mathrm{bias}_{\ell(B)}^2$: larger budgets select lists that are noisier but less exposed to floor and ceiling effects. If the control count is independent of (Z_i, D_i), then $|\,\mathrm{bias}_\ell\,| \leq \max(\phi_{\mathrm{c}}, \phi_{\mathrm{f}})\, \pi\, e_\ell$ (split) or $\leq \max(\phi_{\mathrm{c}}, \phi_{\mathrm{f}})\, \pi^H\, e_\ell$ (paired).

(d) Structure of the control list. *Suppose the control count is independent of (Z_i, D_i).*

(i) For $J \geq 2$ independent control items with common prevalence p, the exposure $p^J + (1 - p)^J$ is uniquely minimised at $p = 1/2$, where the variance $Jp(1 - p)$ is maximised.

(ii) Let $J = 2m$ items form m independent pairs, each item with prevalence $1/2$ and within-pair correlation $\rho \in [-1, 1]$. Then

$$\sigma_{\mathsf{C}}^2 = \frac{m(1 + \rho)}{2}, \qquad \mathbb{P}(W_i = 0) = \mathbb{P}(W_i = J) = \left(\frac{1 + \rho}{4}\right)^m, \tag{B.6}$$

and both are strictly increasing in ρ. Hence V_ℓ^*, e_ℓ and $\mathrm{MSE}_\ell(B)$ are all minimised by the most negative admissible ρ.

(iii) At $\rho = -1$ the count is constant and the treatment count reveals Z_i (Proposition 5.1(b)), so ρ must be restricted by a disclosure requirement. For any closed set $R \subset (-1, 1]$ of admissible correlations, for example those meeting a bound on the mutual information $I(Y_i; Z_i \mid \mathsf{T})$ or on the share of treated respondents whose posterior $\mathbb{P}(Z_i = 1 \mid Y_i, \mathsf{T})$ exceeds a threshold, the MSE-optimal pair design uses $\rho^* = \min R$. Under the variance floor $\sigma_{\mathsf{C}}^2 \geq v$ with $0 < v \leq m$, $\rho^* = 2v/m - 1$, $\sigma_{\mathsf{C}}^2 = v$, and the exposure is $2\{v/(2m)\}^m$, which decreases in m.

Part (a) is a Neyman-type allocation. For the inputs of the illustration in Section 8.4 ($\sigma_{\mathsf{C}}^2 = \sigma_{\mathsf{T}}^2 = 1$, $\pi^D = 0.2$, equal costs), equal allocation gives $n\,\mathrm{Var} = 6.48$ against 5.76 at the optimum, with shares $(1/6, 5/12, 5/12)$. Equal count variances implicitly assume that the control items covary slightly negatively with the sensitive item ($2\sum_j \gamma_j = -\pi(1-\pi)$). With the independent items of Section 5.2 and $\pi = 0.2$, $\sigma_{\mathsf{T}}^2 = 1.16$; the shares become $(0.16, 0.40, 0.43)$, and $n\,\mathrm{Var}$ is 6.14 against 6.96 under equal allocation.

Part (b) has a sharper message. In the paired design every list respondent also answers the direct question, so the direct-only arm only refines $\widehat{P}_0$. Its contribution $\nu_0 = (\pi^H)^2 P_1/P_0$ is second order in π^H. The direct-only arm is used if and only if $c_{\mathsf{D}}/(c_{\mathsf{L}} - c_{\mathsf{D}}) < \nu_0/\bar{b}^2$. By Minkowski's inequality $\sqrt{v_{\mathsf{T}0}} + \sqrt{v_{\mathsf{C}0}} \geq \{\Delta_0(1-\Delta_0)\}^{1/2}$, so for every control list

$$\frac{\nu_0}{\bar{b}^2} \leq \frac{\pi^H P_1}{P_0 - \pi^H},$$

and if $v_{\mathsf{C}0} \geq v > 0$ the threshold is at most $\Delta_0^2 P_1/v = O\{(\pi^H)^2\}$, or $\Delta_0^2 P_1/(4v)$ if also $v_{\mathsf{T}0} \geq v$. With $\pi^H = 0.05$, $\pi = 0.2$, $\sigma_{\mathsf{C}}^2 = 1$ and a control count independent of (Z_i, D_i), the universal bound is 0.0094 and the exact threshold is 1.3×10^{-4}. A direct-only interview would have to cost about 10^{-4} times the extra cost of the list module for the arm to pay. A small direct arm may still be fielded to test treatment independence (Corollary B.3(c)), but not for precision.

Part (d) needs a caveat. Neither the variance floor nor the exposure e_ℓ measures disclosure: e_ℓ is the risk of floor and ceiling responses, and a small σ_{C}^2 is compatible with counts that are very informative about Z_i. With $\pi = 0.2$, the two negatively correlated pairs of Section 5.2 cut L^* from about 27 to 12. They also double the mutual information between the treatment count and Z_i, from 0.11 to 0.24 bits, and raise the share of treated respondents whose posterior $\mathbb{P}(Z_i = 1 \mid Y_i, \mathsf{T})$ is at least 0.5 from 11% to 30%. At a fixed floor $v = 1$, lengthening the list lowers the exposure but leaves the mutual information essentially unchanged (0.11 bits for $m = 2, \ldots, 5$). The mutual information also rises as $\rho \to 1$, because the parity of the count then becomes informative, so the admissible set R in (d)(iii) is in general an interval. Negatively correlated lists trade disclosure for efficiency. The plan therefore bounds disclosure measures directly, not only the variance of the control count (Section 8.4).

Corollary B.6 (Paired versus split). *Let c_{L}' be the cost of a list interview without the direct question, and suppose the control count is independent of (Z_i, D_i), Assumption A5 holds and $\pi \leq 1/2$. Then, for any $c_{\mathsf{D}} \geq 0$,*

$$\frac{V^*_{\text{paired}}}{V^*_{\text{split}}} \leq \frac{c_{\mathsf{L}}}{c_{\mathsf{L}}'}\left\{P_0 + \frac{(\pi^H)^2 P_1}{P_0(\sigma_{\mathsf{T}} + \sigma_{\mathsf{C}})^2}\right\}.$$

The paired design therefore wins whenever the direct question adds less than a fraction of about P_1/P_0 to the cost of a list interview. In the illustration above, with nearly free direct interviews and equal list costs in the two designs, the ratio of optimal variances is about 0.81. The cost that the corollary does not price is

the risk to Assumption A4(iv). A direct question asked after the list may be answered differently, and a list answered after a direct denial may be pulled towards consistency. This is why the plan of Section 8.4 keeps the split design as primary and pairs only a subsample at the end of the questionnaire.

Proof of Theorem B.5. Throughout, the quantities $v_0, b_{\mathsf{T}}, b_{\mathsf{C}}, v_a$ are taken to be positive, and $c_{\mathsf{D}} > 0$.

Part (a). For any feasible allocation, by the Cauchy–Schwarz inequality,

$$K_{\mathrm{S}}^2 = \Big(\sum_a \sqrt{v_a/n_a}\,\sqrt{c_a n_a}\Big)^2 \le \Big(\sum_a v_a/n_a\Big)\Big(\sum_a c_a n_a\Big) = B\ \mathrm{Var}\big(\widehat{\mathrm{SDB}}_{\mathrm{S}}\big),$$

with equality if and only if $\sqrt{v_a/n_a}$ is proportional to $\sqrt{c_a n_a}$, that is, $n_a \propto \sqrt{v_a/c_a}$. The budget fixes the constant, which gives (B.3). With equal costs, $n_a \propto \sqrt{v_a}$.

Part (b). Use the variables $(n, n_{\mathsf{T}}, n_{\mathsf{C}})$, with $n_{\mathsf{D}} = n - n_{\mathsf{T}} - n_{\mathsf{C}}$. The problem is to minimise $\Psi = v_0/n + b_{\mathsf{T}}/n_{\mathsf{T}} + b_{\mathsf{C}}/n_{\mathsf{C}}$ subject to $c_{\mathsf{D}} n + c_+(n_{\mathsf{T}} + n_{\mathsf{C}}) = B$ and $n \ge n_{\mathsf{T}} + n_{\mathsf{C}}$. The objective is strictly convex on the positive orthant and the constraints are linear, so there is a unique minimiser. If $c_+ = 0$, the budget fixes $n = B/c_{\mathsf{D}}$ and Ψ decreases in n_{T} and n_{C}, so the constraint $n \ge n_{\mathsf{T}} + n_{\mathsf{C}}$ binds. If $c_+ > 0$, drop that constraint. The relaxed problem has the form of part (a) with costs $(c_{\mathsf{D}}, c_+, c_+)$ attached to $(n, n_{\mathsf{T}}, n_{\mathsf{C}})$, so its unique minimiser is (B.4) with value K_{P}^2/B. This minimiser satisfies $n > n_{\mathsf{T}} + n_{\mathsf{C}}$ if and only if $\sqrt{v_0/c_{\mathsf{D}}} > (\sqrt{b_{\mathsf{T}}} + \sqrt{b_{\mathsf{C}}})/\sqrt{c_+}$, that is, $v_0 c_+ > c_{\mathsf{D}} \bar{b}^2$. In that case it is feasible for the original problem and hence optimal. Otherwise the original minimiser has $n = n_{\mathsf{T}} + n_{\mathsf{C}}$. If it did not, the constraint would be slack in a neighbourhood, so the point would be a local and, by convexity, a global minimiser of the relaxed problem; this contradicts uniqueness. On the face $n = n_{\mathsf{T}} + n_{\mathsf{C}} =: n_{\mathsf{L}}$ the budget gives $n_{\mathsf{L}} = B/c_{\mathsf{L}}$. By Cauchy–Schwarz, $b_{\mathsf{T}}/n_{\mathsf{T}} + b_{\mathsf{C}}/n_{\mathsf{C}} \ge \bar{b}^2/n_{\mathsf{L}}$, with equality if and only if $n_{\mathsf{T}}/n_{\mathsf{C}} = \sqrt{b_{\mathsf{T}}/b_{\mathsf{C}}}$. The minimum is therefore $(v_0 + \bar{b}^2)/n_{\mathsf{L}} = c_{\mathsf{L}}(v_0 + \bar{b}^2)/B$.

Part (c). Under the floor-and-ceiling model, the treatment-arm response is $Y_i = W_i + Z_i - \mathbf{1}\{Z_i = 1, W_i = J\}\,\chi_i^{\mathrm{c}} - \mathbf{1}\{Z_i = 1, W_i = 0\}\,\chi_i^{\mathrm{f}}$, where $\chi_i^{\mathrm{c}} \sim \mathrm{Bernoulli}(\phi_{\mathrm{c}})$ and $\chi_i^{\mathrm{f}} \sim \mathrm{Bernoulli}(\phi_{\mathrm{f}})$ are independent of everything else. Control and direct responses are unchanged. Hence $\mathbb{E}(\overline{Y}_{\mathsf{T}} - \overline{Y}_{\mathsf{C}} - \overline{D}_{\mathsf{D}}) = \mathrm{SDB} - \phi_{\mathrm{c}}\mathbb{P}(W_i = J, Z_i = 1) - \phi_{\mathrm{f}}\mathbb{P}(W_i = 0, Z_i = 1)$, which is (B.5) for the split design. In the paired design the same computation conditional on $D_i = 0$ shifts the probability limit of $\widehat{\Delta}_0$ by $-\phi_{\mathrm{c}}\mathbb{P}(W_i = J, Z_i = 1 \mid D_i = 0) - \phi_{\mathrm{f}}\mathbb{P}(W_i = 0, Z_i = 1 \mid D_i = 0)$. Multiplying by P_0 and using $\mathbb{P}(Z_i = 1, D_i = 0) = \pi^H$ gives the stated form. For the comparative statics, let ℓ be optimal at B and ℓ' optimal at $B' > B$. Optimality gives

$$\frac{V_\ell^*}{B} + \mathrm{bias}_\ell^2 \le \frac{V_{\ell'}^*}{B} + \mathrm{bias}_{\ell'}^2, \qquad \frac{V_{\ell'}^*}{B'} + \mathrm{bias}_{\ell'}^2 \le \frac{V_\ell^*}{B'} + \mathrm{bias}_\ell^2 .$$

Adding the two inequalities gives $(V_\ell^* - V_{\ell'}^*)(1/B - 1/B') \le 0$, so $V_{\ell'}^* \ge V_\ell^*$. The second inequality then gives $\mathrm{bias}_{\ell'}^2 \le \mathrm{bias}_\ell^2 + (V_\ell^* - V_{\ell'}^*)/B' \le \mathrm{bias}_\ell^2$. If the control count is independent of (Z_i, D_i), the conditional probabilities in (B.5) equal $\mathbb{P}(W_i = J)$ and $\mathbb{P}(W_i = 0)$, whose sum is e_ℓ (the events are disjoint because $J \ge 1$), and the bound follows.

Part (d). (i) The exposure $p^J + (1-p)^J$ is symmetric about $1/2$, and for $J \ge 2$ its second derivative $J(J-1)\{p^{J-2} + (1-p)^{J-2}\}$ is positive, so $p = 1/2$ is its unique minimiser. The variance $Jp(1-p)$ is maximised at $p = 1/2$. (ii) For a pair (X, X') with prevalences $1/2$ and correlation ρ, $\mathbb{P}(X = X' = 1) = \mathbb{E}XX' = 1/4 + \rho/4 =: p_{11}$, so $\mathbb{P}(X = 1, X' = 0) = \mathbb{P}(X = 0, X' = 1) = 1/2 - p_{11}$ and $\mathbb{P}(X = X' = 0) = p_{11}$. The pair sum takes the values $0, 1, 2$ with probabilities $p_{11}, 1 - 2p_{11}, p_{11}$, and has variance $2p_{11} = (1 + \rho)/2$. Summing m independent pairs gives (B.6). Both expressions are strictly increasing in ρ. With the control count independent of (Z_i, D_i), the arm variances that enter V_ℓ^* are

$v_{\mathsf{C}} = \sigma_{\mathsf{C}}^2$ and $v_{\mathsf{T}} = \sigma_{\mathsf{C}}^2 + \pi(1-\pi)$ (split), or $v_{\mathsf{C}0} = \sigma_{\mathsf{C}}^2$ and $v_{\mathsf{T}0} = \sigma_{\mathsf{C}}^2 + \Delta_0(1-\Delta_0)$ (paired). For every fixed allocation the variance is therefore strictly increasing in σ_{C}^2. If $\rho < \rho'$, with optimal allocations $x(\rho)$ and $x(\rho')$, then $V^*(\rho) \le V(x(\rho');\rho) < V(x(\rho');\rho') = V^*(\rho')$. The bias $|\,\mathrm{bias}_\ell\,| = (\phi_{\mathrm{c}} + \phi_{\mathrm{f}})\,\varpi\, p_{11}^m$, with $\varpi = \pi$ (split) or π^H (paired), is non-decreasing in ρ. So $\mathrm{MSE}_\ell(B)$ is strictly increasing in ρ. (iii) At $\rho = -1$, $p_{11} = 0$, so every pair sums to one and $W_i \equiv m$. By (ii), $\mathrm{MSE}_\ell(B)$ is strictly increasing in ρ, so over any closed admissible set $R \subset (-1, 1]$ it is minimised at $\min R$. The floor $m(1+\rho)/2 \ge v$ gives $R = [2v/m - 1, 1]$ with $2v/m - 1 \in (-1, 1]$, so $\rho^* = 2v/m - 1$. There $\sigma_{\mathsf{C}}^2 = v$, and the exposure is $2p_{11}^m$ with $p_{11} = (1+\rho^*)/4 = v/(2m)$. Finally, $\frac{\mathrm{d}}{\mathrm{d}m}\, m \log\{v/(2m)\} = \log\{v/(2m)\} - 1 < 0$ because $v \le m$, so the exposure decreases in m. □

Corollary B.6. The allocation with an empty direct arm is always feasible in the paired design, so $V^*_{\text{paired}} \le c_{\mathsf{L}}(\nu_0 + \bar{b}^2)$ by part (b). In the split design, part (a) with list cost c'_{L} gives $V^*_{\text{split}} = \{\sqrt{v_{\mathsf{D}} c_{\mathsf{D}}} + \sqrt{c'_{\mathsf{L}}}(\sigma_{\mathsf{T}} + \sigma_{\mathsf{C}})\}^2 \ge c'_{\mathsf{L}}(\sigma_{\mathsf{T}} + \sigma_{\mathsf{C}})^2$. With the control count independent of (Z_i, D_i), $v_{\mathsf{C}0} = \sigma_{\mathsf{C}}^2$, $v_{\mathsf{T}0} = \sigma_{\mathsf{C}}^2 + \Delta_0(1-\Delta_0)$ and $\sigma_{\mathsf{T}}^2 = \sigma_{\mathsf{C}}^2 + \pi(1-\pi)$. Under Assumption A5, $\Delta_0 = (\pi - \pi^D)/(1-\pi^D) \le \pi \le 1/2$, and $x \mapsto x(1-x)$ is increasing on $[0, 1/2]$, so $v_{\mathsf{T}0} \le \sigma_{\mathsf{T}}^2$. Hence $\bar{b} = \sqrt{P_0}(\sqrt{v_{\mathsf{T}0}} + \sqrt{v_{\mathsf{C}0}}) \le \sqrt{P_0}(\sigma_{\mathsf{T}} + \sigma_{\mathsf{C}})$. Dividing, and using $\nu_0 = \Delta_0^2 P_0 P_1 = (\pi^H)^2 P_1 / P_0$, gives the bound. □

B.5 Linking the list experiment to the measurement model

Proof of Proposition 5.5. (a) Since $\delta_j \ge 0$, $\{Y^*_{ij} > b_{jc^*} + \delta_j\} \subseteq \{Y^*_{ij} > b_{jc^*}\}$, so $D_{ij} = 1$ implies $Z_{ij} = 1$, which is Assumption A5. For each c, $\mathbb{P}(Y_{ij} \ge c \mid \boldsymbol{\theta}_i) = \mathbb{P}(\varepsilon_{ij} > b_{jc} + \delta_j - \eta_{ij}) = F(\eta_{ij} - b_{jc} - \delta_j)$ by the symmetry of F. This is the graded response model with the same loadings and thresholds $b_{jc} + \delta_j$, which remain ordered. The same computation without δ_j gives $\mathbb{P}(Z_{ij} = 1 \mid \boldsymbol{\theta}_i) = F(\eta_{ij} - b_{jc^*})$. Finally, $\pi_j^H(\boldsymbol{\theta}_i) = \mathbb{P}(b_{jc^*} - \eta_{ij} < \varepsilon_{ij} \le b_{jc^*} + \delta_j - \eta_{ij} \mid \boldsymbol{\theta}_i)$, which is (5.12), and $\mathrm{SDB}_j = \mathbb{P}(Z_{ij} = 1, D_{ij} = 0) = \mathbb{E}\,\pi_j^H(\boldsymbol{\theta}_i)$ by Theorem 5.3(a) under Assumption A5.

(b) The list experiment identifies π_j (Proposition 5.1). The law of η_{ij} is identified from the loadings and the law of $\boldsymbol{\theta}_i$, and $b^D_{jc^*}$ is identified. By (a), $\pi_j = \mathbb{E}F(\eta_{ij} - b_{jc^*}) = \psi(\delta_j)$ with $\psi(\delta) = \mathbb{E}F(\eta_{ij} - b^D_{jc^*} + \delta)$. The function ψ is continuous and strictly increasing, because F is, and by dominated convergence it tends to 0 and 1 as $\delta \to \mp\infty$. Since $0 < \pi_j < 1$, the root of $\psi(\delta) = \pi_j$ exists and is unique, and it equals δ_j. The remaining quantities are explicit functions of δ_j and identified parameters, and $\mathrm{SDB}_j(x) = \mathbb{E}\{\pi_j^H(\boldsymbol{\theta}_i) \mid X_i = x\}$ uses the identified law of $\boldsymbol{\theta}_i$ given X_i. A negative root contradicts $\delta_j \ge 0$.

(c) If δ_j is common to all groups, then by (a) the direct item has the same loadings in every group as the private item and thresholds shifted by the same δ_j in every group, so it is invariant whenever the private item is. ECV_j and $\mathrm{ECV}_j^{(g)}$ in (3.8) depend only on the loadings and the factor variances, which are therefore the same in the two models. If $\delta_j = \delta_j^{(g)}$ in group g, the direct thresholds are $b_{jc} + \delta_j^{(g)}$, so the between-group difference in every threshold is $\delta_j^{(g)} - \delta_j^{(g')}$, with equal loadings: uniform DIF. Applying (b) within group g identifies $\delta_j^{(g)}$ when the group's latent law is identified. □

Proposition B.7 (Overidentification and non-uniform shifts). *Under Definition 5.4 and the conditions of Proposition 5.5(b), the following hold.*

(d) Overidentification. *With $m \ge 2$ covariate values, the m equations $\mathbb{E}\{F(\eta_{ij} - b^D_{jc^*} + \delta) \mid X_i = x_l\} = \pi_j(x_l)$ share one unknown, which gives $m - 1$ testable restrictions of the uniform shift.*

(e) Non-uniform shift. *Suppose instead that* $Z_{ij} = \mathbf{1}\{\upsilon\eta_{ij} + \varepsilon_{ij} \geq b\}$ *with* $\upsilon > 0$*, that* ε_{ij} *is standard normal (the normal-ogive version), and that* $\eta_{ij} \mid X_i = x_l \sim \mathrm{N}(\bar{\eta}_l, s_\eta^2)$ *for* $l = 1, 2$ *with* $\bar{\eta}_1 \neq \bar{\eta}_2$*. Then* (υ, b) *is identified by*

$$\frac{\upsilon}{\sqrt{1+\upsilon^2 s_\eta^2}} = \frac{\Phi^{-1}\{\pi_j(x_1)\} - \Phi^{-1}\{\pi_j(x_2)\}}{\bar{\eta}_1 - \bar{\eta}_2}, \qquad b = \upsilon\bar{\eta}_1 - \sqrt{1+\upsilon^2 s_\eta^2}\,\Phi^{-1}\{\pi_j(x_1)\}, \quad \text{(B.7)}$$

provided the right-hand side of the first equation lies in $(0, 1/s_\eta)$*; otherwise the model is rejected. Assumption A5 then holds for every* $\boldsymbol{\theta}_i$ *only if* $\upsilon = 1$*, and the private and direct versions of* ECV_j *coincide because the loadings are rescaled jointly.*

Proof. (d) Applying Proposition 5.5(b) conditionally on $X_i = x_l$ gives a unique root $\delta(x_l)$ for each l. The uniform shift implies $\delta(x_1) = \cdots = \delta(x_m)$, which is $m - 1$ restrictions on identified quantities.

(e) With standard normal ε_{ij} and $\eta_{ij} \mid X_i = x_l \sim \mathrm{N}(\bar{\eta}_l, s_\eta^2)$, $\pi_j(x_l) = \mathbb{P}(\varepsilon_{ij} \geq b - \upsilon\eta_{ij}) = \mathbb{P}(\varepsilon_{ij} + \upsilon\eta_{ij} \geq b) = \Phi\{(\upsilon\bar{\eta}_l - b)/\sqrt{1+\upsilon^2 s_\eta^2}\}$, since $\varepsilon_{ij} + \upsilon\eta_{ij} \sim \mathrm{N}(\upsilon\bar{\eta}_l, 1 + \upsilon^2 s_\eta^2)$. Taking Φ^{-1} and differencing over l gives the first equation of (B.7). The map $\upsilon \mapsto \upsilon/\sqrt{1+\upsilon^2 s_\eta^2}$ is continuous and strictly increasing from $(0, \infty)$ onto $(0, 1/s_\eta)$, since its derivative is $(1+\upsilon^2 s_\eta^2)^{-3/2} > 0$. So υ is identified when the right-hand side lies in that interval, and b follows from the equation for $l = 1$. For monotonicity, $\{\eta + \varepsilon \geq b^D\} \subseteq \{\upsilon\eta + \varepsilon \geq b\}$ for all (η, ε) requires $b^D - \eta \geq b - \upsilon\eta$ for all $\eta \in \mathbb{R}$. This holds only if $\upsilon = 1$ (and then if and only if $b \leq b^D$). The private item has loadings $\upsilon(a_{jG}, a_{js})$, so its ECV_j is unchanged. □

B.6 Supplementary results and proofs for Section 6

Proposition B.8 (Count indices under non-invariance and nonlinearity). *In the setting of Proposition 6.3, the following hold.*

(c) Uniform DIF. *In a location family with* $\mu_1 > \mu_2$*, invariant slopes* $a_j > 0$*, and* $b_j^{(1)} = b_j^{(2)} + \Delta b_j$*: if* $\Delta b_j = a_j(\mu_1 - \mu_2)$ *for all* j*, the responses* $(U_{i1}, \ldots, U_{iM})$ *have the same law in both groups; if* $\Delta b_j \geq a_j(\mu_1 - \mu_2)$ *for all* j*, with strict inequality for some* j*, then* $P_1(q) < P_2(q)$ *for every* q*.*

(d) Non-uniform DIF. *Let* $M \geq 2$*, let* F *and* Γ_0 *be symmetric about zero with* Γ_0 *continuous, and let* $b_j^{(g)} = 0$ *for all* j *and* g*. For every* $\mu_1 > 0 = \mu_2$ *there are slopes* $a^{(1)} > 0$ *(small) and* $a^{(2)} > 0$ *(large), common to all items within a group, such that* $P_1(1) > P_2(1)$ *but* $P_1(M) < P_2(M)$*. The sign of the index difference then depends on* q*, although the thresholds are invariant.*

(e) Nonlinearity. *Under full invariance with all* $a_j > 0$*, in a location family,* $\mu \mapsto P(q; \mu)$ *is strictly increasing from* 0 *to* 1*. Hence for any* $\Delta_A > \Delta_B > 0$ *there are baselines* μ_A, μ_B *with* $P(q; \mu_A + \Delta_A) - P(q; \mu_A) < P(q; \mu_B + \Delta_B) - P(q; \mu_B)$*: a larger latent change can produce a smaller change in the index.*

Part (c) shows that a uniform shift of all thresholds by $a_j(\mu_1 - \mu_2)$ is observationally equivalent to equal latent means; this is the index counterpart of Theorem 4.5(d), and it is the indeterminacy that anchor items resolve (Proposition 4.6). Slightly larger shifts reverse the sign of every index comparison. Part (d) shows that differences in discrimination alone, with identical thresholds, can make the low-threshold and high-threshold versions of the index disagree about which group has more agreement. Part (e) concerns changes. If two groups start from different baselines, the index can rank their latent changes in the wrong order even under full invariance.

Example B.9 (Numerical instances). Take $M = 10$, logistic F and normal Γ_g. (i) For Proposition 6.3(b), let all $a_j = 1.5$, let the b_j be equally spaced on $[0.5, 3.5]$, and let $\Gamma_1 = \mathrm{N}(0.2, 0.6^2)$ and $\Gamma_2 = \mathrm{N}(0, 1.4^2)$. Then $P_1(q) > P_2(q)$ for $q \le 2$ and $P_1(q) < P_2(q)$ for $q \ge 3$; at $q = 5$, $P_1 = 0.10$ and $P_2 = 0.21$. (ii) For part (d) of Proposition B.8, let $b_j = 0$, $a^{(1)} = 0.25$, $a^{(2)} = 3$, $\mu_1 = 0.3$, $\mu_2 = 0$ and unit variances. Then $P_1(1) = 0.999$ and $P_2(1) = 0.806$, but $P_1(10) = 0.002$ and $P_2(10) = 0.194$; the sign changes between $q = 5$ and $q = 6$. (iii) For part (e), in the limit of perfectly discriminating statements with common location t and $\Gamma_0 = \mathrm{N}(0, 1)$, $P(q; \mu) = \Phi(\mu - t)$. A group starting at $\mu = t - 3$ whose latent mean rises by 0.5 moves the index by 0.005; a group starting at $\mu = t$ whose mean rises by 0.25 moves it by 0.099.

Poststratification identities. Let a member of area r be drawn at random. Its cell k has probability w_k, and given k its latent vector has mean $\boldsymbol{\mu}_k$ and covariance $\boldsymbol{\Sigma}_k$. The law of total expectation gives $\mathbb{E}_r \varphi(\boldsymbol{\theta}) = \sum_k w_k \mathbb{E}[\varphi(\boldsymbol{\theta}) \mid k]$ for every integrable φ; with $\varphi(\boldsymbol{\theta}) = \boldsymbol{\theta}$ this is (6.2), and with $\varphi(\boldsymbol{\theta}) = \mathbf{1}\{\theta_G > t\}$ and Gaussian cells it is the tail-share formula. The law of total covariance gives $\boldsymbol{\Sigma}_r = \mathbb{E}_r\{\mathrm{Var}(\boldsymbol{\theta} \mid k)\} + \mathrm{Var}_r\{\mathbb{E}(\boldsymbol{\theta} \mid k)\}$, which is (6.3). For a finite population the same identities hold exactly for the empirical distribution of the N_r members, with realised cell means and covariances (divisor N_k) in place of $\boldsymbol{\mu}_k$ and $\boldsymbol{\Sigma}_k$. For Remark 6.1: if the members of cell k are independent draws from the superpopulation, the realised cell means are independent with covariance $\boldsymbol{\Sigma}_k / N_k$. So $\sum_k w_k(\bar{\boldsymbol{\theta}}_k^{\mathrm{pop}} - \boldsymbol{\mu}_k)$ has covariance $\sum_k w_k^2 \boldsymbol{\Sigma}_k / N_k = N_r^{-1} \sum_k w_k \boldsymbol{\Sigma}_k$, which is of order N_r^{-1}.

Proposition 6.2. Under the prior $\mathrm{N}(m_0, \sigma_0^2)$ and $\hat{\theta}_i \mid \theta_i \sim \mathrm{N}(\theta_i, s^2)$, the posterior mean is $\tilde{\theta}_i = m_0 + \varrho(\hat{\theta}_i - m_0)$. Since $\mathbb{E}(\hat{\theta}_i \mid k) = \mathbb{E}\{\mathbb{E}(\hat{\theta}_i \mid \theta_i) \mid k\} = \mu_k$, linearity gives (6.5). The two-step estimator averages $\tilde{\theta}_i$ within cells, which converges to $\mathbb{E}(\tilde{\theta}_i \mid k)$, and then poststratifies. Its limit is $\sum_k w_k\{m_0 + \varrho(\mu_k - m_0)\} = m_0 + \varrho(\bar{\theta}_r - m_0)$, and the difference between two areas is $\varrho(\bar{\theta}_r - \bar{\theta}_{r'})$. In the joint model (6.1), μ_k is a parameter of a correctly specified regular model, and as $n_k \to \infty$ its posterior concentrates at the true value (van der Vaart, 1998, ch. 10), so its posterior mean is consistent. In finite cells, partial pooling shrinks it towards the regression prediction by an amount that vanishes as n_k grows, whereas the two-step limit is attenuated by ϱ for every n_k.

Proposition 6.3. Let $V_1, \dots, V_M$ be independent uniform variables, independent of θ_i, and set $U_{ij} = \mathbf{1}\{V_j \le F(a_j\theta_i - b_j)\}$. This reproduces (6.6) under full invariance. Write $p_q(\theta) = \mathbb{P}(U_{i+} \ge q \mid \theta_i = \theta)$.

Step 1: p_q is continuous and strictly increasing for $1 \le q \le M$ when all $a_j > 0$. Continuity follows from that of F. If $\theta < \theta'$, each U_{ij} is pointwise no larger at θ than at θ', so U_{i+} is too and $p_q(\theta) \le p_q(\theta')$. Moreover, $0 < F < 1$ on $\mathbb{R}$ because F is strictly increasing. So with positive probability exactly $q - 1$ of the V_j, $j \ne 1$, fall below $F(a_j\theta - b_j)$, the others exceed $F(a_j\theta' - b_j)$, and $F(a_1\theta - b_1) < V_1 \le F(a_1\theta' - b_1)$. On that event $U_{i+} < q$ at θ and $U_{i+} \ge q$ at θ', so $p_q(\theta) < p_q(\theta')$.

Part (a). Let Γ_g^{-1} be the quantile functions and Υ a uniform variable. Stochastic dominance is equivalent to $\Gamma_1^{-1}(u) \ge \Gamma_2^{-1}(u)$ for all $u \in (0, 1)$, and $P_g(q) = \mathbb{E}\, p_q\{\Gamma_g^{-1}(\Upsilon)\}$. Step 1 gives $P_1(q) \ge P_2(q)$. If $\Gamma_1 \ne \Gamma_2$, the two quantile functions differ at some u_0. Both are left-continuous and non-decreasing, so they differ on a set of positive Lebesgue measure; otherwise they would agree on a dense set and hence everywhere. On that set $p_q\{\Gamma_1^{-1}(u)\} > p_q\{\Gamma_2^{-1}(u)\}$, which makes the inequality strict. In a location family $\Gamma_1^{-1}(u) - \Gamma_2^{-1}(u) = \mu_1 - \mu_2$ for all u, which gives the sign statement.

Part (b). Set $a_j = a$ and $b_j = at$ for all j. For $\theta > t$, $p_q(\theta) \ge \mathbb{P}(U_{i+} = M \mid \theta) = F\{a(\theta - t)\}^M \to 1$ as $a \to \infty$. For $\theta < t$, $p_q(\theta) \le \mathbb{P}(U_{i+} \ge 1 \mid \theta) \le MF\{a(\theta - t)\} \to 0$. Since $\mathbb{P}(\theta_i = t) = 0$, dominated convergence gives $P_g(q) \to 1 - \Phi\{(t - \mu_g)/\sigma_g\}$ for every $q \in \{1, \dots, M\}$. Now $(t - \mu_1)/\sigma_1 > (t - \mu_2)/\sigma_2$ if and only if $t(\sigma_2 - \sigma_1) > \mu_1\sigma_2 - \mu_2\sigma_1$, that is, $t > t_\times$. So the limit satisfies $P_1(q) < P_2(q)$ for $t > t_\times$ and $P_1(q) > P_2(q)$ for $t < t_\times$, and both strict inequalities persist for all sufficiently large finite a. The largeness condition matters: for moderate a the sign at a given t can depend on q.

Proposition B.8, part (c). Write $\theta_i = \mu_g + \xi_i$, with $\xi_i \sim \Gamma_0$ in both groups. In group 1 the linear predictor of item j is $a_j(\mu_1 + \xi_i) - b_j^{(2)} - \Delta b_j = \{a_j(\mu_2 + \xi_i) - b_j^{(2)}\} + \{a_j(\mu_1 - \mu_2) - \Delta b_j\}$. If $\Delta b_j = a_j(\mu_1 - \mu_2)$ for all j, the conditional law of $(U_{i1}, \ldots, U_{iM})$ given ξ_i is the same in both groups, and so is the law of ξ_i. If $\Delta b_j \geq a_j(\mu_1 - \mu_2)$ for all j, each item probability in group 1 is no larger, given ξ_i, than in group 2, and it is strictly smaller for some j^*. Now $\mathbb{P}(U_{i+} \geq q) = p_j \mathbb{P}(U_{i,-j} \geq q - 1) + (1 - p_j)\mathbb{P}(U_{i,-j} \geq q)$ is affine in each item probability p_j with slope $\mathbb{P}(U_{i,-j} = q - 1) \geq 0$, where $U_{i,-j}$ omits item j. That slope is positive when all other item probabilities lie in $(0, 1)$. Hence $\mathbb{P}(U_{i+} \geq q \mid \xi_i, g = 1) < \mathbb{P}(U_{i+} \geq q \mid \xi_i, g = 2)$ for every ξ_i, and integrating over Γ_0 gives $P_1(q) < P_2(q)$.

Part (d). With $b_j = 0$ and common slope $a^{(g)}$ in group g, $P_g(q)$ is continuous in $a^{(g)} \in [0, \infty)$ by dominated convergence. At $a^{(1)} = 0$, the item probabilities equal $F(0) = 1/2$ by symmetry, so $U_{i+} \mid g = 1 \sim \text{Binomial}(M, 1/2)$, $P_1(1) = 1 - 2^{-M}$ and $P_1(M) = 2^{-M}$. As $a^{(2)} \to \infty$, the argument of part (b) with $t = 0$ gives $P_2(q) \to \mathbb{P}(\theta_i > 0 \mid g = 2) = 1/2$ for every q, because Γ_0 is continuous and symmetric and $\mu_2 = 0$. For $M \geq 2$, $1 - 2^{-M} > 1/2 > 2^{-M}$. By continuity, $P_1(1) > P_2(1)$ and $P_1(M) < P_2(M)$ for all sufficiently small $a^{(1)} > 0$ and sufficiently large $a^{(2)}$.

Part (e). $P(q; \mu) = \mathbb{E}\, p_q(\mu + \xi_i)$ is strictly increasing in μ by Step 1. As $\mu \to -\infty$, $p_q(\mu + \xi_i) \leq \mathbb{P}(U_{i+} \geq 1 \mid \mu + \xi_i) \leq \sum_j F\{a_j(\mu + \xi_i) - b_j\} \to 0$ pointwise, and as $\mu \to \infty$, $p_q(\mu + \xi_i) \geq \prod_j F\{a_j(\mu + \xi_i) - b_j\} \to 1$. Dominated convergence gives the limits 0 and 1. Fix μ_B and let $c = P(q; \mu_B + \Delta_B) - P(q; \mu_B) > 0$. Choose μ_A with $P(q; \mu_A + \Delta_A) < c$. Then $P(q; \mu_A + \Delta_A) - P(q; \mu_A) < c$.

Several factors. If U_{ij} depends on $\boldsymbol{\theta}_i \in \mathbb{R}^3$ through $F(\boldsymbol{a}_j^\top \boldsymbol{\theta}_i - b_j)$ with $\boldsymbol{a}_j \geq \mathbf{0}$ componentwise, the coupling of Step 1 shows that p_q is non-decreasing in each coordinate. The usual multivariate stochastic order is defined by $\mathbb{E} f(\boldsymbol{\theta}^{(1)}) \geq \mathbb{E} f(\boldsymbol{\theta}^{(2)})$ for all bounded non-decreasing f, so $P_1(q) \geq P_2(q)$ follows directly. □

Appendix C Simulation details

This appendix documents the simulation study of Section 7: software, seeds and runs (Appendix C.1), configurations (Appendix C.2), estimator settings (Appendix C.3), Monte Carlo standard errors (Appendix C.4), exact checks and recomputed illustrations (Appendix C.5), further results (Appendix C.6) and reproduction (Appendix C.7). All data are synthetic. Every configuration value below is stored with the results in `results/sim_results.json`, and the per-replication values are in `results/raw/X1.json`, ..., `results/raw/X9.json`, with the near-proportional design in `X1b.json` and the stereotype-anchor experiment in `X2b.json`. `results/SUMMARY-numbers.md` and `results/SUMMARY-new-experiments.md` print the main numbers from the results file.

C.1 Software, seeds and runs

Experiment labels. The experiments are labelled X1 to X9, with X1b (near-proportional design) and X2b (DIF on stereotype anchors). X1 to X5 and X1b were labelled E1 to E5 and E1b when they were run. They were renamed, in the code, the results file, the raw files and the tables, to avoid a clash with the estimands (E1)–(E9) of Section 8.6. The renaming changed no seed and no number: the seed of each replication depends on a numerical experiment index, which is unchanged, and re-running replications of X1 and X2 with the renamed code reproduces the stored values exactly. The results file records the mapping and, for each renamed experiment, its old label.

Software. The experiments are run by `code/run_experiments.py`, which calls the package `code/attsim/`: data generation (`generate.py`, `designs.py`), polychoric correlations (`polychoric.py`,

`bvn.py`), bifactor and correlated-factors fits (`fit.py`), fits with arbitrary loading patterns such as cross-loadings and method factors (`genfit.py`), boundary estimands and intervals (`boundary.py`), DIF procedures (`dif.py`), tetrad tests (`tetrad.py`), list experiments and their link to the measurement model (`listexp.py`, `listirt.py`), MRP (`mrp.py`) and count indices (`index.py`). The figures and the tables `tables/tab-sim-*.tex` are written by `code/make_figures.py`, and the summary of the new experiments by `code/make_summary.py`. The recorded environment is Python 3.13.18, NumPy 2.5.3, SciPy 1.18.1 and Matplotlib 3.11.2 on Linux x86_64 (glibc 2.39). No other packages are used. The unit tests in `code/tests/` (32 tests, run with `pytest`) check the generators, including the cross-loading and acquiescence generators of X6 and the equivalence of the general generator with the bifactor generator; the polychoric estimator and its standard errors against Monte Carlo; the analytic Jacobians; exact population fits of the bifactor model and of the models with cross-loadings or a method factor; the tetrad statistics at null and alternative populations; the rank deficiency of a two-item specific factor; the list-experiment estimators, the allocations and the root of (5.13); the mixed model and the poststratification identity; the two-step EAP shrinkage; the index quadrature; and the identities of Theorem 4.5(d)(i) and Remark 4.7.

Seeds. The base seed is 20261103. Every replication has its own NumPy generator, created from a seed sequence `SeedSequence([20261103, i, keys..., r])`, where i is the experiment index and r the replication.

- X1 and X1b ($i = 1$): keys (design index, n), with design indices $0, \dots, 4$ for $|\mathcal{I}| = 6, 3, 2$, proportional and near-proportional. The sequence is split in two, one stream for the data and one for the bootstrap and the random starts. The weights of the population fits come from one sample of 100,000 respondents per design.
- X2 ($i = 2$): key 100κ, rounded; the weights of the population fits come from samples of 60,000 per group with keys $(999,\ 100\kappa)$. X7 regenerates the X2 data sets from the same seeds.
- X2b ($i = 22$): keys $(100\kappa_I,\ 100\kappa_S)$; population weights from 60,000 per group with keys $(999,\ 100\kappa_I,\ 100\kappa_S)$.
- X3 ($i = 3$): keys (cell index), with cells numbered in the order of Table 14 (parts A and B) and then Table 15 (part C), 32 cells in all.
- X4 ($i = 4$): keys (n); the sequence is split into a population stream and a sample stream.
- X5 ($i = 5$): keys $(100s,\ 100|\Delta\mu_G|)$ for the random uniform DIF, $(700 + 100s_a,\ 100\Delta\mu_G)$ for the non-uniform DIF, and $(500 + 100s,\ 100\Delta\mu_G,\ r)$ for the finite-sample check, each rounded to an integer.
- X6 ($i = 6$): keys (cell index), split into a data stream and a stream for the random starts; population weights from a sample of 60,000 with keys $(999,$ cell index$)$.
- X8 ($i = 8$): keys (scenario index). X9 ($i = 9$): keys (configuration index, n).

Because each replication has its own stream, the results do not depend on the number of worker processes or on the order in which tasks run.

Runs. All runs were on 3 November 2026, with two worker processes, `OPENBLAS_NUM_THREADS=1` and `nice`, on a shared four-core machine. Run times are therefore conservative.

(1) X1 ran from 21:49:41 UTC, with a wall time of 1,276 s and 2,547 s of CPU time for 870 fits with bootstrap, under a load average of 11 to 14.

(2) The replications of X1b were computed in a run started at 22:11:23 UTC (wall time 101 s), which later failed while summarising X2. The run started at 22:15:16 UTC reused these replications unchanged and computed X2, X3, X4 and X5 (wall times 58, 16, 5 and 22 s). At 22:17:13 UTC the summaries of X1, X1b and X2 were recomputed from the raw replications with the then final version of `run_experiments.py`. The X1 replications were produced by an earlier version of that script (SHA-256 prefix `ca450e64a827b7ce`) whose generation and estimation modules are byte-identical to those of the later one; X1b to X5 were produced by version `cd237869adbe67af` and the summaries by `2729078df210b26b`.

(3) The new experiments ran from 23:26:48 UTC with version `ad5429c1a40083a8`: X2b, X6, X7, X8 and X9, with wall times of 242, 367, 28, 20 and 146 s. Afterwards the block of disputed Israel-related items was renamed from IP to ID in the code, the one-item IT×S check, then proposed in the draft plan, and a second identification diagnostic were added to X6, and the X2b bound was extended to a stated maximum DIF of 0.4. With version `748bb62959101007`, X6 was then re-run with the same seeds from 23:41:57 UTC (wall time 574 s), which reproduces every quantity of the first X6 run and adds the new ones, and the population fits and summary of X2b were recomputed from its stored replications (option `-resummarize`). The generation and estimation code of X2b, X7, X8 and X9 is the same in the two versions.

The results file records, for each experiment, the SHA-256 prefixes of every source file that produced it and of the code that recomputed its summary. An earlier full run of X2 to X5 (21:41:53 UTC, in `results/run_log.txt`) was superseded and is not used.

C.2 Configurations

Table 9 summarises the experiments.

X1. All items have five categories with standardised thresholds giving marginal shares 0.30, 0.30, 0.20, 0.12 and 0.08, shifted by equally spaced amounts from −0.3 to 0.3 across the items of a design. The link is the normal ogive. The stereotype items S1–S8 have ECVs 0.85, 0.45, 0.75, 0.55, 0.90, 0.50, 0.80, 0.60 and communalities 0.55, 0.50, 0.60, 0.50, 0.55, 0.60, 0.50, 0.55. In the proportional design every stereotype item has ECV 0.70 and communality 0.55. In the near-proportional design (X1b) the ECVs are 0.74, 0.66, 0.72, 0.68, 0.75, 0.65, 0.73, 0.67 with communality 0.55. The Israel-related items I1–I6 have ECVs 0.25, 0.75, 0.40, 0.60, 0.15, 0.85 and communality 0.5. Standardised loadings are $\lambda_{jG} = (\mathrm{ECV}_j h_j^2)^{1/2}$ and $\lambda_{js} = \{(1 - \mathrm{ECV}_j)h_j^2\}^{1/2}$, converted to slopes and thresholds by Lemma 4.1. $n \in \{500, 1000, 3000\}$; $\tau = 0.5$. The pseudo-true ρ_{SI} is 0.621, 0.589, 0.591, 0.590 and 0.590 for $|\mathcal{I}| = 6$, 3, 2, proportional and near-proportional.

X2. The six-item design of X1, two groups of 1,500, focal latent means $(0.3, 0, 0.3)$ and focal covariance $\boldsymbol{I}_3$. Uniform DIF on I1 and I2 sets $b_{jc}^{(1)} = b_{jc} - \kappa\,(1 + a_{jG}^2 + a_{js}^2)^{1/2}$; non-uniform DIF on I3 sets $a_{jG}^{(1)} = (1 - 1.25\kappa)a_{jG}$. The leave-one-anchor-out refits are run at $\kappa = 0.4$ in all 100 replications. Anchor sets: correct S1–S4, I5, I6; one contaminated S1–S4, I1, I6; two contaminated S1–S4, I1, I2; stereotype-only S1–S4. DIF screen: Wald test of equal slopes and thresholds (6 degrees of freedom), Benjamini–Hochberg at $q = 0.05$ over the tested items, and, in the drafted screen only, an effect-size threshold of 0.25 at the focal mean of G.

X2b. As X2, with uniform DIF on two of the four stereotype anchors: in the focal group the thresholds of S1 and S2 are lowered by $\kappa_S \in \{0.1, 0.2, 0.3\}$ standard deviations of their latent responses, in the same

Table 9. Design of the simulation study (synthetic data only). R is the number of Monte Carlo replications. X1–X5 were labelled E1–E5 when they were run.

Exp.	Results tested	Truth and estimators	R
X1	Theorem 4.2 and Propositions 3.2 and 4.8; Remark A.5	One group; 8 stereotype items and $\lvert\mathcal{I}\rvert \in \{2, 3, 6\}$ Israel-related items; heterogeneous, nearly proportional (X1b) or proportional stereotype loadings; $n \in \{500, 1000, 3000\}$; bifactor and correlated-factors fits	100 ($\lvert\mathcal{I}\rvert = 6, 3$); 50 ($\lvert\mathcal{I}\rvert = 2$); 40 (others)
X2	Theorem 4.5 and Proposition 4.6; Remarks 4.7 and A.9	Two groups of 1,500; uniform DIF on I1, I2 and non-uniform DIF on I3 of magnitude $\kappa \in \{0, 0.2, 0.4, 0.6\}$; ignoring DIF, correct and contaminated anchors, stereotype-only anchors, and the drafted DIF screen	100 per κ
X2b	Proposition 4.6(c); Remark 4.7	As X2, with uniform DIF of $\kappa_S \in \{0.1, 0.2, 0.3\}$ on the stereotype anchors S1, S2, with or without the Israel-related DIF; default and stereotype-only anchors; the bound (4.12)	100 per cell
X3	Proposition 5.1 and Theorems 5.3 and B.5	One sensitive item with a threshold shift (Definition 5.4); n = 3,000; split and paired designs; equal, optimal and pilot-based allocations; control lists with $J \in \{2, \ldots, 6\}$, with and without floor and ceiling effects	2,000 (allocation, pairing); 1,000 (list choice)
X4	Equation (6.2) and Proposition 6.2	50 areas, 32 demographic cells each, differential non-response; $n \in \{1000, 3000, 6000\}$; MRP, MRP on two-step EAP scores, and direct estimates	200 per n
X5	Propositions 6.3 and B.8	Index $P_g(q) = \mathbb{P}(U_{i+} \geq q \mid g)$ for the 8 stereotype items, $q \in \{2, 4, 6\}$; random uniform and non-uniform DIF; unequal latent variances; exact computation	1,000 configurations per cell
X6	Cross-loadings and acquiescence (Section 8.7)	Registered sizes (10 S, 8 ID, 6 IT items), n = 6,000; IT cross-loading on S; acquiescence with registered or balanced keying; standard fit and four remedies	100 per cell
X7	The revised DIF screen (Section 8.3)	The X2 data sets; Benjamini–Hochberg screen without effect-size requirement, then refit	100 per κ
X8	Proposition 5.5(b),(c); Proposition B.7	Sensitive item in the six-item bank, two groups of 3,000, direct and list arms; δ_j from (5.13); common, group-specific and non-uniform shifts	300 per scenario
X9	Rule D2(ii); Assumption A3 and Lemmas A.3 and A.4	Registered sizes (10 and 14 items per domain), $n \in \{1500, 6000\}$; joint tetrad test, intersection–union test, earlier Bonferroni rule	400 (n = 6,000); 200 (n = 1,500)

direction, which is the pattern ideology-type DIF would produce if one group were more ready to agree with some stereotype statements at a given level of G. The Israel-related DIF of X2 is absent ($\kappa_I = 0$) or present at $\kappa_I = 0.4$. Strategies: the oracle (all DIF-free items as anchors); the default anchors S1–S4, I5, I6, with a free baseline and with the revised screen and refit; and stereotype-only anchors S1–S4 with the focal μ_I and σ_I^2 fixed. Leave-one-anchor-out refits are run for both anchor sets in every replication. $R = 100$ per cell.

X6. The registered sizes of Table 6: 10 stereotype items with the ECVs of X1 extended by 0.70 and 0.65 (communalities 0.50 to 0.60), 8 ID items with ECVs 0.10, 0.15, 0.20, 0.25, 0.30, 0.40, 0.60, 0.70 and 6 IT items with ECVs 0.35, 0.40, 0.45, 0.60, 0.75, 0.85 (communality 0.5), five categories, $n = 6{,}000$. In part (a) every IT item also loads on S with standardised loading $c \in \{0, 0.1, 0.2, 0.3\}$; its loadings on G and I, and so its ECV, are unchanged. In part (b) an acquiescence factor, orthogonal to the content factors, has slope $+\alpha$ on every positively worded item and $-\alpha$ on every reverse-keyed item after reverse scoring, $\alpha \in \{0.25, 0.40, 0.55\}$ (standardised loadings 0.16 to 0.17, 0.25 to 0.27 and 0.33 to 0.36). The registered keying reverses S2, S7, ID2 and ID6; balanced keying reverses half of every block (5 S, 4 ID and 3 IT items). One further cell has registered keying and item-specific slopes α times 0.5 to 1.5. The true ECV is the content ECV $a_{jG}^2/(a_{jG}^2 + a_{js}^2)$ of the item's own domain. $R = 100$ per cell.

X7. The 400 data sets of X2, regenerated from the same seeds, analysed with the revised screen: the free-baseline fit with the anchor set, a Wald test of each non-anchor item, Benjamini–Hochberg at $q = 0.05$, every significant item freed, and the partial-invariance refit. The expected-score difference averaged over the focal latent distribution (20-point Gauss–Hermite quadrature in each of θ_G and the item's specific factor) is recorded as the secondary description.

X8. The six-item bank of X1 with I4 (ECV 0.60) as the sensitive item; agreement is one of the top two categories. Two groups (or covariate values) of 3,000, with $\boldsymbol{\theta} \sim \mathrm{N}(\boldsymbol{\mu}_x, \boldsymbol{I})$, $\boldsymbol{\mu}_0 = \mathbf{0}$ and $\boldsymbol{\mu}_1 = (0.5, 0, 0.3)$. Every respondent is randomised with equal probability to the direct arm or to one of the two list arms, which use four control items in two pairs with $\rho = -0.6$; I4 is missing by design in the list arms. Scenarios: a common shift $\delta = 0.4$ latent-response standard deviations; group-specific shifts 0.4 and 0.8; and a private response with slope multiplied by $\upsilon = 0.5$ (Proposition B.7(e)), its threshold chosen so that group 0 has the same private prevalence as under the common shift. $R = 300$ per scenario.

X9. The registered sizes, $|\mathcal{S}| = 10$ and $|\mathcal{I}| = 14$, $n \in \{1{,}500, 6{,}000\}$. Configurations: proportional (all stereotype ECVs 0.70, all Israel-related ECVs 0.45); all but one proportional (one item per domain with ECV 0.20 or 0.90); two lines (two items per domain with ECV 0.30 or 0.85, the rest as in the proportional configuration); near-proportional (stereotype ECVs 0.65 to 0.75, Israel-related 0.38 to 0.52); and the heterogeneous bank of X6. The first two violate Assumption A3 in both domains; the other three satisfy it in both. $R = 400$ at $n = 6{,}000$ and 200 at $n = 1{,}500$.

X3. The sensitive item's latent response is $0.75\,\theta_G + \varepsilon$ with $\varepsilon \sim \mathrm{N}(0, 1)$. The private threshold gives $\pi = 0.30$ and the direct threshold, shifted by δ_j, gives $\pi^D = 0.20$. False endorsement, when present, sets $D_i = 1$ with probability $0.02/0.7$ when $Z_i = 0$, so $\pi^F = 0.02$. Control items have prevalence $1/2$. They form $\lfloor J/2 \rfloor$ independent pairs, each with $\mathbb{P}(1, 1) = \mathbb{P}(0, 0) = (1 + \rho)/4$, plus one independent item if J is odd. $n = 3{,}000$ and costs are equal. The pilot-based allocation uses a pilot of 300 respondents with equal arms, which is not reused. The covariate-adjusted estimator uses a score of θ_G with reliability 0.85 and separate-slope regression adjustment of each arm mean, with HC2 variances. Floor and ceiling effects: in arm T, a respondent with $Z_i = 1$ and $W_i = J$ reports J with probability ϕ_{c}, and one with $Z_i = 1$

and $W_i = 0$ reports 0 with probability ϕ_{f}. The theoretical MSE-optimal list is computed on the menu $J \in \{2, \ldots, 6\} \times \rho \in \{0, -0.6\}$ for $n \in \{300, 10^3, 3 \times 10^3, \ldots, 10^6\}$. $R = 2{,}000$ in parts A and B and 1,000 in part C.

X4. Population of 10^7 in 50 areas with sizes proportional to $\exp(Z_a)$, $Z_a \sim \mathrm{N}(0, 1)$. Within each area, the 32 cells (age 4 × sex 2 × education 4) have a Dirichlet mix with concentration 30 per cell around a uniform national mix. Cell means of G are $b_{\text{age}} + b_{\text{sex}} + b_{\text{edu}} + 0.25\, z_a + u_a$ with $b_{\text{age}} = (0.25, 0.05, -0.10, -0.20)$, $b_{\text{sex}} = (0.05, -0.05)$, $b_{\text{edu}} = (0.20, 0.10, -0.10, -0.20)$, $z_a \sim \mathrm{N}(0, 1)$ and $u_a \sim \mathrm{N}(0, 0.2^2)$. The within-cell standard deviation of G is 0.85. The relative response propensity is the product of an age factor $(0.6, 0.8, 1.1, 1.4)$ and an education factor $(0.6, 0.9, 1.1, 1.4)$. Respondents are drawn with probability proportional to cell size times propensity, and each contributes $y_i = \theta_{iG} + e_i$ with $e_i \sim \mathrm{N}(0, 0.4^2)$. $n \in \{1000, 3000, 6000\}$, $R = 200$, with a new population in each replication.

X5. The eight stereotype items of X1. Agreement is a response in one of the top two categories, and U_{i+} is the number of statements agreed with. The population count distribution is computed by tensor Gauss–Hermite quadrature with 40 nodes per dimension over (θ_G, θ_S), and the Poisson-binomial distribution of the count given the latent vector. Uniform DIF adds $\mathrm{N}(0, s^2)$ times the latent-response standard deviation of each item to its agreement threshold, with $s \in \{0, 0.1, \ldots, 0.5\}$, $|\Delta\mu_G| \in \{0.05, 0.1, 0.2, 0.3\}$ with random sign, $q \in \{2, 4, 6\}$ and 1,000 configurations per cell. Unequal variances: $\sigma_G^{(1)} \in \{0.6, 0.8, 1.25, 1.5\}$ with $\Delta\mu_G \in \{0.1, 0.2\}$. Non-uniform DIF: general slopes multiplied by $\exp\{\mathrm{N}(0, s_a^2)\}$, $s_a \in \{0.2, 0.4, 0.6\}$, $\Delta\mu_G = 0.1$, 1,000 configurations each. Finite-sample check: 1,000 respondents per group, $q = 4$, $\Delta\mu_G = 0.1$, $s \in \{0, 0.3\}$, 300 replications each.

C.3 Estimator settings

Polychorics and weights. Thresholds are normal quantiles of the cumulative marginal shares. Each polychoric correlation maximises the bivariate ordinal-probit likelihood of its table with the thresholds fixed, by vectorised Fisher scoring (Olsson, 1979). The asymptotic covariance $\boldsymbol{\Gamma}$ of thresholds and correlations is the empirical second moment of per-respondent influence functions, including the effect of the estimated thresholds. The weights are $\boldsymbol{V} = \mathrm{diag}(\boldsymbol{\Gamma})^{-1}$, and the sandwich covariance is $(\boldsymbol{\Delta}^\top \boldsymbol{V} \boldsymbol{\Delta})^{-1} \boldsymbol{\Delta}^\top \boldsymbol{V} \boldsymbol{\Gamma} \boldsymbol{V} \boldsymbol{\Delta} (\boldsymbol{\Delta}^\top \boldsymbol{V} \boldsymbol{\Delta})^{-1}$, where $\boldsymbol{\Delta}$ is the Jacobian of the model-implied statistics. In one group only the correlations are fitted, since the thresholds are saturated. In two groups thresholds and correlations are fitted jointly, with the reference group's means set to zero and its variances to one.

Optimisation. A projected Levenberg–Marquardt algorithm with analytic Jacobian, at most 200 iterations (300 in X6), and bounds $|a_{jf}| \le 12$ (a standardised loading of 0.997), focal log-variances in $[-3, 3]$ and thresholds in $[-40, 40]$. Factors are reflected so that the reference group's loadings on each factor sum to a positive number, and the method slope is reported as non-negative. Starts: a default ($\lambda_{jG} = 0.6$, $\lambda_{js} = 0.4$ for every item; cross-loadings 0; method slope 0.2) and random starts with $\lambda_{jG} \sim \mathrm{U}(0.2, 0.8)$ and $\lambda_{js} \sim \mathrm{U}(0.1, 0.6)$ (four in X1, two in X6). The fit with the smallest objective is kept. The *start range* of ECV_j is its range over the starts whose objective is within a relative 10^{-6} of the best. A fit is counted as *at a bound* if any slope has $|a_{jf}| \ge 0.999 \times 12$. The identification diagnostic is $\sigma_{\min}/\sigma_{\max}$ of $\boldsymbol{V}^{1/2}\boldsymbol{\Delta}$ at the estimate.

Intervals. Wald intervals for ECV_j are formed on the logit scale from the delta-method (sandwich) standard error, with $z = 1.96$, or $z = \Phi^{-1}\{1 - 0.05/(2|\mathcal{I}|)\}$ for the Bonferroni intervals. The bootstrap of X1 draws $B = 100$ samples of respondents with replacement, re-estimates the polychorics (at most 30 scoring iterations, started at the full-sample values), and refits the bifactor model from the full-sample estimate

with the full-sample weights (at most 60 iterations); percentile intervals are reported. The classification is "in" if the lower limit is at least τ, "out" if the upper limit is below τ, and "undetermined" otherwise. The correlated-factors model is fitted by the same weighted least squares with $\rho_{SI} = \tanh z$.

The anchor bound (X2b). For the limited-information estimator the role of the Fisher information in Proposition 4.6(c) is played by $\boldsymbol{\Delta}^\top \boldsymbol{V} \boldsymbol{\Delta}$. We add focal intercept shifts ι_j for the anchors (all thresholds of anchor j in the focal group moved by $-\iota_j$) and take $\boldsymbol{W}$ to be the Schur complement of their block of $\boldsymbol{\Delta}^\top \boldsymbol{V} \boldsymbol{\Delta}$ with respect to all parameters of the anchored model other than the focal latent means, at the population fit without anchor DIF. The derivative of the statistics with respect to the focal means equals that with respect to $\boldsymbol{\iota}$ along $\boldsymbol{A}_{\mathcal{A}}$ plus a combination of the other parameters' columns, to within 3×10^{-15}, so the first-order bias is $(\boldsymbol{A}_{\mathcal{A}}^\top \boldsymbol{W} \boldsymbol{A}_{\mathcal{A}})^{-1} \boldsymbol{A}_{\mathcal{A}}^\top \boldsymbol{W} \boldsymbol{d}_{\mathcal{A}}$, as in (4.11). For a stated maximum DIF δ, the anchor DIF ranges over $|d_j| \le \delta\,(1 + a_{jG}^2 + a_{js}^2)^{1/2}$. The largest first-order bias of $\hat{\mu}_G$ is then $\delta \sum_j |c_j| (1 + a_{jG}^2 + a_{js}^2)^{1/2}$, where c_j is the entry of the first row of $(\boldsymbol{A}_{\mathcal{A}}^\top \boldsymbol{W} \boldsymbol{A}_{\mathcal{A}})^{-1} \boldsymbol{A}_{\mathcal{A}}^\top \boldsymbol{W}$ for anchor j, and the two norms of (4.12) are maximised over the vertices of the box. With stereotype-only anchors $\boldsymbol{A}_{\mathcal{A}}$ has the columns of G and S only. All of these are linear in δ.

Misspecification models (X6). The fitted models are single-group, correlation-only weighted least squares fits whose slope matrix is linear in the parameters: the standard bifactor model; the same with a free loading on S for every IT item (cross-loading model); the same with a fourth, orthogonal factor whose slopes are $+m$ for positively worded and $-m$ for reverse-keyed items according to the instrument's keying, with one free parameter m (method factor; Maydeu-Olivares and Coffman, 2006); and the standard model fitted to the S and ID items alone. The mean-residual check averages the 60 IT×S residual correlations $r_{jk} - \hat{\rho}_{jk}$ and divides by its standard error from $\boldsymbol{P} \boldsymbol{\Gamma} \boldsymbol{P}^\top$, with $\boldsymbol{P} = \boldsymbol{I} - \boldsymbol{\Delta}(\boldsymbol{\Delta}^\top \boldsymbol{V} \boldsymbol{\Delta})^{-1} \boldsymbol{\Delta}^\top \boldsymbol{V}$; it rejects one-sidedly at 5%. The two-stage rule of the plan (Section 8.7) uses the cross-loading model when this check rejects and the standard model otherwise. The one-item check, proposed in an earlier draft of the plan, tests each of the 60 residual correlations two-sidedly with the same covariance, applies Benjamini–Hochberg at $q = 0.05$, and fits, for each IT item in turn, the standard model plus that item's loading on S, started at the standard fit. An IT item is flagged if any of its residual correlations is significant or its cross-loading has a Wald interval excluding zero. A flagged item is classified only if the standard fit and the Bonferroni interval for its share of common variance due to G in the one-item model, $a_{jG}^2 / (a_{jG}^2 + a_{jS}^2 + a_{jI}^2)$, place it on the same side of τ; otherwise it is undetermined. Because the method slope m enters the correlations only through m^2, its information is zero at $m = 0$, and the identification diagnostic is also computed over the content parameters alone.

Tetrad tests (X9). For a set of $m \ge 4$ items with reference triple $(1, 2, 3)$, the non-redundant tetrads are $r_{1k} r_{23} - r_{2k} r_{13}$ and $r_{1k} r_{23} - r_{3k} r_{12}$ for $k \ge 4$, and $r_{kl} r_{23} - r_{2k} r_{3l}$ for $4 \le k < l$, $m(m-3)/2$ in all. Each introduces one new correlation, so their Jacobian $\boldsymbol{T}$ has full row rank when $r_{12} r_{13} r_{23} \ne 0$. The statistic $\boldsymbol{t}^\top (\boldsymbol{T} \boldsymbol{\Gamma} \boldsymbol{T}^\top)^{-1} \boldsymbol{t}$ is referred to $\chi^2_{m(m-3)/2}$ (Bollen and Ting, 1993). The reference triple is the three items with the largest mean absolute correlation with the others. A residual-based test of one-factor structure, which does not depend on how the restrictions are written, gave rejection rates within 4.0 percentage points of the tetrad test in every cell (`results/SUMMARY-new-experiments.md`). The intersection–union test applies the tetrad test to every $\mathcal{D} \setminus \{k\}$ (Berger, 1982). The earlier rule computes a Wald z-statistic for each of the three tetrads of every quadruple, $3\binom{10}{4} + 3\binom{14}{4} = 3{,}633$ in all, and rejects if any exceeds the Bonferroni critical value at 5%.

Lists, MRP and the threshold shift. The list estimators are those of Theorem 5.3, with plug-in variances from (5.6), (5.9) and (5.11). The combined estimator is set to 0 if a stratum $D_i = 0$ of a list arm has

fewer than two respondents. In X8, $\hat{\delta}_x = \hat{b}^D_{jc^*} - \hat{\boldsymbol{a}}_j^\top \hat{\boldsymbol{\mu}}_x + \hat{s}_x \Phi^{-1}(\hat{\pi}_x)$ with $\hat{s}_x = (1 + \hat{\boldsymbol{a}}_j^\top \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{a}}_j)^{1/2}$, from the two-group fit to the direct arm with item j free across groups. Its variance adds the delta-method term from the sandwich covariance of that fit (numerical gradient) and $\hat{s}_x^2 \operatorname{Var}(\hat{\pi}_x)/\phi\{\Phi^{-1}(\hat{\pi}_x)\}^2$; the two are independent because the arms are disjoint. The test of a common shift is the z-test of $\hat{\delta}_1 - \hat{\delta}_0$ with the shared measurement parameters in its variance. MRP uses restricted maximum likelihood for the ratio of the area variance to the residual variance, generalised least squares for the fixed effects, best linear unbiased predictors for the area effects, and the prediction-error variance from Henderson's mixed-model equations. The two-step scores use the prior $\mathrm{N}(m_0, \sigma_0^2)$ with $m_0 = \bar{y}$ and $\sigma_0^2 = \widehat{\operatorname{Var}}(y) - 0.4^2$, so that $\hat{\varrho} = \sigma_0^2/(\sigma_0^2 + 0.4^2)$.

C.4 Monte Carlo standard errors

For a mean the MCSE is the replication SD divided by $\sqrt{R}$; for a proportion p it is $\{p(1-p)/R\}^{1/2}$; for an RMSE it is the delta-method value $\mathrm{SD}(e^2)/(2\,\mathrm{RMSE}\sqrt{R})$; and for a median it is $1.2533 \times \mathrm{IQR}/(1.349\sqrt{R})$. A coverage near 95% therefore has an MCSE of about 2.2 percentage points with $R = 100$ and 3.4 with $R = 40$. Coverages and classification rates far from 95% have larger MCSEs: in the X1 designs with $R = 40$, the item-level Wald coverages have MCSEs of 5.2 to 7.9 points and the Bonferroni set coverages 4.7 to 7.1 points, about 7 points in round terms. Pooled X1 quantities treat the $|\mathcal{I}|$ items of a replication as independent, which understates their MCSE. In X2 the MCSE of the bias of $\hat{\mu}_G$ is 0.005 to 0.020; in X2b it is 0.005 to 0.026 for the default anchors and up to 1.7 for the stereotype-only fits at $\kappa_S = 0.3$, which break down. In X3 the MCSE of the bias is 0.0005 to 0.0017, in X4 the MCSE of the RMSE is at most 0.004, and in X5 a reversal rate from 1,000 configurations has an MCSE of at most 1.6 points. In X6 and X7 a rate from $R = 100$ has an MCSE of at most 5 points. In X8 the MCSE of the bias of $\hat{\delta}_j$ is 0.006 to 0.009 latent-response SDs and that of a coverage near 95% about 1.3 points. In X9 a size near 5% has an MCSE of 1.1 points at $n = 6{,}000$ and 1.5 at $n = 1{,}500$. All MCSEs are stored in the results file next to the quantities they refer to.

C.5 Exact checks and recomputed illustrations

Three identities of Section 4 were checked on exact population statistics in X2. Uniform DIF $-\boldsymbol{a}_j^\top \boldsymbol{v}$ on every item, with $\boldsymbol{v} = (0.25, -0.10, 0.15)$, is fitted exactly by the invariant model with latent mean $\boldsymbol{\mu} + \boldsymbol{v} = (0.55, -0.10, 0.45)$ (Theorem 4.5(d)(i)). Three anchors (S1, S2, I1) with DIF $-\boldsymbol{A}_{\mathcal{A}}\boldsymbol{v}$ give the same shift, and every non-anchor item acquires spurious DIF equal to $\boldsymbol{a}_k^\top \boldsymbol{v}$ to within 2.3×10^{-9} (Proposition 4.6(b)). Stereotype-only anchors recover μ_G and μ_S to within 2×10^{-13} and every focal ECV to within 2.1×10^{-14} under the Israel-related DIF of X2 at $\kappa = 0.4$ (Remark 4.7). The unit tests check that the cross-loading and method-factor models of X6 fit the population statistics of their own data-generating processes exactly, and that the tetrad statistics of X9 vanish at the two null populations.

The results file also recomputes, independently, the illustrative numbers quoted in Sections 5, 6 and 8. For four independent control items with prevalence 1/2 and $\pi = 0.2$, $L^* = 26.96$; for two pairs with $\rho = -0.6$, $L^* = 11.92$ (Proposition 5.1). The thresholds of Remark 5.2 are 0.0456 ($n = 2{,}000$) and 0.0263 ($n = 6{,}000$). For Theorem B.5(a) with $\sigma_{\mathsf{C}}^2 = \sigma_{\mathsf{T}}^2 = 1$ and $\pi^D = 0.2$, $n\,\mathrm{Var}$ is 6.48 under equal allocation and 5.76 at the optimum, with shares $(1/6, 5/12, 5/12)$. The direct-arm cost threshold of Theorem B.5(b) is 1.26×10^{-4}, and the paired-to-split variance ratio of Corollary B.6 is 0.810. The arm size for the power calculation of Section 8.4 is 1,695 for SDB = 0.10 and 6,781 for SDB = 0.05. Example B.9 gives $P_1(5) = 0.100$ and $P_2(5) = 0.214$ in (i), $P_1(1) = 0.999$, $P_2(1) = 0.806$, $P_1(10) = 0.002$ and $P_2(10) = 0.194$ in (ii), and changes of 0.0049 and 0.0987 in (iii).

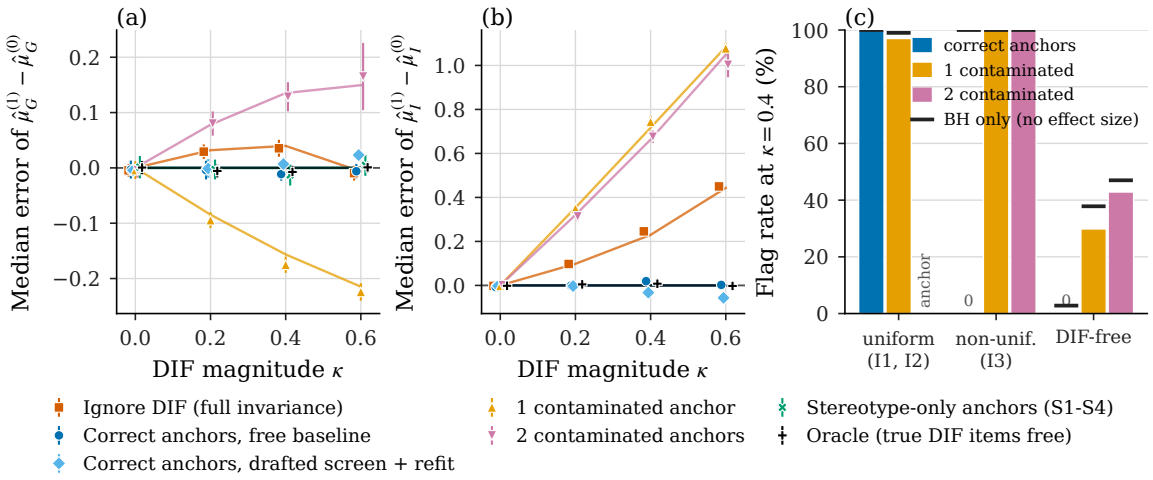


Figure 5. X2, group comparisons under DIF on the Israel-related items (synthetic data; 1,500 respondents per group; $R = 100$ per κ). (a), (b) Median error of the focal-minus-reference difference in the means of G and I (±2 robust MCSE bars); lines are probability limits. Medians are shown because 2% of the fits that ignore DIF at $\kappa = 0.6$ reach a variance bound. (c) Flag rates at $\kappa = 0.4$ of the drafted screen (bars: Benjamini–Hochberg *and* an expected-score difference of at least 0.25) and of Benjamini–Hochberg alone (black ticks), which is the revised screen of X7. With the correct anchors the drafted screen never flags the item with non-uniform DIF.

C.6 Further results

Figure 5 and Tables 10 to 13 give further results of X2, X2b, X7, X6 and X8, and Figures 6 to 8 and Tables 14 to 17 the full results of X3, X4 and X5 that Sections 7.3 to 7.6 summarise.

C.7 Reproduction

From `code/`, the commands

```
export OPENBLAS_NUM_THREADS=1
nice -n 10 python run_experiments.py --only X1
nice -n 10 python run_experiments.py --only X1b X2 X3 X4 X5
python run_experiments.py --resummarize X1 X1b X2
nice -n 10 python run_experiments.py --only X2b X6 X7 X8 X9
python make_figures.py
python make_summary.py
python -m pytest tests/
```

regenerate `results/sim_results.json`, `results/raw/`, the figures `figures/fig-sim-*.pdf`, the tables `tables/tab-sim-*.tex`, the summary of the new experiments and the test report. A single call without `-only` runs all experiments, and `-quick` runs a small smoke test. X7 reads `results/raw/X2.json` to compare its flags with those of X2, so X2 must be run first. On the shared machine described above, the full study takes about 40 minutes of wall time with two workers, half of it in X1. The figures take a few seconds and the tests about eight seconds. Because each replication has its own seed, the numbers do not depend on the number of workers. Results can differ in the last digits across platforms or library versions, through floating-point differences in the optimiser.

Table 10. X2b, DIF on two of the four stereotype anchors (synthetic data; 1,500 per group; $R = 100$ per row). Bias (median over replications, because the stereotype-only fits break down at $\kappa_S = 0.3$) and coverage (%) of $\hat{\mu}_G^{(1)} - \hat{\mu}_G^{(0)}$ (true 0.3) from the free-baseline fits. First order: the first-order bias at the actual anchor DIF. Bound: the largest first-order |bias| of $\hat{\mu}_G$ over all uniform anchor DIF of at most κ_S latent-response SDs on every anchor (Appendix C.3). LOO: mean range of $\hat{\mu}_G$ over the leave-one-anchor-out refits (0.043 with clean anchors in X2).

		Default anchors S1–S4, I5, I6					Stereotype-only anchors S1–S4				
κ_I	κ_S	Bias	Cov.	First ord.	Bound	LOO	Bias	Cov.	First ord.	Bound	LOO
0	0.1	+0.023	96	+0.026	0.35	0.165	+0.066	84	+0.058	0.44	0.464
0	0.2	+0.045	82	+0.052	0.69	0.766	+0.233	32	+0.114	0.89	1.410
0	0.3	+0.024	28	+0.076	1.03	12.034	+1.118	22	+0.170	1.33	5.857
0.4	0.1	+0.033	89	+0.026	0.34	0.166	+0.078	75	+0.058	0.44	0.469
0.4	0.2	+0.063	72	+0.052	0.69	0.525	+0.236	36	+0.115	0.89	1.946
0.4	0.3	+0.008	42	+0.076	1.03	11.872	+1.250	20	+0.170	1.33	5.387

Table 11. X7, the revised DIF screen (Benjamini–Hochberg at $q = 0.05$, no effect-size requirement, then refit) with the correct anchors on the X2 data sets (synthetic data; 1,500 per group; $R = 100$ per row). Flag rates of the items with uniform DIF (mean of I1 and I2), of I3 (non-uniform DIF) and of the DIF-free items, and the share of replications with at least one DIF-free item flagged; bias and coverage (%) of the latent mean differences; bias of the reference and focal ECV of I3.

	Flag rate (%)				$\mu_G^{(1)} - \mu_G^{(0)}$		$\mu_I^{(1)} - \mu_I^{(0)}$		ECV_{I3} bias	
κ	I1, I2	I3	DIF-free	Any false	Bias	Cov.	Bias	Cov.	Ref.	Focal
0	–	–	0.8	5	−0.003	95	−0.003	97	−0.001	+0.000
0.2	96	41	3.6	17	−0.004	96	−0.002	87	−0.020	+0.042
0.4	100	100	2.8	13	−0.006	98	+0.008	95	+0.009	−0.002
0.6	100	100	1.6	8	+0.000	95	+0.000	94	+0.000	+0.002

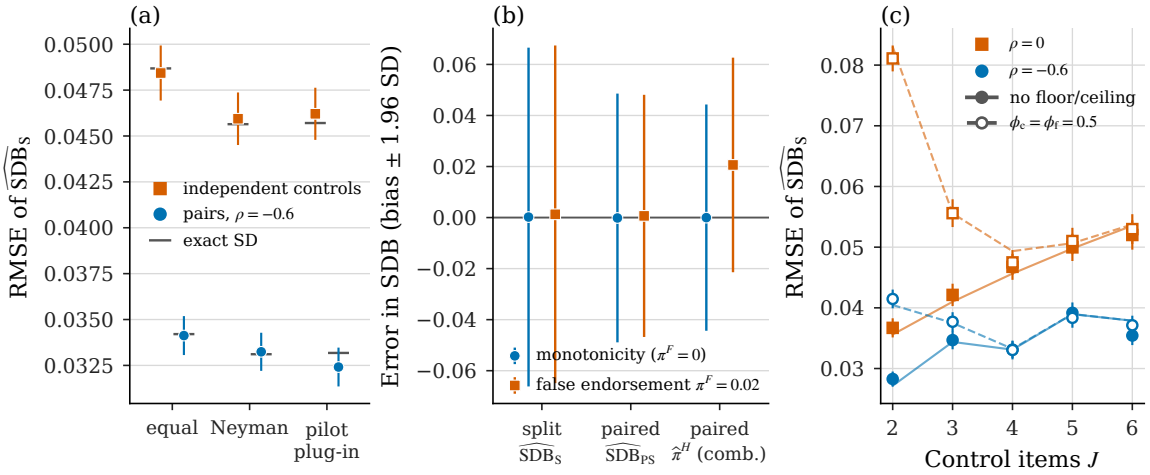


Figure 6. X3, list experiments (synthetic data; $n = 3{,}000$). (a) RMSE of the split-design estimator under three allocations, with independent controls or negatively correlated pairs ($R = 2{,}000$; ±2 MCSE bars); ticks are the exact SDs from (5.6). (b) Error of three estimators of SDB (bias ±1.96 Monte Carlo SD) under monotonicity ($\pi^F = 0$) and with false endorsement ($\pi^F = 0.02$). (c) RMSE of the split-design estimator against the number of control items, with independent ($\rho = 0$) or paired ($\rho = -0.6$) controls, without and with floor and ceiling effects ($\phi_c = \phi_f = 0.5$; $R = 1{,}000$); lines are the theoretical values from (5.6) and (B.5).

Table 12. X6, misspecification at the registered sizes (synthetic data; n = 6,000; R = 100 per row). Flag: % of replications in which the one-item IT×S check of an earlier draft flags at least one IT item. IT, ID: mean bias of $\widehat{\mathrm{ECV}}_j$ in the block under the standard fit. Max: largest |Monte Carlo bias| over the Israel-related items in the model (the ID items for the ID-only refit). Wrong: % of those items on the wrong side of $\tau = 0.5$ by Bonferroni intervals. IT wrong: % of IT items on the wrong side under the standard fit (Std.) and under the one-item check (1-item). The two-stage rule of the plan is reported in Section 7.4. reg., bal.: registered and balanced keying; item-spec.: acquiescence slopes varying between items.

		Standard bifactor				Cross-loadings		Method factor		$\mathcal{I}$ = ID only		IT wrong	
Cell	Flag	IT	ID	Max	Wrong	Max	Wrong	Max	Wrong	Max	Wrong	Std.	1-item
cross-loading $c = 0$	26	+0.003	+0.001	0.005	0.0	0.006	0.0	0.005	0.0	0.004	0.0	0.0	0.0
cross-loading $c = 0.1$	100	+0.018	−0.041	0.071	0.0	0.005	0.0	0.071	0.0	0.003	0.0	0.0	0.0
cross-loading $c = 0.2$	100	+0.051	−0.066	0.117	0.1	0.007	0.0	0.117	0.1	0.006	0.0	0.3	0.0
cross-loading $c = 0.3$	100	+0.089	−0.080	0.141	7.1	0.006	0.0	0.141	7.1	0.006	0.0	14.2	1.5
reg. $\alpha = 0.25$	42	+0.023	+0.014	0.030	0.0	0.039	0.0	0.003	0.0	0.030	0.0	0.0	0.0
reg. $\alpha = 0.40$	98	+0.015	+0.000	0.132	0.0	0.139	0.0	0.003	0.0	0.164	0.0	0.0	0.0
reg. $\alpha = 0.55$	100	+0.050	+0.010	0.312	0.6	0.317	2.7	0.004	0.0	0.322	0.0	1.3	1.2
bal. $\alpha = 0.25$	100	−0.099	−0.064	0.121	0.1	0.195	6.6	0.005	0.0	0.128	0.1	0.2	0.2
bal. $\alpha = 0.40$	100	−0.128	−0.082	0.170	3.9	0.255	8.1	0.006	0.0	0.168	1.4	5.0	5.0
bal. $\alpha = 0.55$	100	+0.103	+0.256	0.433	45.0	0.672	54.4	0.003	0.0	0.173	1.1	36.0	32.7
reg. $\alpha = 0.40$, item-spec.	100	−0.019	−0.020	0.112	0.0	0.117	0.0	0.043	0.0	0.123	0.0	0.0	0.0

Table 13. X8, the threshold shift δ_j identified by the list experiment (synthetic data; two groups of 3,000, each split equally into direct, list-control and list-treatment arms; R = 300 per scenario). δ_j in standard deviations of the item's reference latent response: true value, bias, Monte Carlo SD, mean standard error and coverage (%) of the 95% interval; true SDB_j by group and the bias of its model-based estimate; rejection rate (%) of the 5% test of a common shift. Under the non-uniform scenario the uniform shift is misspecified and δ_j has no true value.

		δ_j (latent-response SDs)					SDB_j		Common
Scenario	Group	True	Bias	SD	SE	Cov.	True	Bias	shift rej.
Common uniform shift	$x = 0$	0.40	+0.000	0.153	0.149	94.3	0.073	+0.0010	3.0
	$x = 1$	0.40	−0.003	0.109	0.109	95.7	0.112	−0.0003	
Group-specific shifts	$x = 0$	0.40	+0.001	0.150	0.148	95.7	0.073	+0.0020	58.0
	$x = 1$	0.80	−0.003	0.108	0.115	96.7	0.186	−0.0018	
Non-uniform ($\upsilon = 0.5$)	$x = 0$	–	–	0.153	0.148	–	–	–	14.0
	$x = 1$	–	–	0.116	0.119	–	–	–	

Table 14. X3 parts A and B, estimation of social-desirability bias (synthetic data; n = 3,000; R = 2,000 per row). Shares (r_D, r_C, r_T) of the direct, list-control and list-treatment arms (means over replications for the pilot-based allocation). SD (theory): exact standard deviation from (5.6), or asymptotic from (5.9) and (5.11), at the allocation used. Bias and RMSE are as estimators of SDB, which is 0.10 when $\pi^F = 0$ and 0.08 when $\pi^F = 0.02$. Cov.: coverage (%) of the 95% Wald interval with the plug-in standard error; SE/SD: mean standard error over the Monte Carlo SD.

Design and allocation	Estimator	(r_D, r_C, r_T)	SD (theory)	Bias	SD	RMSE	Cov.	SE/SD
A. Split design, $J = 4$, $\pi^F = 0$								
indep. controls, equal	$\widehat{\mathrm{SDB}}_{\mathrm{S}}$	(0.33, 0.33, 0.33)	0.0487	+0.0010	0.0484	0.0484	95.5	1.00
indep. controls, Neyman	$\widehat{\mathrm{SDB}}_{\mathrm{S}}$	(0.16, 0.40, 0.44)	0.0456	+0.0008	0.0459	0.0459	94.3	0.99
indep. controls, pilot plug-in	$\widehat{\mathrm{SDB}}_{\mathrm{S}}$	(0.16, 0.40, 0.44)	0.0457	+0.0016	0.0462	0.0462	95.1	0.99
pairs $\rho = -0.6$, equal	$\widehat{\mathrm{SDB}}_{\mathrm{S}}$	(0.33, 0.33, 0.33)	0.0342	−0.0007	0.0341	0.0341	95.2	1.00
pairs $\rho = -0.6$, Neyman	$\widehat{\mathrm{SDB}}_{\mathrm{S}}$	(0.22, 0.35, 0.43)	0.0331	+0.0008	0.0332	0.0332	95.2	1.00
pairs $\rho = -0.6$, pilot plug-in	$\widehat{\mathrm{SDB}}_{\mathrm{S}}$	(0.22, 0.35, 0.43)	0.0332	+0.0008	0.0324	0.0324	95.2	1.02
B. Paired vs split, $J = 4$, pairs $\rho = -0.6$								
split, Neyman, $\pi^F = 0$	$\widehat{\mathrm{SDB}}_{\mathrm{S}}$	(0.22, 0.35, 0.43)	0.0331	+0.0002	0.0339	0.0339	94.4	0.98
paired, optimal, $\pi^F = 0$	$\widehat{\mathrm{SDB}}_{\mathrm{PS}}$	(0.00, 0.47, 0.53)	0.0243	−0.0002	0.0249	0.0249	94.5	0.98
	$\widehat{\pi}^H$	(0.00, 0.47, 0.53)	0.0220	+0.0000	0.0226	0.0226	94.8	0.97
paired, equal, $\pi^F = 0$	$\widehat{\mathrm{SDB}}_{\mathrm{PS}}$	(0.33, 0.33, 0.33)	0.0298	−0.0002	0.0296	0.0296	95.5	1.01
	$\widehat{\pi}^H$	(0.33, 0.33, 0.33)	0.0270	−0.0001	0.0268	0.0268	95.5	1.01
split, Neyman, $\pi^F = 0.02$	$\widehat{\mathrm{SDB}}_{\mathrm{S}}$	(0.23, 0.35, 0.43)	0.0334	+0.0013	0.0337	0.0338	94.5	0.99
paired, optimal, $\pi^F = 0.02$	$\widehat{\mathrm{SDB}}_{\mathrm{PS}}$	(0.00, 0.47, 0.53)	0.0246	+0.0007	0.0242	0.0242	95.2	1.02
	$\widehat{\pi}^H$	(0.00, 0.47, 0.53)	0.0218	+0.0206	0.0214	0.0297	85.3	1.01
paired, equal, $\pi^F = 0.02$	$\widehat{\mathrm{SDB}}_{\mathrm{PS}}$	(0.33, 0.33, 0.33)	0.0301	−0.0002	0.0302	0.0302	95.0	1.00
	$\widehat{\pi}^H$	(0.33, 0.33, 0.33)	0.0267	+0.0198	0.0263	0.0329	89.8	1.01

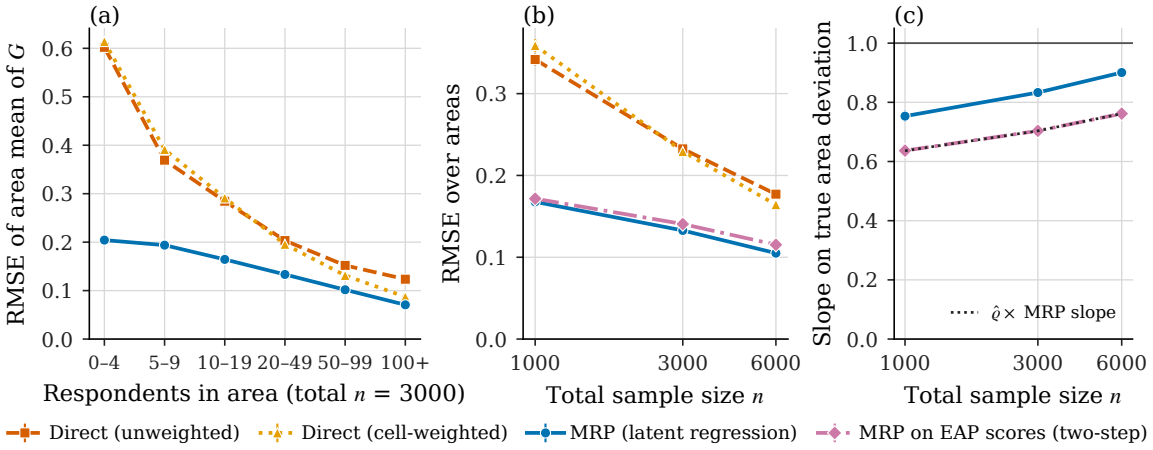


Figure 7. X4, small-area estimates of the mean of G (synthetic data; 50 areas; R = 200 per n). (a) RMSE by the realised number of respondents in the area, at n = 3,000. (b) RMSE over areas against the total sample size (±2 MCSE bars). (c) Slope of estimated on true area deviations from the national mean; the dotted line is $\hat{\varrho}$ of Proposition 6.2 times the MRP slope. The measurement step is abstracted: each respondent contributes an unbiased score of G, not item responses.

Table 15. X3 part C, choice of control list (synthetic data; split design, Neyman allocation, n = 3,000; R = 1,000 per row). Exposure is $\mathbb{P}(W_i \in \{0, J\})$; ϕ_c and ϕ_f are the ceiling and floor probabilities. Theoretical bias from (B.5) and SD from (5.6); Cov.: coverage (%) of the 95% Wald interval.

				Bias		SD			
J	ρ	$\phi_c = \phi_f$	Exposure	Theory	Sim.	Theory	Sim.	RMSE	Cov.
2	0	0	0.500	+0.0000	+0.0000	0.0356	0.0367	0.0367	94.6
2	−0.6	0	0.200	+0.0000	−0.0005	0.0272	0.0283	0.0283	94.0
3	0	0	0.250	+0.0000	−0.0007	0.0410	0.0421	0.0421	94.2
3	−0.6	0	0.100	+0.0000	+0.0008	0.0344	0.0347	0.0347	95.0
4	0	0	0.125	+0.0000	+0.0004	0.0456	0.0468	0.0468	94.2
4	−0.6	0	0.020	+0.0000	−0.0008	0.0331	0.0331	0.0331	96.1
5	0	0	0.062	+0.0000	+0.0007	0.0498	0.0500	0.0500	95.6
5	−0.6	0	0.010	+0.0000	+0.0027	0.0390	0.0391	0.0392	95.0
6	0	0	0.031	+0.0000	+0.0001	0.0535	0.0520	0.0520	95.3
6	−0.6	0	0.002	+0.0000	−0.0001	0.0379	0.0355	0.0354	96.4
2	0	0.5	0.500	−0.0750	−0.0726	0.0356	0.0361	0.0811	44.1
2	−0.6	0.5	0.200	−0.0300	−0.0313	0.0272	0.0272	0.0415	78.4
3	0	0.5	0.250	−0.0375	−0.0365	0.0410	0.0419	0.0556	85.8
3	−0.6	0.5	0.100	−0.0150	−0.0145	0.0344	0.0348	0.0377	92.4
4	0	0.5	0.125	−0.0187	−0.0180	0.0456	0.0440	0.0475	95.0
4	−0.6	0.5	0.020	−0.0030	−0.0033	0.0331	0.0329	0.0331	94.4
5	0	0.5	0.062	−0.0094	−0.0084	0.0498	0.0503	0.0510	94.7
5	−0.6	0.5	0.010	−0.0015	−0.0007	0.0390	0.0384	0.0383	95.2
6	0	0.5	0.031	−0.0047	−0.0045	0.0535	0.0528	0.0530	95.3
6	−0.6	0.5	0.002	−0.0003	+0.0013	0.0379	0.0371	0.0371	95.2

Table 16. X4, area means of G (synthetic data; 50 areas; R = 200 populations and samples per n). Empty: mean number of areas without respondents. Cov.: coverage (%) of 95% intervals. Slope: regression of estimated on true area deviations from the national mean (1 means no attenuation). Direct estimators are evaluated on areas with at least two respondents.

		MRP			MRP on EAP scores			Direct, unweighted		Direct, wtd.
n	Empty	RMSE	Cov.	Slope	RMSE	Cov.	Slope	RMSE	Bias	RMSE
1000	0.8	0.168	92.0	0.75	0.172	85.3	0.64	0.342	−0.099	0.359
3000	0.1	0.133	93.7	0.83	0.141	86.0	0.70	0.232	−0.098	0.229
6000	0.0	0.105	94.7	0.90	0.115	84.0	0.76	0.177	−0.094	0.164

Table 17. X5, sign reversals of stereotype-count indices (synthetic data; exact population computations). Percentage of 1,000 random uniform-DIF configurations per cell in which $P_1(q) - P_0(q)$ has the opposite sign to $\Delta\mu_G$. Columns: $|\Delta\mu_G|$ within each index threshold q (of $M = 8$); rows: standard deviation s of the threshold shifts, in standard deviations of each item's latent response.

	$q = 2$				$q = 4$				$q = 6$			
s	0.05	0.1	0.2	0.3	0.05	0.1	0.2	0.3	0.05	0.1	0.2	0.3
0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.1	21.1	5.6	0.0	0.0	20.1	5.1	0.0	0.0	20.1	5.2	0.0	0.0
0.2	33.5	21.1	6.5	0.7	32.4	18.9	5.5	0.3	33.0	21.4	5.5	0.3
0.3	41.3	30.2	14.6	7.8	39.7	28.4	12.7	4.6	39.4	28.7	14.5	4.2
0.4	44.4	36.7	23.5	14.8	41.6	36.8	18.3	9.4	43.0	37.6	20.8	10.2
0.5	47.4	41.1	32.1	24.5	46.2	38.3	25.2	17.5	46.6	39.3	25.8	18.1

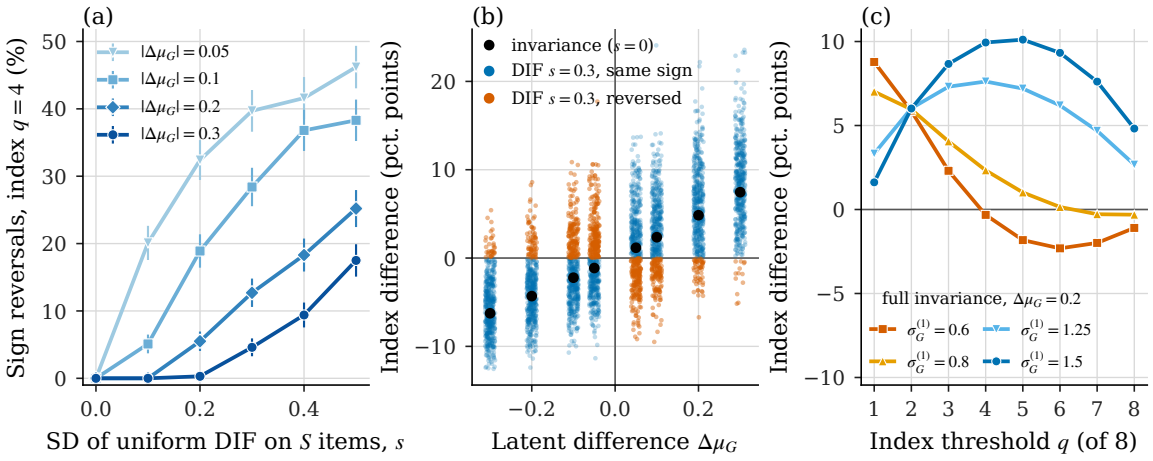


Figure 8. X5, stereotype-count indices (synthetic data; exact population computations; 1,000 random DIF configurations per point). (a) Share of configurations in which the difference in the index "agrees with at least $q = 4$ of 8" has the opposite sign to the latent difference $\Delta\mu_G$, against the standard deviation s of random uniform DIF (±2 MCSE bars). (b) Index difference against $\Delta\mu_G$ under invariance (black) and with $s = 0.3$, coloured by whether the sign is reversed. (c) Full invariance with $\Delta\mu_G = 0.2$ and unequal standard deviations $\sigma_G^{(1)}$ of G: the index difference by threshold q (Proposition 6.3(b)).